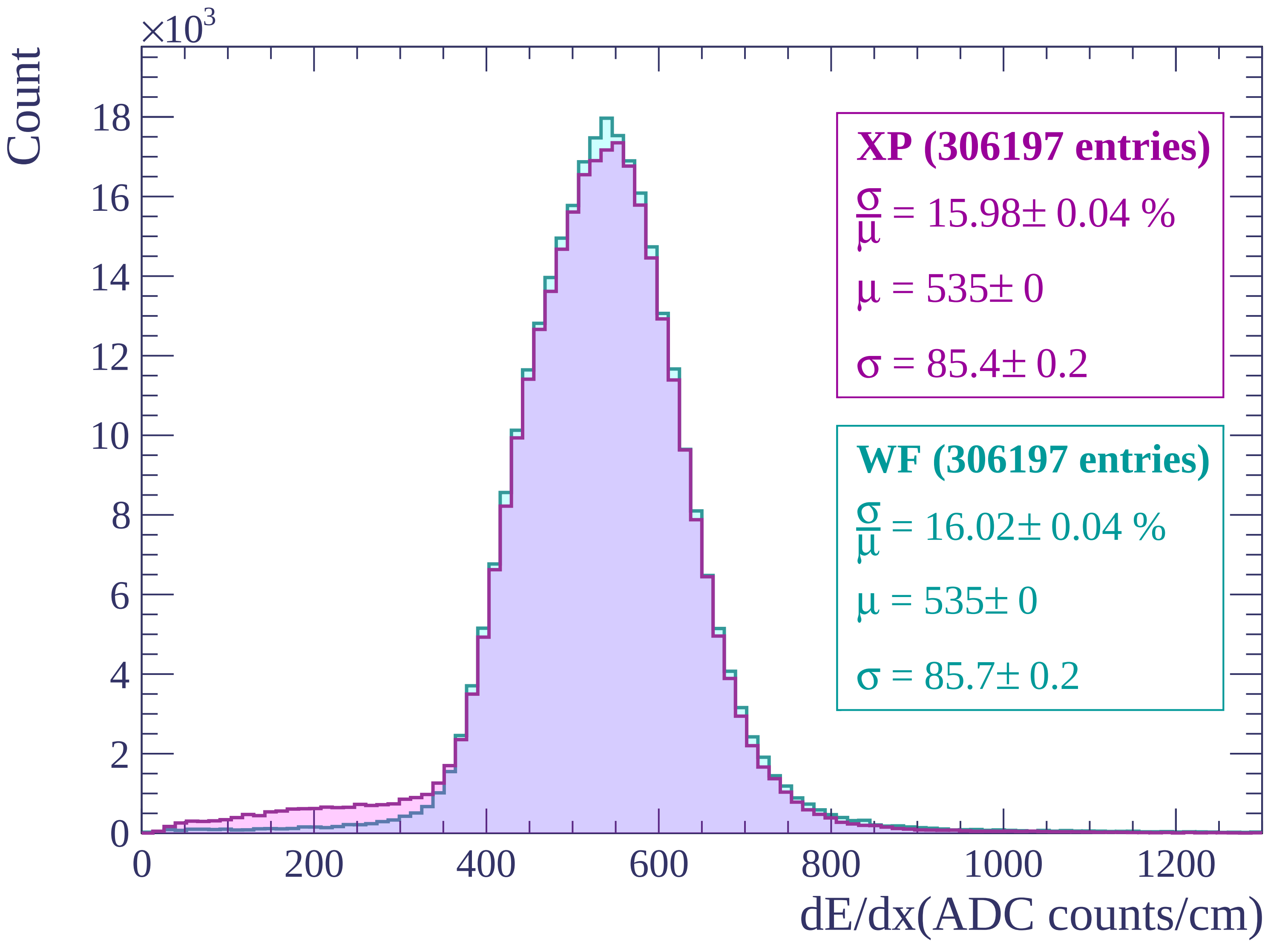
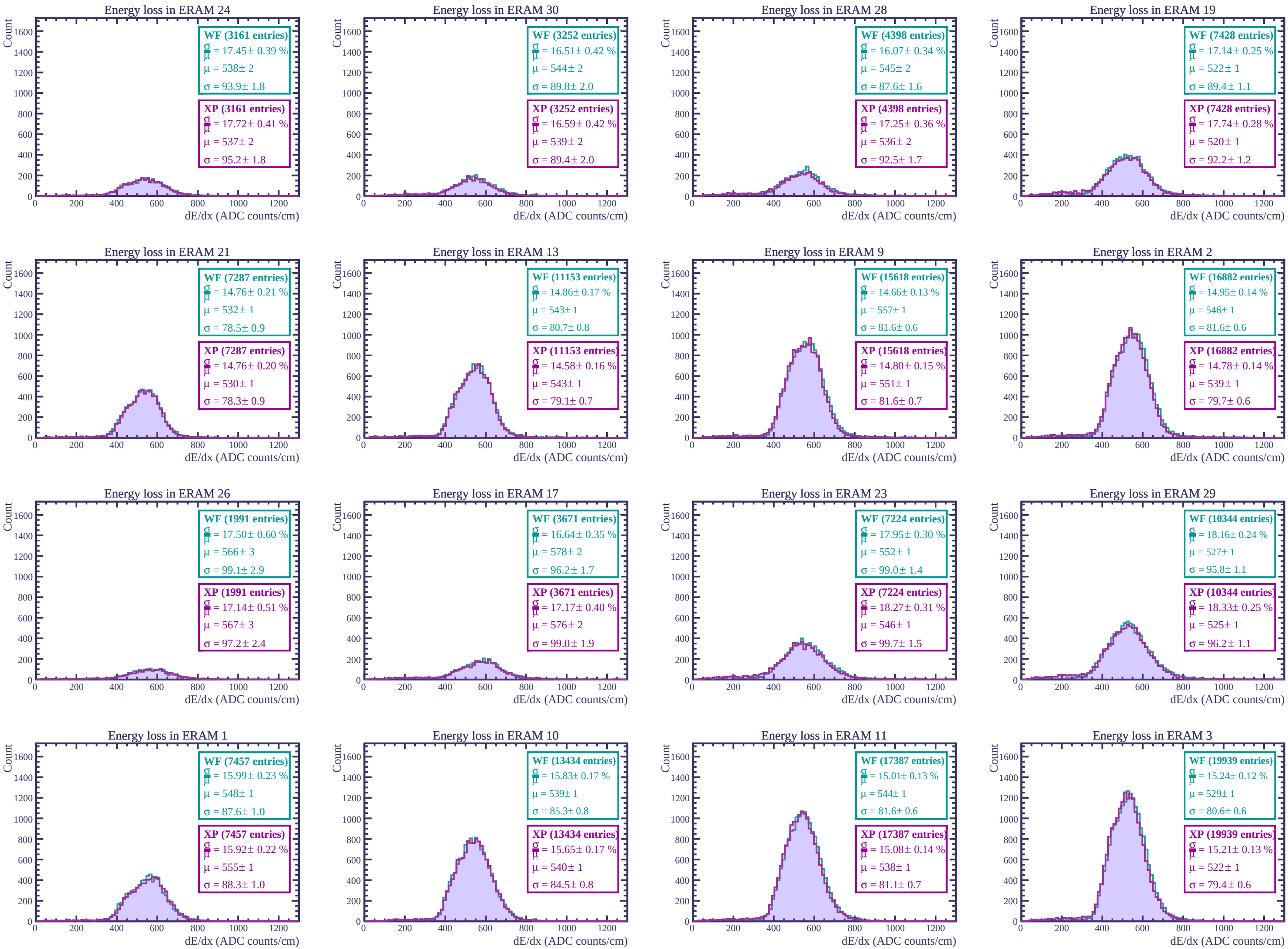
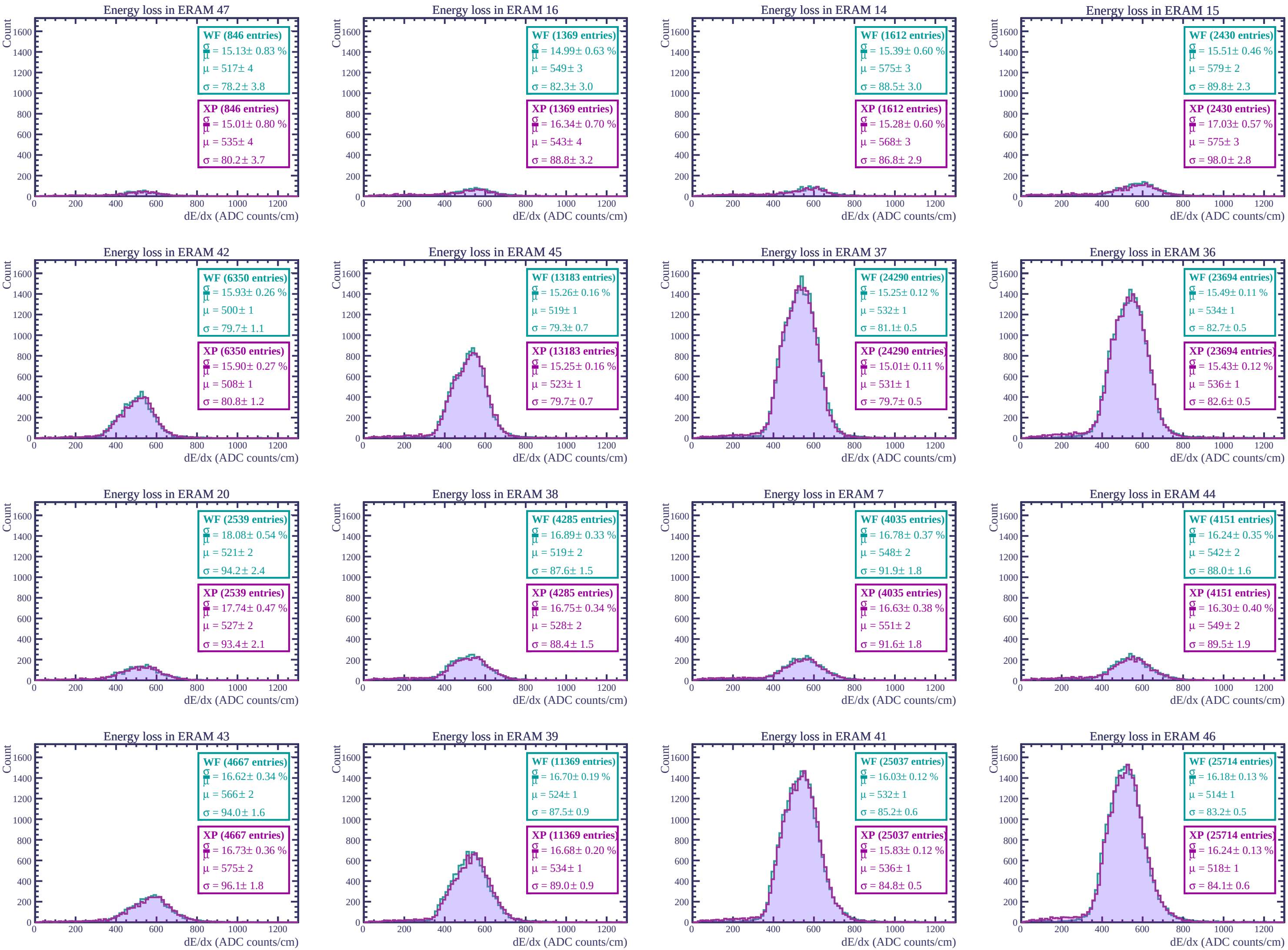
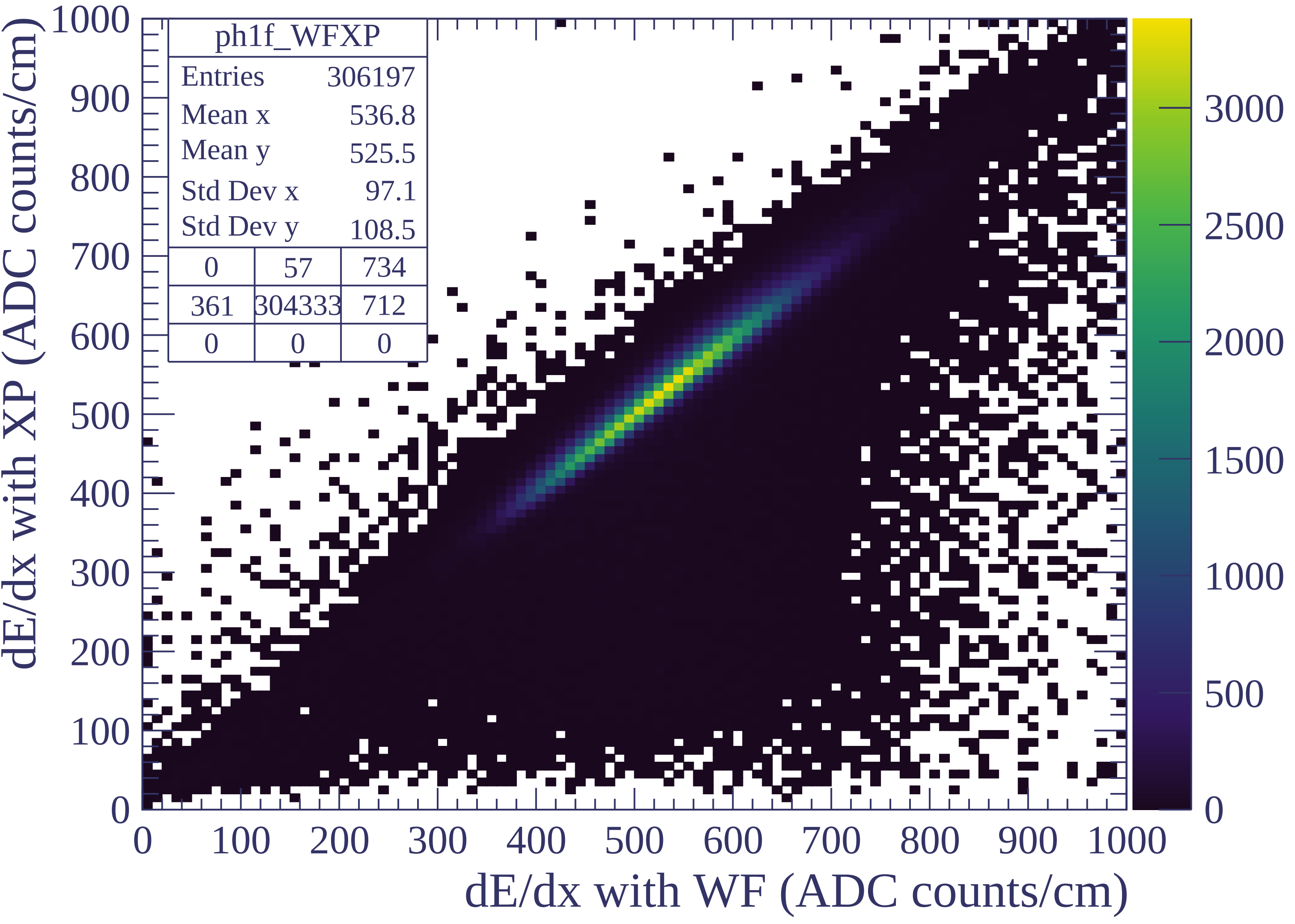


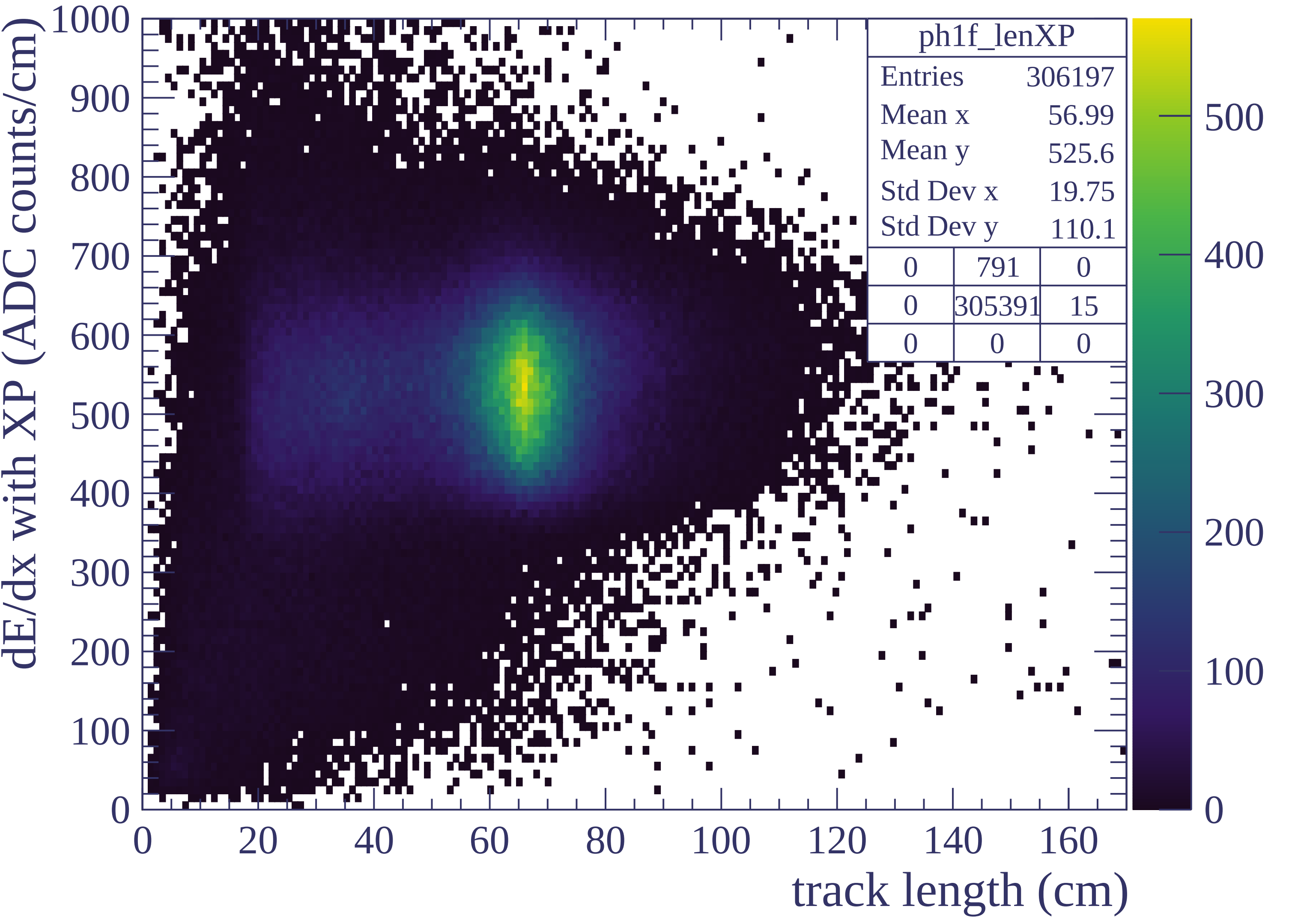
×10^3
Count
XP (306197 entries)
σ/μ = 15.98± 0.04 %
μ = 535± 0
σ = 85.4± 0.2
WF (306197 entries)
σ/μ = 16.02± 0.04 %
μ = 535± 0
σ = 85.7± 0.2
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)

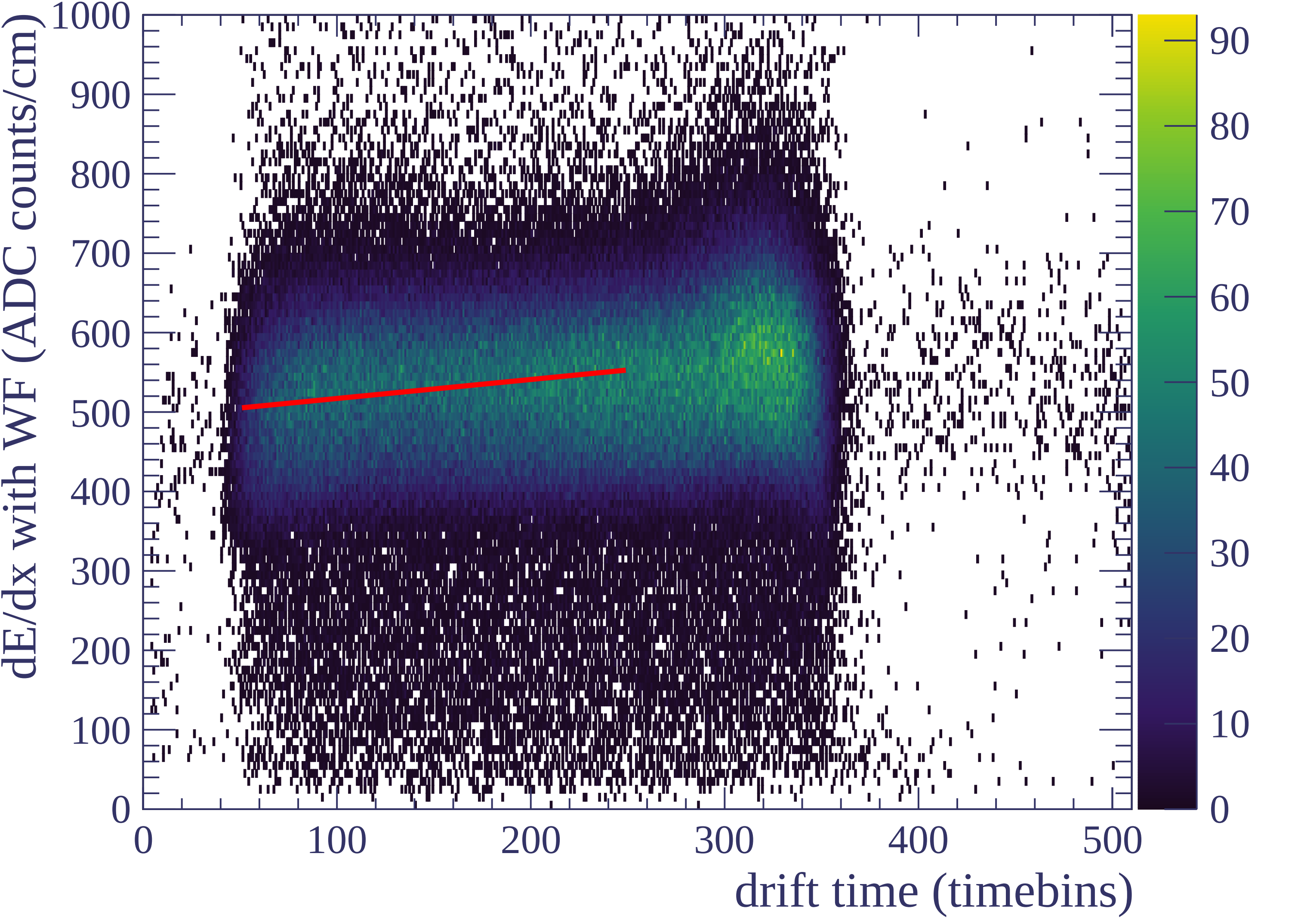

Energy loss in ERAM 24
WF (3161 entries)
σ/μ = 17.45± 0.39 %
μ = 538± 2
σ = 93.9± 1.8
XP (3161 entries)
σ/μ = 17.72± 0.41 %
μ = 537± 2
σ = 95.2± 1.8
Energy loss in ERAM 30
WF (3252 entries)
σ/μ = 16.51± 0.42 %
μ = 544± 2
σ = 89.8± 2.0
XP (3252 entries)
σ/μ = 16.59± 0.42 %
μ = 539± 2
σ = 89.4± 2.0
Energy loss in ERAM 28
WF (4398 entries)
σ/μ = 16.07± 0.34 %
μ = 545± 2
σ = 87.6± 1.6
XP (4398 entries)
σ/μ = 17.25± 0.36 %
μ = 536± 2
σ = 92.5± 1.7
Energy loss in ERAM 19
WF (7428 entries)
σ/μ = 17.14± 0.25 %
μ = 522± 1
σ = 89.4± 1.1
XP (7428 entries)
σ/μ = 17.74± 0.28 %
μ = 520± 1
σ = 92.2± 1.2
Energy loss in ERAM 21
WF (7287 entries)
σ/μ = 14.76± 0.21 %
μ = 532± 1
σ = 78.5± 0.9
XP (7287 entries)
σ/μ = 14.76± 0.20 %
μ = 530± 1
σ = 78.3± 0.9
Energy loss in ERAM 13
WF (11153 entries)
σ/μ = 14.86± 0.17 %
μ = 543± 1
σ = 80.7± 0.8
XP (11153 entries)
σ/μ = 14.58± 0.16 %
μ = 543± 1
σ = 79.1± 0.7
Energy loss in ERAM 9
WF (15618 entries)
σ/μ = 14.66± 0.13 %
μ = 557± 1
σ = 81.6± 0.6
XP (15618 entries)
σ/μ = 14.80± 0.15 %
μ = 551± 1
σ = 81.6± 0.7
Energy loss in ERAM 2
WF (16882 entries)
σ/μ = 14.95± 0.14 %
μ = 546± 1
σ = 81.6± 0.6
XP (16882 entries)
σ/μ = 14.78± 0.14 %
μ = 539± 1
σ = 79.7± 0.6
Energy loss in ERAM 26
WF (1991 entries)
σ/μ = 17.50± 0.60 %
μ = 566± 3
σ = 99.1± 2.9
XP (1991 entries)
σ/μ = 17.14± 0.51 %
μ = 567± 3
σ = 97.2± 2.4
Energy loss in ERAM 17
WF (3671 entries)
σ/μ = 16.64± 0.35 %
μ = 578± 2
σ = 96.2± 1.7
XP (3671 entries)
σ/μ = 17.17± 0.40 %
μ = 576± 2
σ = 99.0± 1.9
Energy loss in ERAM 23
WF (7224 entries)
σ/μ = 17.95± 0.30 %
μ = 552± 1
σ = 99.0± 1.4
XP (7224 entries)
σ/μ = 18.27± 0.31 %
μ = 546± 1
σ = 99.7± 1.5
Energy loss in ERAM 29
WF (10344 entries)
σ/μ = 18.16± 0.24 %
μ = 527± 1
σ = 95.8± 1.1
XP (10344 entries)
σ/μ = 18.33± 0.25 %
μ = 525± 1
σ = 96.2± 1.1
Energy loss in ERAM 1
WF (7457 entries)
σ/μ = 15.99± 0.23 %
μ = 548± 1
σ = 87.6± 1.0
XP (7457 entries)
σ/μ = 15.92± 0.22 %
μ = 555± 1
σ = 88.3± 1.0
Energy loss in ERAM 10
WF (13434 entries)
σ/μ = 15.83± 0.17 %
μ = 539± 1
σ = 85.3± 0.8
XP (13434 entries)
σ/μ = 15.65± 0.17 %
μ = 540± 1
σ = 84.5± 0.8
Energy loss in ERAM 11
WF (17387 entries)
σ/μ = 15.01± 0.13 %
μ = 544± 1
σ = 81.6± 0.6
XP (17387 entries)
σ/μ = 15.08± 0.14 %
μ = 538± 1
σ = 81.1± 0.7
Energy loss in ERAM 3
WF (19939 entries)
σ/μ = 15.24± 0.12 %
μ = 529± 1
σ = 80.6± 0.6
XP (19939 entries)
σ/μ = 15.21± 0.13 %
μ = 522± 1
σ = 79.4± 0.6
Count
dE/dx (ADC counts/cm)

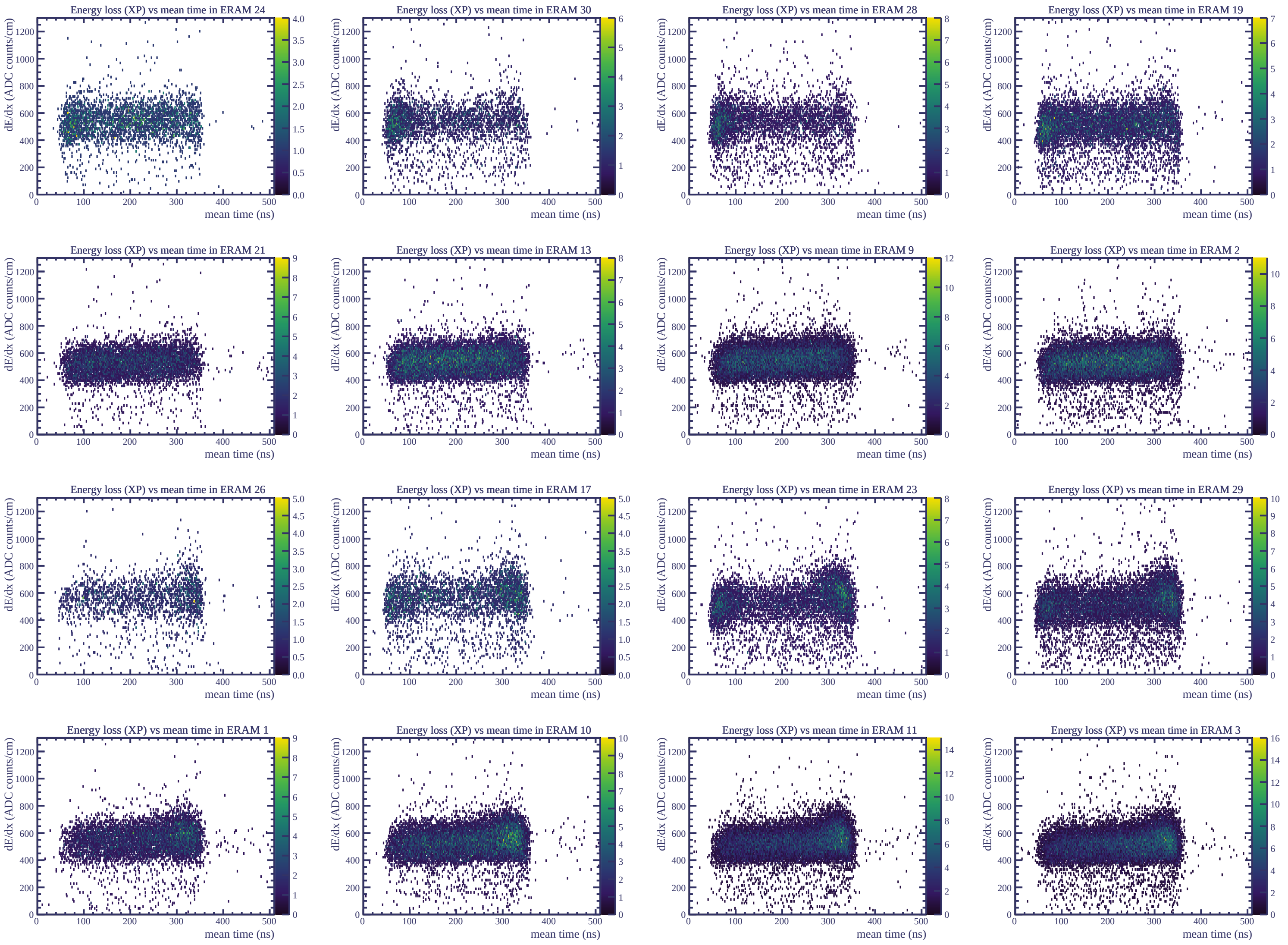


Energy loss in ERAM 47
WF (846 entries)
σ/μ = 15.13± 0.83 %
μ = 517± 4
σ = 78.2± 3.8
XP (846 entries)
σ/μ = 15.01± 0.80 %
μ = 535± 4
σ = 80.2± 3.7
Energy loss in ERAM 16
WF (1369 entries)
σ/μ = 14.99± 0.63 %
μ = 549± 3
σ = 82.3± 3.0
XP (1369 entries)
σ/μ = 16.34± 0.70 %
μ = 543± 4
σ = 88.8± 3.2
Energy loss in ERAM 14
WF (1612 entries)
σ/μ = 15.39± 0.60 %
μ = 575± 3
σ = 88.5± 3.0
XP (1612 entries)
σ/μ = 15.28± 0.60 %
μ = 568± 3
σ = 86.8± 2.9
Energy loss in ERAM 15
WF (2430 entries)
σ/μ = 15.51± 0.46 %
μ = 579± 2
σ = 89.8± 2.3
XP (2430 entries)
σ/μ = 17.03± 0.57 %
μ = 575± 3
σ = 98.0± 2.8
Energy loss in ERAM 42
WF (6350 entries)
σ/μ = 15.93± 0.26 %
μ = 500± 1
σ = 79.7± 1.1
XP (6350 entries)
σ/μ = 15.90± 0.27 %
μ = 508± 1
σ = 80.8± 1.2
Energy loss in ERAM 45
WF (13183 entries)
σ/μ = 15.26± 0.16 %
μ = 519± 1
σ = 79.3± 0.7
XP (13183 entries)
σ/μ = 15.25± 0.16 %
μ = 523± 1
σ = 79.7± 0.7
Energy loss in ERAM 37
WF (24290 entries)
σ/μ = 15.25± 0.12 %
μ = 532± 1
σ = 81.1± 0.5
XP (24290 entries)
σ/μ = 15.01± 0.11 %
μ = 531± 1
σ = 79.7± 0.5
Energy loss in ERAM 36
WF (23694 entries)
σ/μ = 15.49± 0.11 %
μ = 534± 1
σ = 82.7± 0.5
XP (23694 entries)
σ/μ = 15.43± 0.12 %
μ = 536± 1
σ = 82.6± 0.5
Energy loss in ERAM 20
WF (2539 entries)
σ/μ = 18.08± 0.54 %
μ = 521± 2
σ = 94.2± 2.4
XP (2539 entries)
σ/μ = 17.74± 0.47 %
μ = 527± 2
σ = 93.4± 2.1
Energy loss in ERAM 38
WF (4285 entries)
σ/μ = 16.89± 0.33 %
μ = 519± 2
σ = 87.6± 1.5
XP (4285 entries)
σ/μ = 16.75± 0.34 %
μ = 528± 2
σ = 88.4± 1.5
Energy loss in ERAM 7
WF (4035 entries)
σ/μ = 16.78± 0.37 %
μ = 548± 2
σ = 91.9± 1.8
XP (4035 entries)
σ/μ = 16.63± 0.38 %
μ = 551± 2
σ = 91.6± 1.8
Energy loss in ERAM 44
WF (4151 entries)
σ/μ = 16.24± 0.35 %
μ = 542± 2
σ = 88.0± 1.6
XP (4151 entries)
σ/μ = 16.30± 0.40 %
μ = 549± 2
σ = 89.5± 1.9
Energy loss in ERAM 43
WF (4667 entries)
σ/μ = 16.62± 0.34 %
μ = 566± 2
σ = 94.0± 1.6
XP (4667 entries)
σ/μ = 16.73± 0.36 %
μ = 575± 2
σ = 96.1± 1.8
Energy loss in ERAM 39
WF (11369 entries)
σ/μ = 16.70± 0.19 %
μ = 524± 1
σ = 87.5± 0.9
XP (11369 entries)
σ/μ = 16.68± 0.20 %
μ = 534± 1
σ = 89.0± 0.9
Energy loss in ERAM 41
WF (25037 entries)
σ/μ = 16.03± 0.12 %
μ = 532± 1
σ = 85.2± 0.6
XP (25037 entries)
σ/μ = 15.83± 0.12 %
μ = 536± 1
σ = 84.8± 0.5
Energy loss in ERAM 46
WF (25714 entries)
σ/μ = 16.18± 0.13 %
μ = 514± 1
σ = 83.2± 0.5
XP (25714 entries)
σ/μ = 16.24± 0.13 %
μ = 518± 1
σ = 84.1± 0.6
Count
dE/dx (ADC counts/cm)

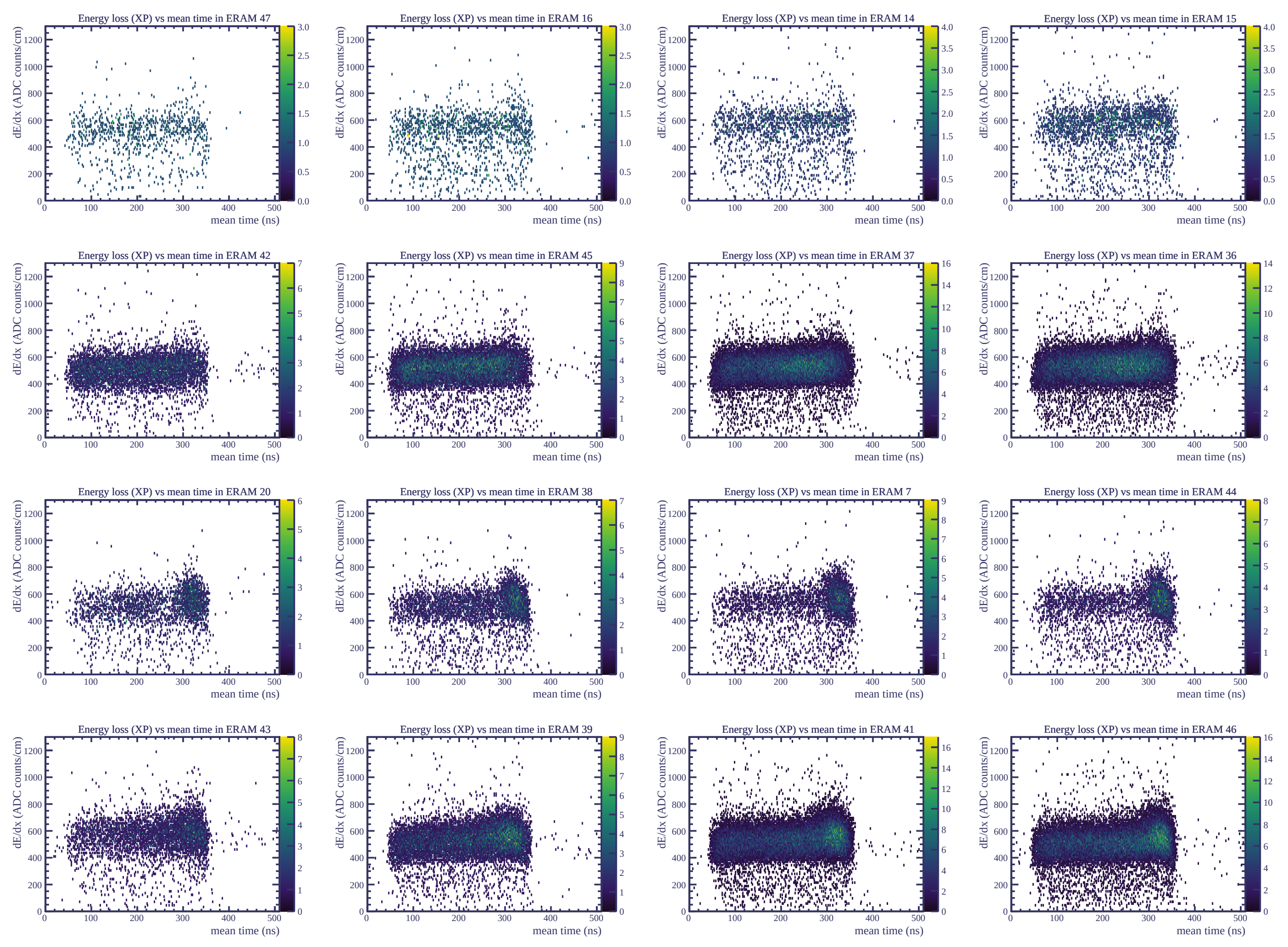

ph1f_WFXP
Entries 306197
Mean x 536.8
Mean y 525.5
Std Dev x 97.1
Std Dev y 108.5
0 57 734
361 304333 712
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500
3000

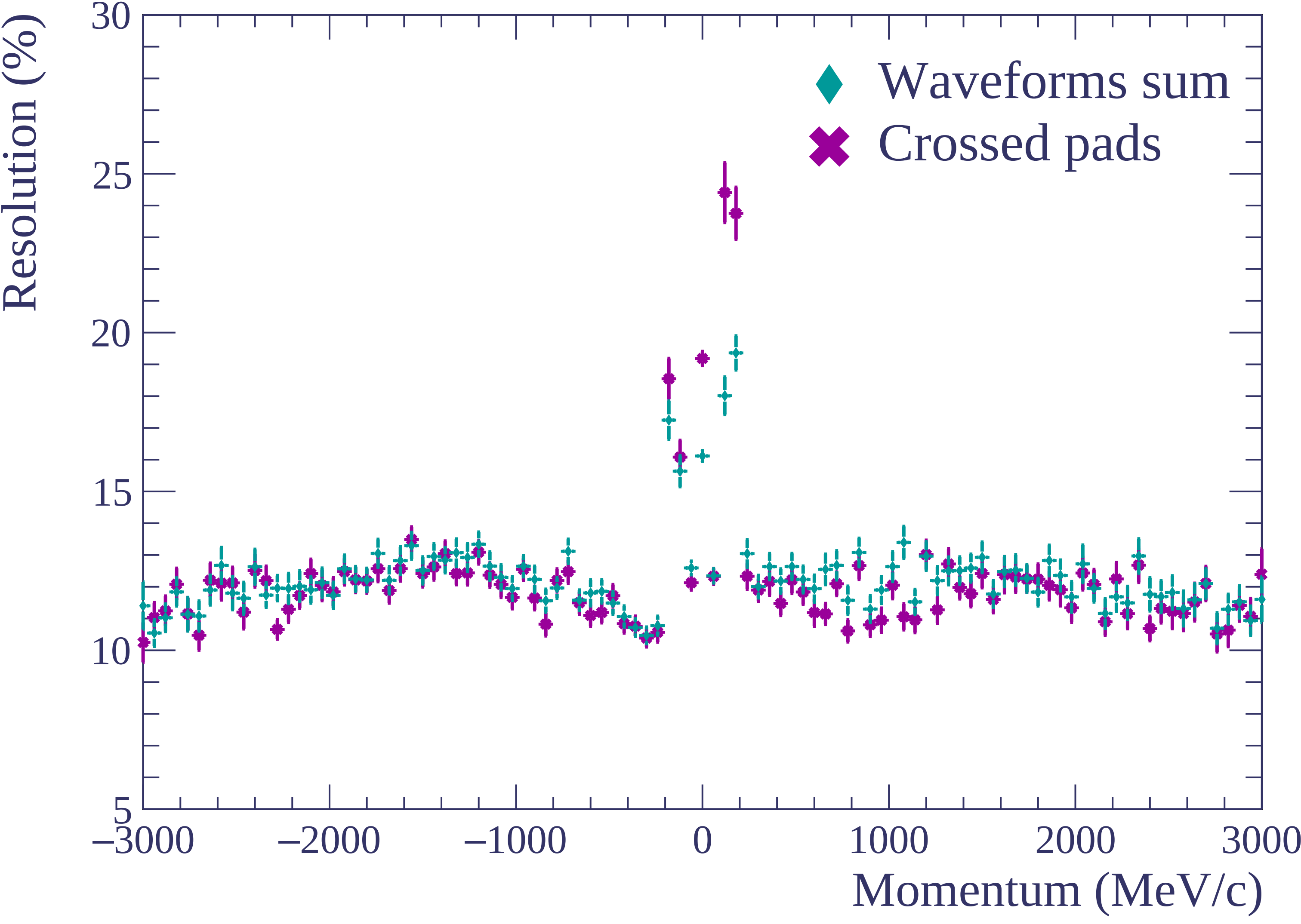

ph1f_lenXP
Entries 306197
Mean x 56.99
Mean y 525.6
Std Dev x 19.75
Std Dev y 110.1
0 791 0
0 305391 15
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
100
200
300
400
500
drift time (timebins)
90
80
70
60
50
40
30
20
10
0

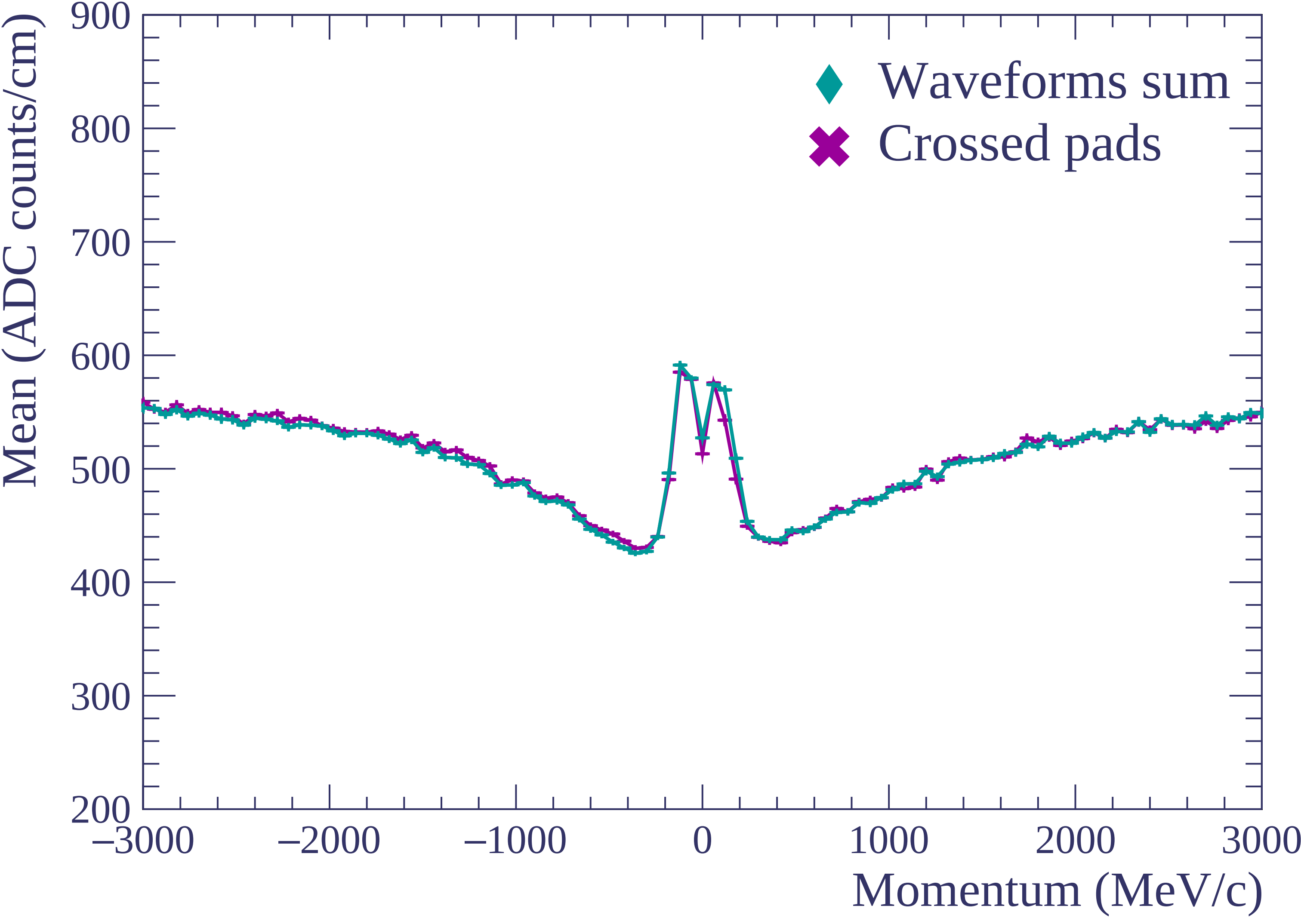


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

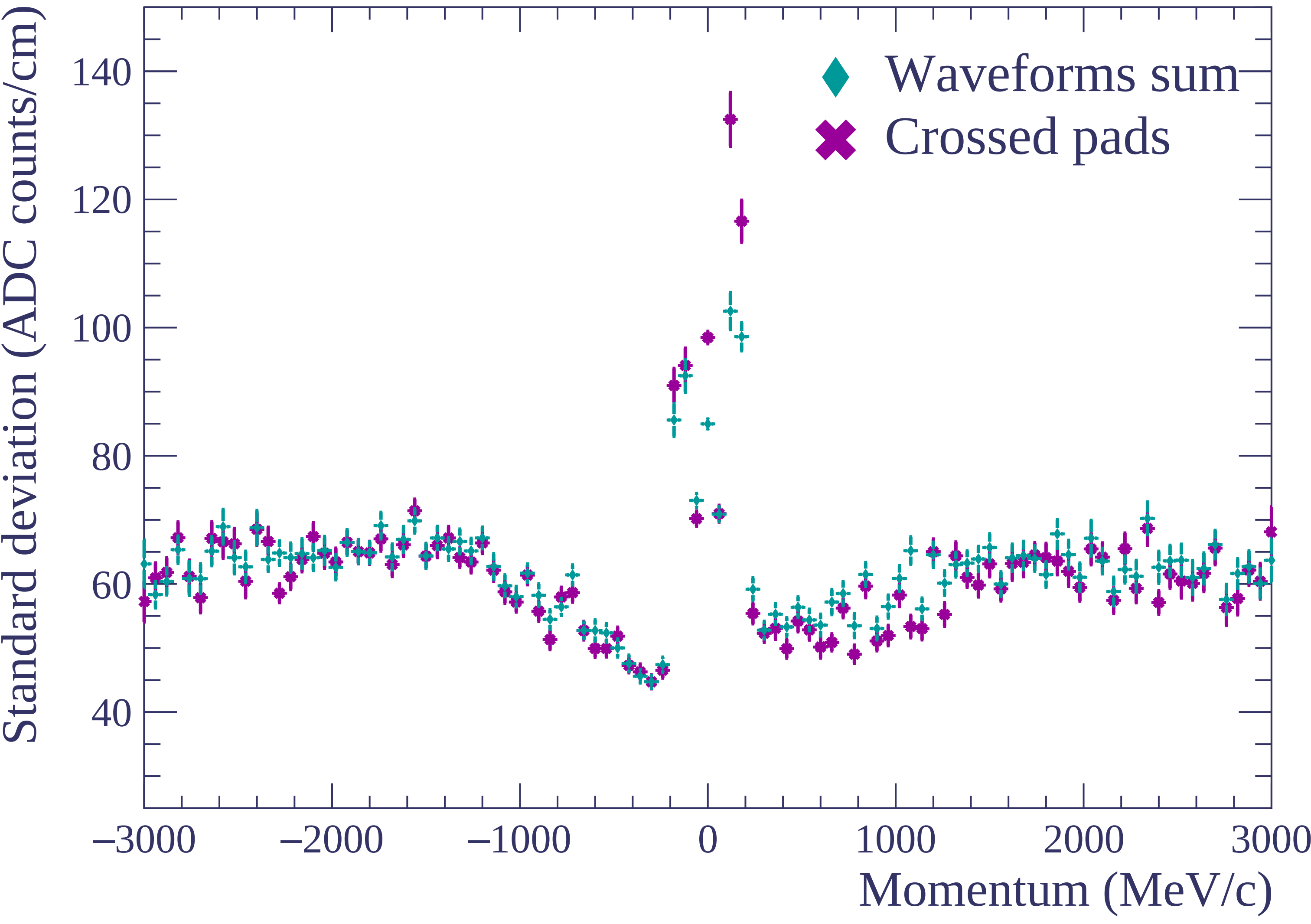

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)



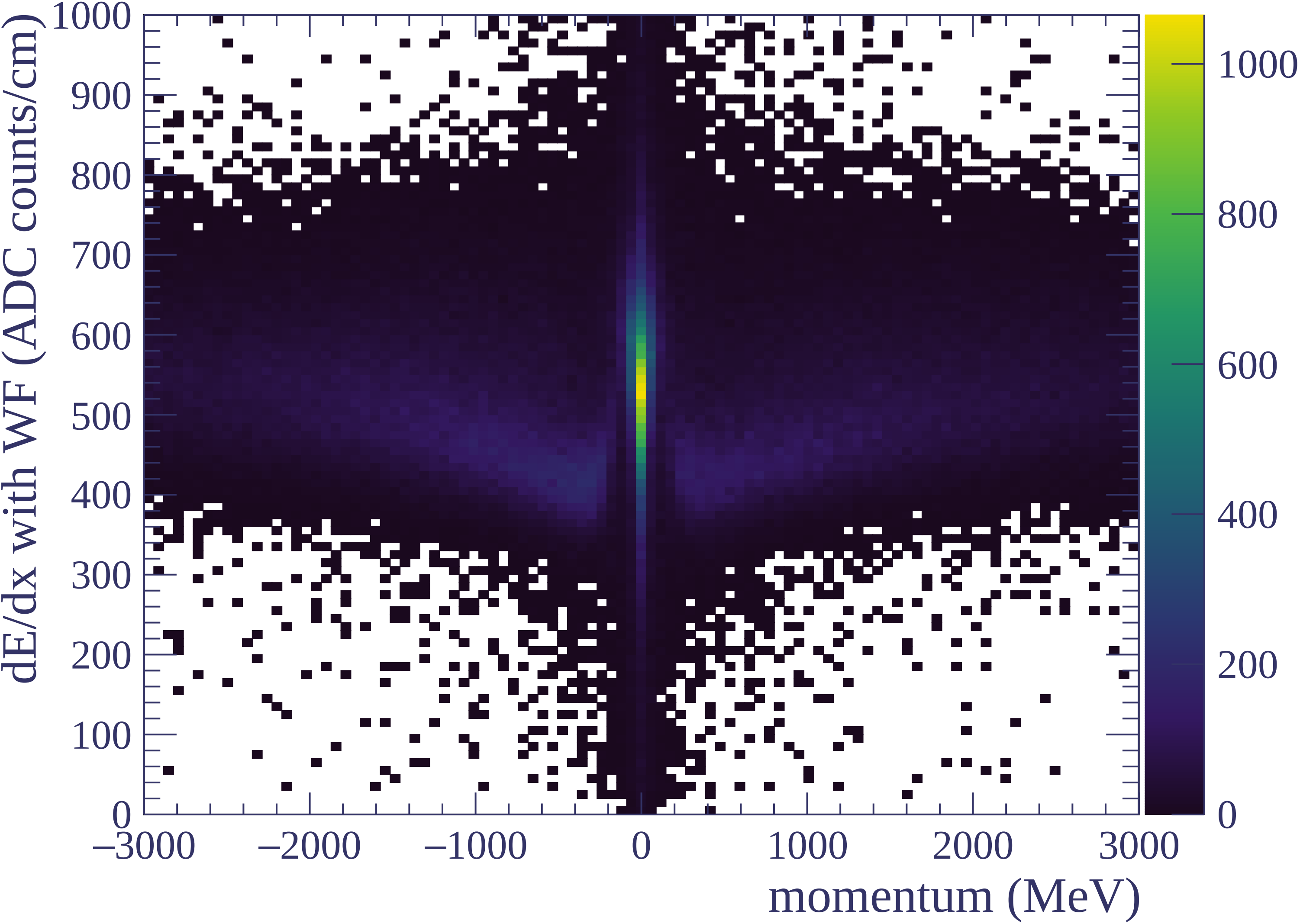
Waveforms sum
Crossed pads
Resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−3000
−2000
−1000
0
1000
2000
3000

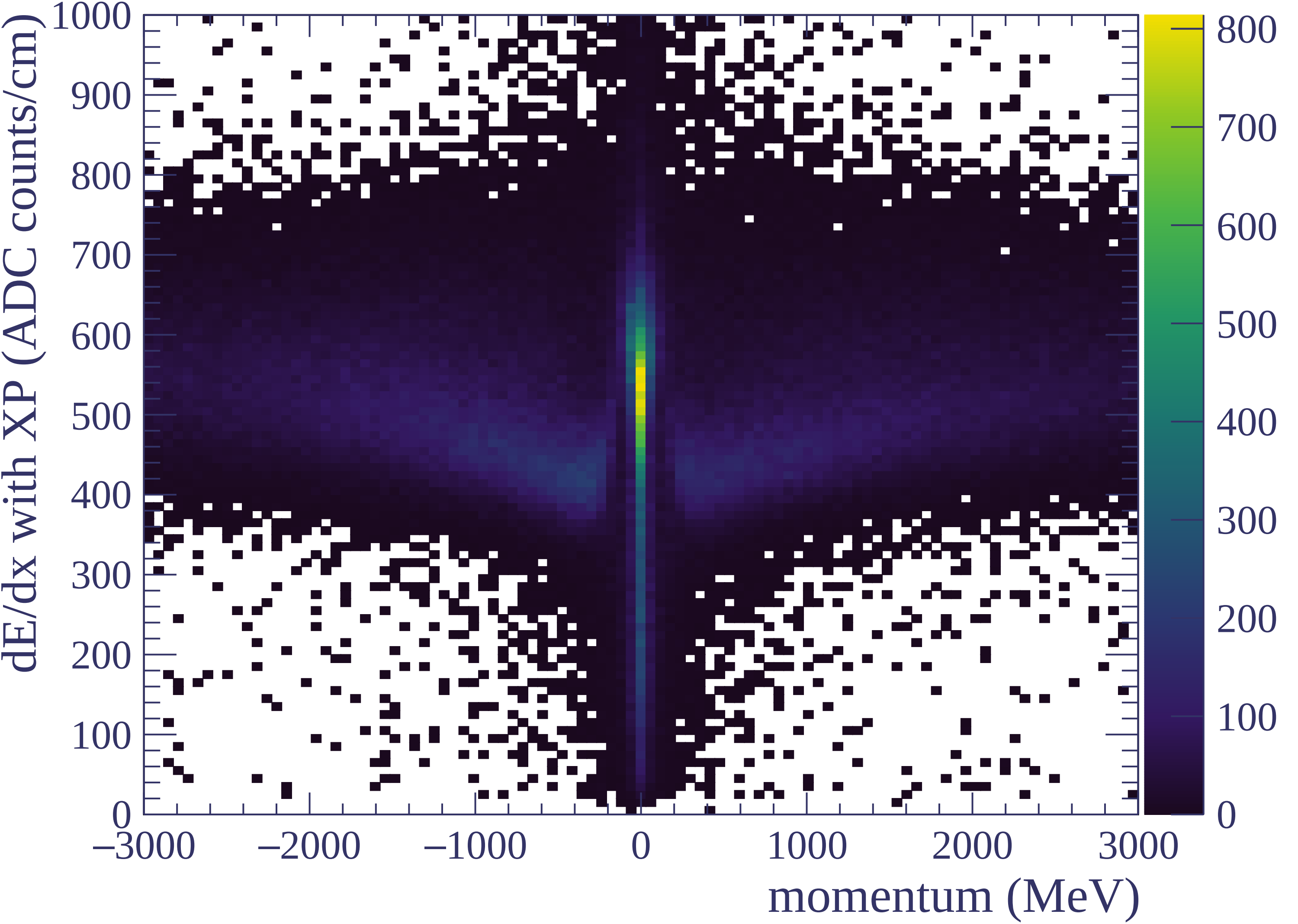


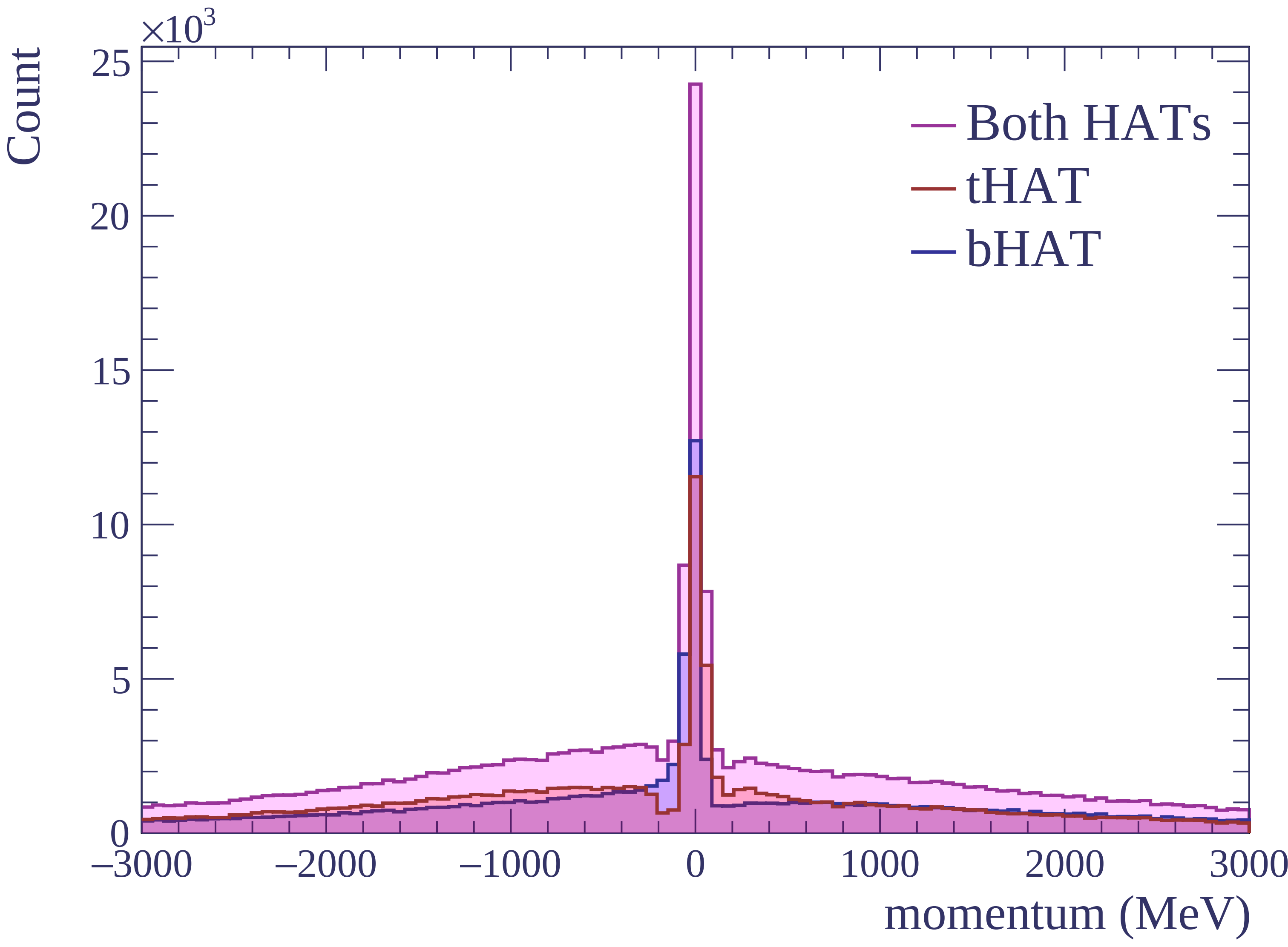
Mean (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
900
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000



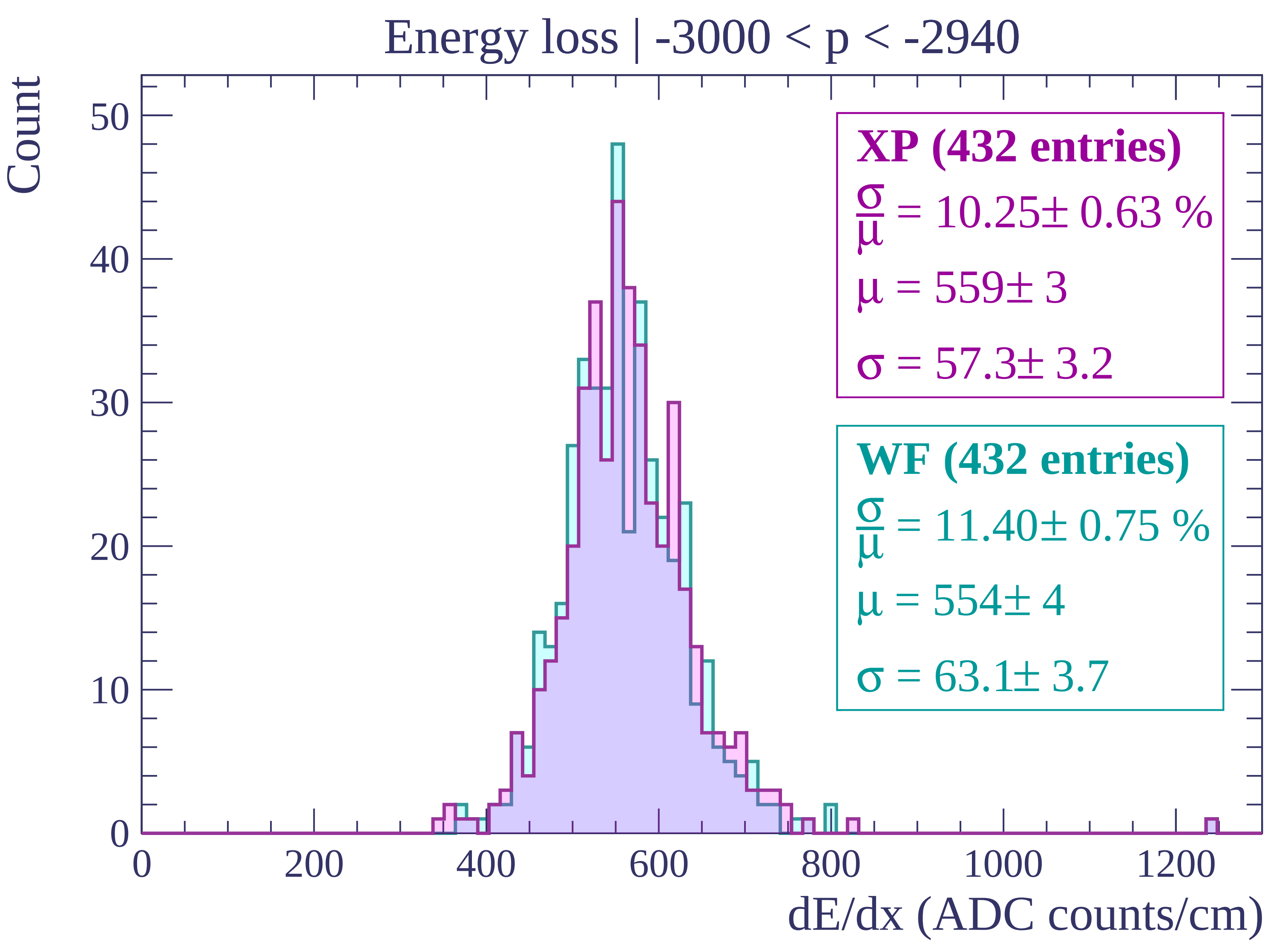
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
140
120
100
80
60
40
−3000
−2000
−1000
0
1000
2000
3000

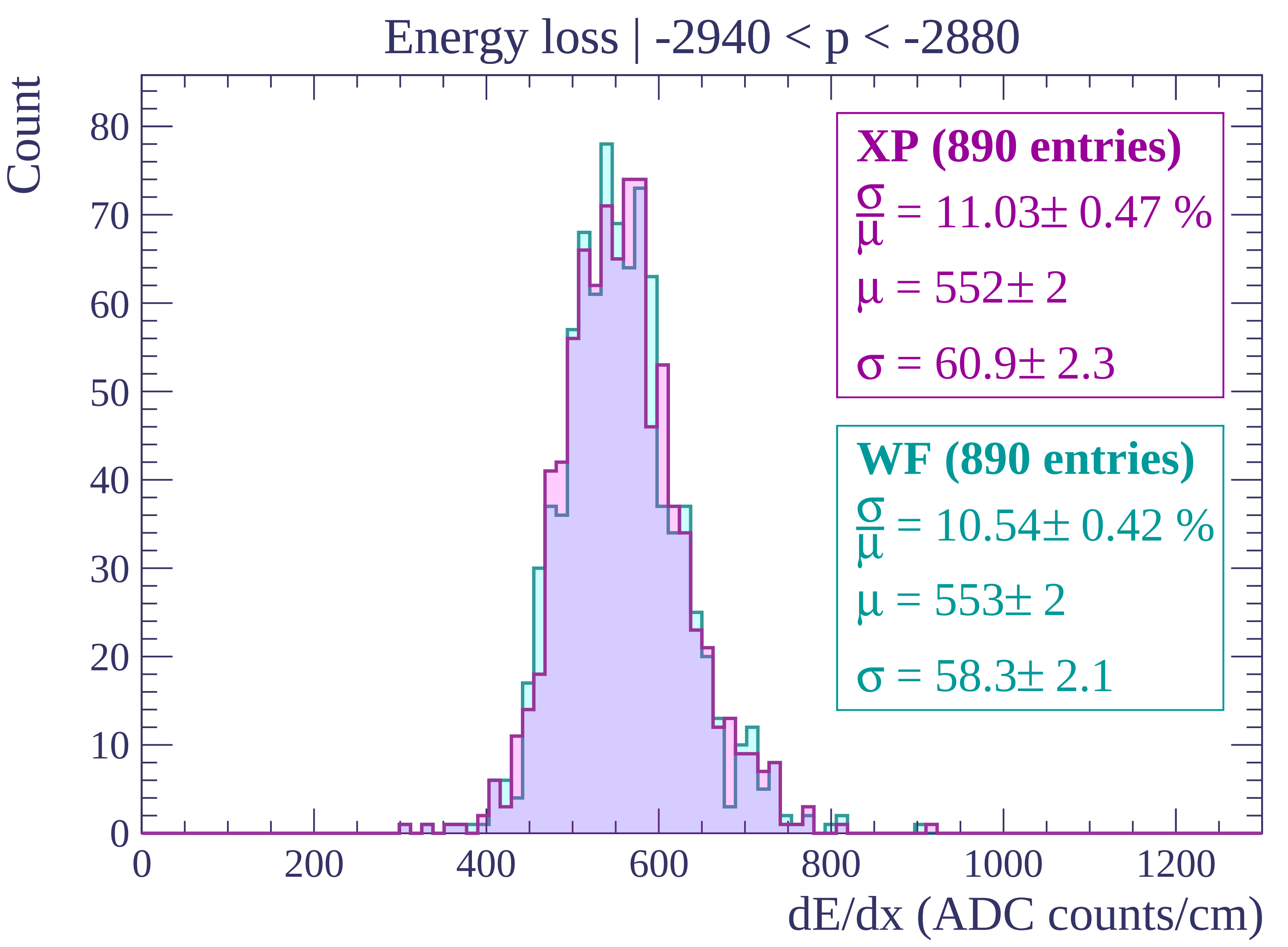

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
1000
800
600
400
200
0


dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
800
700
600
500
400
300
200
100
0

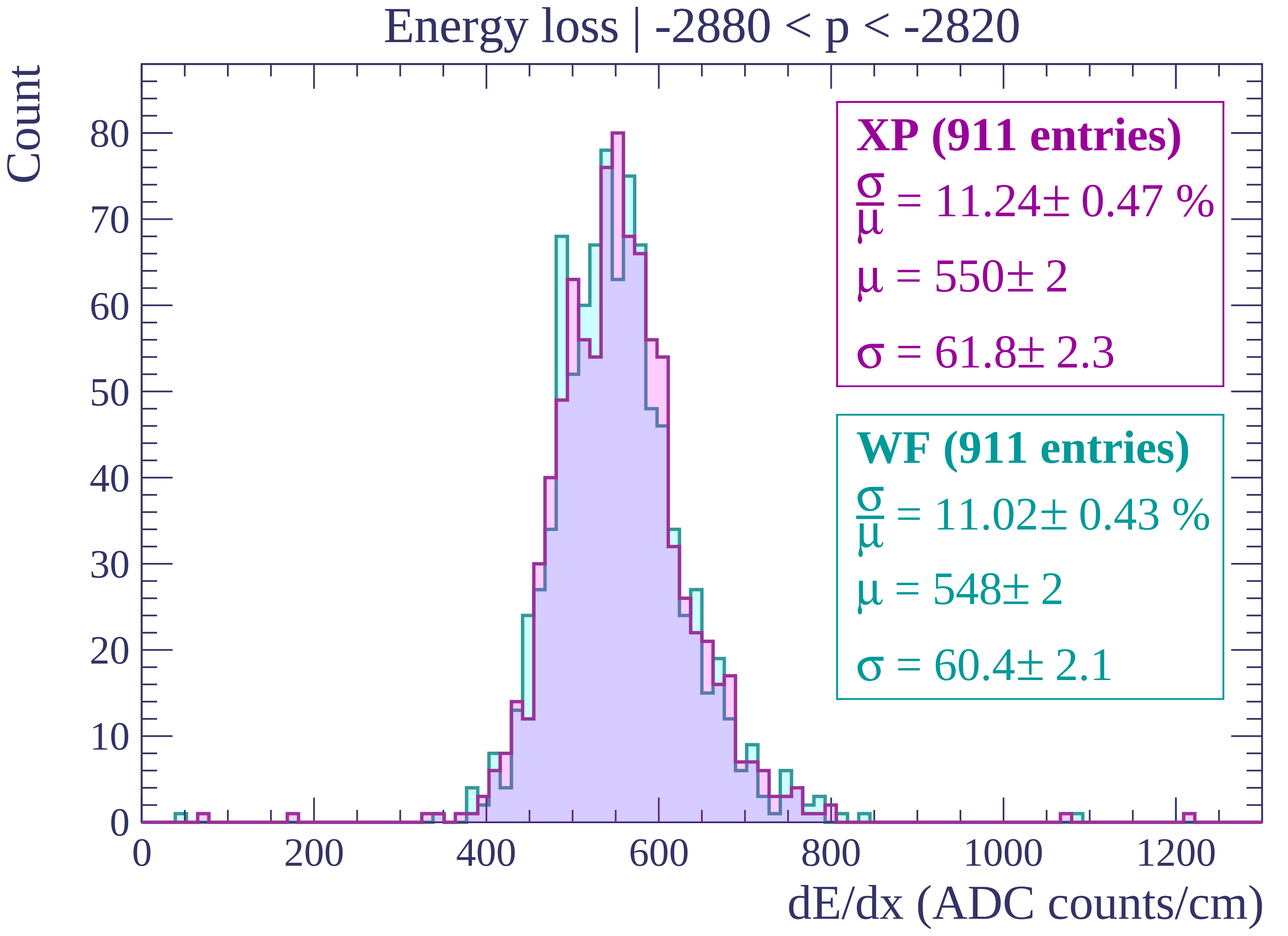


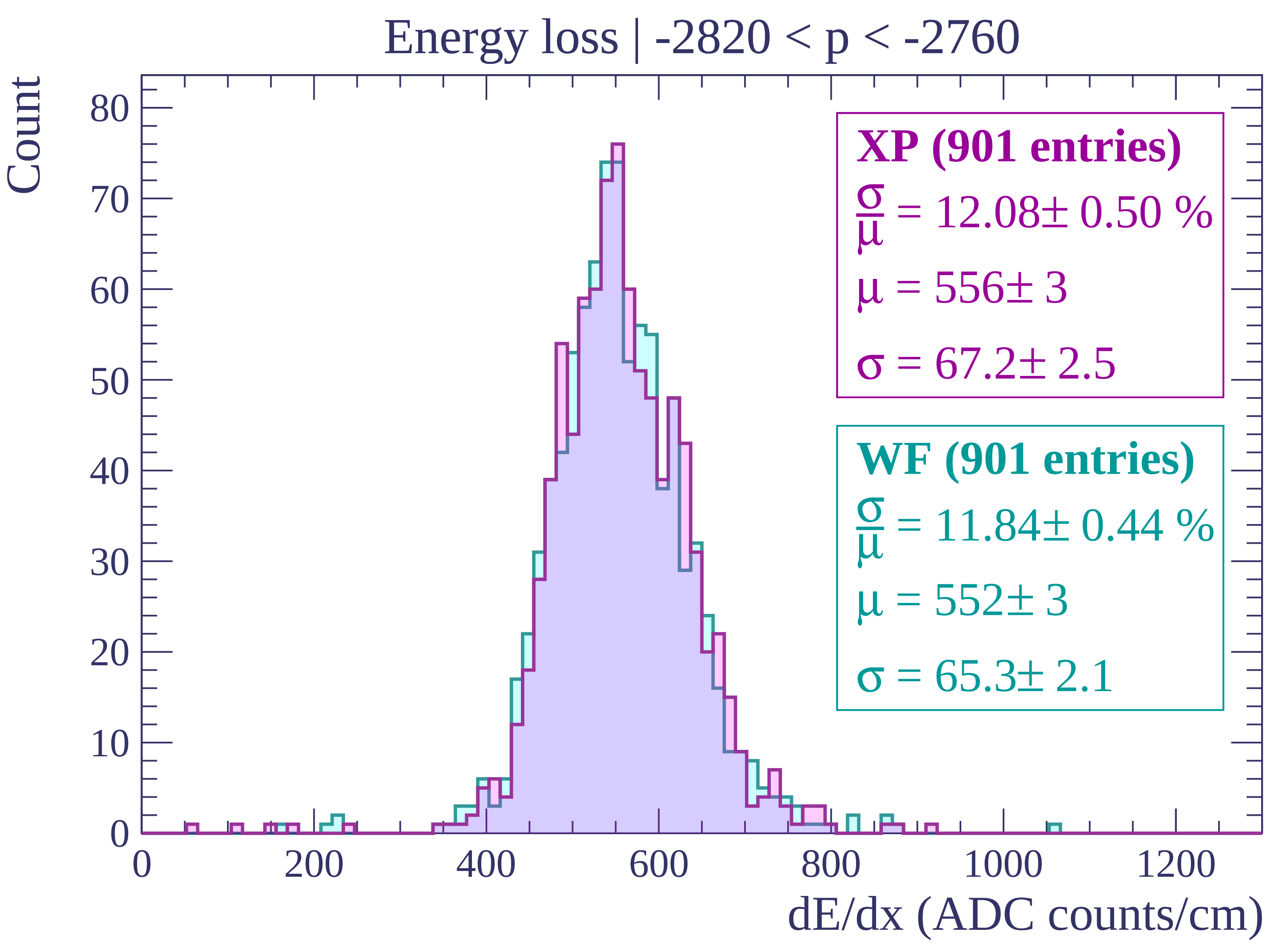
×10^3
Count
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT


Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (432 entries)
σ/μ = 10.25± 0.63 %
μ = 559± 3
σ = 57.3± 3.2
WF (432 entries)
σ/μ = 11.40± 0.75 %
μ = 554± 4
σ = 63.1± 3.7

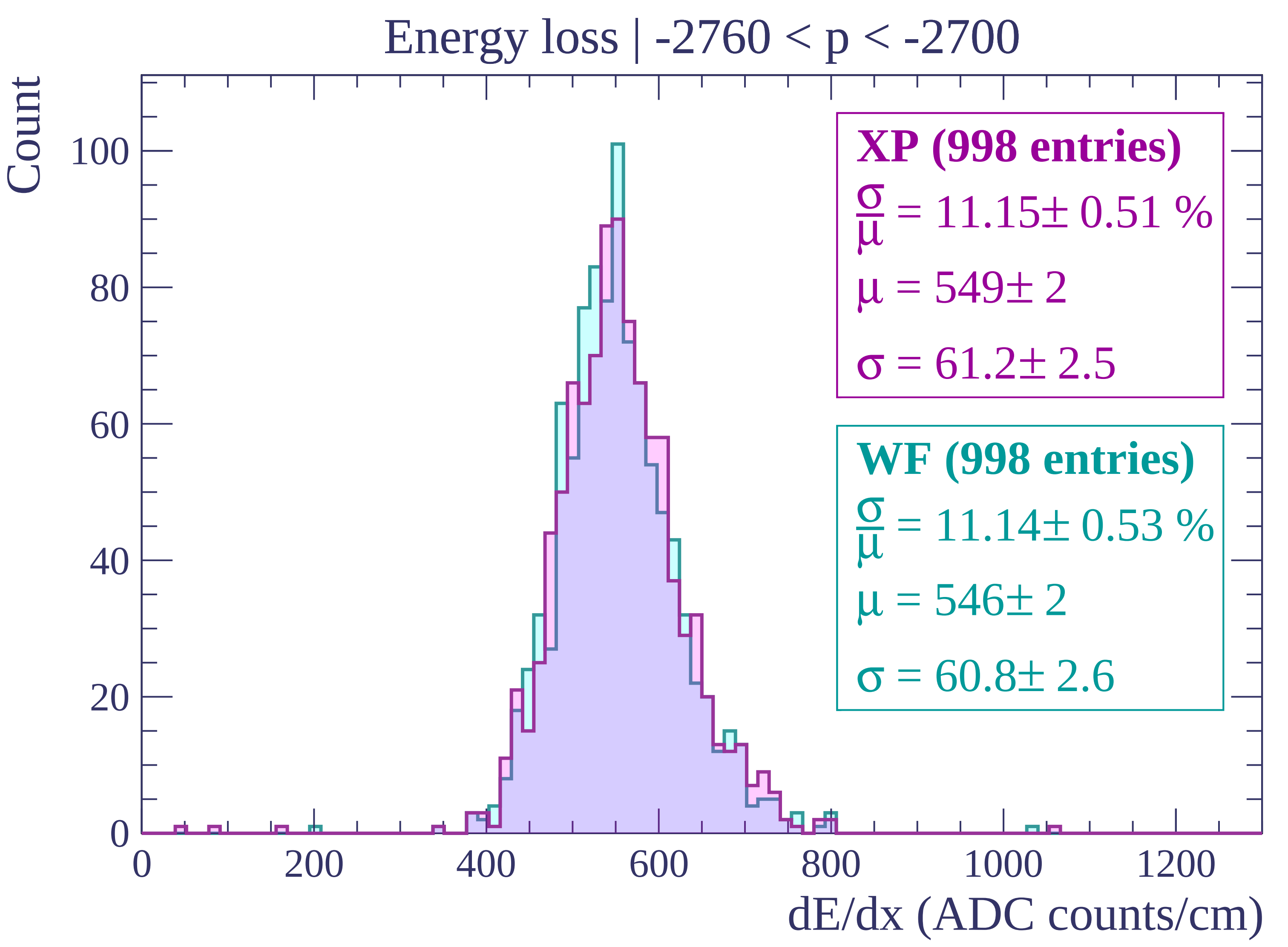


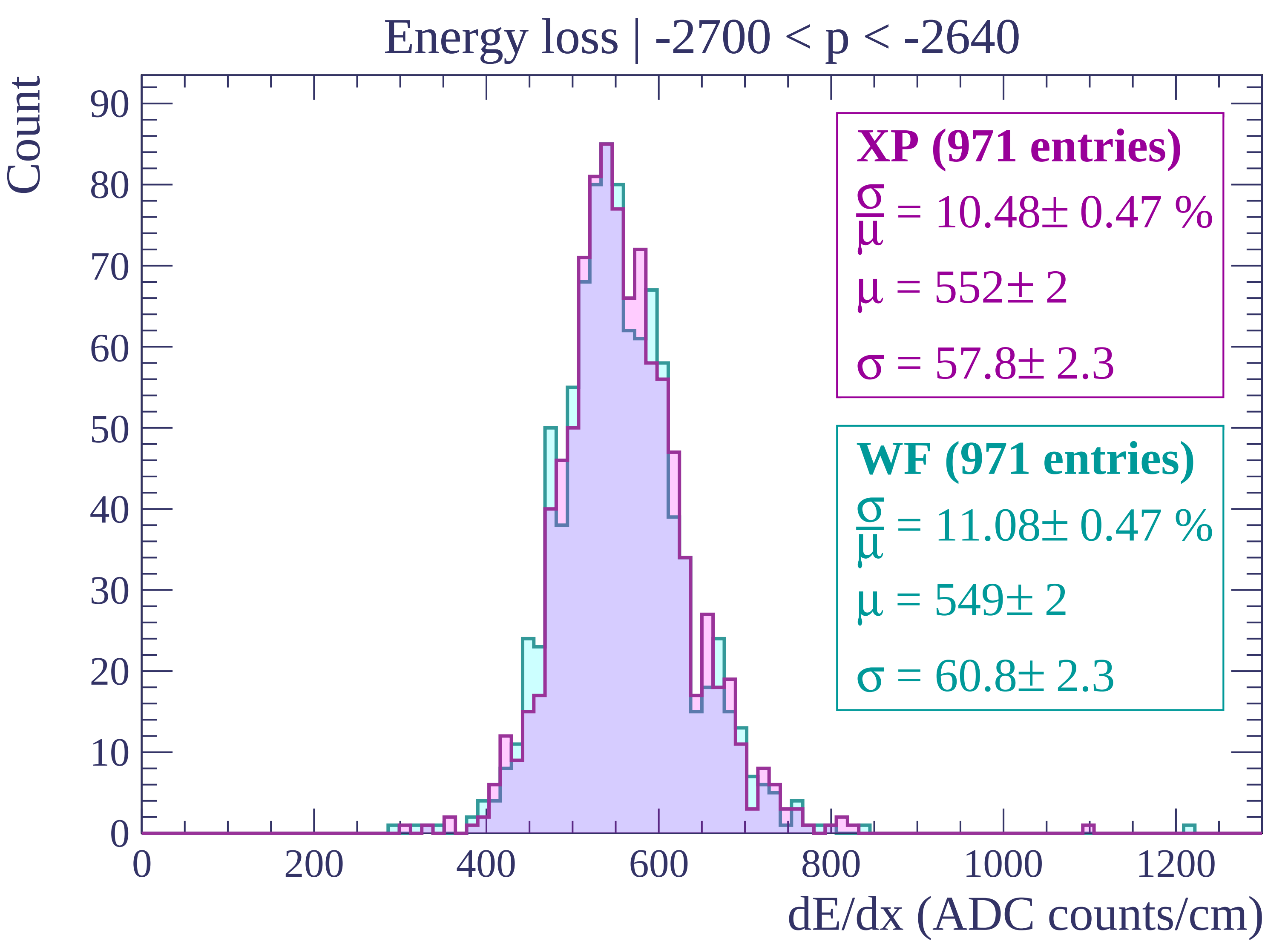
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
XP (890 entries)
σ/μ = 11.03± 0.47 %
μ = 552± 2
σ = 60.9± 2.3
WF (890 entries)
σ/μ = 10.54± 0.42 %
μ = 553± 2
σ = 58.3± 2.1

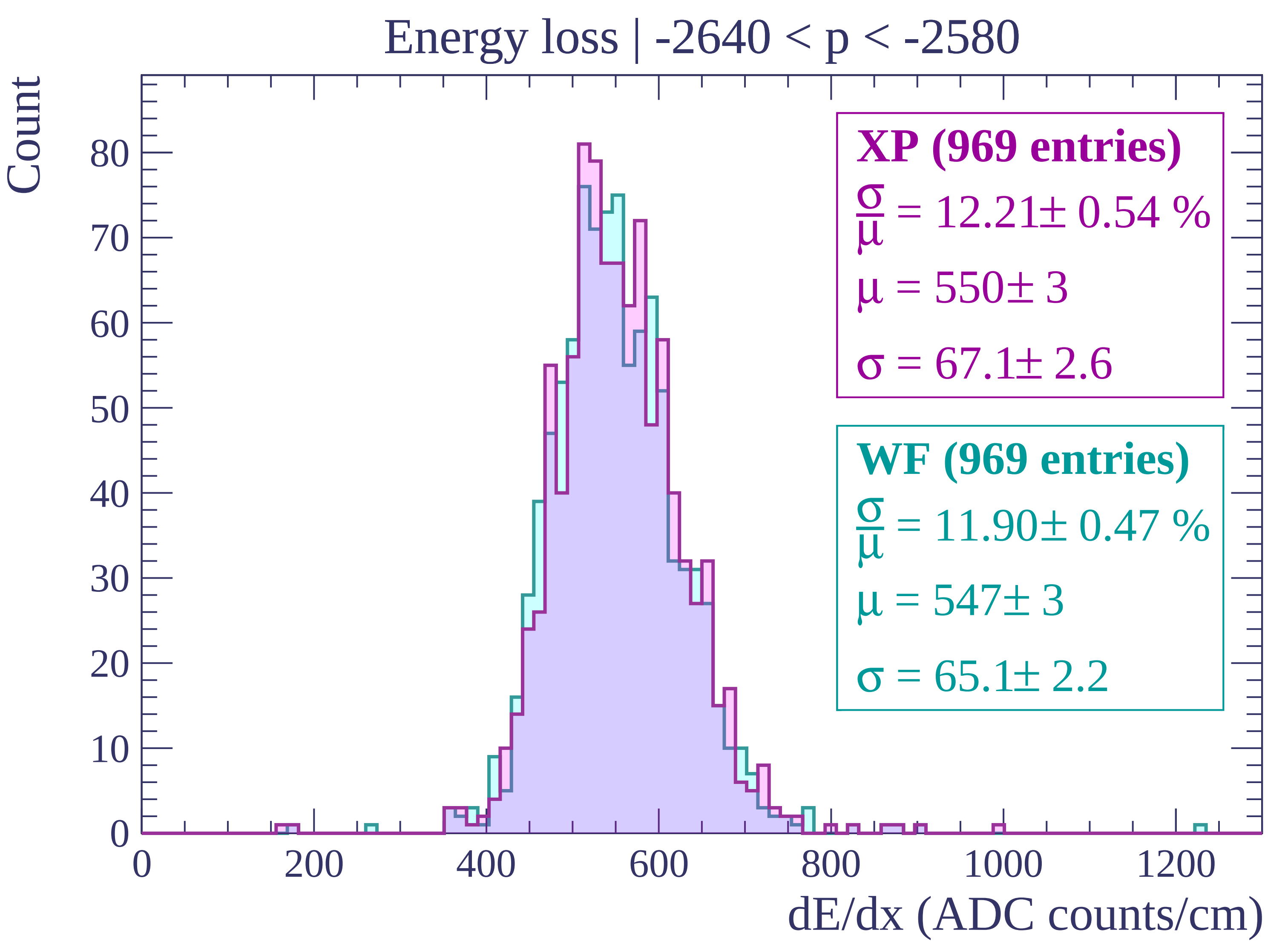

Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
XP (911 entries)
σ/μ = 11.24± 0.47 %
μ = 550± 2
σ = 61.8± 2.3
WF (911 entries)
σ/μ = 11.02± 0.43 %
μ = 548± 2
σ = 60.4± 2.1
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



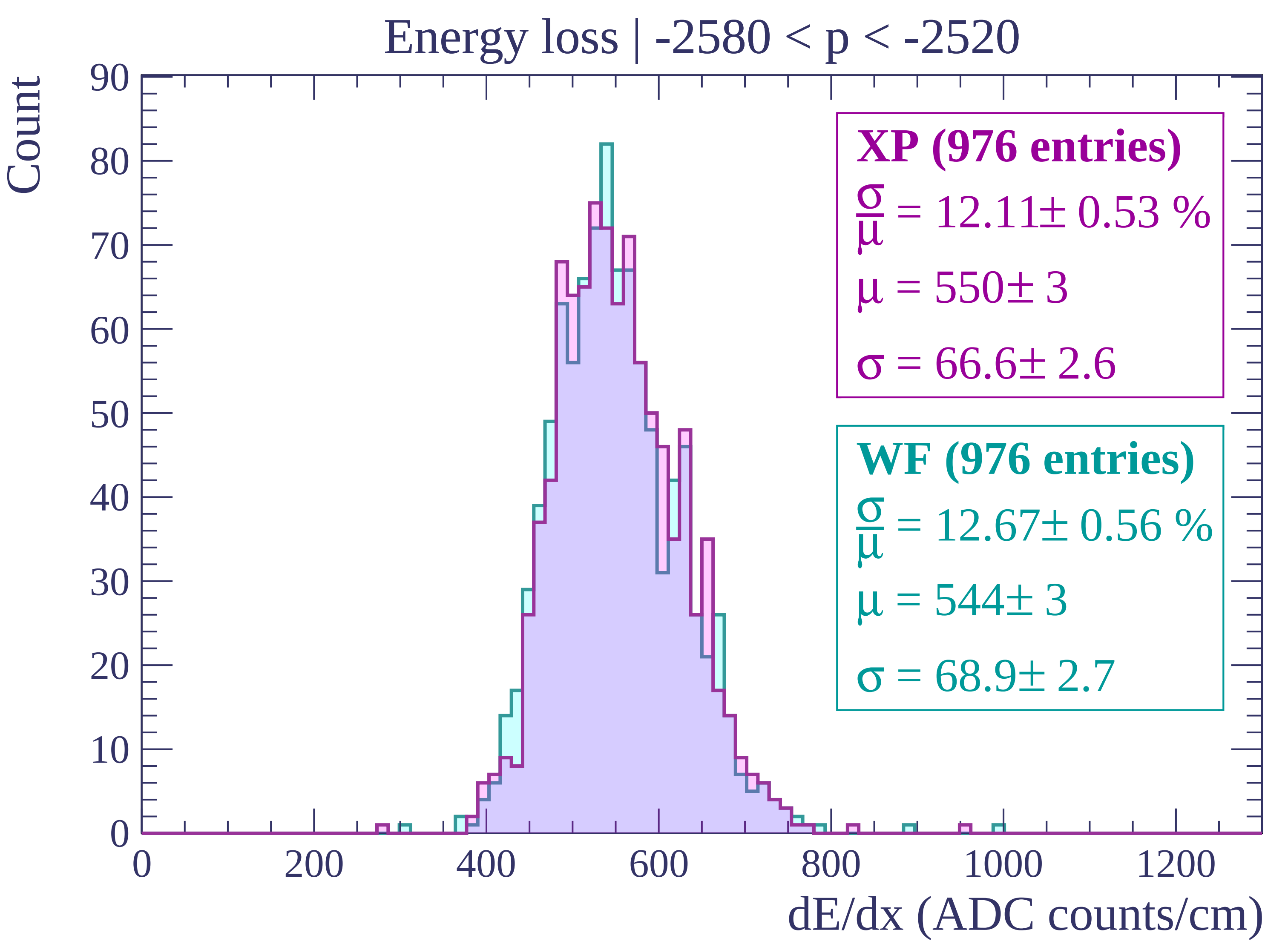
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (901 entries)
σ/μ = 12.08± 0.50 %
μ = 556± 3
σ = 67.2± 2.5
WF (901 entries)
σ/μ = 11.84± 0.44 %
μ = 552± 3
σ = 65.3± 2.1


Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP (998 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.51 %
μ = 549± 2
σ = 61.2± 2.5
WF (998 entries)
$\frac{\sigma}{\mu}$ = 11.14± 0.53 %
μ = 546± 2
σ = 60.8± 2.6
0
20
40
60
80
100
0
200
400
600
800
1000
1200

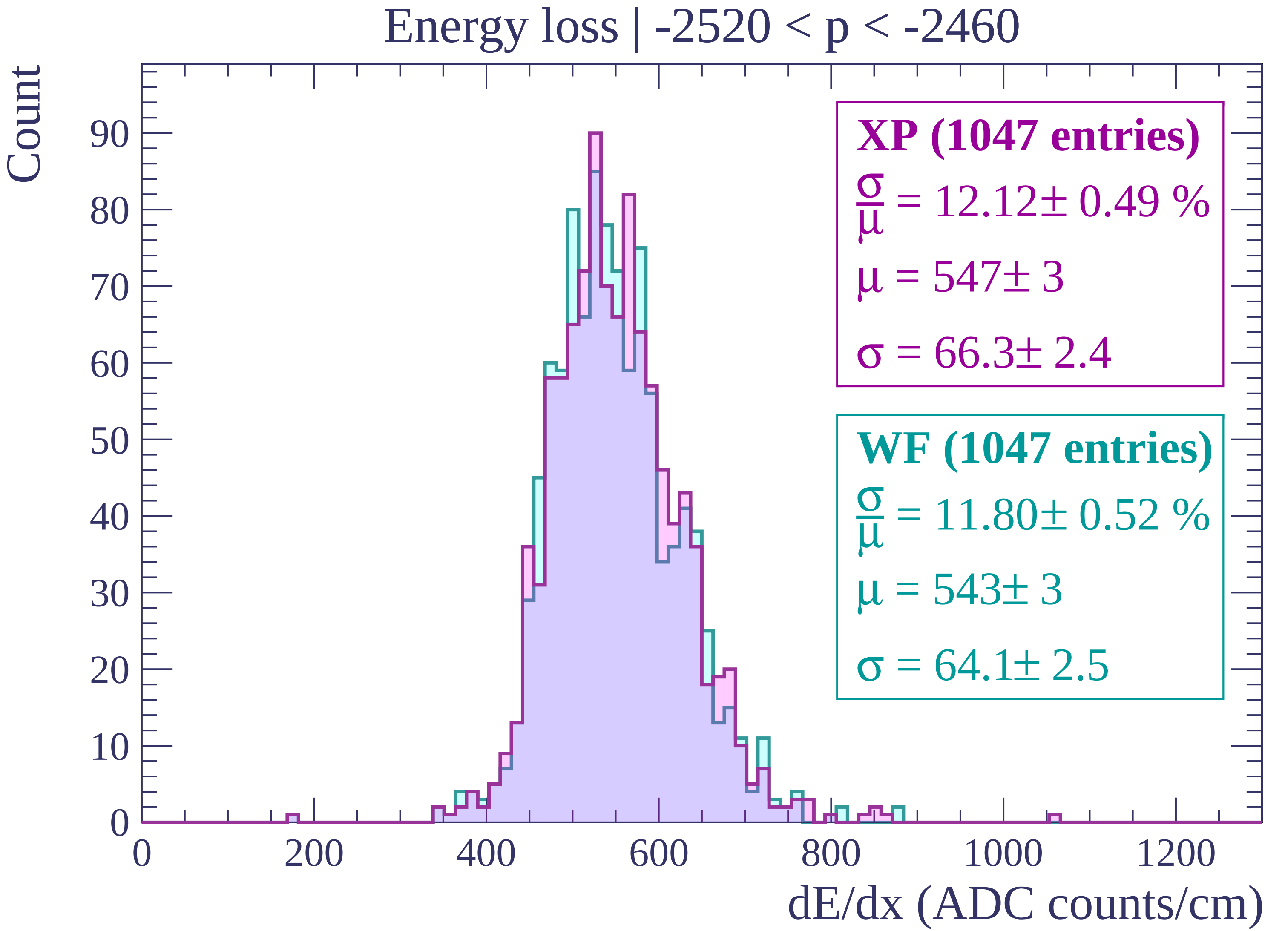

Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (971 entries)
$\frac{\sigma}{\mu}$ = 10.48± 0.47 %
μ = 552± 2
σ = 57.8± 2.3
WF (971 entries)
$\frac{\sigma}{\mu}$ = 11.08± 0.47 %
μ = 549± 2
σ = 60.8± 2.3



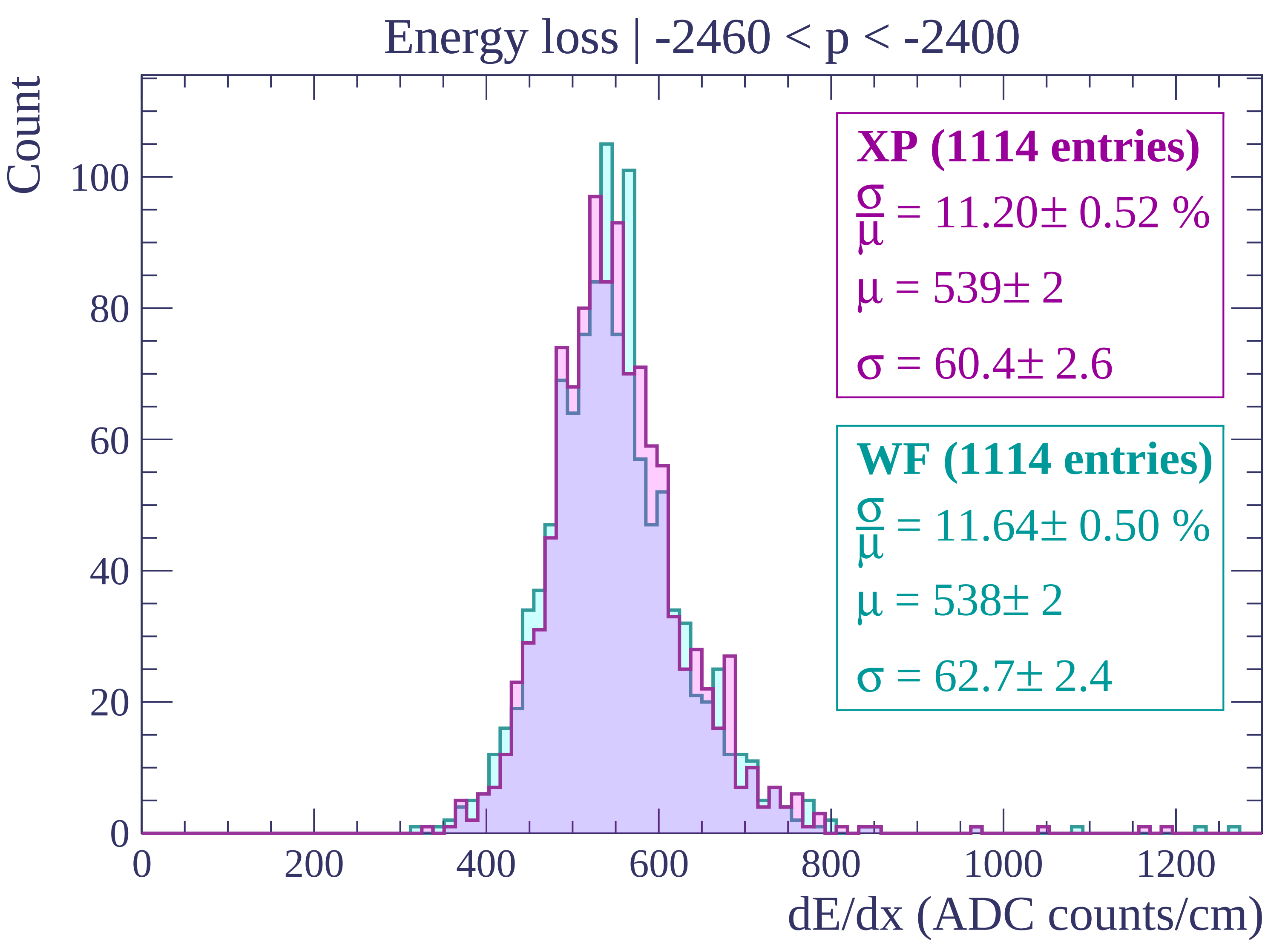
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (969 entries)
σ/μ = 12.21± 0.54 %
μ = 550± 3
σ = 67.1± 2.6
WF (969 entries)
σ/μ = 11.90± 0.47 %
μ = 547± 3
σ = 65.1± 2.2

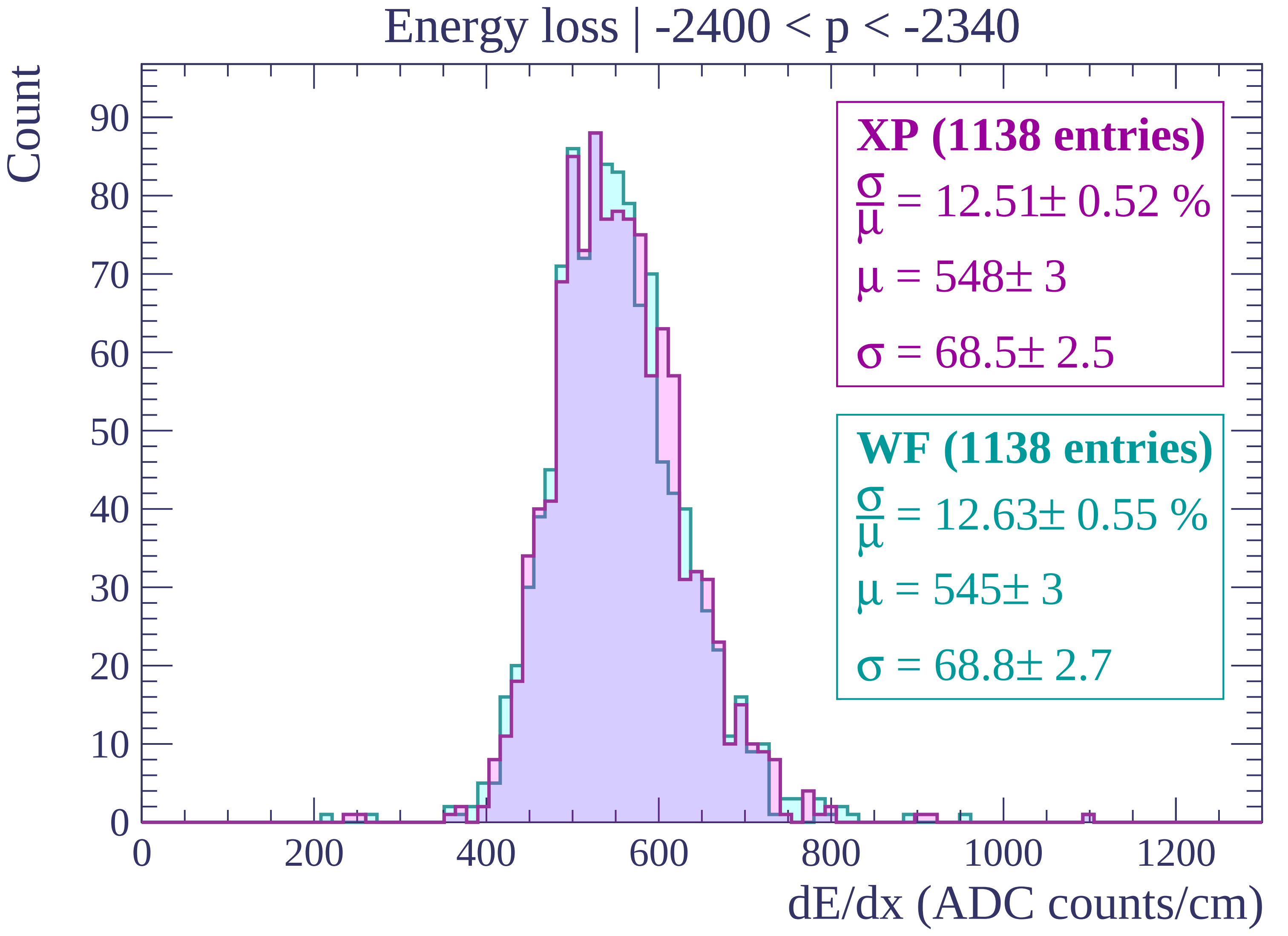


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (976 entries)
σ/μ = 12.11± 0.53 %
μ = 550± 3
σ = 66.6± 2.6
WF (976 entries)
σ/μ = 12.67± 0.56 %
μ = 544± 3
σ = 68.9± 2.7

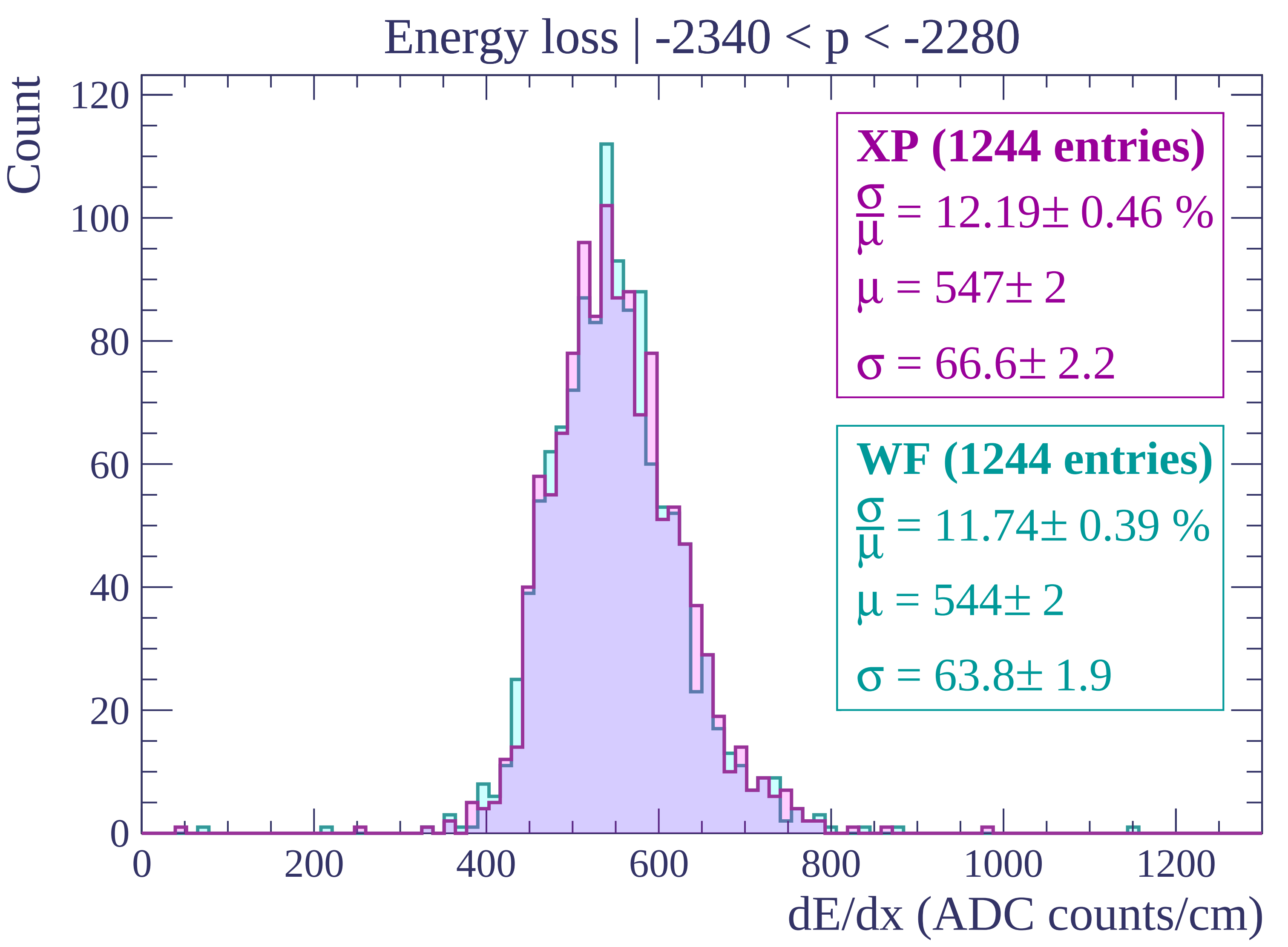


Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
XP (1047 entries)
σ/μ = 12.12± 0.49 %
μ = 547± 3
σ = 66.3± 2.4
WF (1047 entries)
σ/μ = 11.80± 0.52 %
μ = 543± 3
σ = 64.1± 2.5
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200

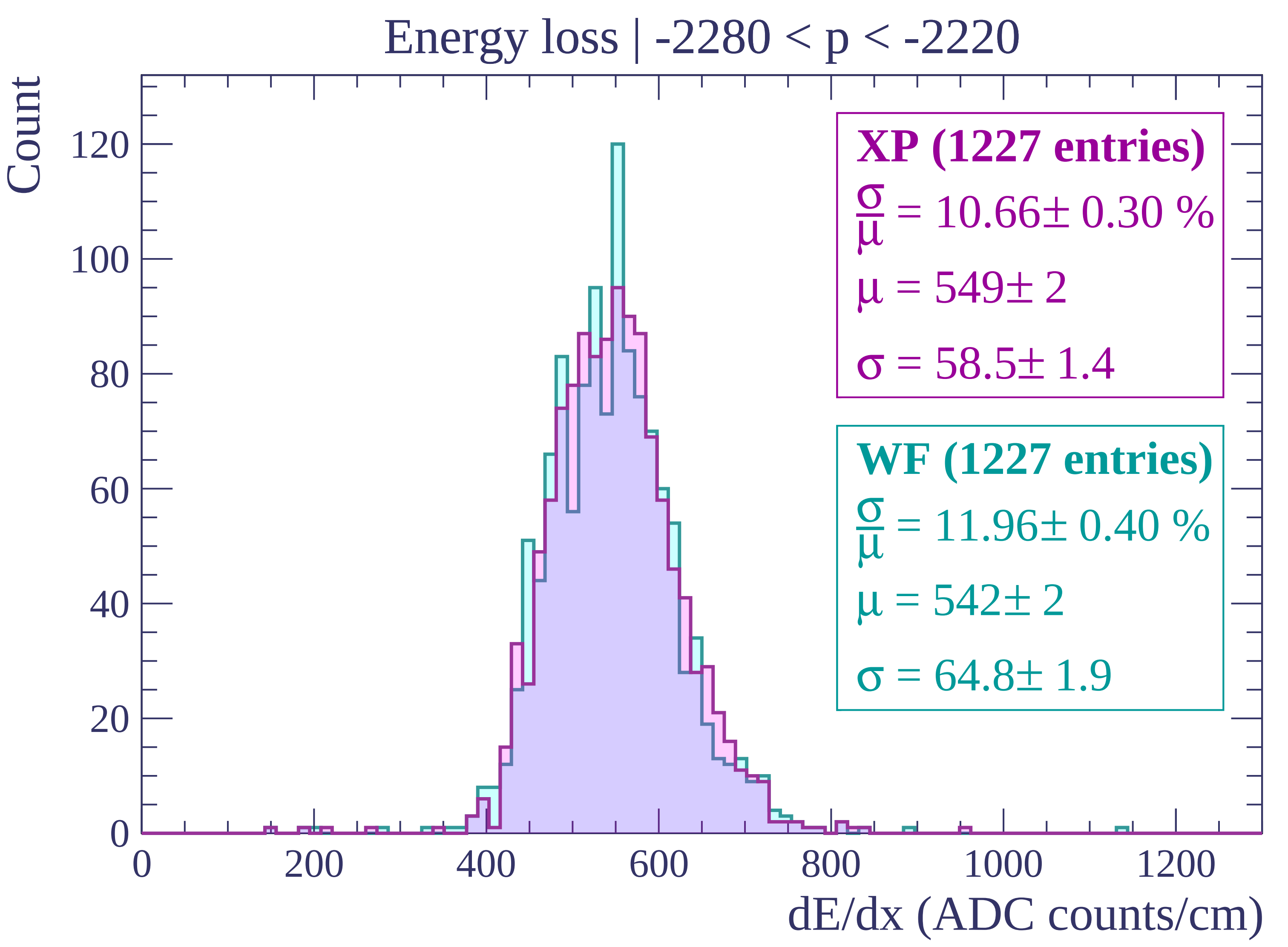

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (1114 entries)
σ/μ = 11.20± 0.52 %
μ = 539± 2
σ = 60.4± 2.6
WF (1114 entries)
σ/μ = 11.64± 0.50 %
μ = 538± 2
σ = 62.7± 2.4



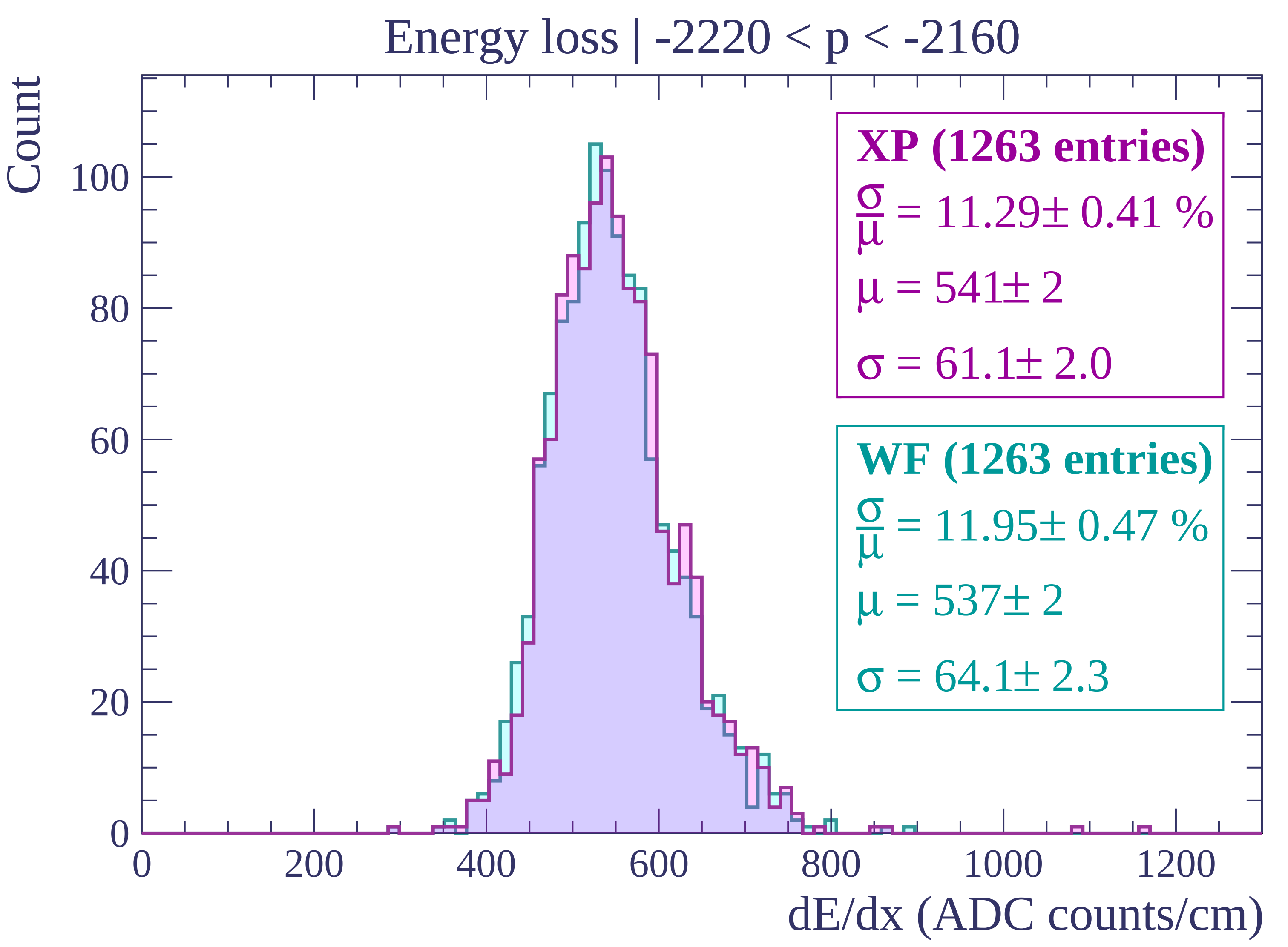
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
XP (1138 entries)
σ/μ = 12.51± 0.52 %
μ = 548± 3
σ = 68.5± 2.5
WF (1138 entries)
σ/μ = 12.63± 0.55 %
μ = 545± 3
σ = 68.8± 2.7
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



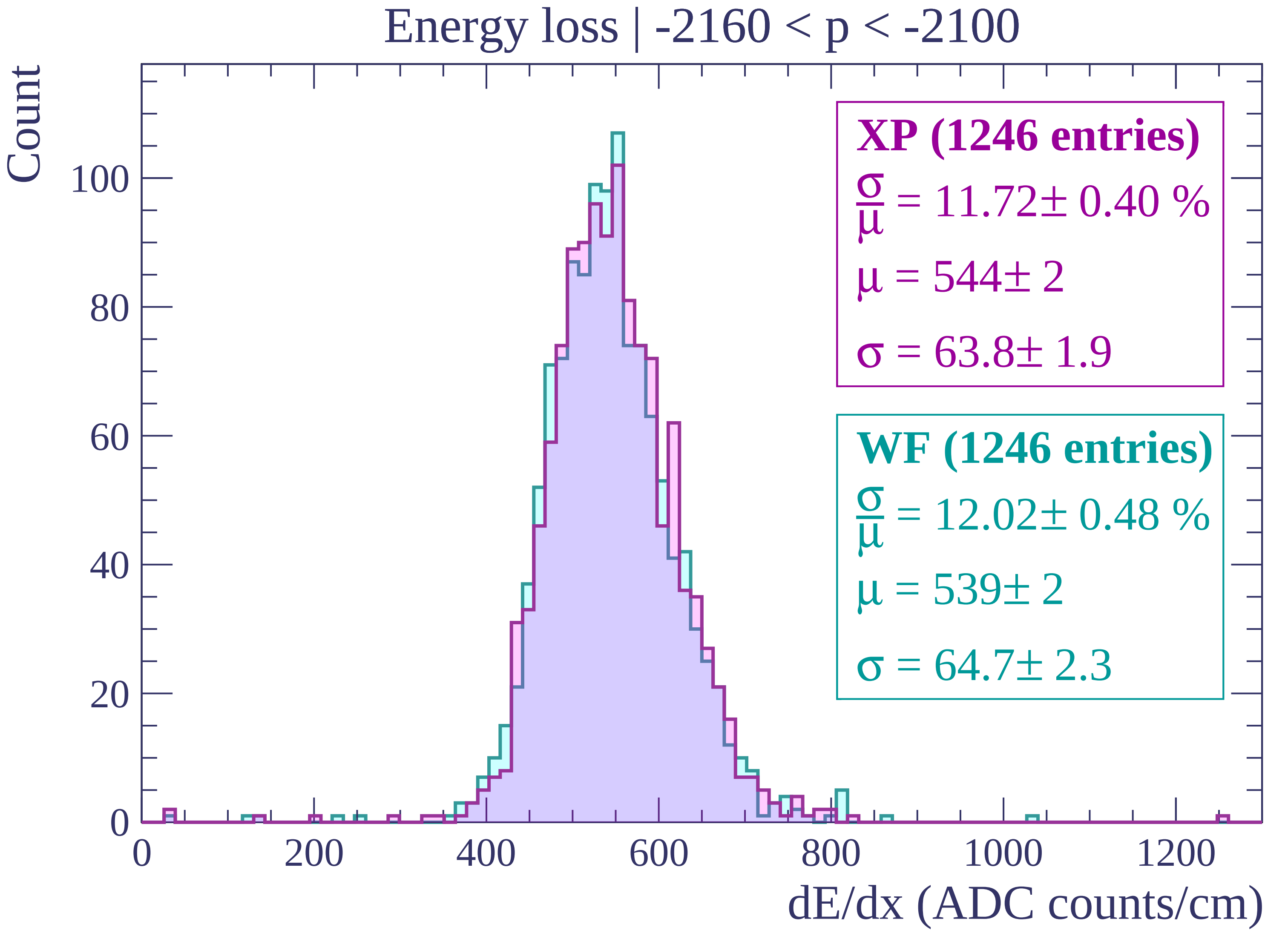
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1244 entries)
σ/μ = 12.19± 0.46 %
μ = 547± 2
σ = 66.6± 2.2
WF (1244 entries)
σ/μ = 11.74± 0.39 %
μ = 544± 2
σ = 63.8± 1.9

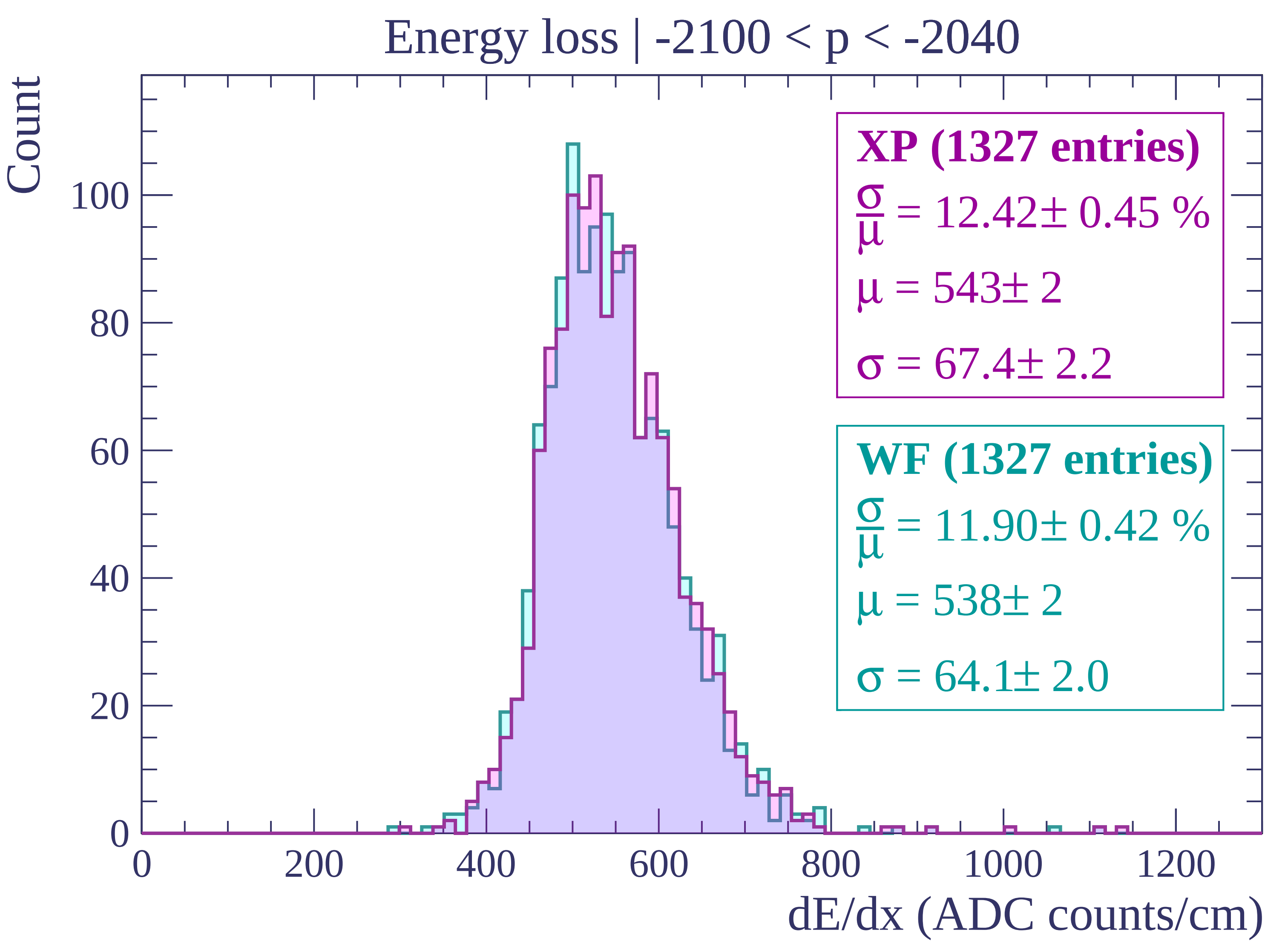


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1227 entries)
σ/μ = 10.66± 0.30 %
μ = 549± 2
σ = 58.5± 1.4
WF (1227 entries)
σ/μ = 11.96± 0.40 %
μ = 542± 2
σ = 64.8± 1.9

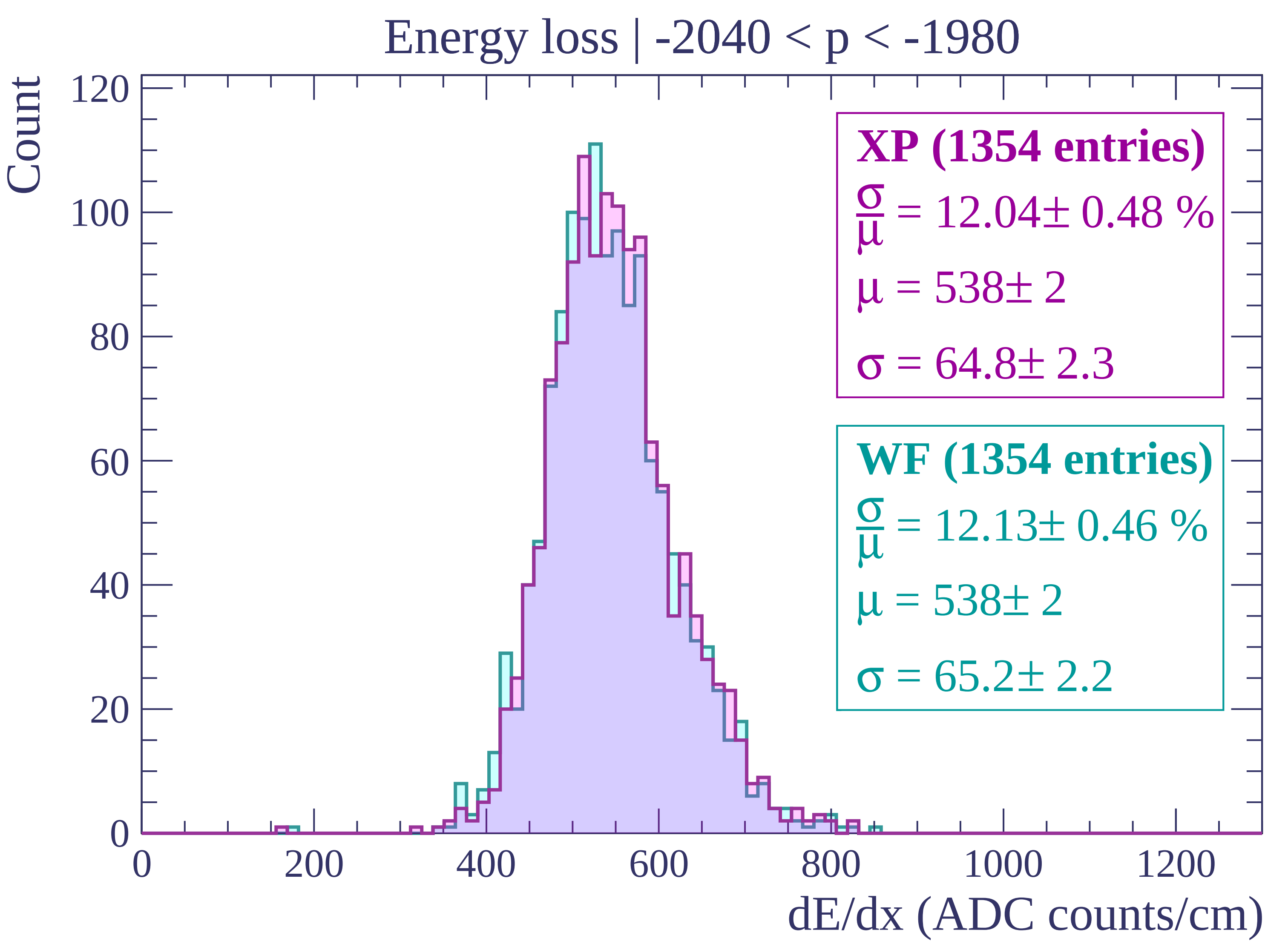


Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1263 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.41 %
μ = 541± 2
σ = 61.1± 2.0
WF (1263 entries)
$\frac{\sigma}{\mu}$ = 11.95± 0.47 %
μ = 537± 2
σ = 64.1± 2.3

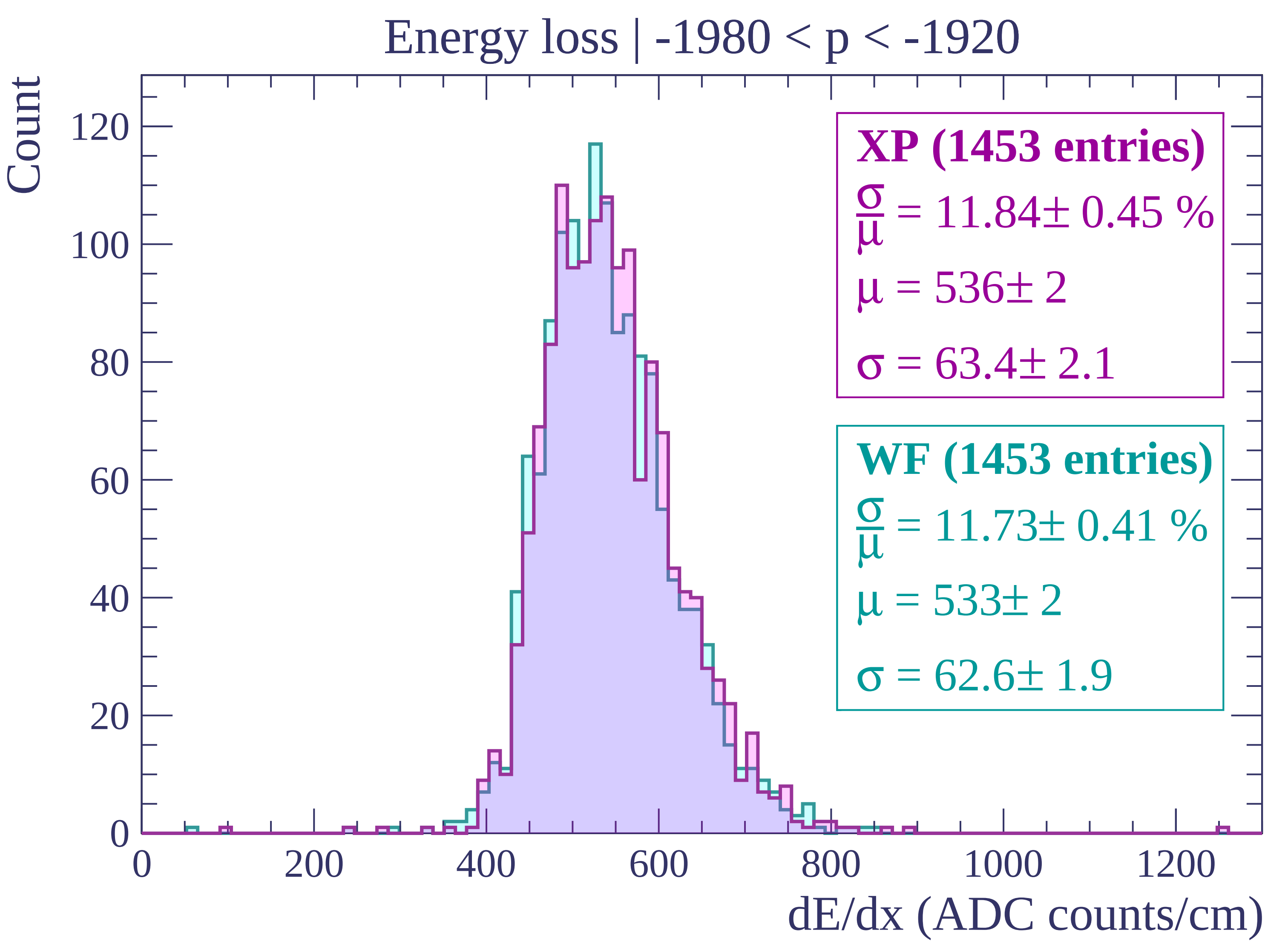

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1246 entries)
$\frac{\sigma}{\mu}$ = 11.72± 0.40 %
μ = 544± 2
σ = 63.8± 1.9
WF (1246 entries)
$\frac{\sigma}{\mu}$ = 12.02± 0.48 %
μ = 539± 2
σ = 64.7± 2.3



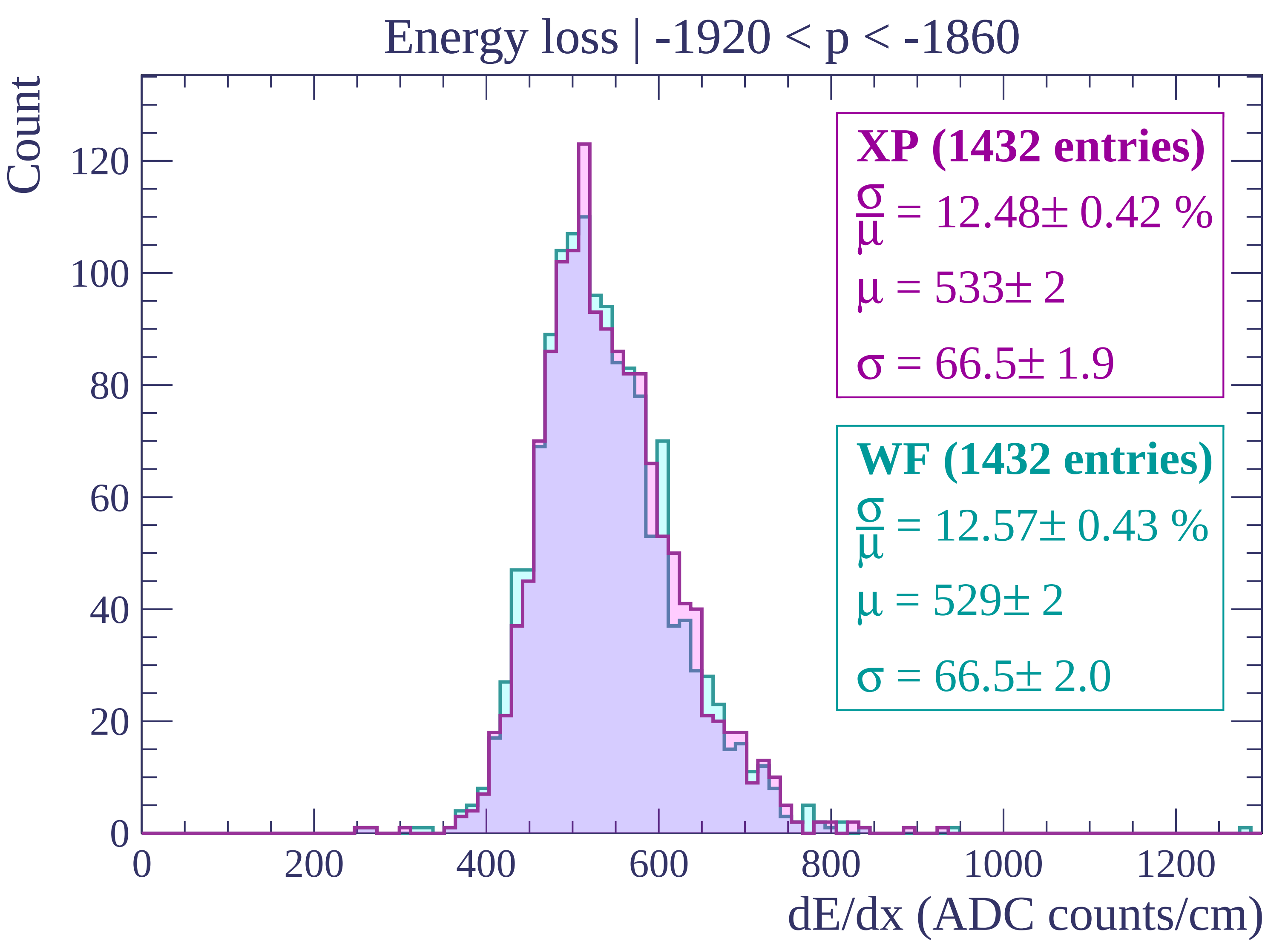
Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (1327 entries)
σ/μ = 12.42± 0.45 %
μ = 543± 2
σ = 67.4± 2.2
WF (1327 entries)
σ/μ = 11.90± 0.42 %
μ = 538± 2
σ = 64.1± 2.0

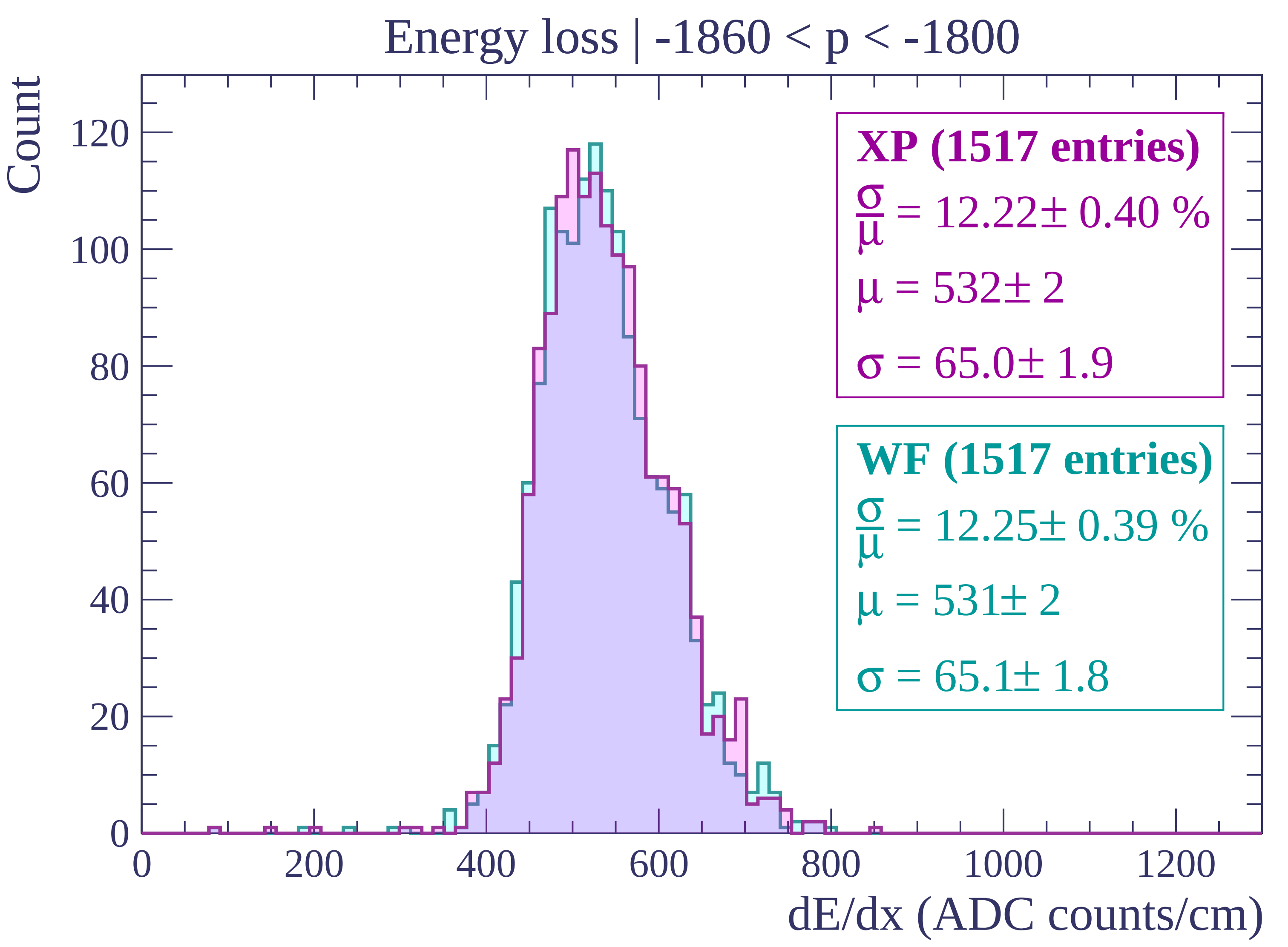


Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1354 entries)
σ/μ = 12.04± 0.48 %
μ = 538± 2
σ = 64.8± 2.3
WF (1354 entries)
σ/μ = 12.13± 0.46 %
μ = 538± 2
σ = 65.2± 2.2

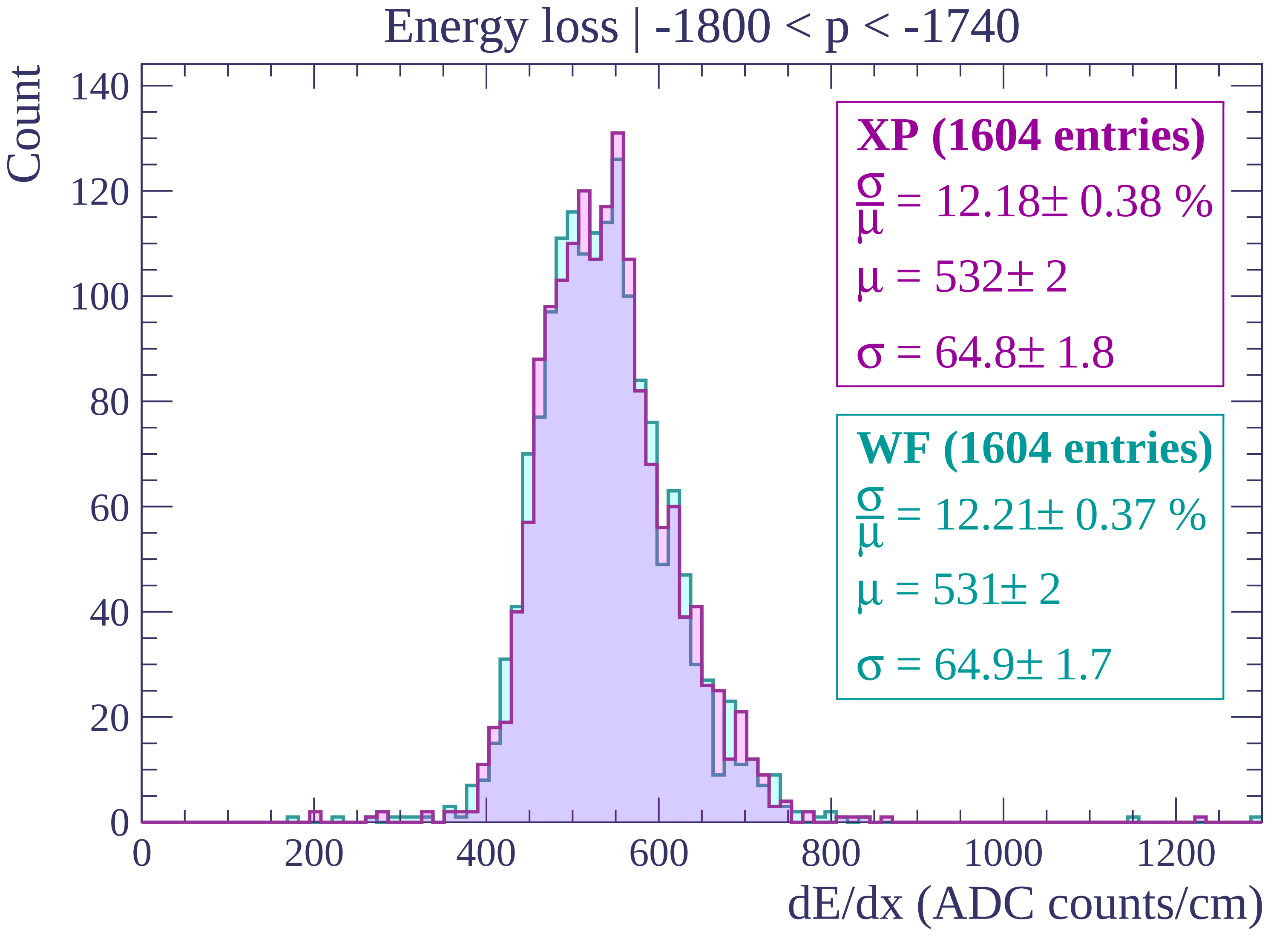


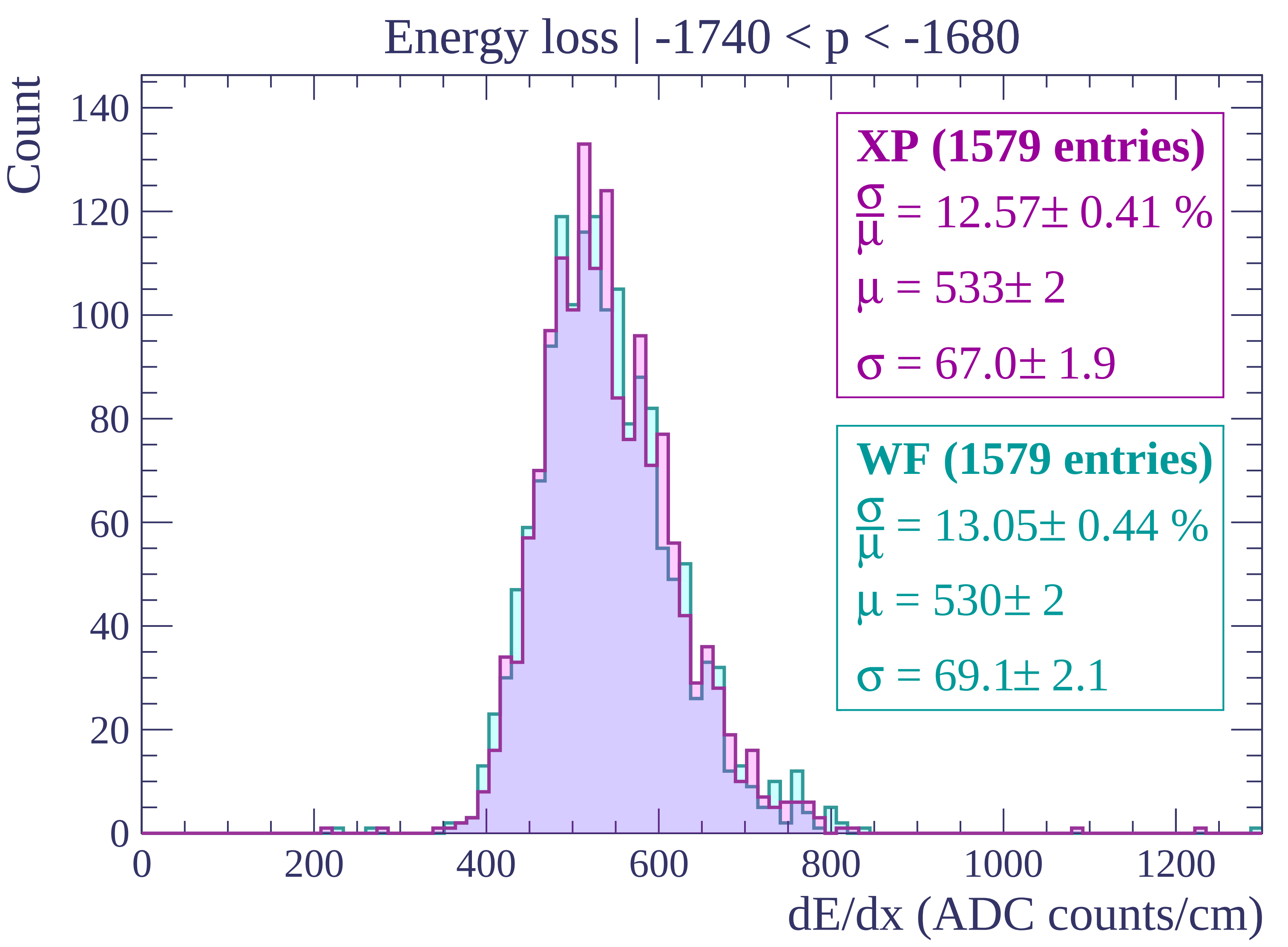
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP (1453 entries)
$\frac{\sigma}{\mu}$ = 11.84± 0.45 %
μ = 536± 2
σ = 63.4± 2.1
WF (1453 entries)
$\frac{\sigma}{\mu}$ = 11.73± 0.41 %
μ = 533± 2
σ = 62.6± 1.9

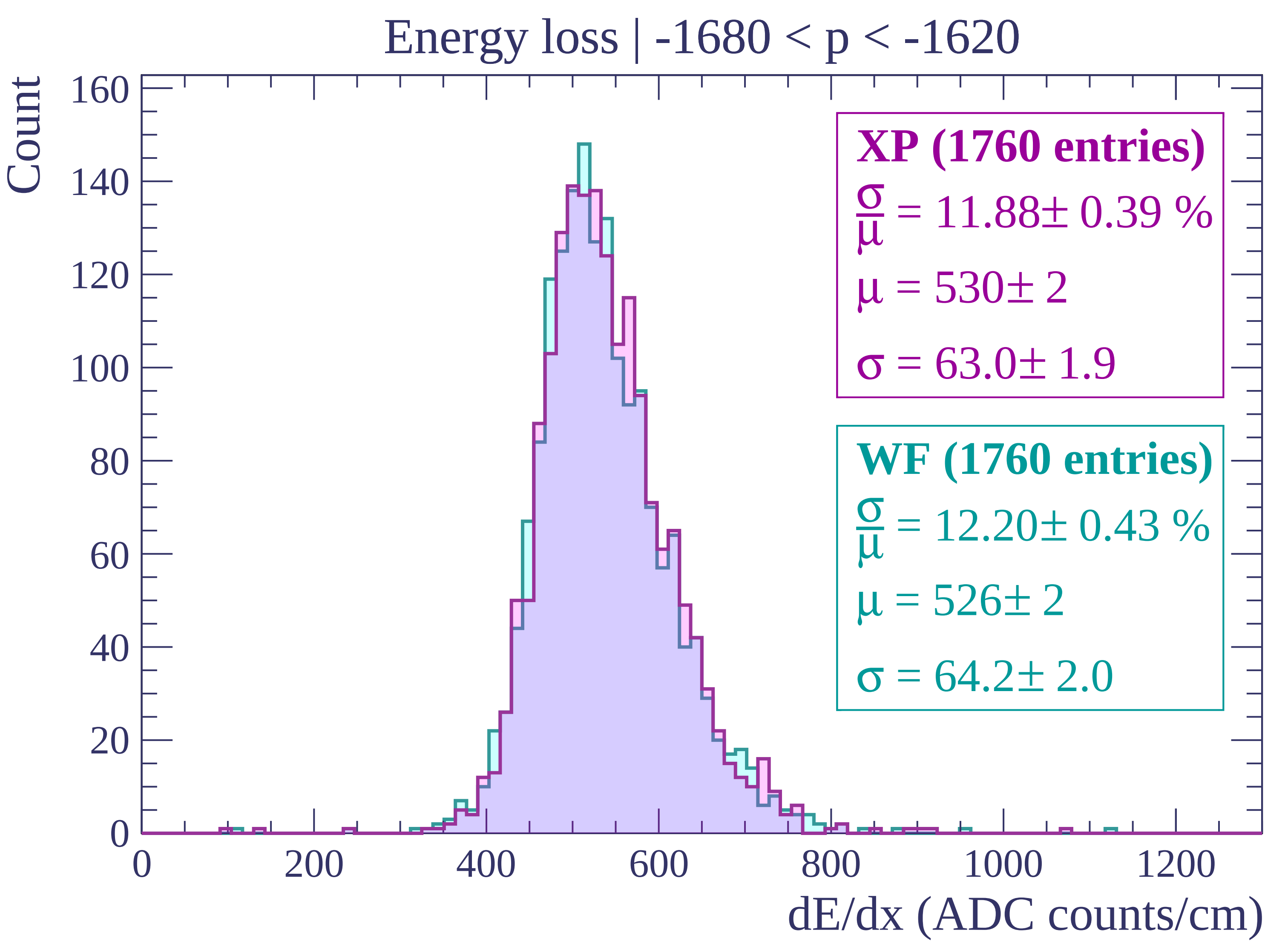

Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP (1432 entries)
σ/μ = 12.48± 0.42 %
μ = 533± 2
σ = 66.5± 1.9
WF (1432 entries)
σ/μ = 12.57± 0.43 %
μ = 529± 2
σ = 66.5± 2.0



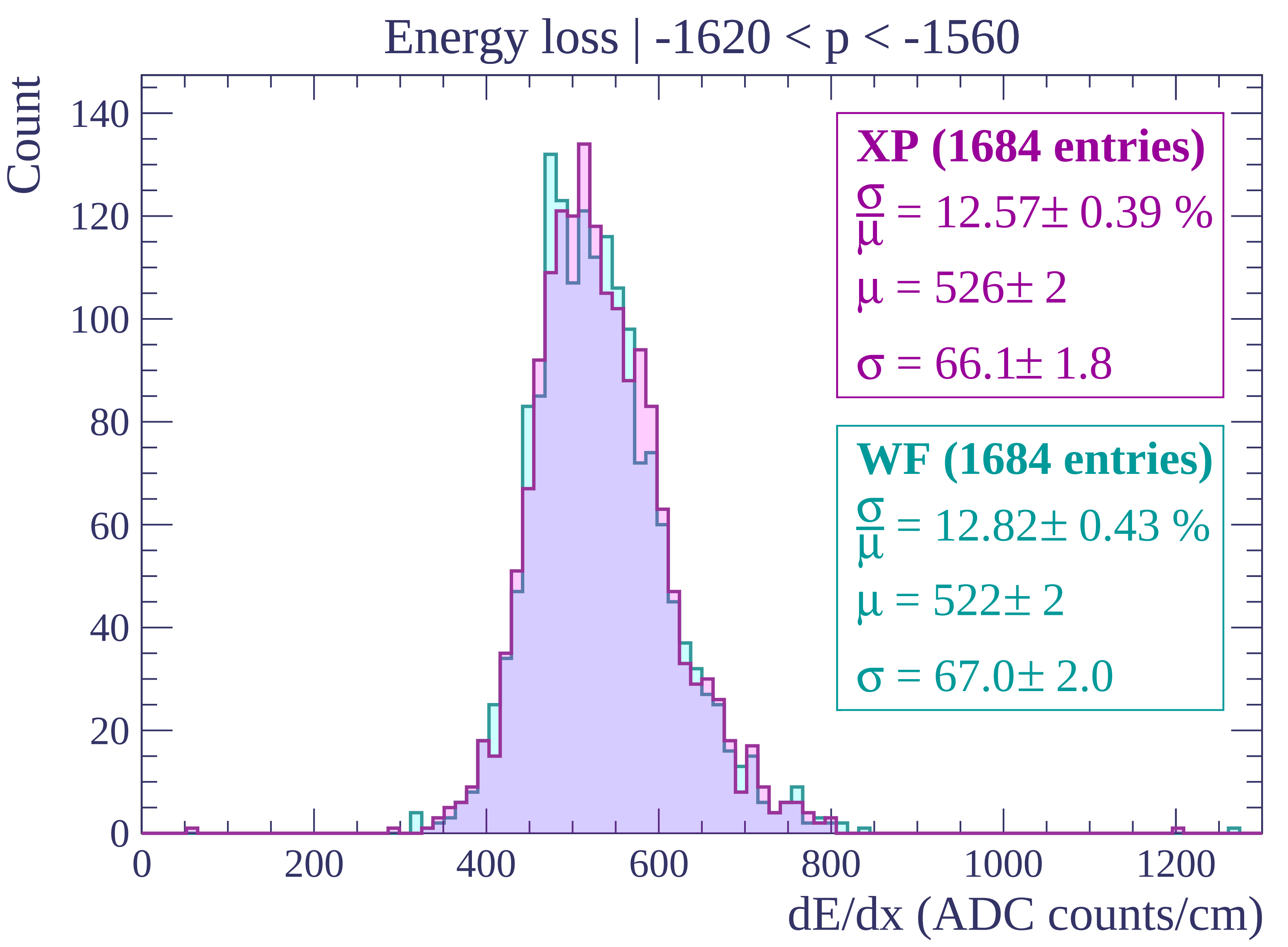
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (1517 entries)
σ/μ = 12.22± 0.40 %
μ = 532± 2
σ = 65.0± 1.9
WF (1517 entries)
σ/μ = 12.25± 0.39 %
μ = 531± 2
σ = 65.1± 1.8

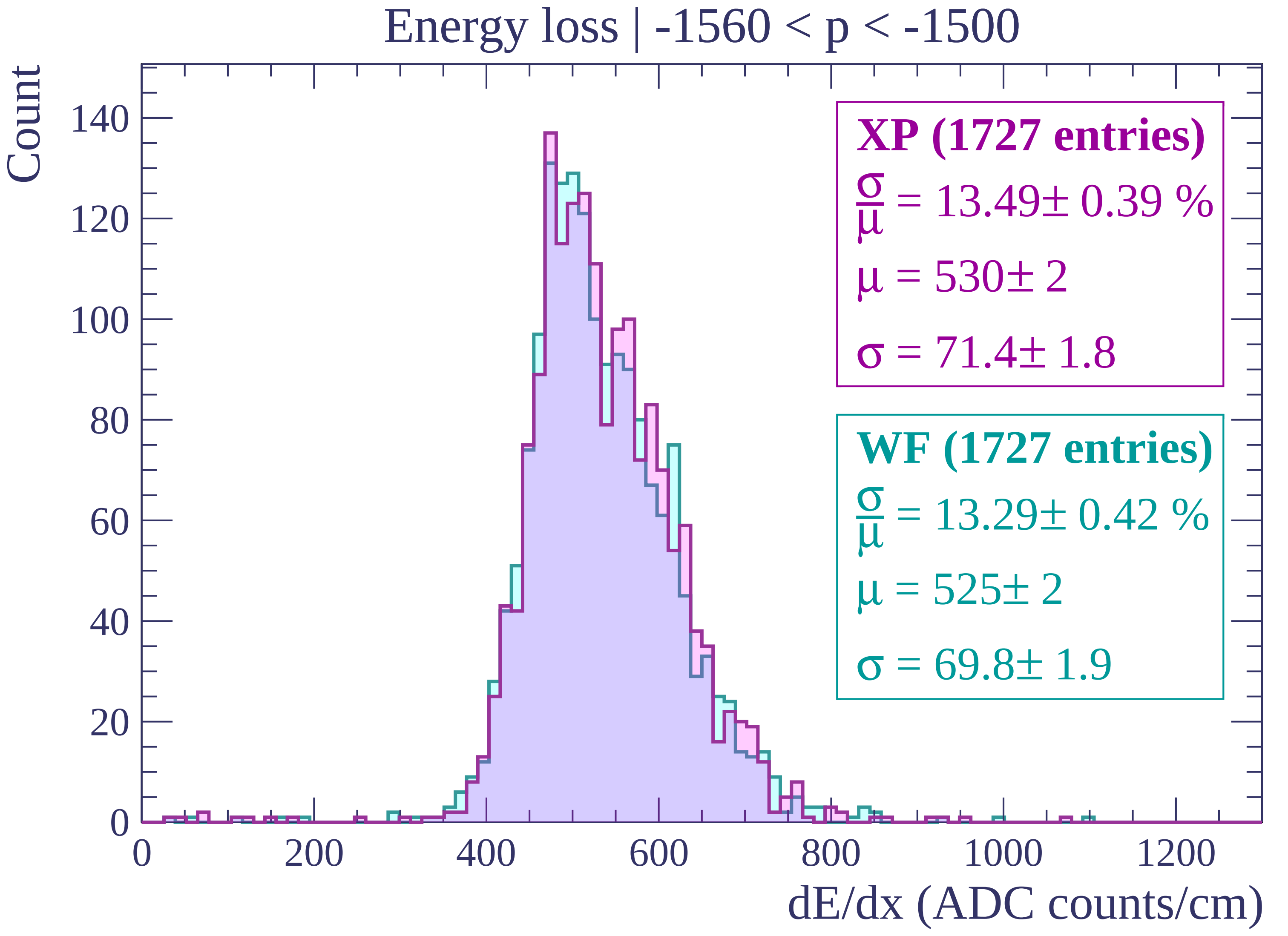

Energy loss | -1800 < p < -1740
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1604 entries)
σ/μ = 12.18± 0.38 %
μ = 532± 2
σ = 64.8± 1.8
WF (1604 entries)
σ/μ = 12.21± 0.37 %
μ = 531± 2
σ = 64.9± 1.7


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP (1579 entries)
σ/μ = 12.57± 0.41 %
μ = 533± 2
σ = 67.0± 1.9
WF (1579 entries)
σ/μ = 13.05± 0.44 %
μ = 530± 2
σ = 69.1± 2.1

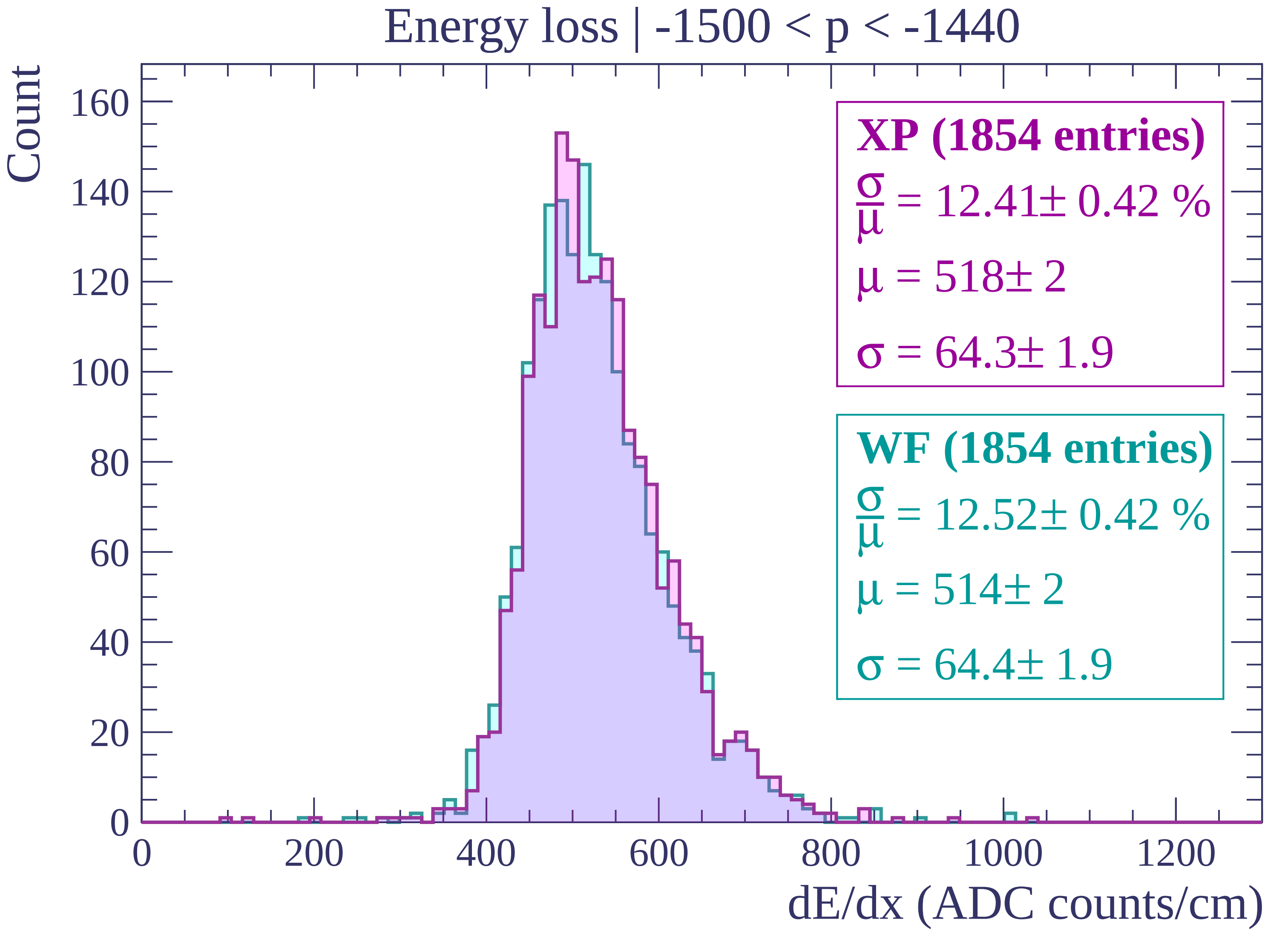


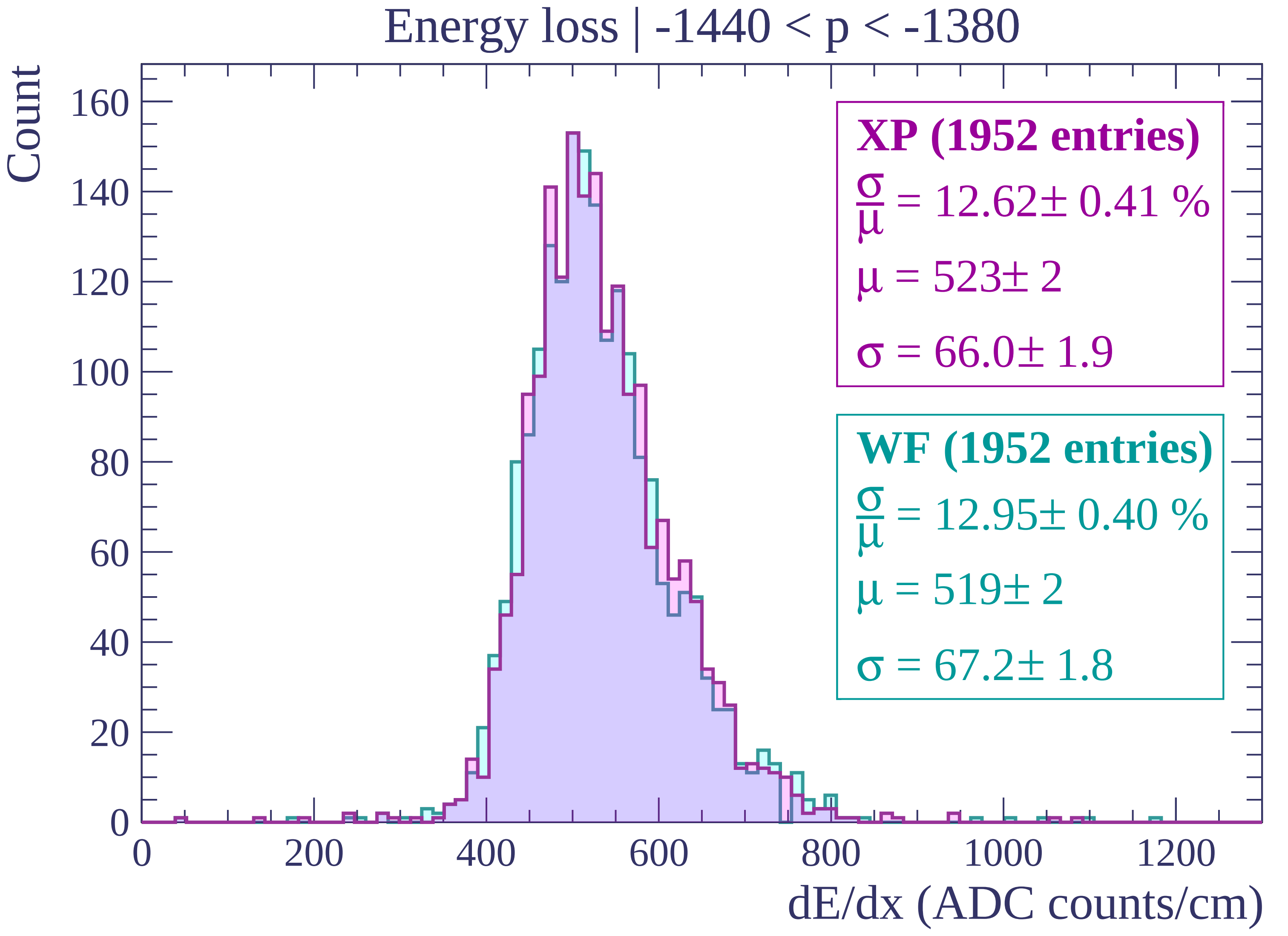
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
XP (1760 entries)
$\frac{\sigma}{\mu}$ = 11.88± 0.39 %
μ = 530± 2
σ = 63.0± 1.9
WF (1760 entries)
$\frac{\sigma}{\mu}$ = 12.20± 0.43 %
μ = 526± 2
σ = 64.2± 2.0

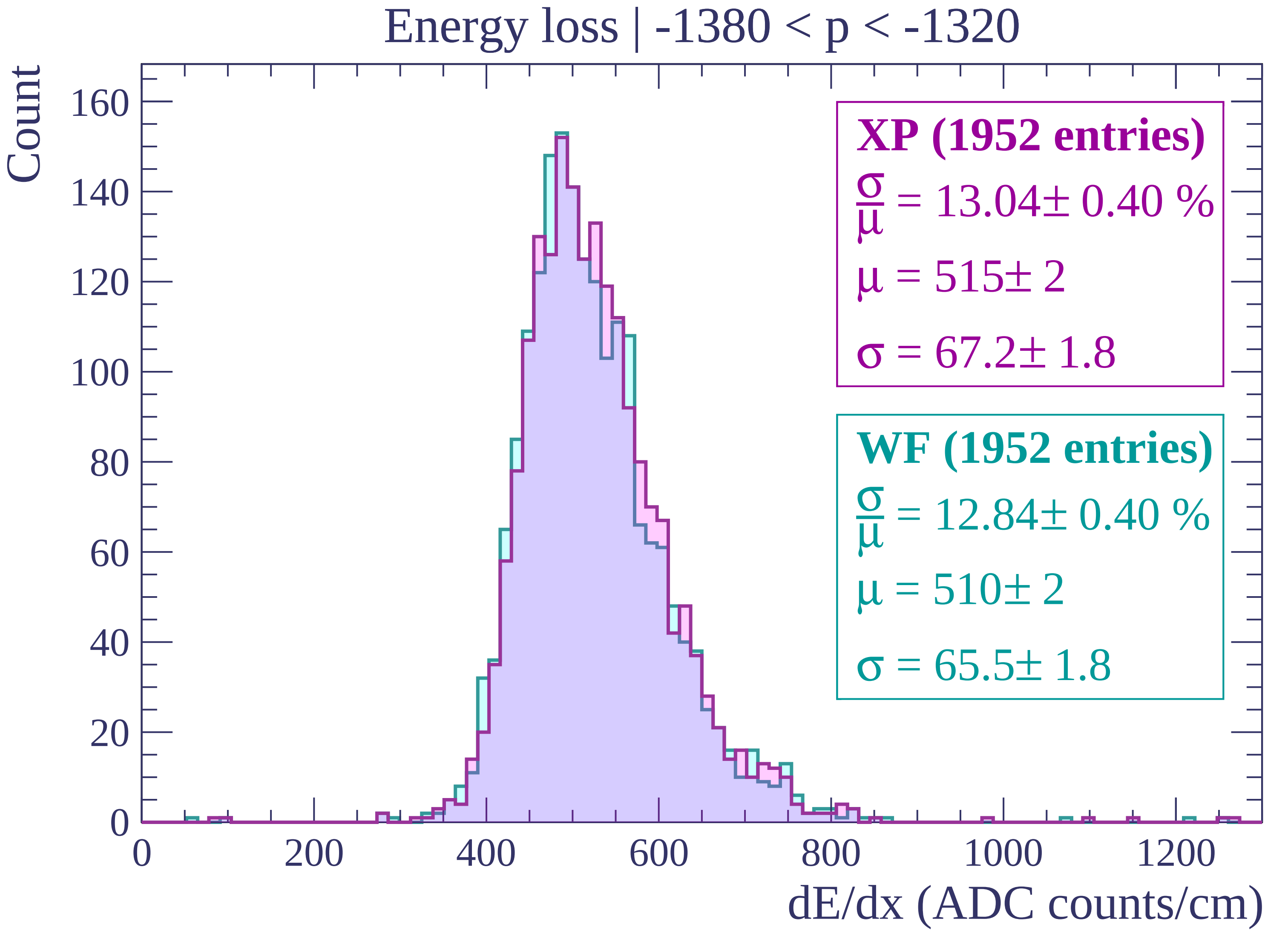

Energy loss | -1620 < p < -1560
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1684 entries)
σ/μ = 12.57± 0.39 %
μ = 526± 2
σ = 66.1± 1.8
WF (1684 entries)
σ/μ = 12.82± 0.43 %
μ = 522± 2
σ = 67.0± 2.0

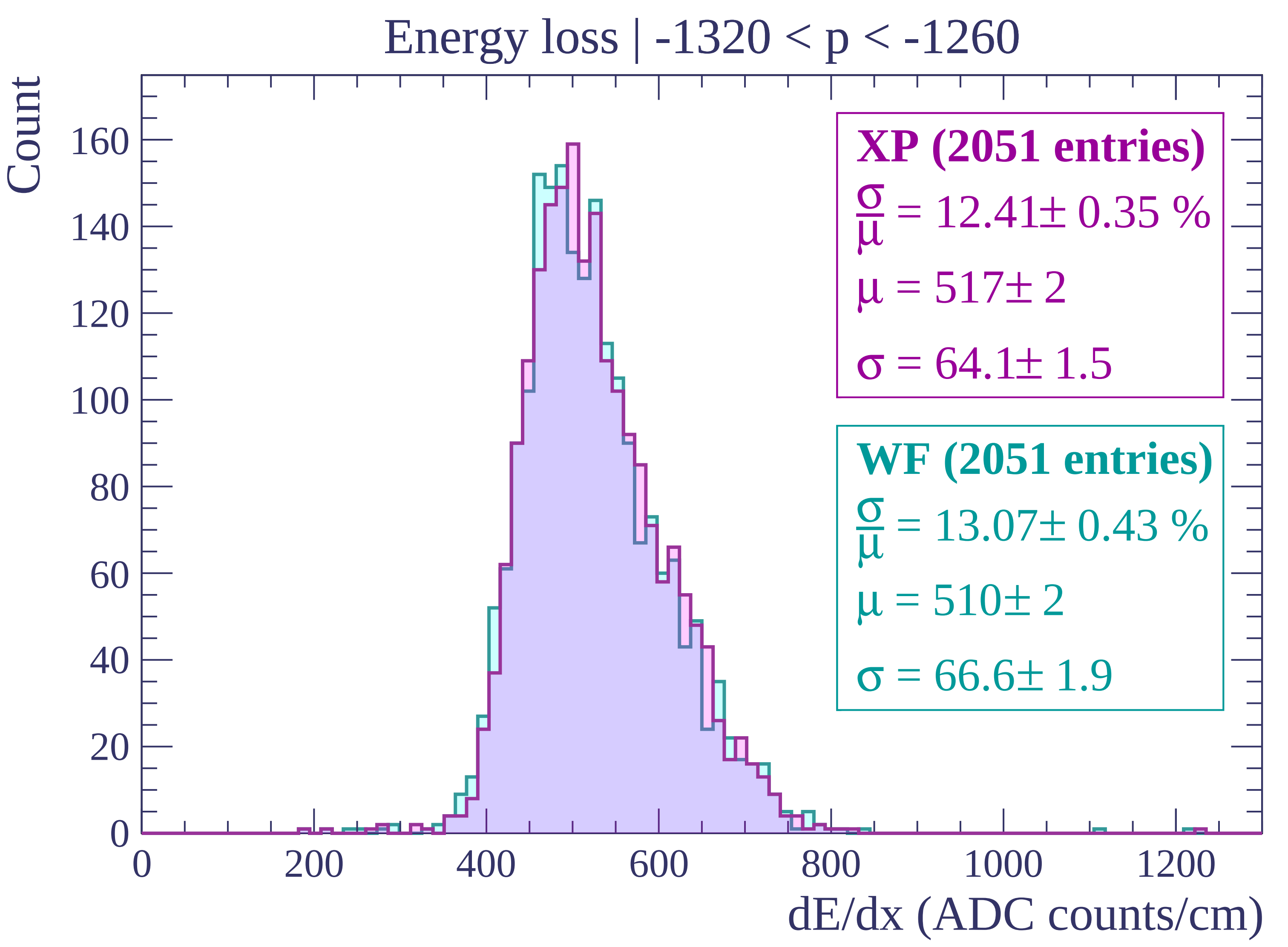


Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1727 entries)
$\frac{\sigma}{\mu}$ = 13.49± 0.39 %
μ = 530± 2
σ = 71.4± 1.8
WF (1727 entries)
$\frac{\sigma}{\mu}$ = 13.29± 0.42 %
μ = 525± 2
σ = 69.8± 1.9

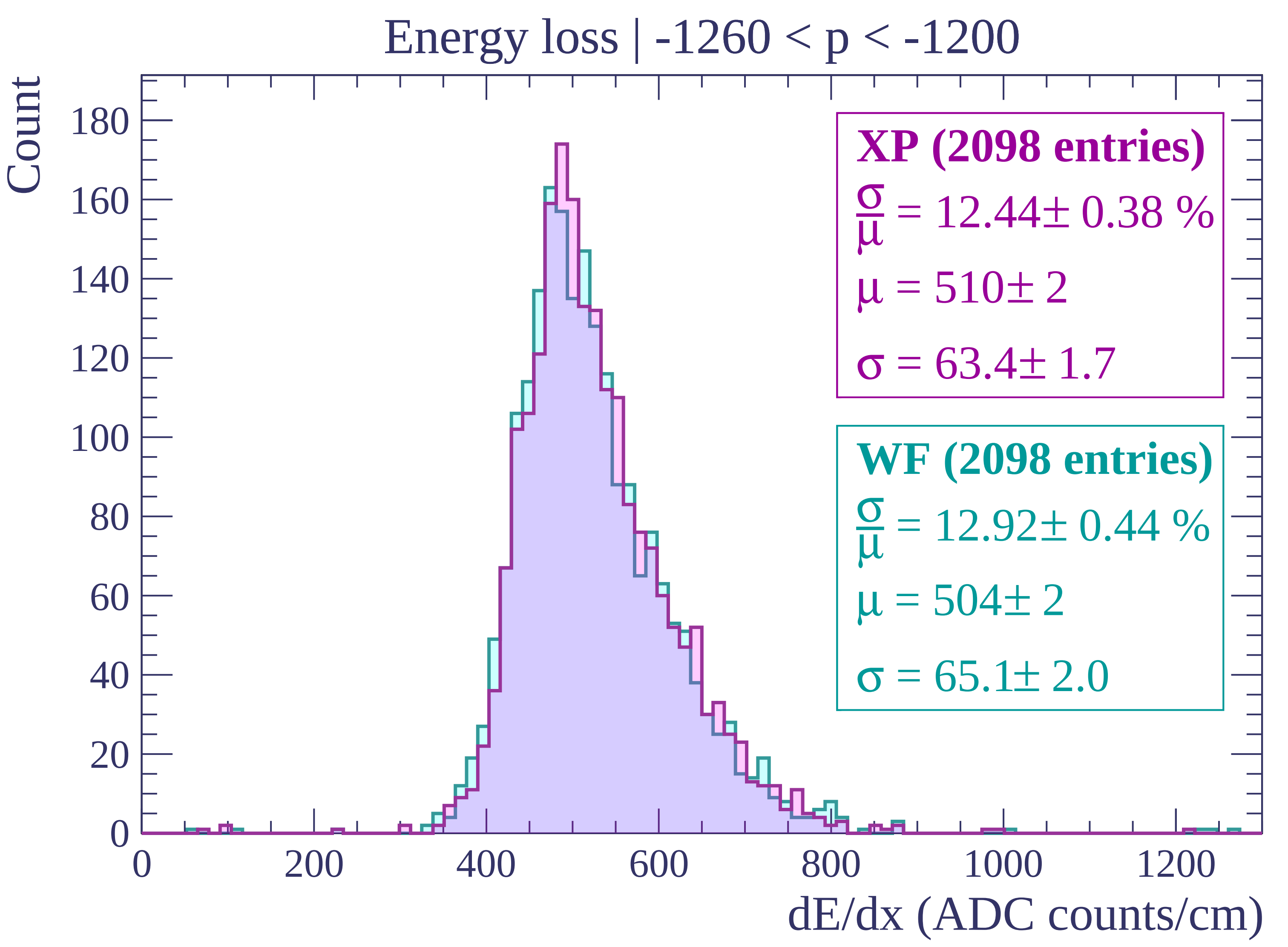

Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
XP (1854 entries)
$\frac{\sigma}{\mu}$ = 12.41± 0.42 %
μ = 518± 2
σ = 64.3± 1.9
WF (1854 entries)
$\frac{\sigma}{\mu}$ = 12.52± 0.42 %
μ = 514± 2
σ = 64.4± 1.9
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160


Energy loss | -1440 < p < -1380
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1952 entries)
σ/μ = 12.62± 0.41 %
μ = 523± 2
σ = 66.0± 1.9
WF (1952 entries)
σ/μ = 12.95± 0.40 %
μ = 519± 2
σ = 67.2± 1.8

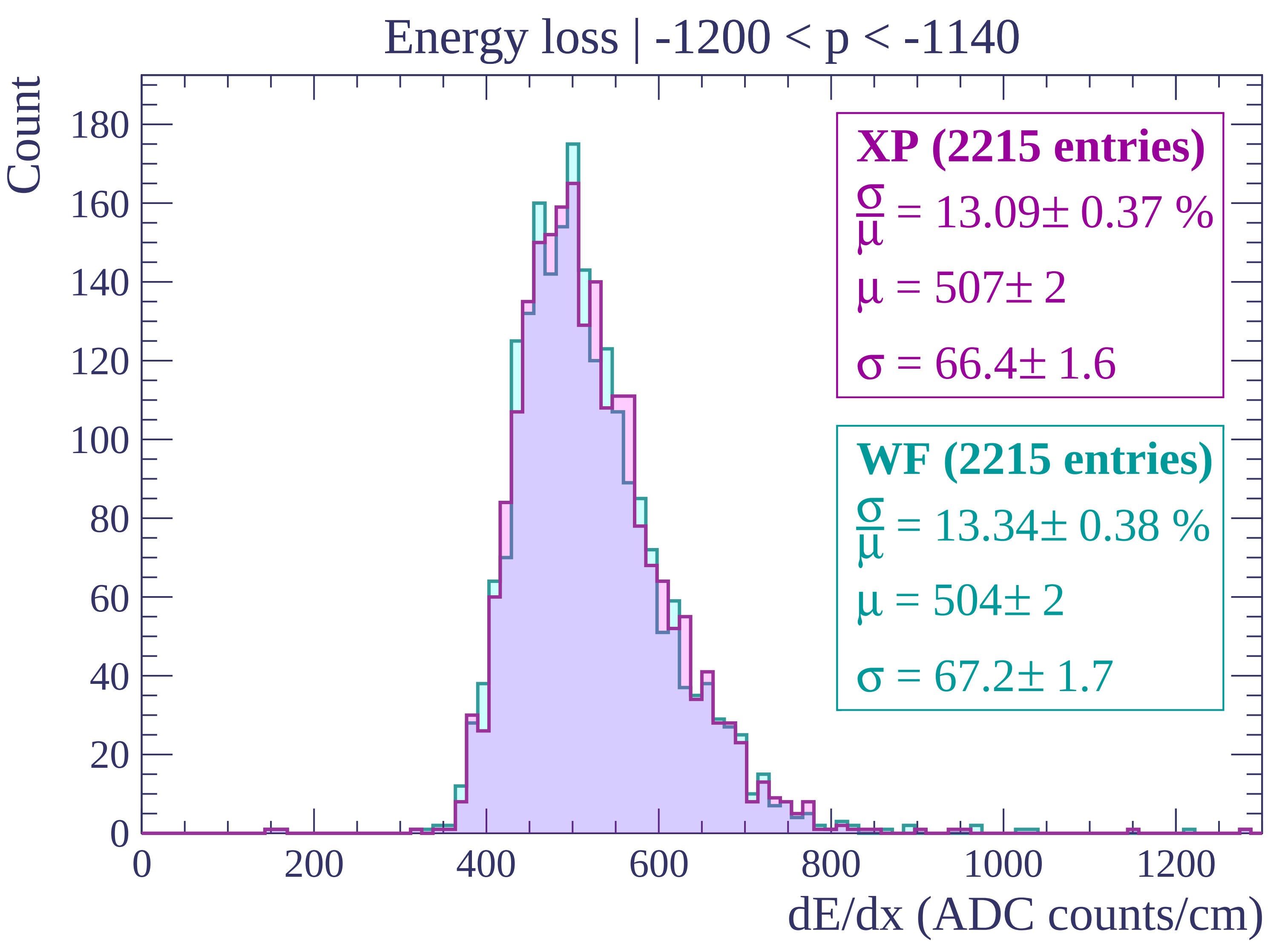

Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (1952 entries)
σ/μ = 13.04± 0.40 %
μ = 515± 2
σ = 67.2± 1.8
WF (1952 entries)
σ/μ = 12.84± 0.40 %
μ = 510± 2
σ = 65.5± 1.8



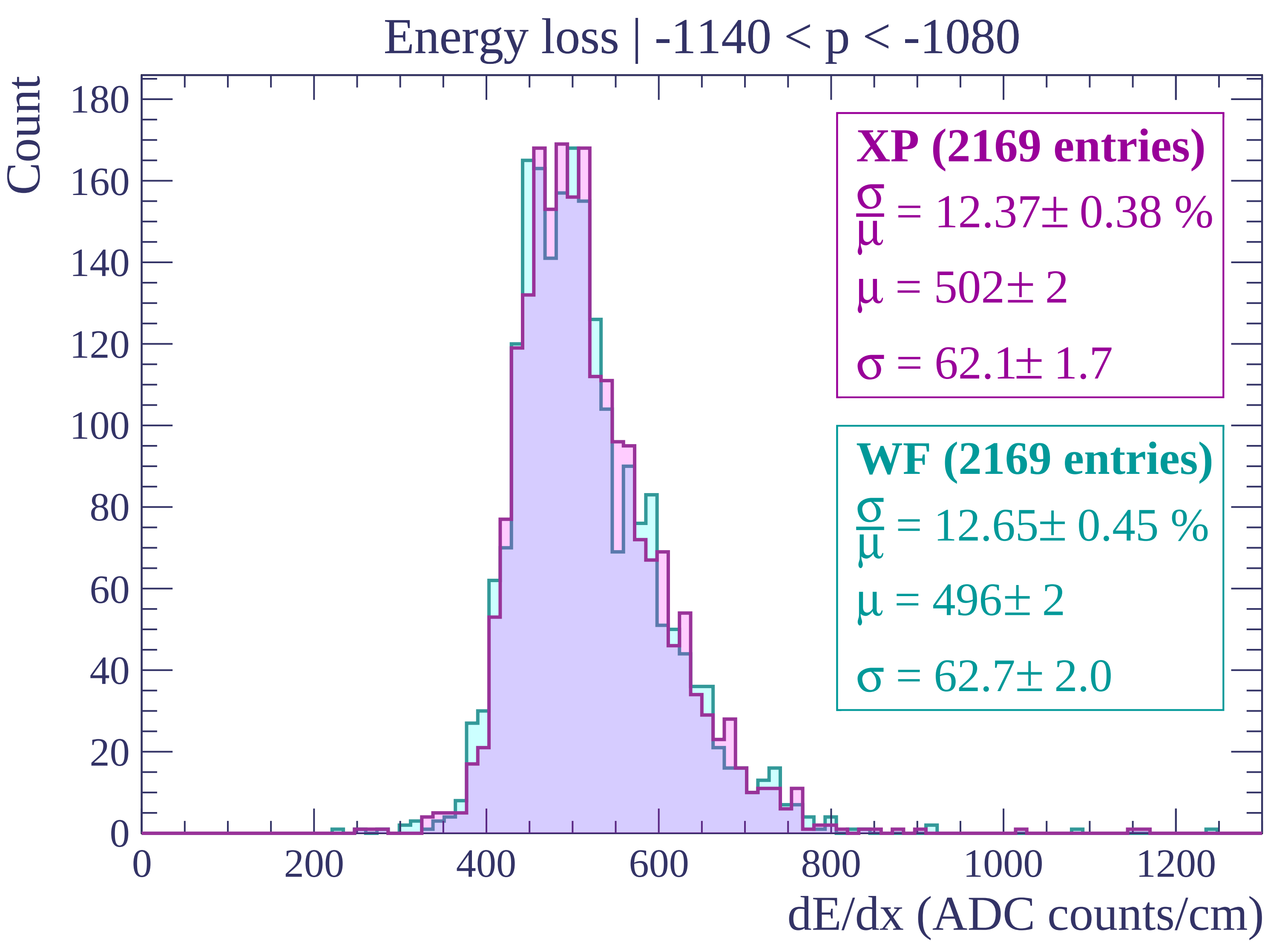
Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (2051 entries)
σ/μ = 12.41± 0.35 %
μ = 517± 2
σ = 64.1± 1.5
WF (2051 entries)
σ/μ = 13.07± 0.43 %
μ = 510± 2
σ = 66.6± 1.9



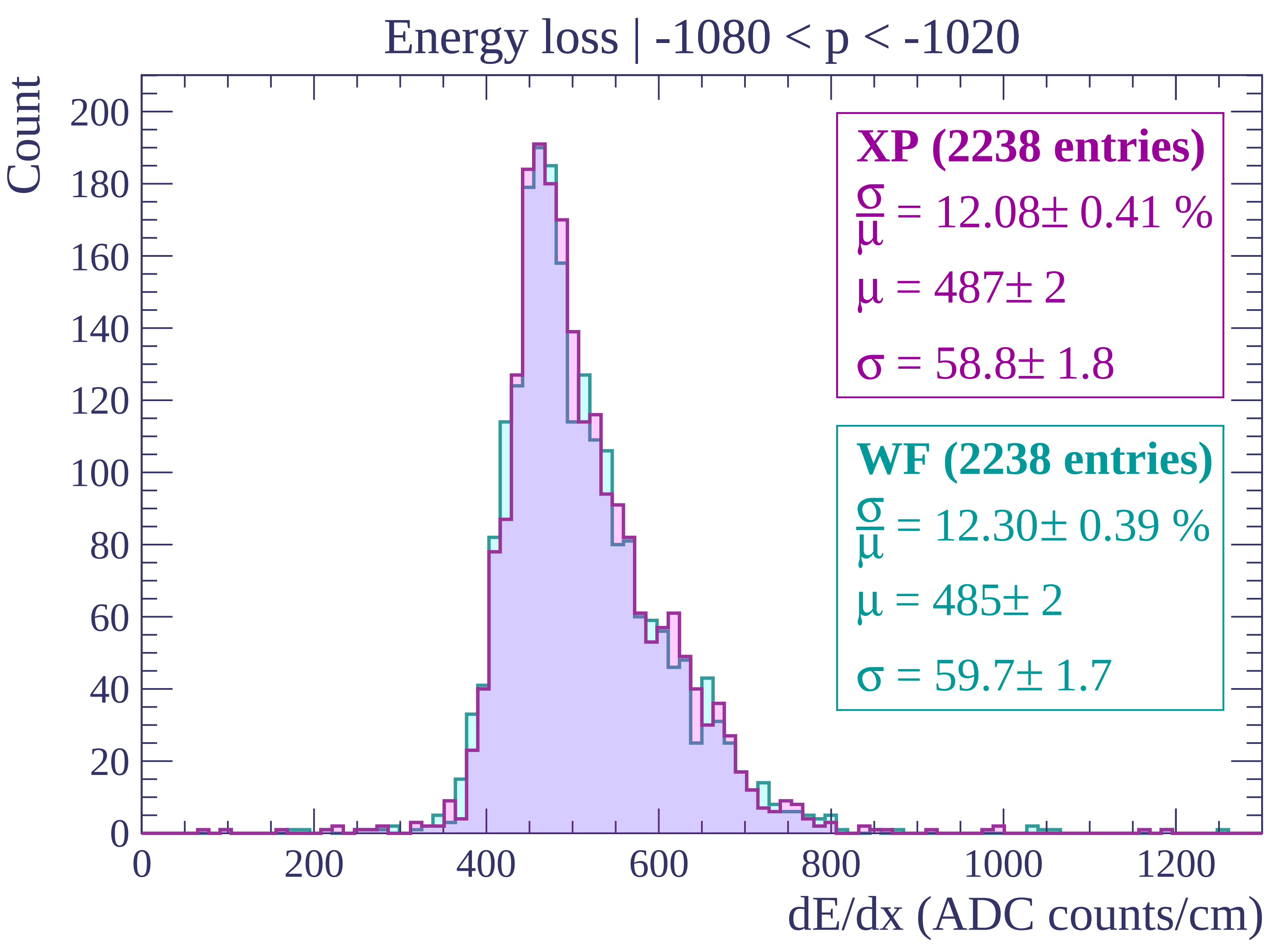
Energy loss | -1260 < p < -1200
Count
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2098 entries)
$\frac{\sigma}{\mu}$ = 12.44± 0.38 %
μ = 510± 2
σ = 63.4± 1.7
WF (2098 entries)
$\frac{\sigma}{\mu}$ = 12.92± 0.44 %
μ = 504± 2
σ = 65.1± 2.0

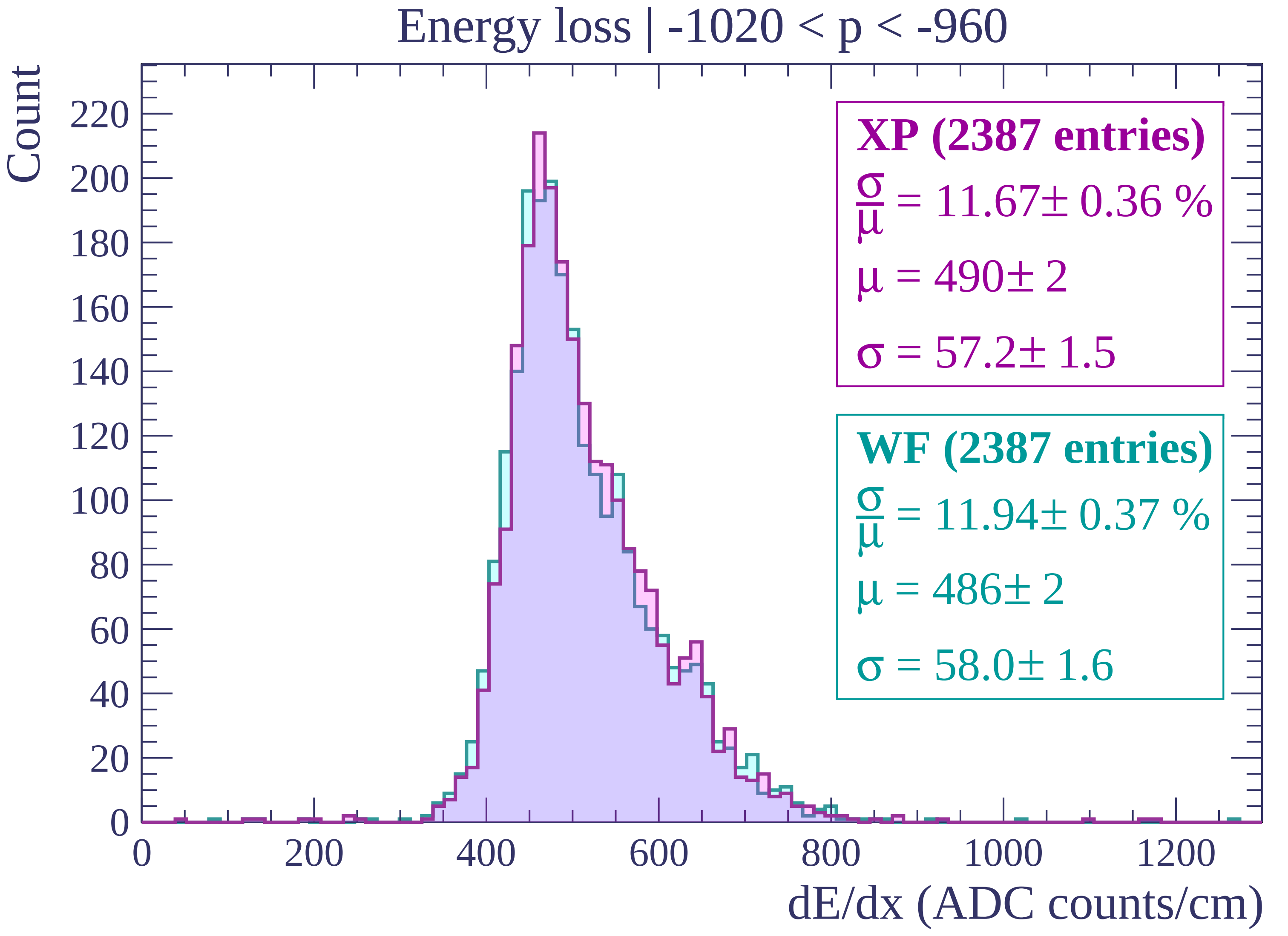


Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (2215 entries)
σ/μ = 13.09± 0.37 %
μ = 507± 2
σ = 66.4± 1.6
WF (2215 entries)
σ/μ = 13.34± 0.38 %
μ = 504± 2
σ = 67.2± 1.7

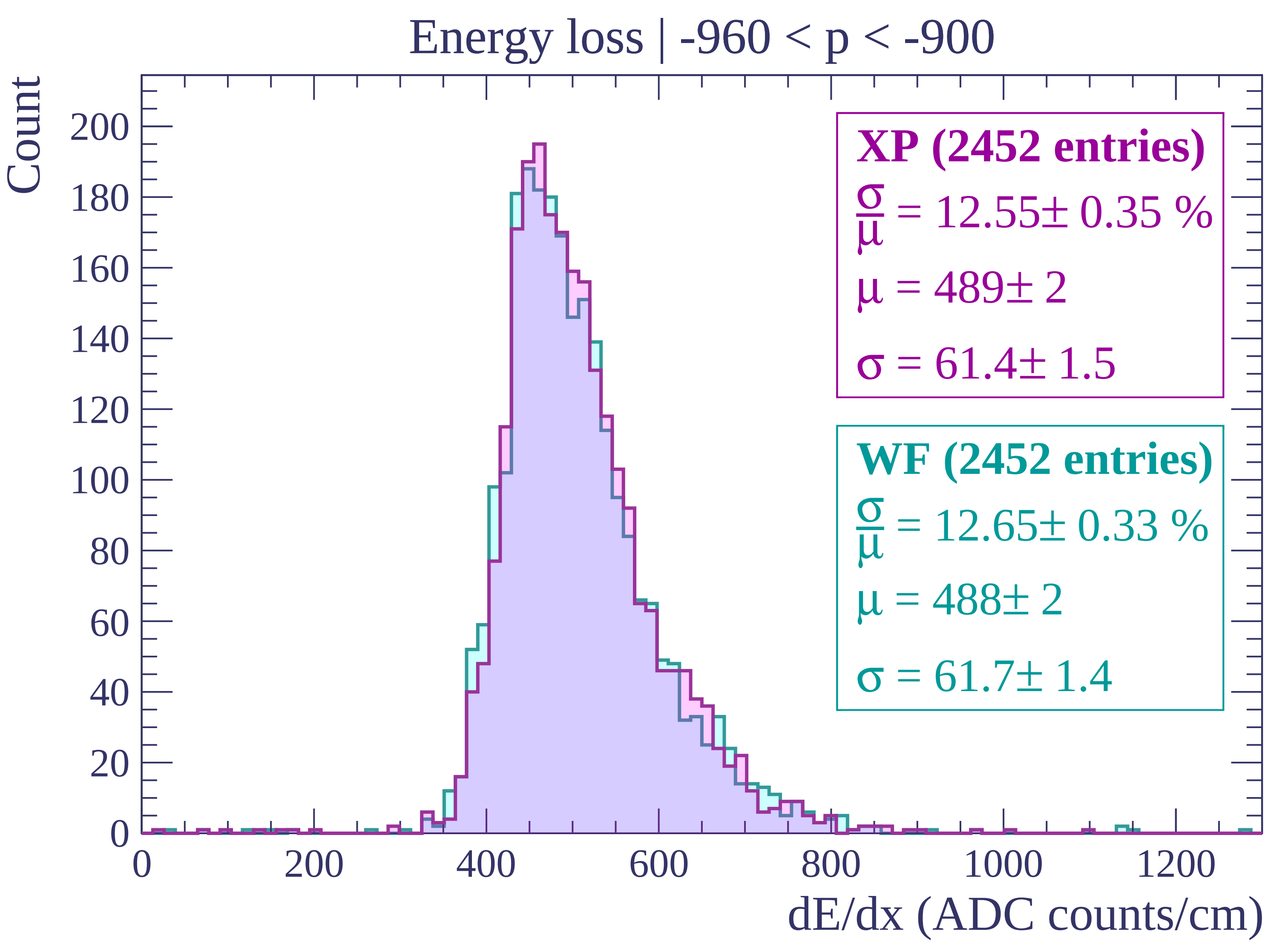


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
XP (2169 entries)
σ/μ = 12.37± 0.38 %
μ = 502± 2
σ = 62.1± 1.7
WF (2169 entries)
σ/μ = 12.65± 0.45 %
μ = 496± 2
σ = 62.7± 2.0

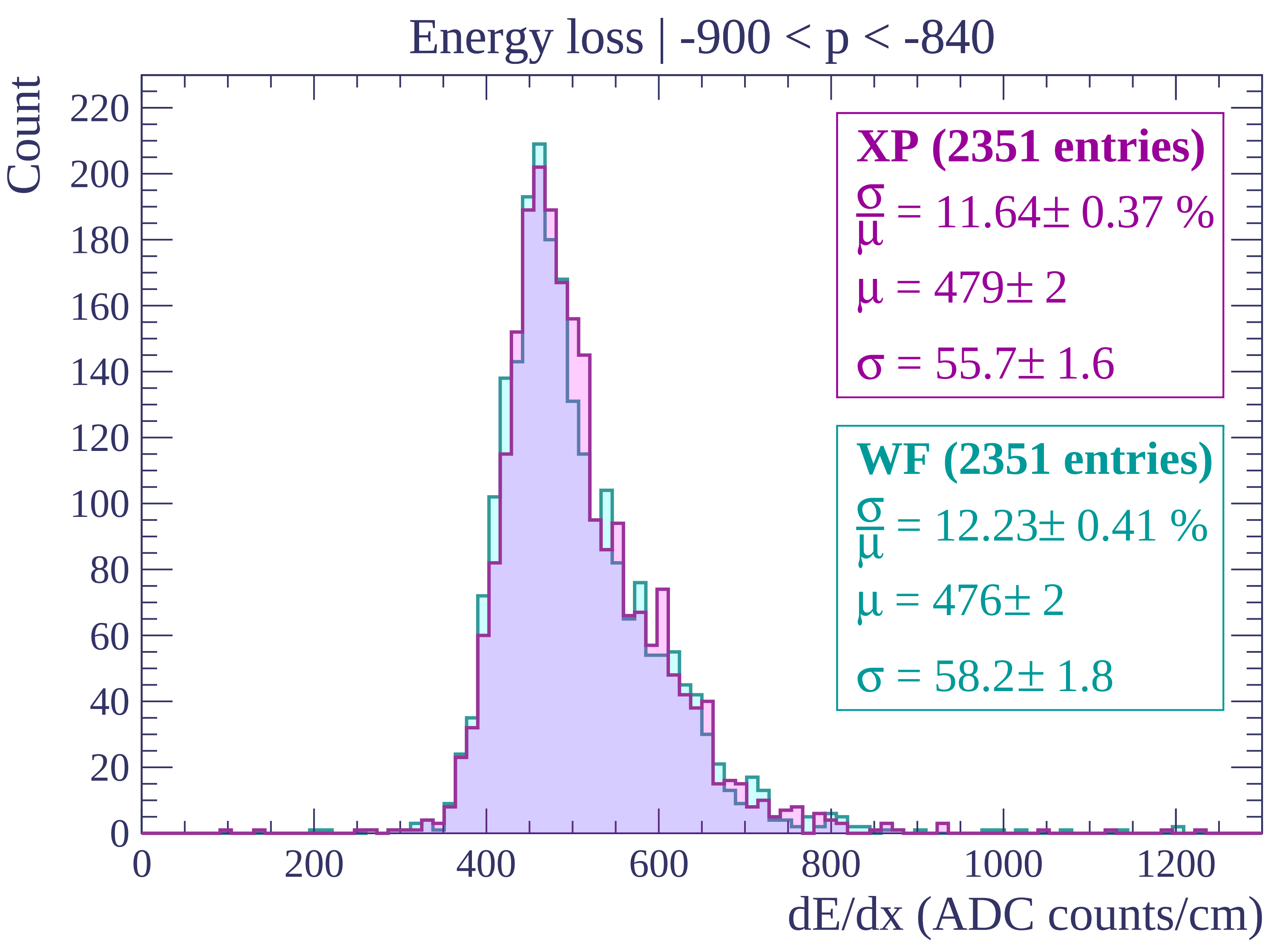

Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (2238 entries)
σ/μ = 12.08± 0.41 %
μ = 487± 2
σ = 58.8± 1.8
WF (2238 entries)
σ/μ = 12.30± 0.39 %
μ = 485± 2
σ = 59.7± 1.7



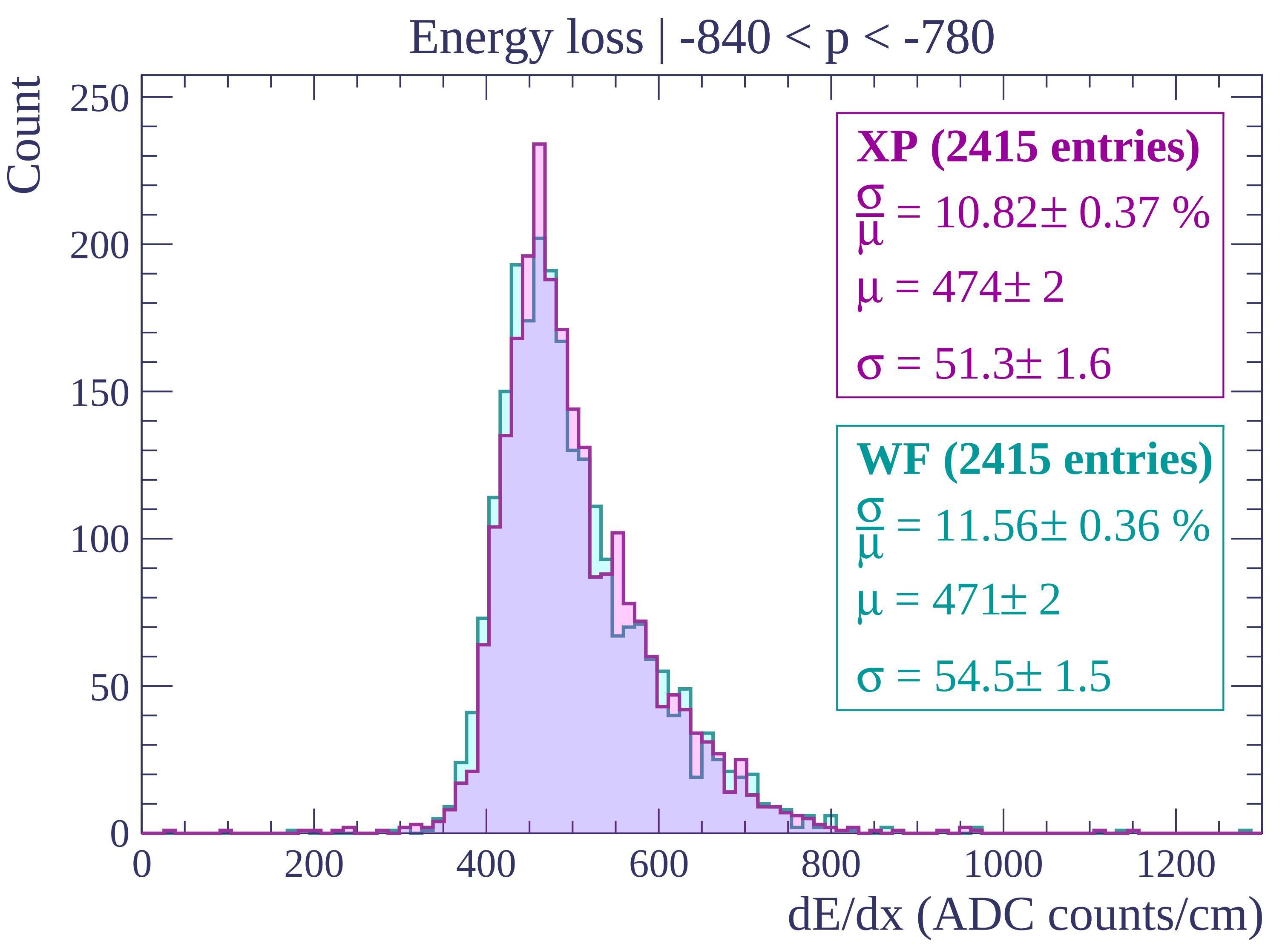
Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP (2387 entries)
$\frac{\sigma}{\mu}$ = 11.67± 0.36 %
μ = 490± 2
σ = 57.2± 1.5
WF (2387 entries)
$\frac{\sigma}{\mu}$ = 11.94± 0.37 %
μ = 486± 2
σ = 58.0± 1.6



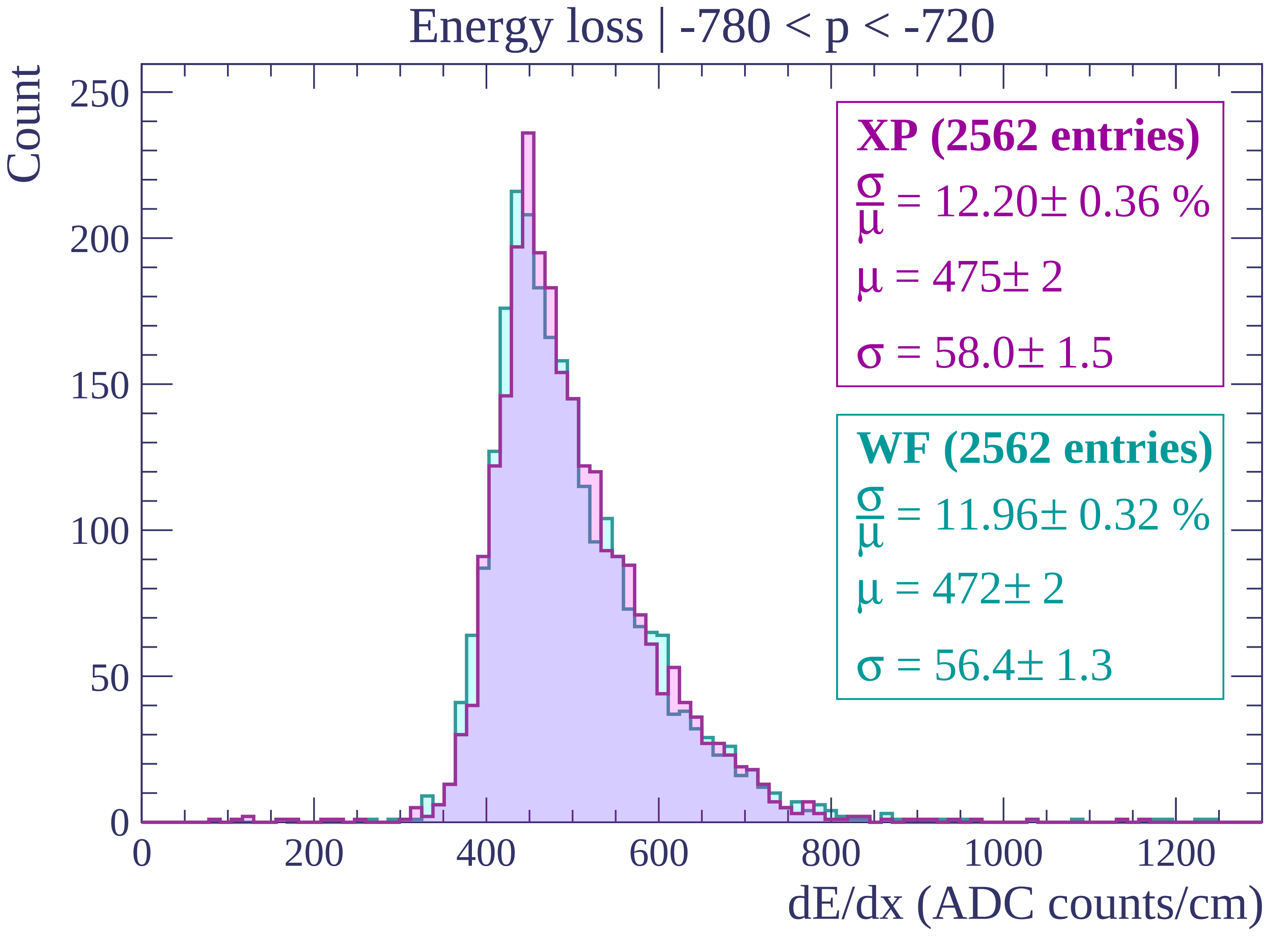
Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (2452 entries)
$\frac{\sigma}{\mu}$ = 12.55± 0.35 %
μ = 489± 2
σ = 61.4± 1.5
WF (2452 entries)
$\frac{\sigma}{\mu}$ = 12.65± 0.33 %
μ = 488± 2
σ = 61.7± 1.4

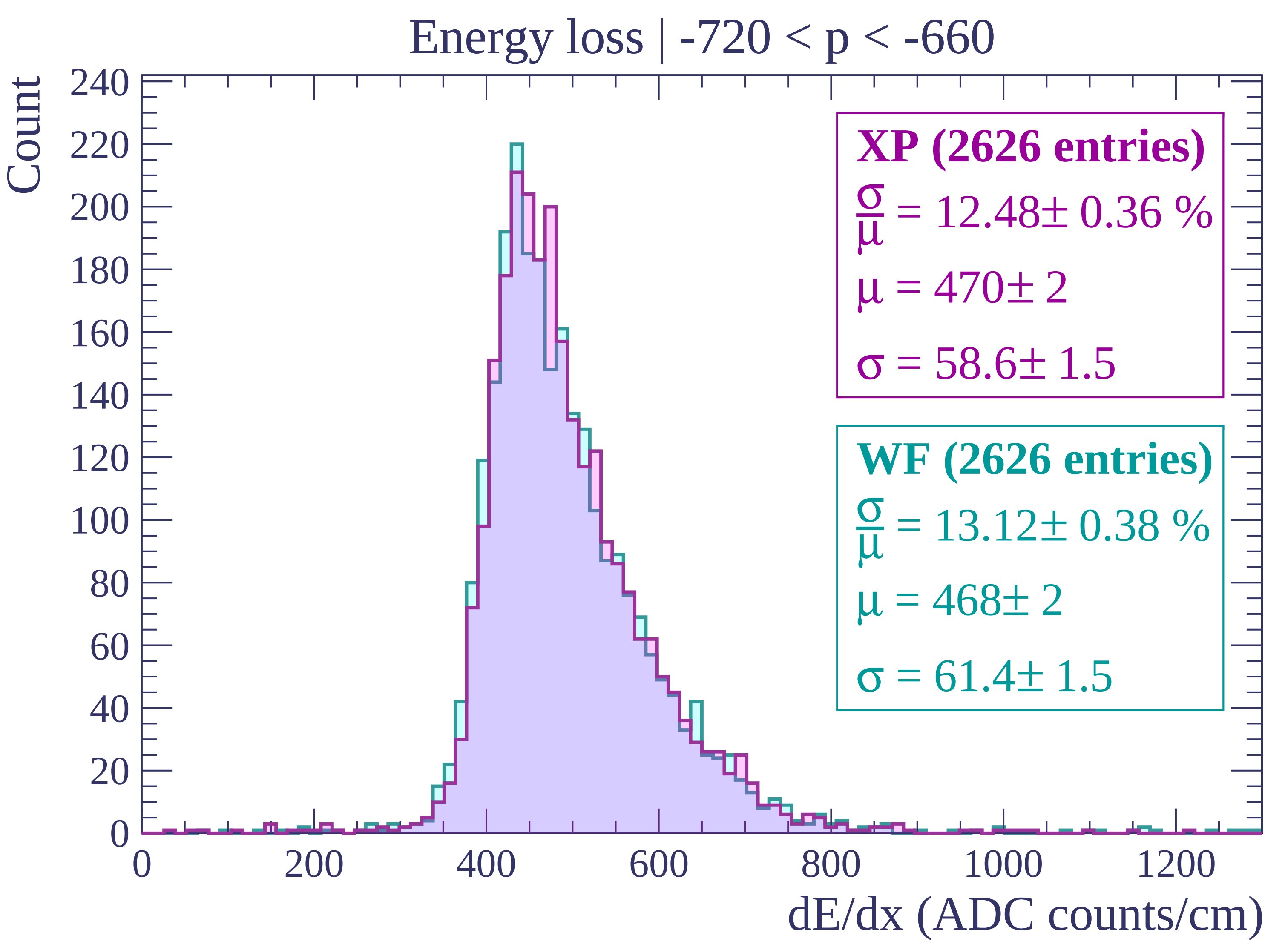


Energy loss | -900 < p < -840
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2351 entries)
σ/μ = 11.64± 0.37 %
μ = 479± 2
σ = 55.7± 1.6
WF (2351 entries)
σ/μ = 12.23± 0.41 %
μ = 476± 2
σ = 58.2± 1.8

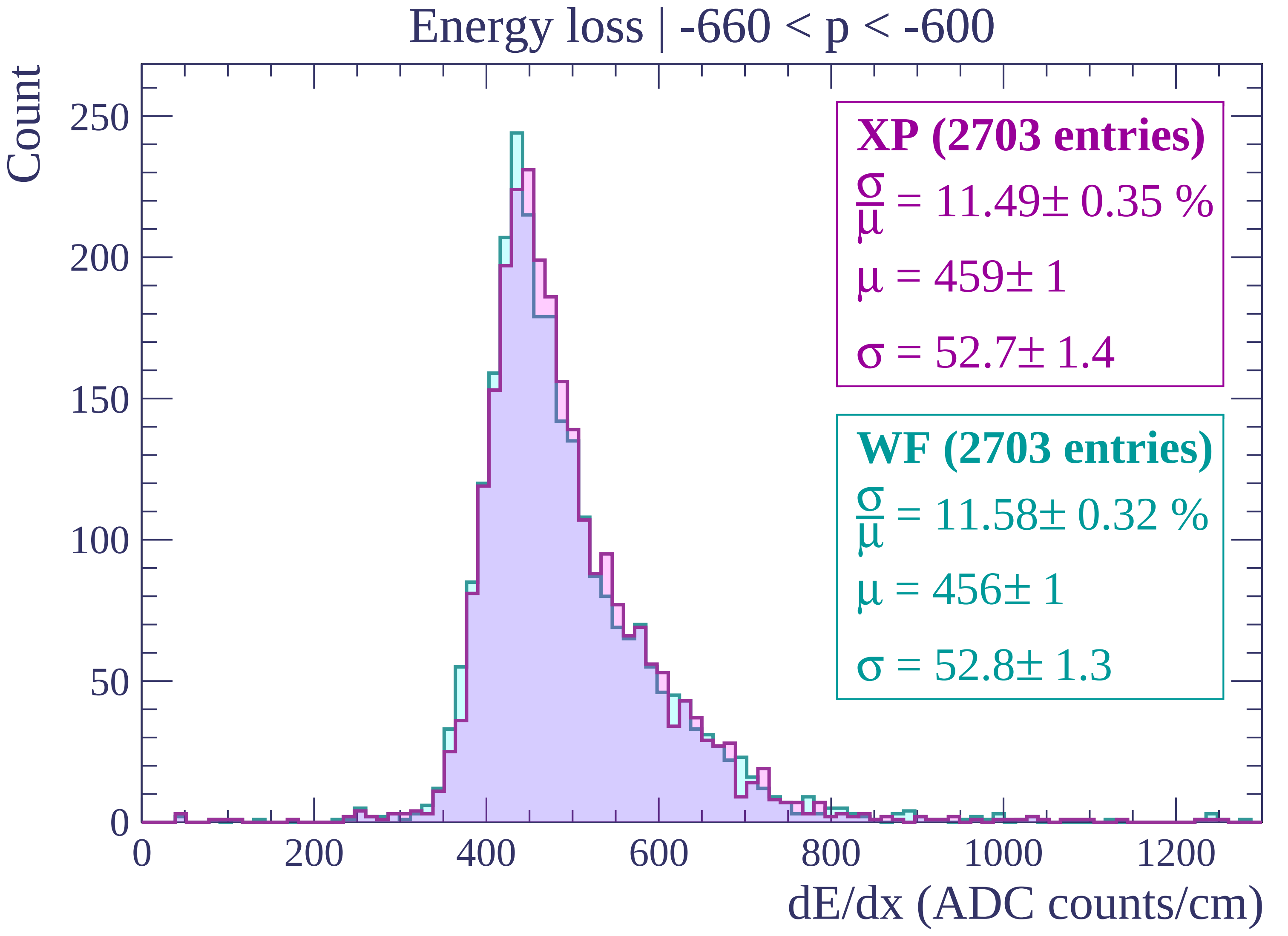


Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2415 entries)
σ/μ = 10.82± 0.37 %
μ = 474± 2
σ = 51.3± 1.6
WF (2415 entries)
σ/μ = 11.56± 0.36 %
μ = 471± 2
σ = 54.5± 1.5

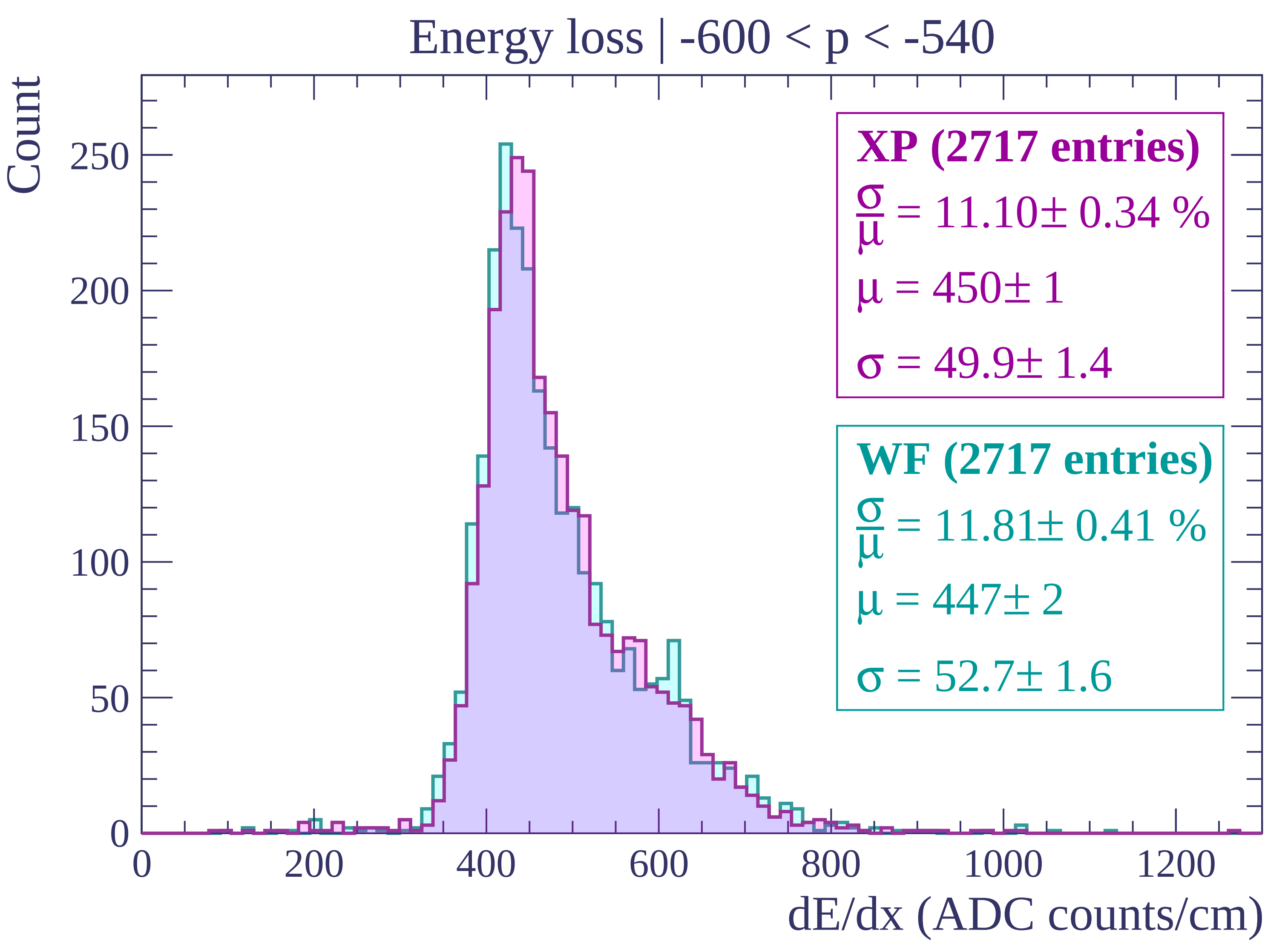

Energy loss | -780 < p < -720
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2562 entries)
$\frac{\sigma}{\mu}$ = 12.20± 0.36 %
μ = 475± 2
σ = 58.0± 1.5
WF (2562 entries)
$\frac{\sigma}{\mu}$ = 11.96± 0.32 %
μ = 472± 2
σ = 56.4± 1.3

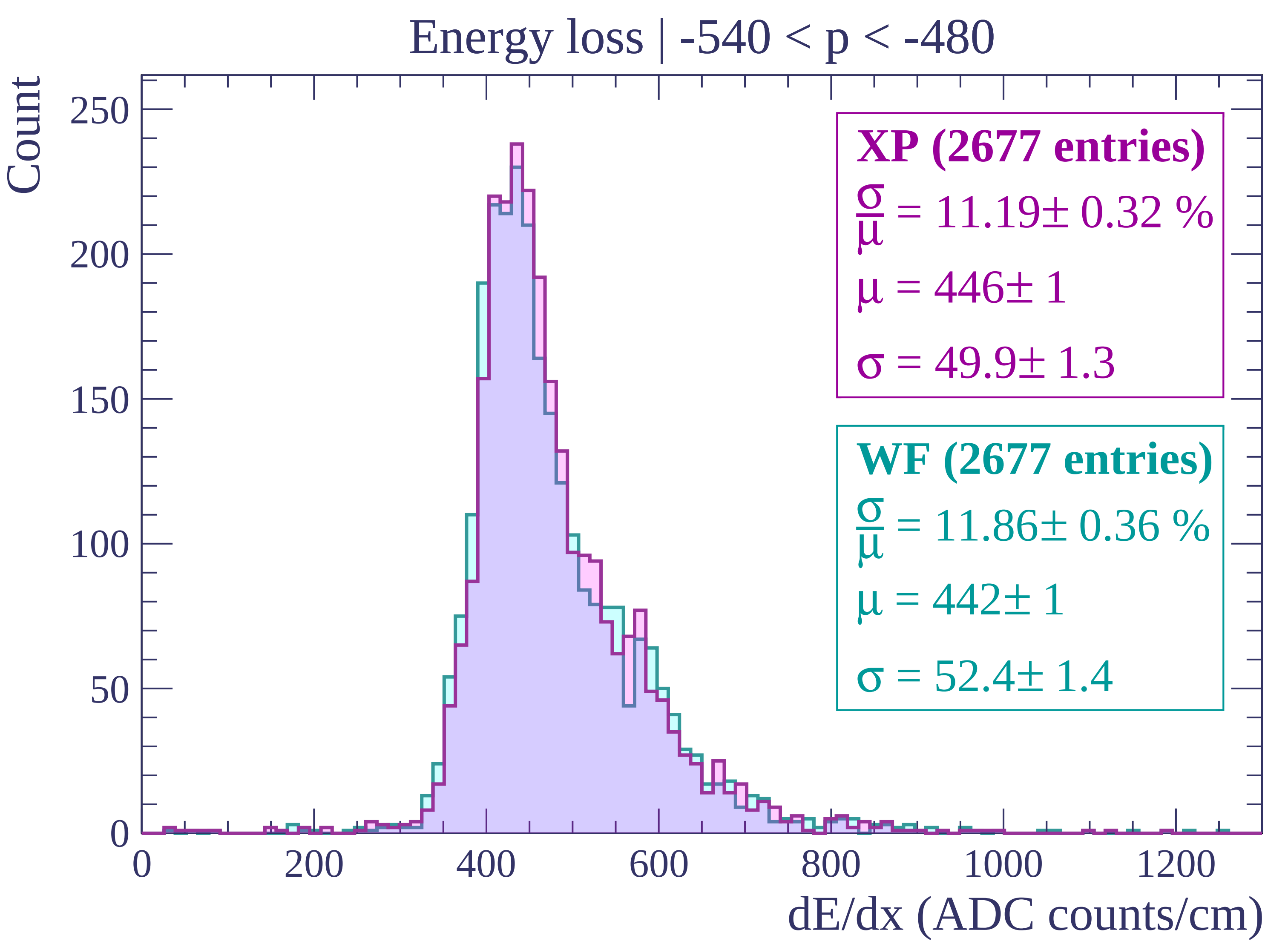


Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP (2626 entries)
σ/μ = 12.48± 0.36 %
μ = 470± 2
σ = 58.6± 1.5
WF (2626 entries)
σ/μ = 13.12± 0.38 %
μ = 468± 2
σ = 61.4± 1.5

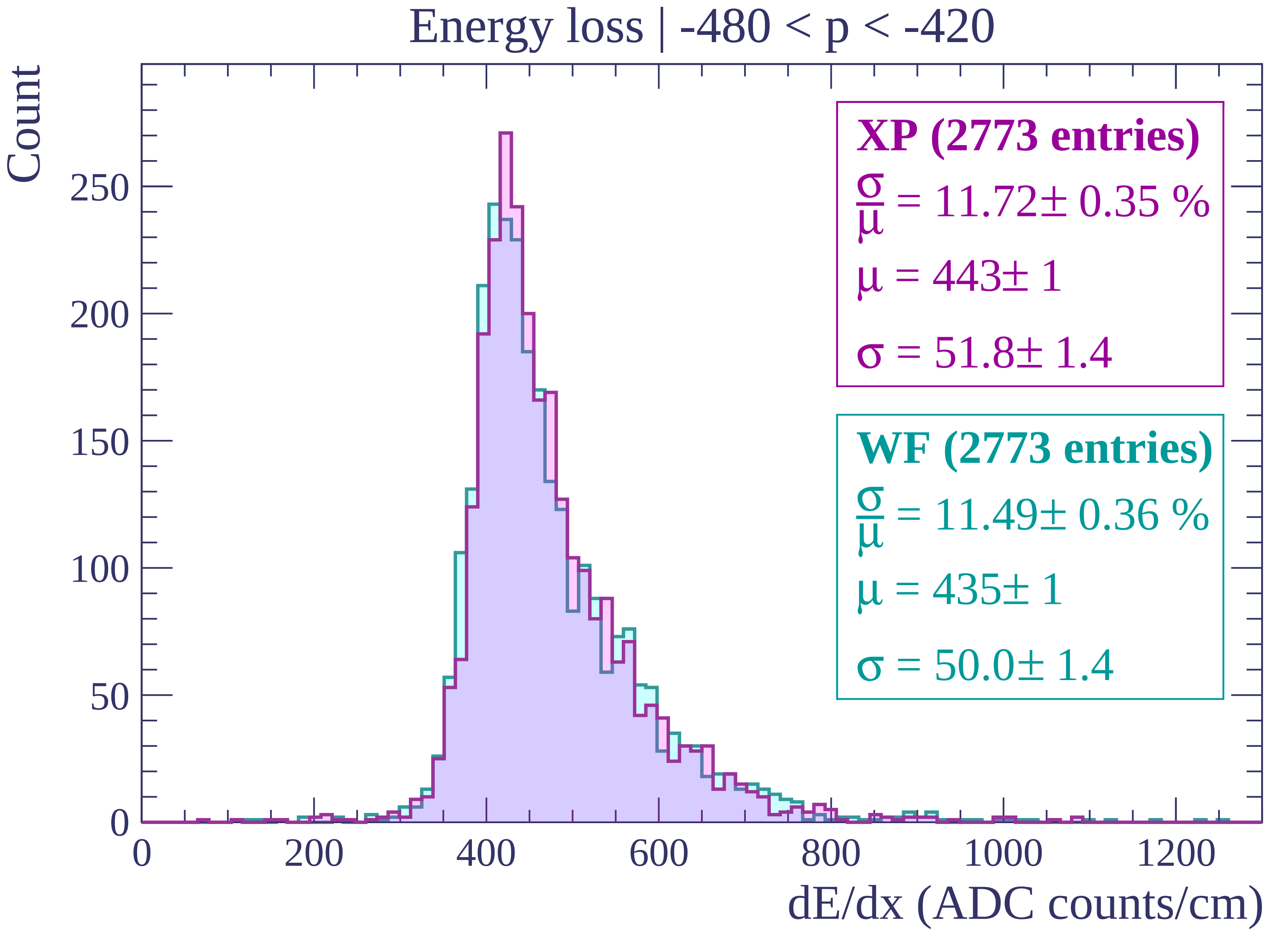


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2703 entries)
$\frac{\sigma}{\mu}$ = 11.49± 0.35 %
μ = 459± 1
σ = 52.7± 1.4
WF (2703 entries)
$\frac{\sigma}{\mu}$ = 11.58± 0.32 %
μ = 456± 1
σ = 52.8± 1.3

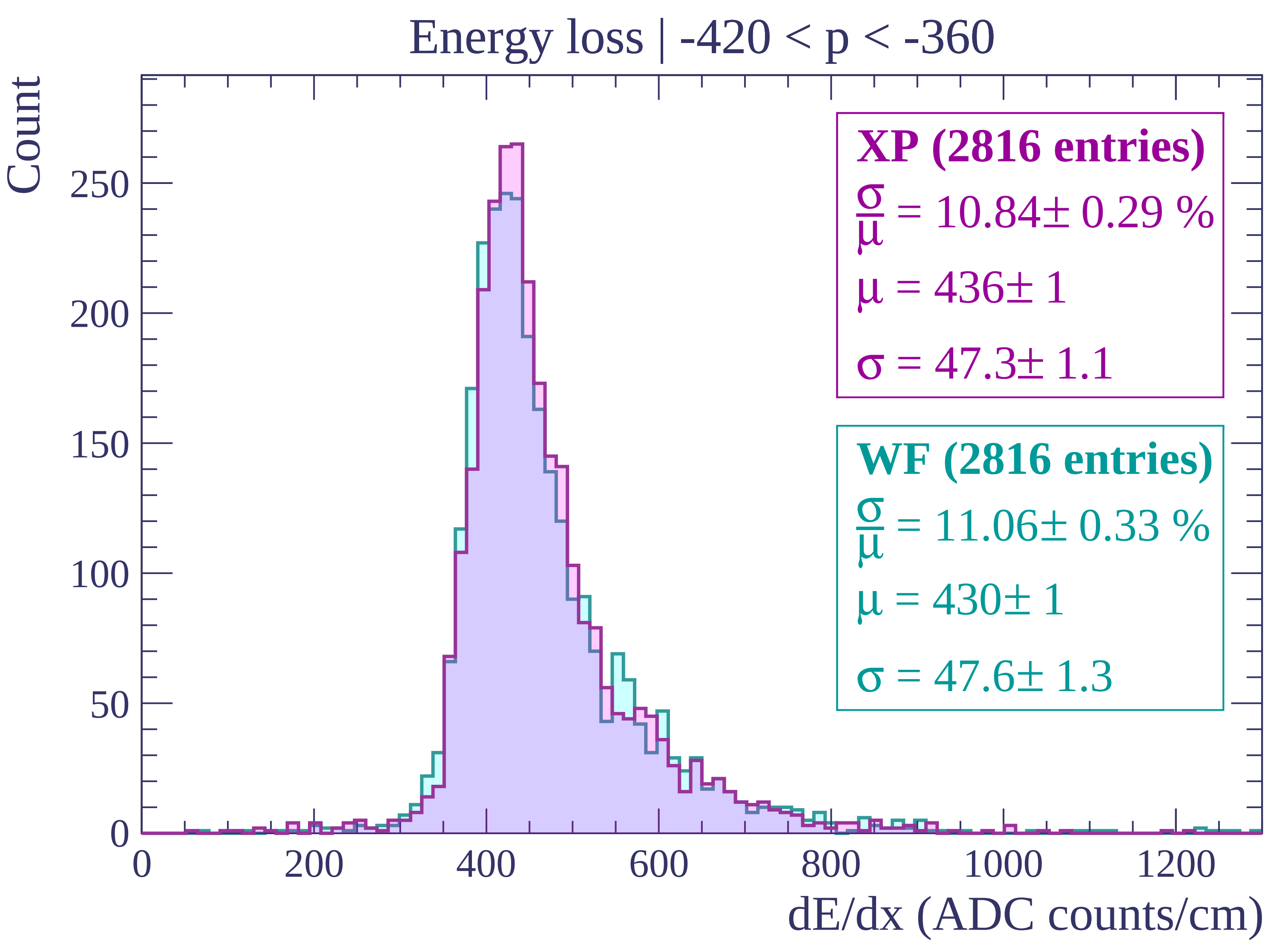

Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2717 entries)
σ/μ = 11.10± 0.34 %
μ = 450± 1
σ = 49.9± 1.4
WF (2717 entries)
σ/μ = 11.81± 0.41 %
μ = 447± 2
σ = 52.7± 1.6



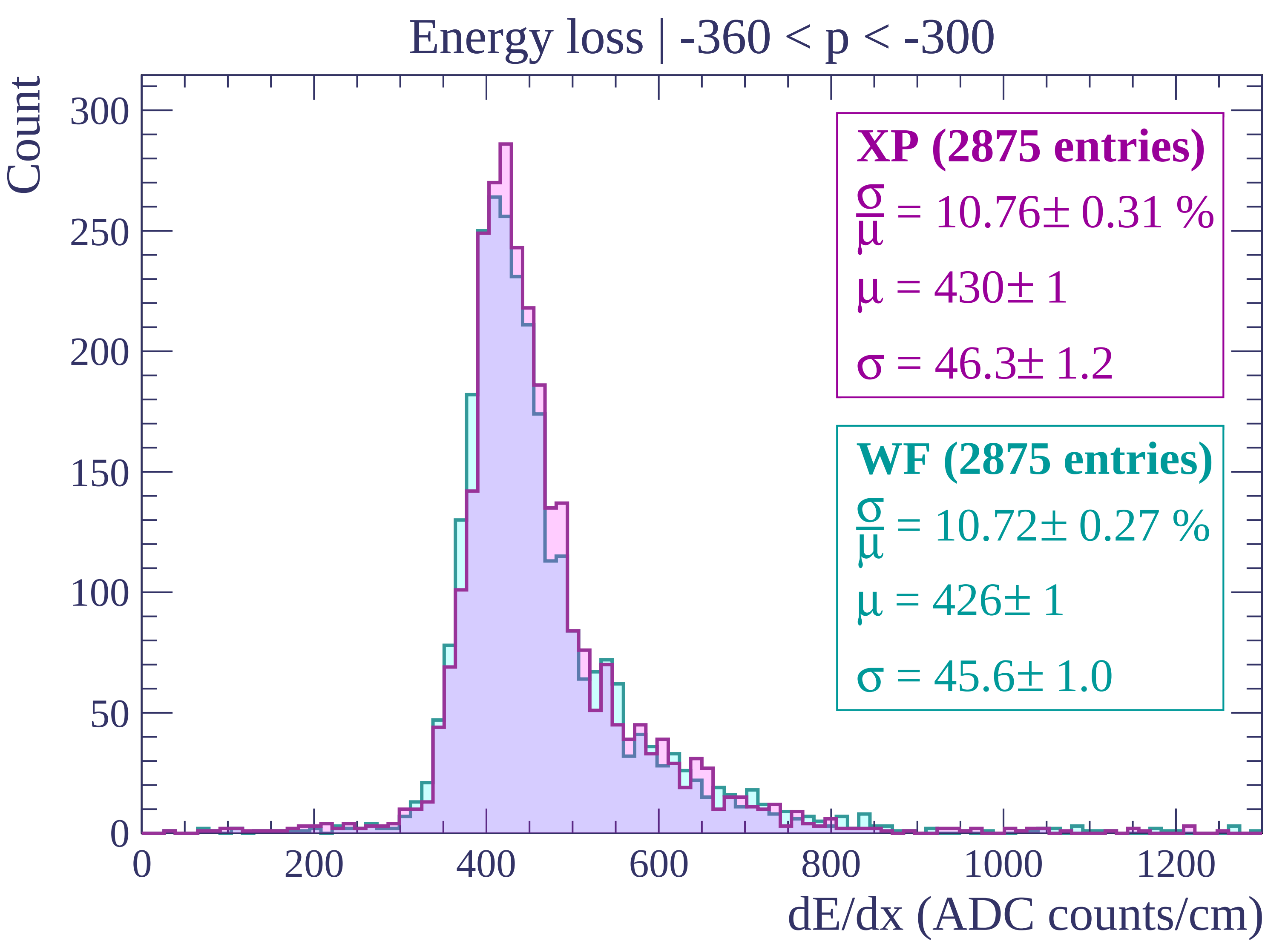
Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2677 entries)
σ/μ = 11.19± 0.32 %
μ = 446± 1
σ = 49.9± 1.3
WF (2677 entries)
σ/μ = 11.86± 0.36 %
μ = 442± 1
σ = 52.4± 1.4

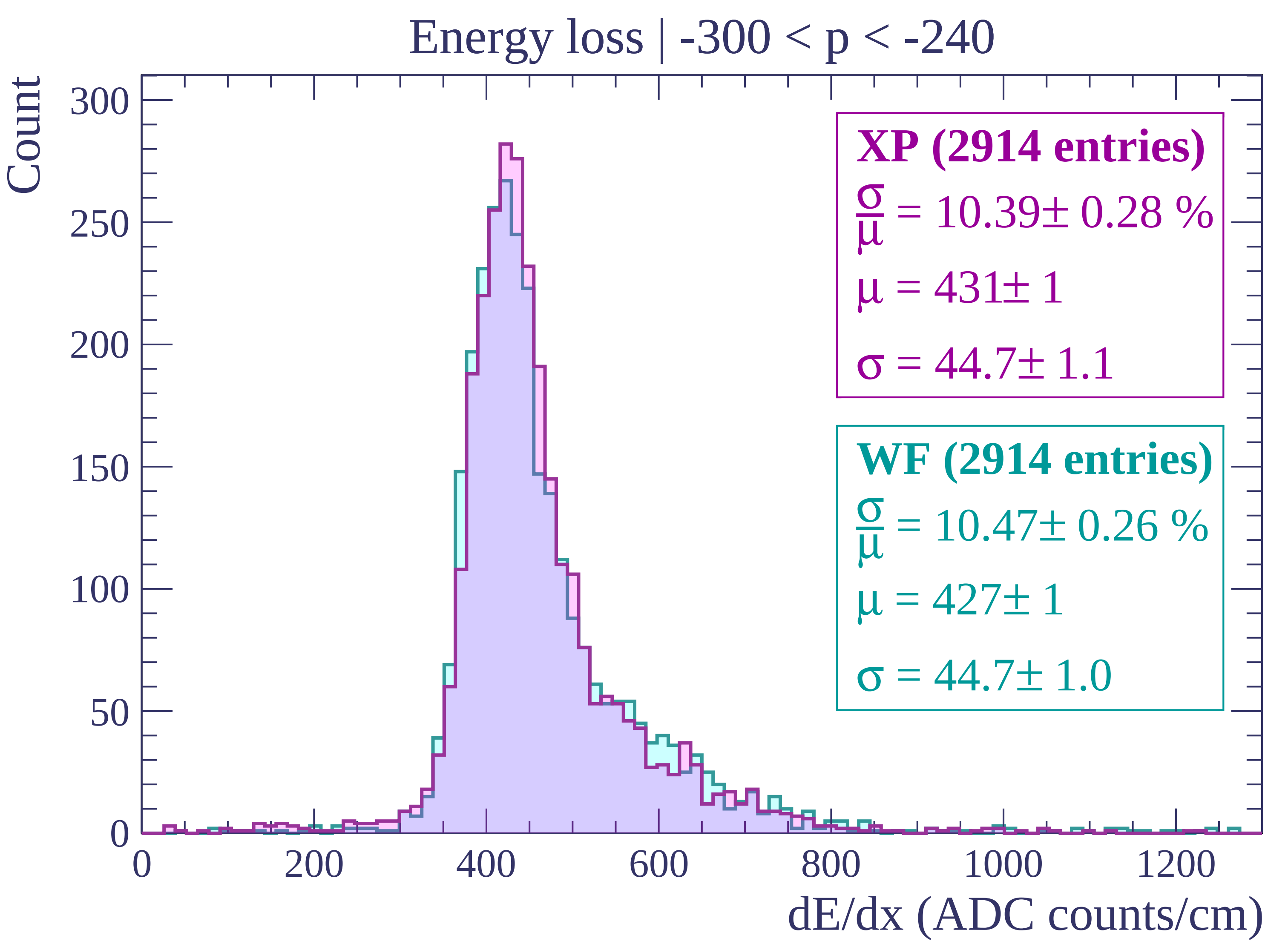


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2773 entries)
σ/μ = 11.72± 0.35 %
μ = 443± 1
σ = 51.8± 1.4
WF (2773 entries)
σ/μ = 11.49± 0.36 %
μ = 435± 1
σ = 50.0± 1.4

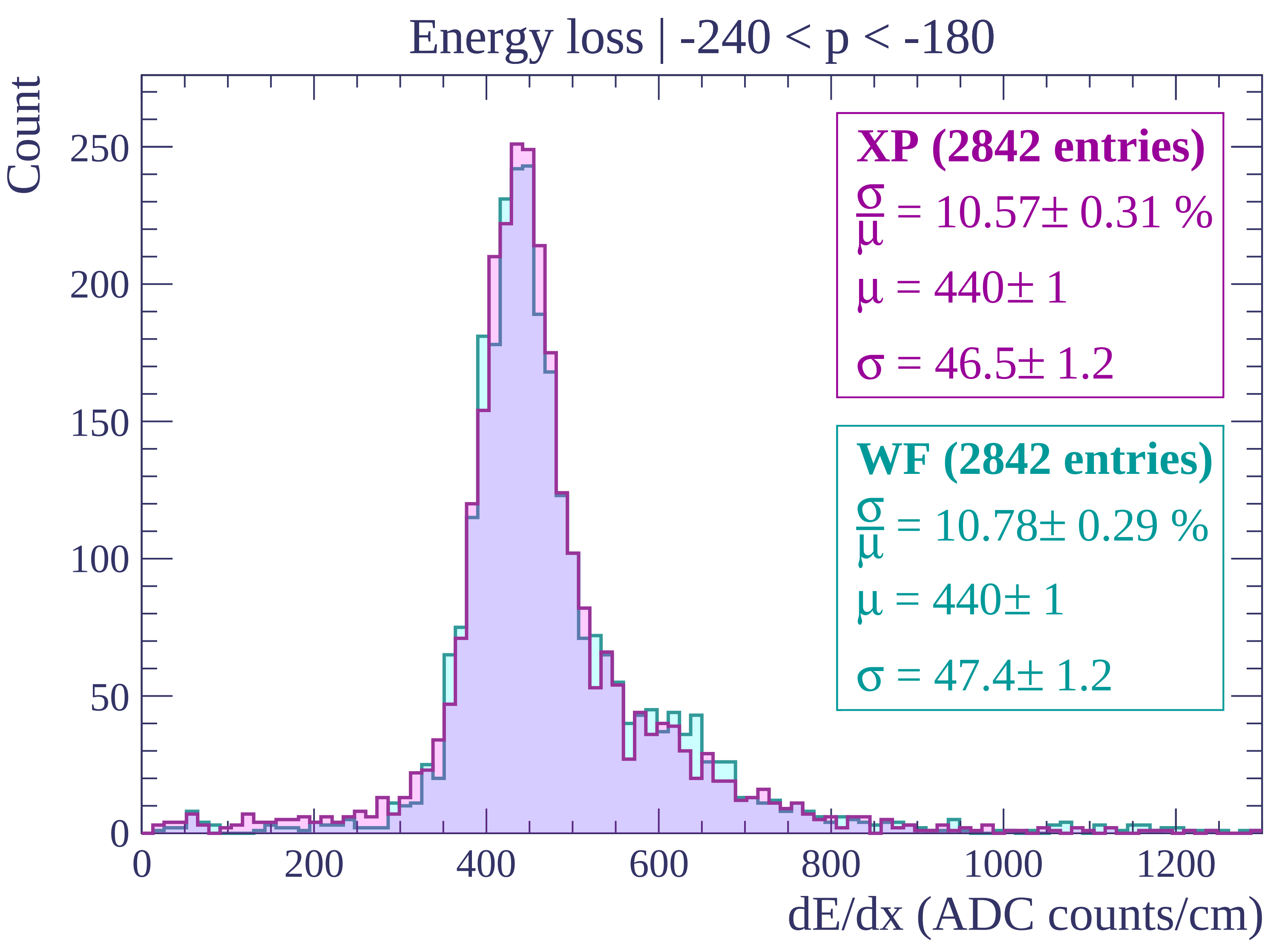


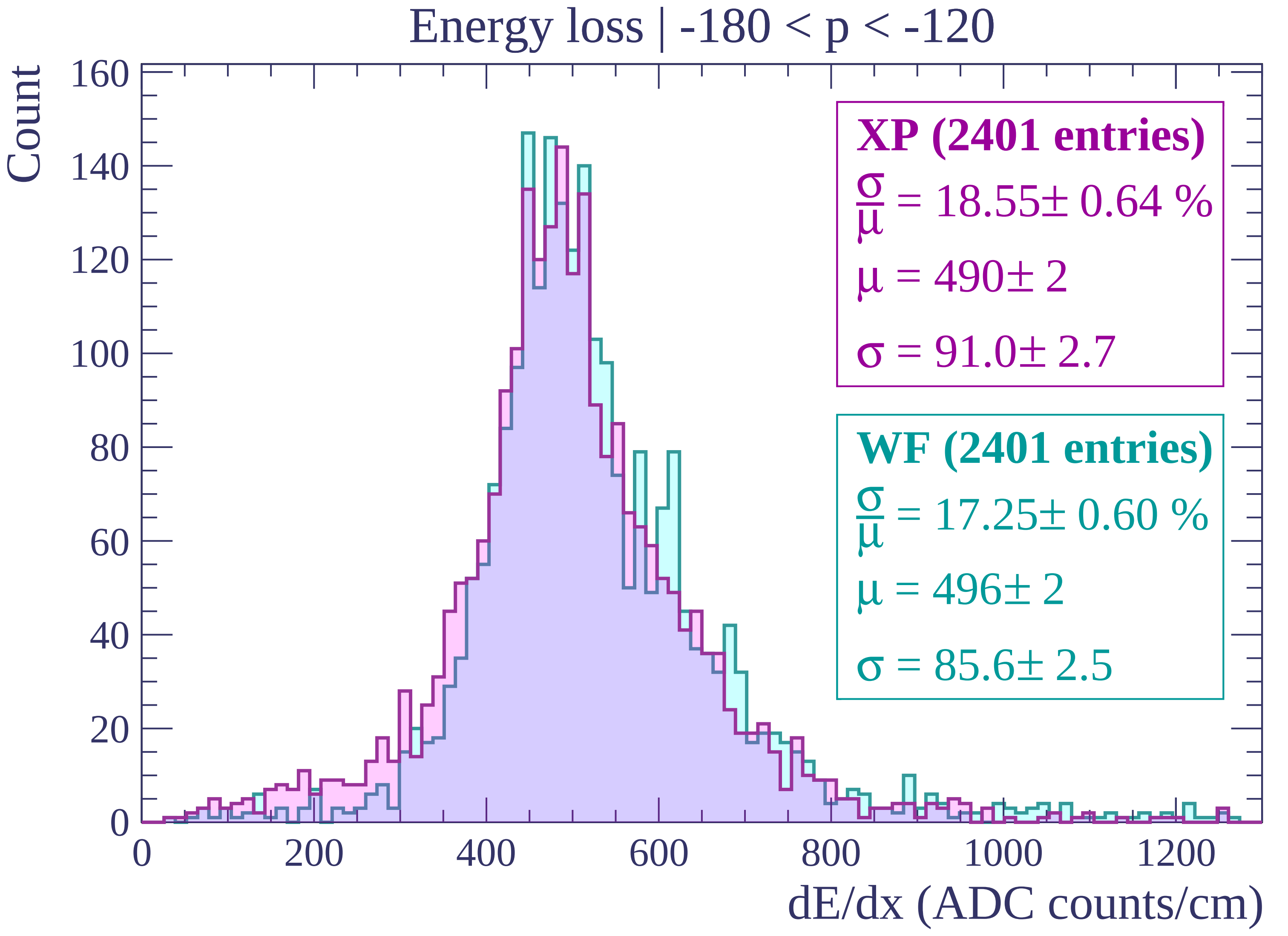
Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2816 entries)
$\frac{\sigma}{\mu}$ = 10.84± 0.29 %
μ = 436± 1
σ = 47.3± 1.1
WF (2816 entries)
$\frac{\sigma}{\mu}$ = 11.06± 0.33 %
μ = 430± 1
σ = 47.6± 1.3


Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP (2875 entries)
σ/μ = 10.76± 0.31 %
μ = 430± 1
σ = 46.3± 1.2
WF (2875 entries)
σ/μ = 10.72± 0.27 %
μ = 426± 1
σ = 45.6± 1.0

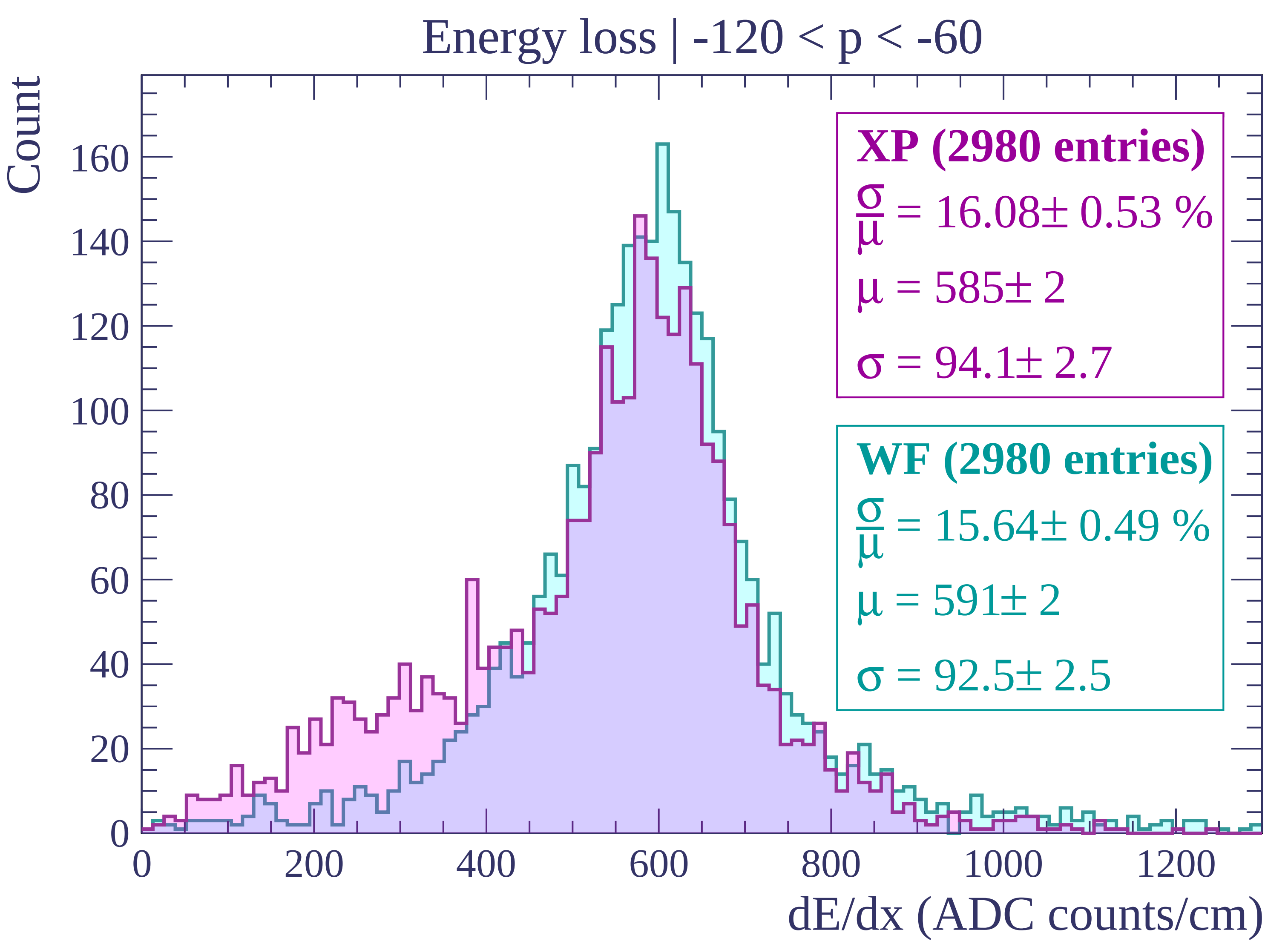


Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2914 entries)
$\frac{\sigma}{\mu}$ = 10.39± 0.28 %
μ = 431± 1
σ = 44.7± 1.1
WF (2914 entries)
$\frac{\sigma}{\mu}$ = 10.47± 0.26 %
μ = 427± 1
σ = 44.7± 1.0

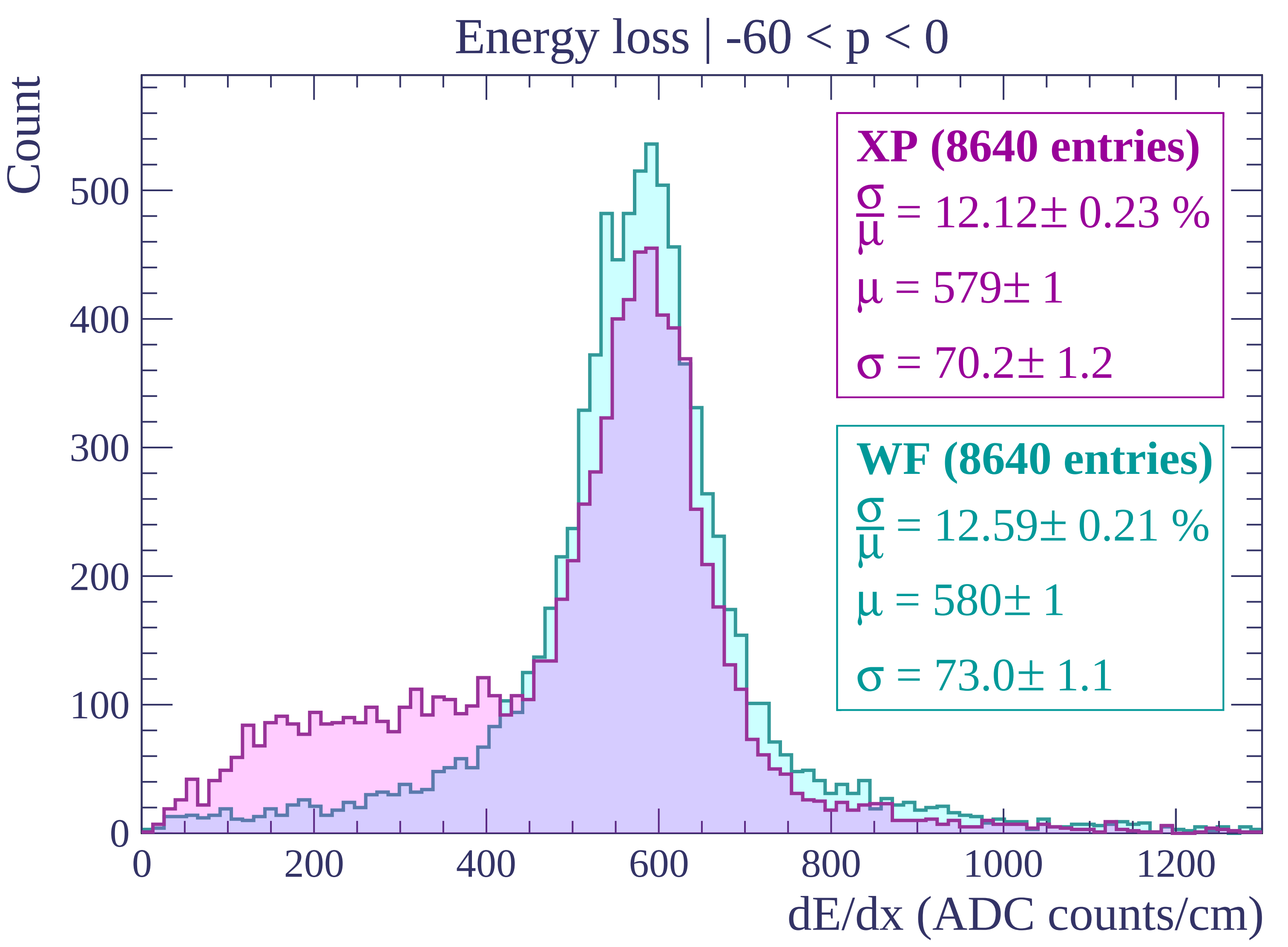

Energy loss | -240 < p < -180
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2842 entries)
σ/μ = 10.57± 0.31 %
μ = 440± 1
σ = 46.5± 1.2
WF (2842 entries)
σ/μ = 10.78± 0.29 %
μ = 440± 1
σ = 47.4± 1.2



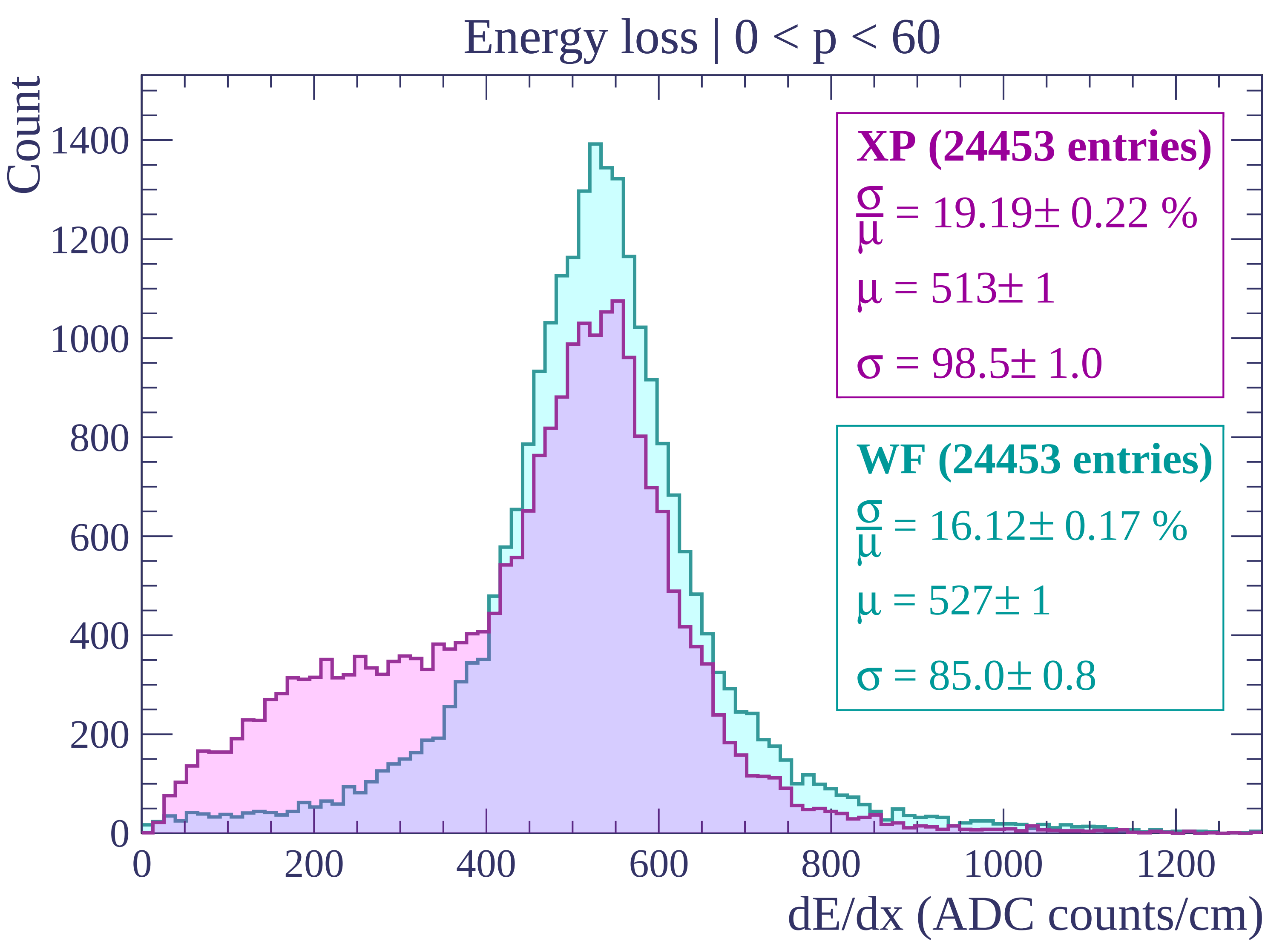
Energy loss | -180 < p < -120
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2401 entries)
σ/μ = 18.55± 0.64 %
μ = 490± 2
σ = 91.0± 2.7
WF (2401 entries)
σ/μ = 17.25± 0.60 %
μ = 496± 2
σ = 85.6± 2.5

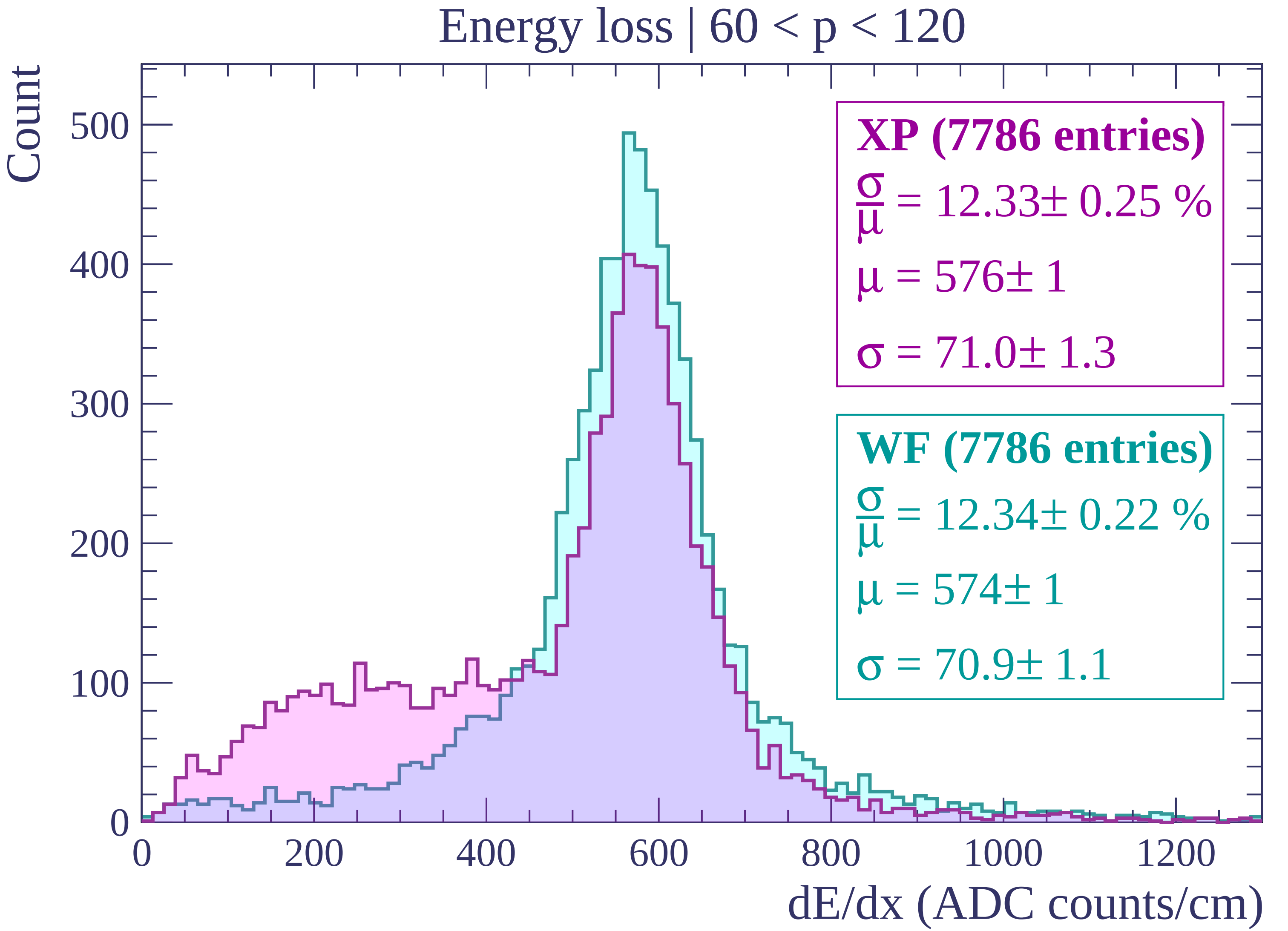


Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (2980 entries)
σ/μ = 16.08± 0.53 %
μ = 585± 2
σ = 94.1± 2.7
WF (2980 entries)
σ/μ = 15.64± 0.49 %
μ = 591± 2
σ = 92.5± 2.5

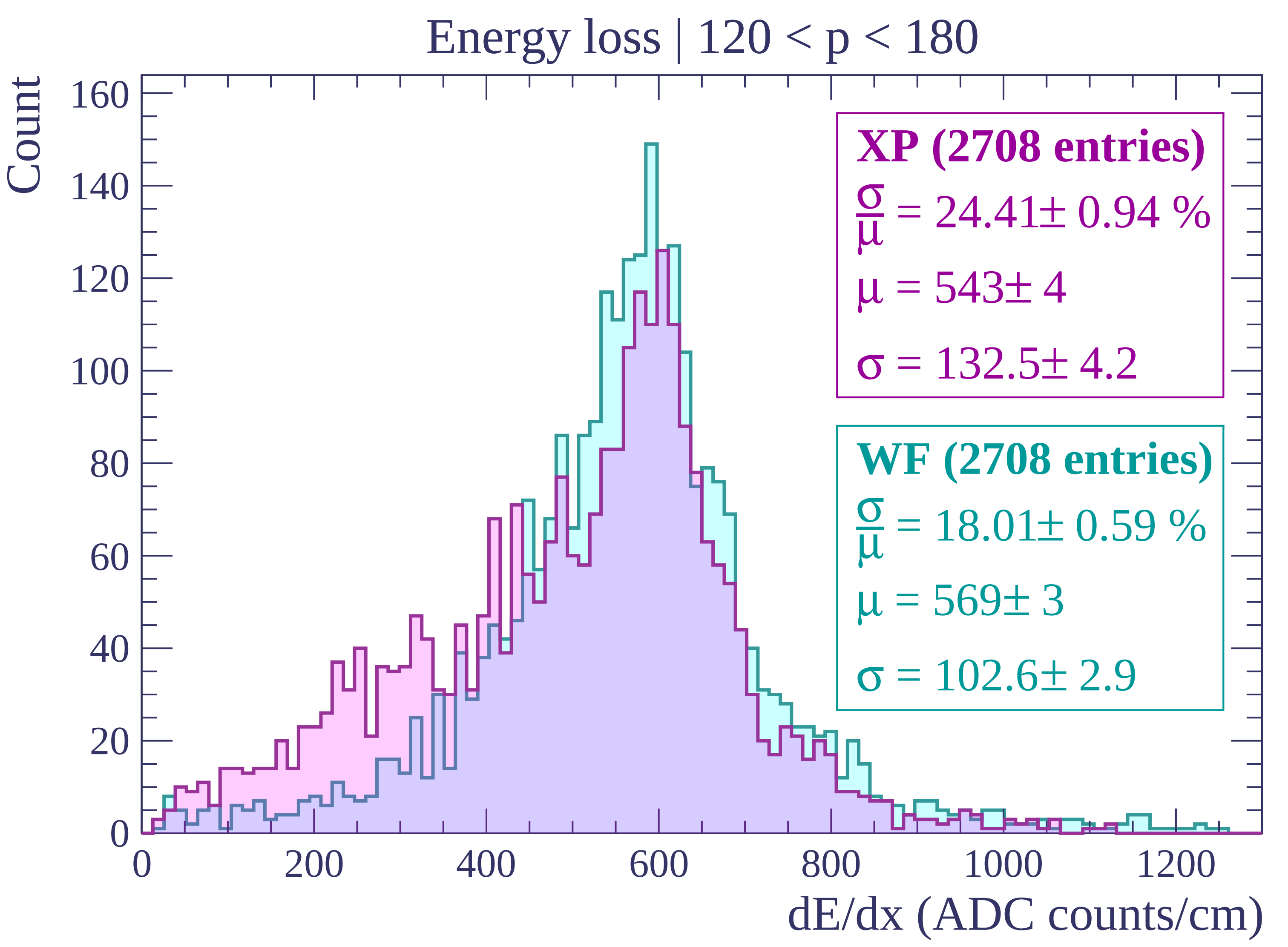


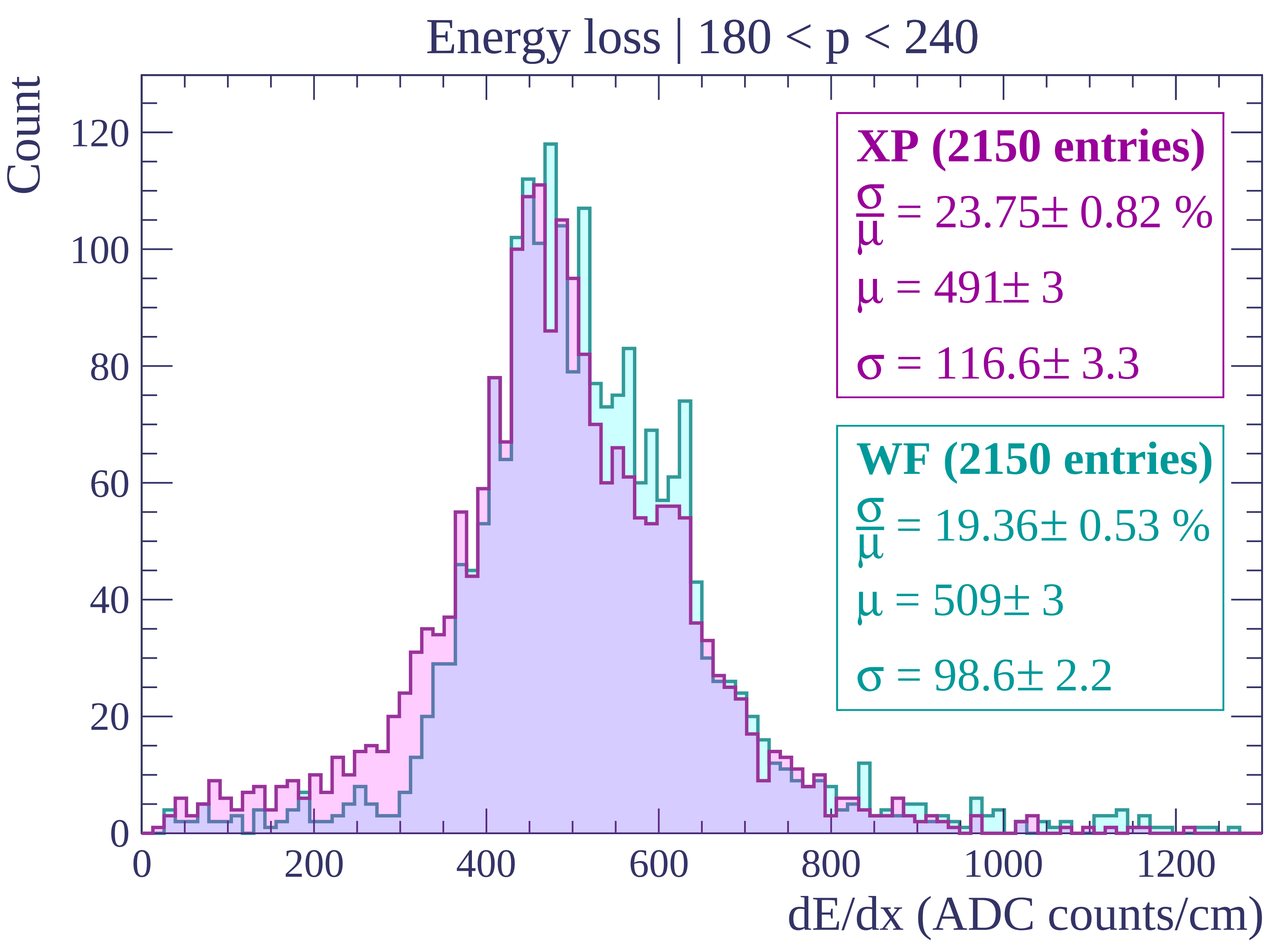
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
XP (8640 entries)
$\frac{\sigma}{\mu}$ = 12.12± 0.23 %
μ = 579± 1
σ = 70.2± 1.2
WF (8640 entries)
$\frac{\sigma}{\mu}$ = 12.59± 0.21 %
μ = 580± 1
σ = 73.0± 1.1
0
100
200
300
400
500
0
200
400
600
800
1000
1200

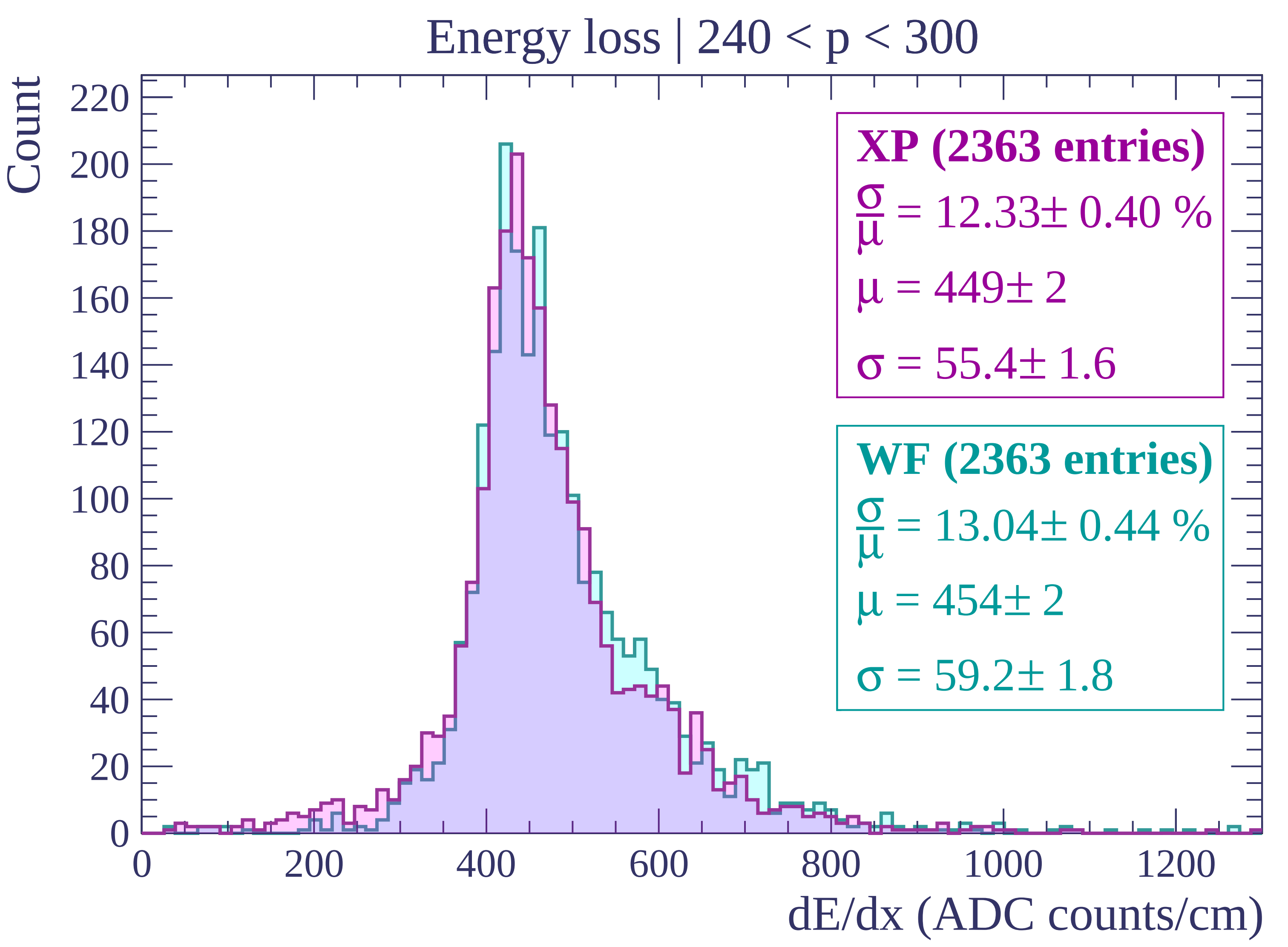

Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP (24453 entries)
$\frac{\sigma}{\mu}$ = 19.19± 0.22 %
μ = 513± 1
σ = 98.5± 1.0
WF (24453 entries)
$\frac{\sigma}{\mu}$ = 16.12± 0.17 %
μ = 527± 1
σ = 85.0± 0.8
0
200
400
600
800
1000
1200
1400

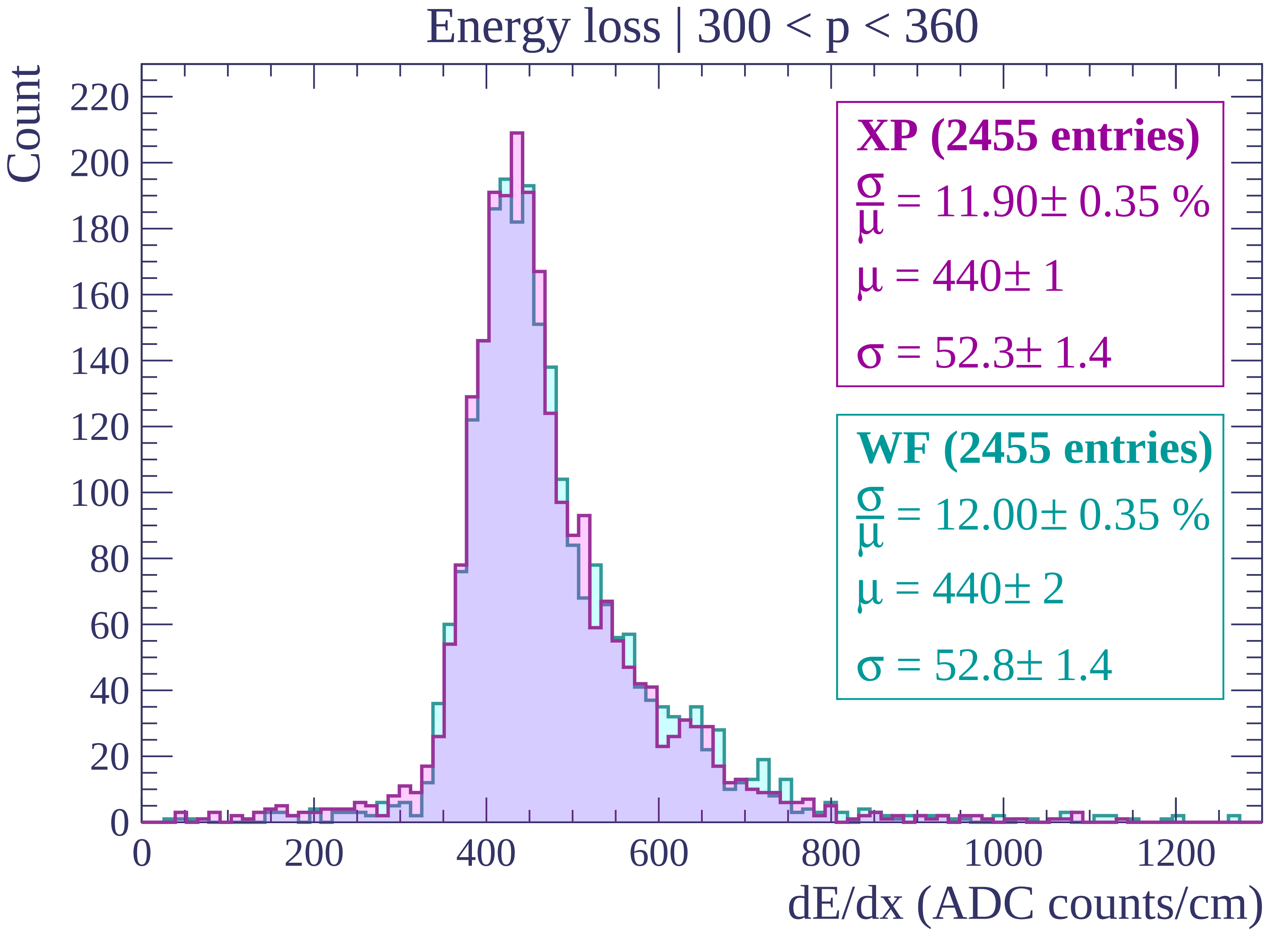


Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (7786 entries)
σ/μ = 12.33± 0.25 %
μ = 576± 1
σ = 71.0± 1.3
WF (7786 entries)
σ/μ = 12.34± 0.22 %
μ = 574± 1
σ = 70.9± 1.1

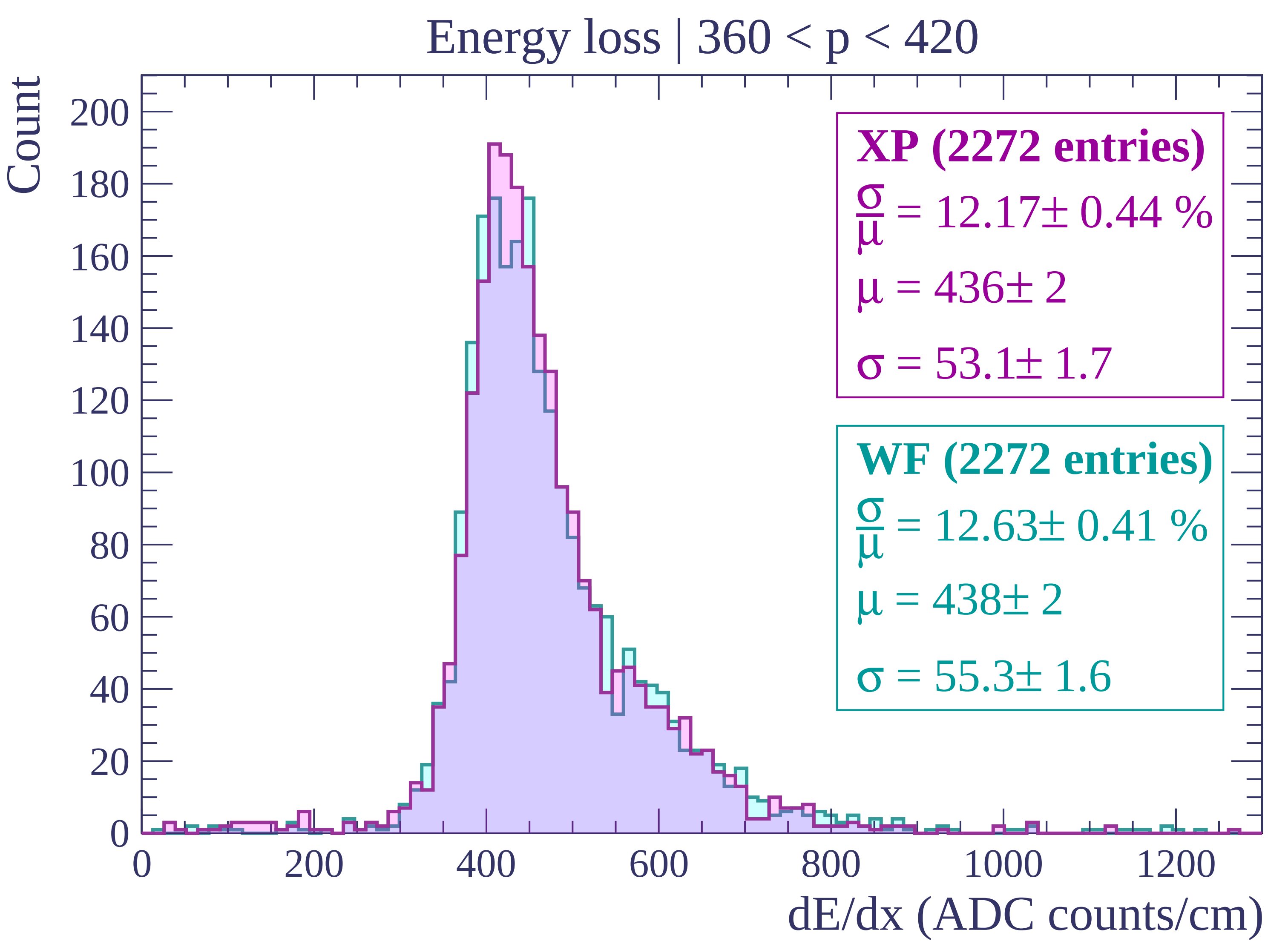

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP (2708 entries)
$\frac{\sigma}{\mu}$ = 24.41± 0.94 %
μ = 543± 4
σ = 132.5± 4.2
WF (2708 entries)
$\frac{\sigma}{\mu}$ = 18.01± 0.59 %
μ = 569± 3
σ = 102.6± 2.9


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (2150 entries)
$\frac{\sigma}{\mu}$ = 23.75± 0.82 %
μ = 491± 3
σ = 116.6± 3.3
WF (2150 entries)
$\frac{\sigma}{\mu}$ = 19.36± 0.53 %
μ = 509± 3
σ = 98.6± 2.2

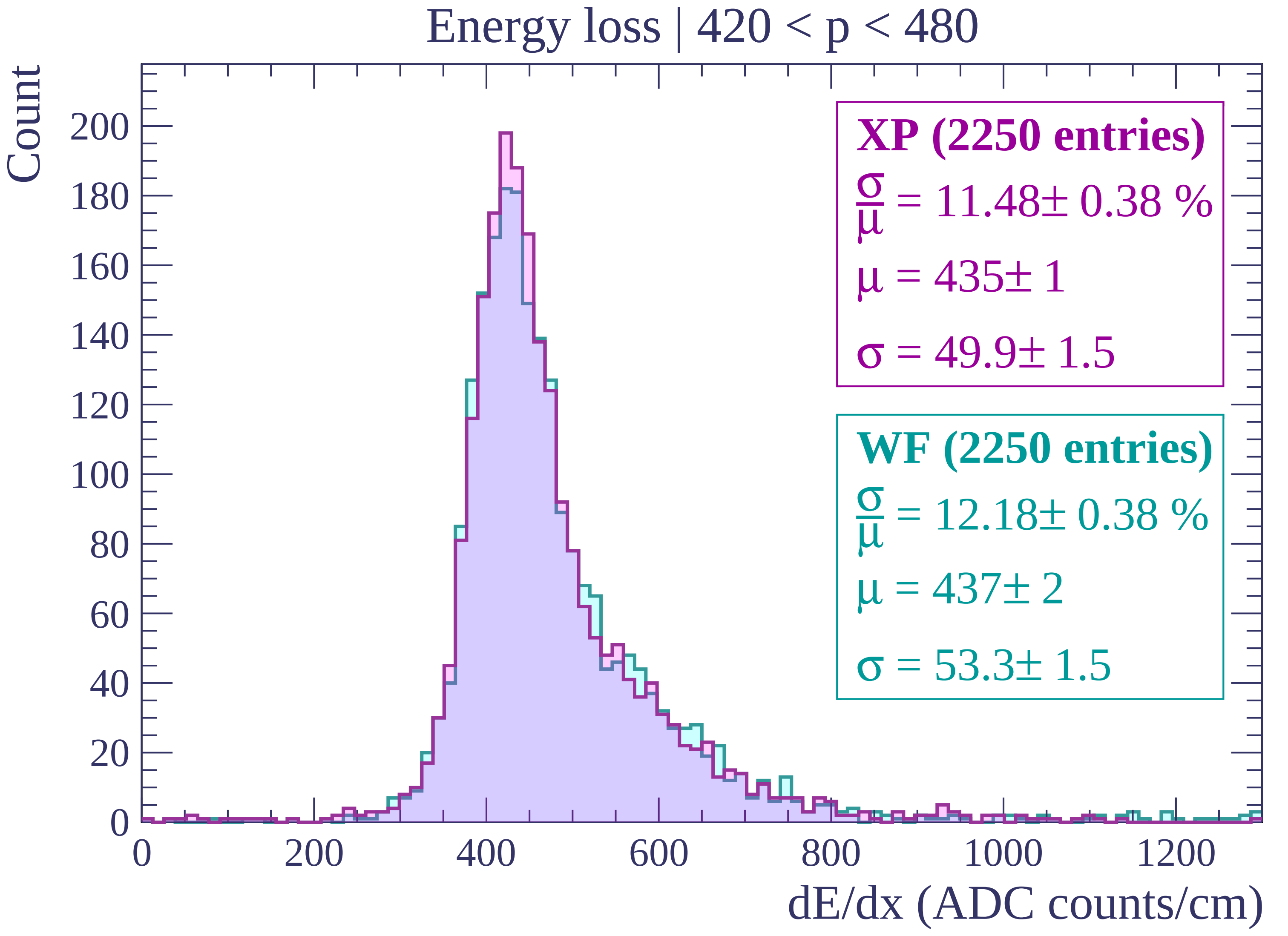

Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2363 entries)
$\frac{\sigma}{\mu}$ = 12.33± 0.40 %
μ = 449± 2
σ = 55.4± 1.6
WF (2363 entries)
$\frac{\sigma}{\mu}$ = 13.04± 0.44 %
μ = 454± 2
σ = 59.2± 1.8

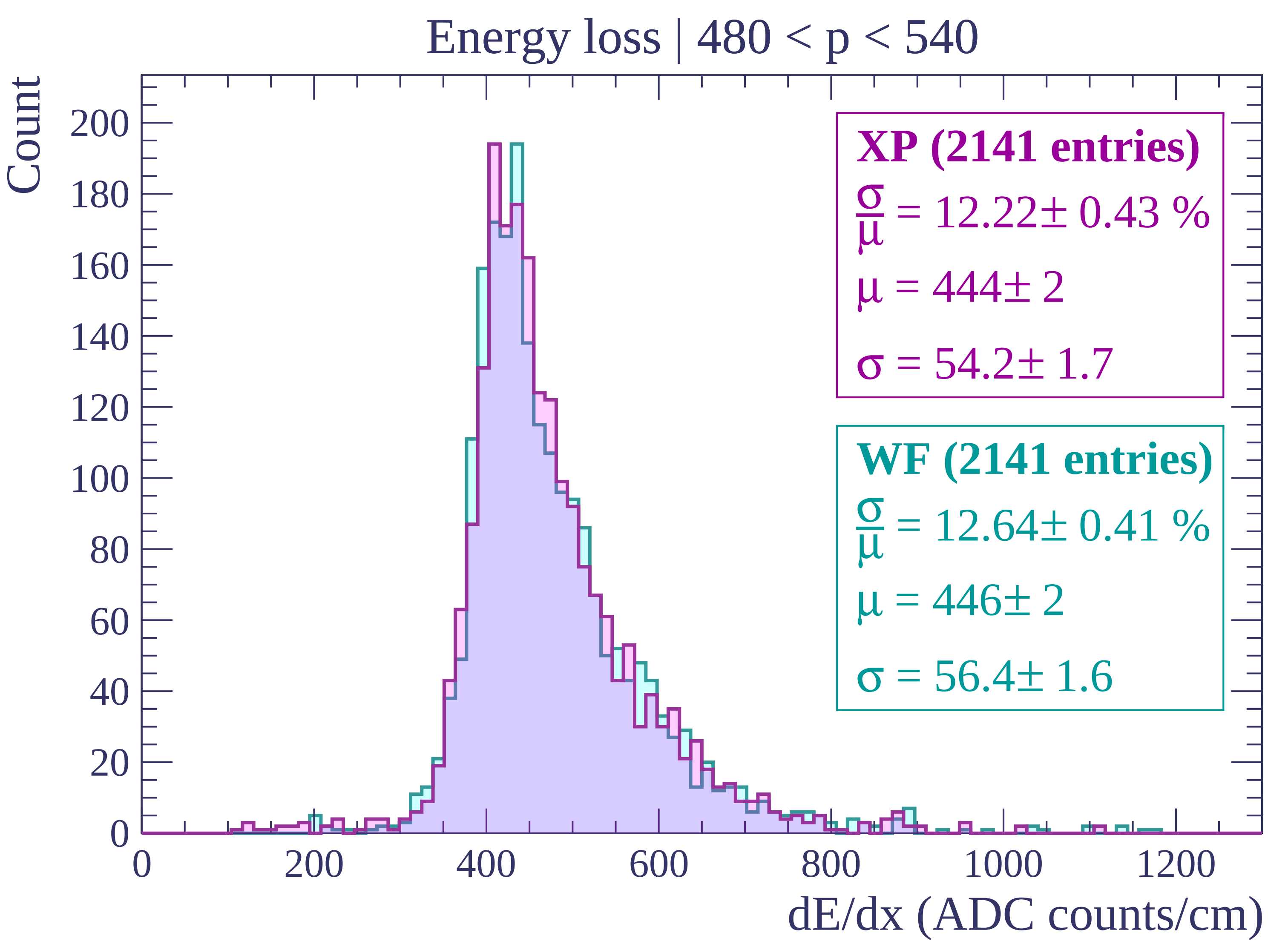


Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2455 entries)
$\frac{\sigma}{\mu}$ = 11.90± 0.35 %
μ = 440± 1
σ = 52.3± 1.4
WF (2455 entries)
$\frac{\sigma}{\mu}$ = 12.00± 0.35 %
μ = 440± 2
σ = 52.8± 1.4

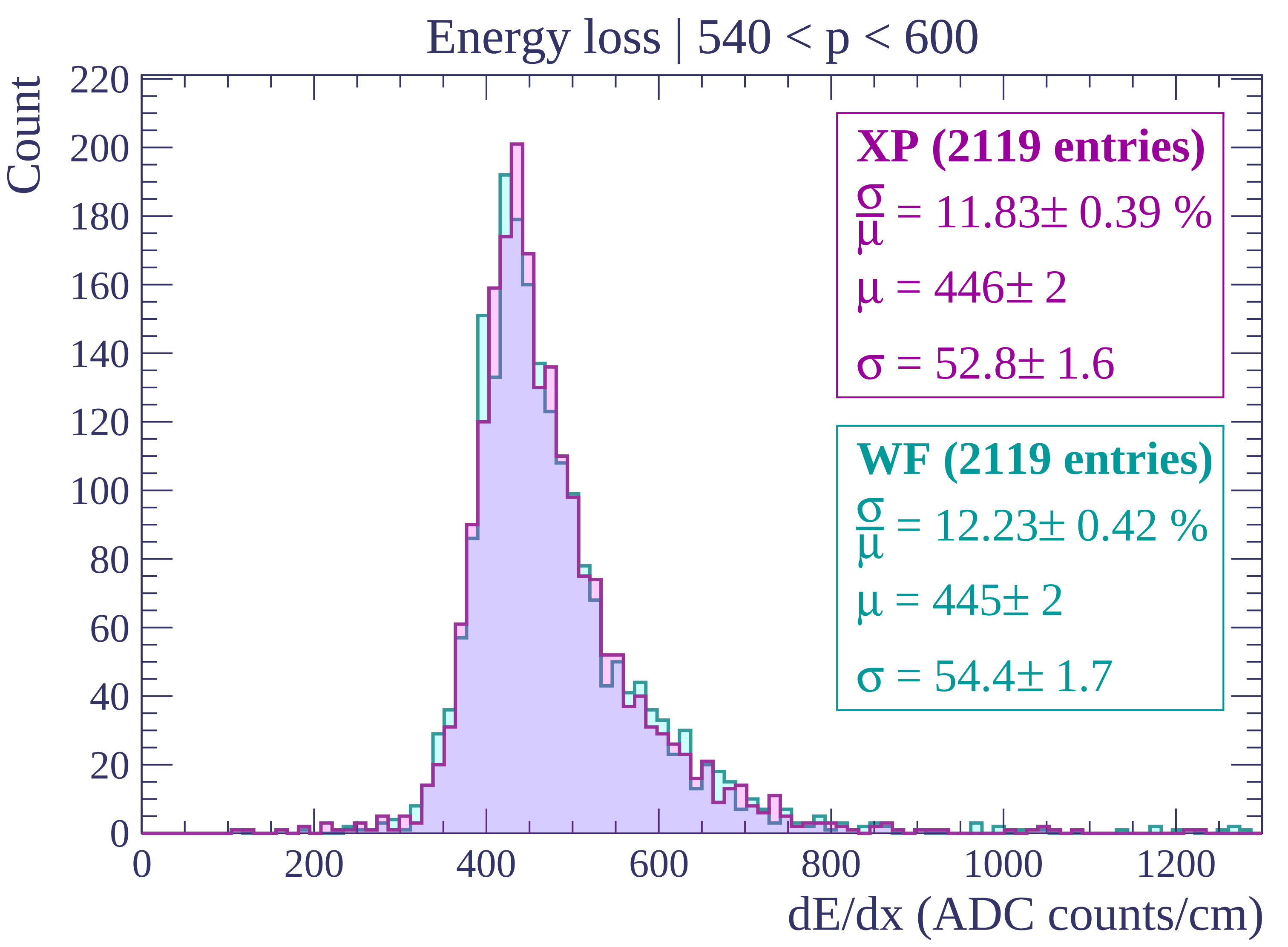


Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (2272 entries)
$\frac{\sigma}{\mu}$ = 12.17± 0.44 %
μ = 436± 2
σ = 53.1± 1.7
WF (2272 entries)
$\frac{\sigma}{\mu}$ = 12.63± 0.41 %
μ = 438± 2
σ = 55.3± 1.6

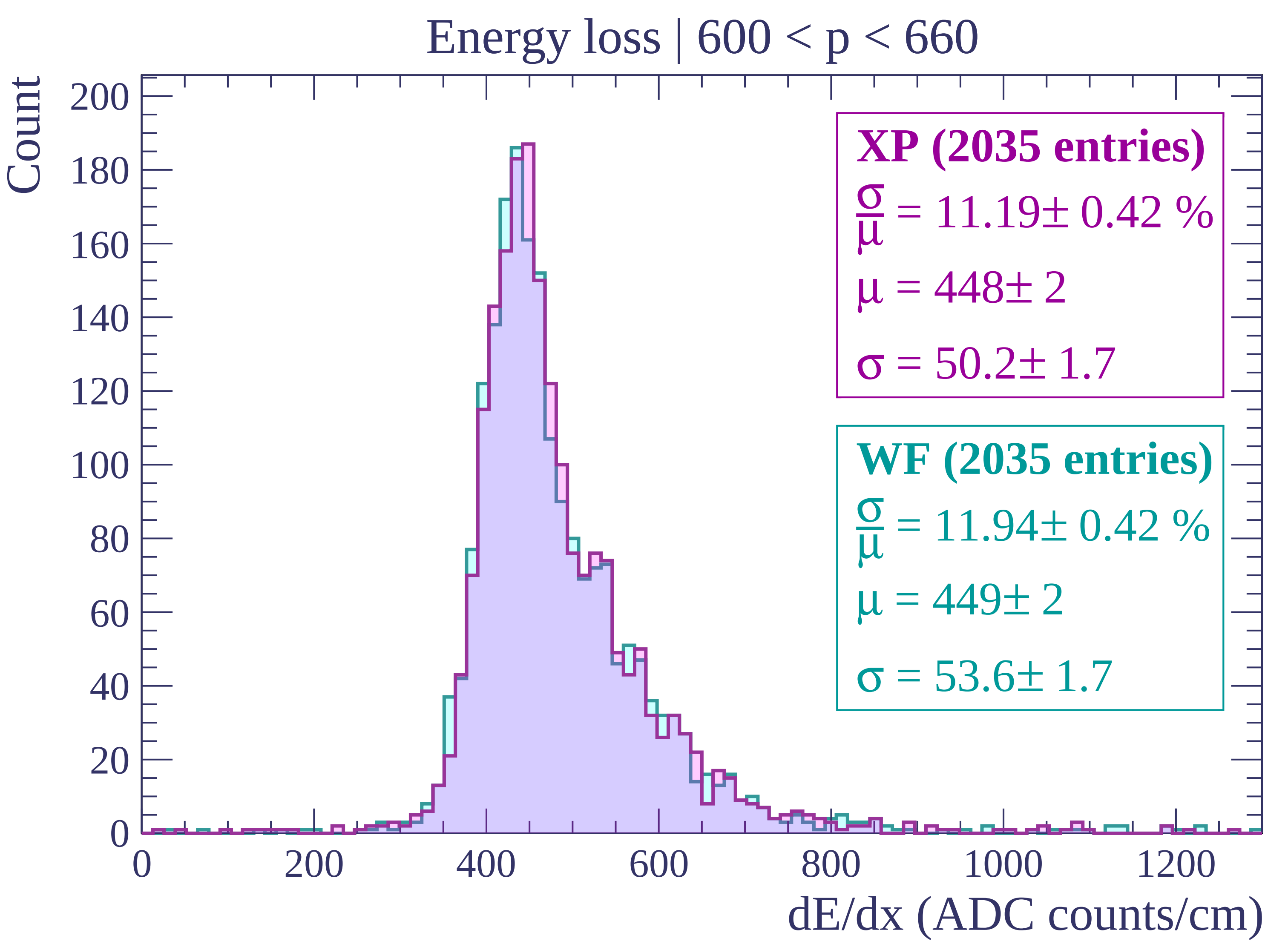

Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
XP (2250 entries)
σ/μ = 11.48± 0.38 %
μ = 435± 1
σ = 49.9± 1.5
WF (2250 entries)
σ/μ = 12.18± 0.38 %
μ = 437± 2
σ = 53.3± 1.5
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



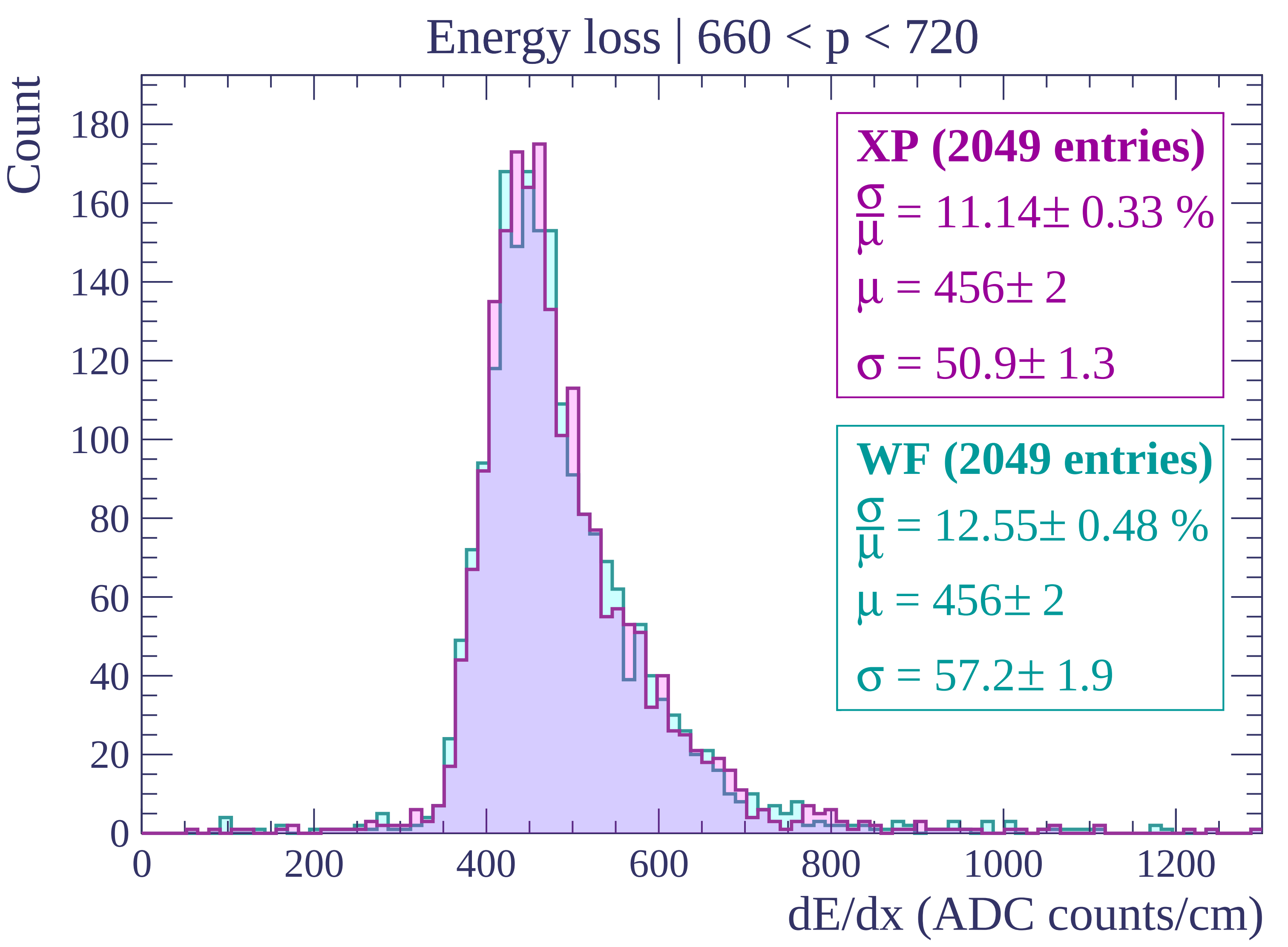
Energy loss | 480 < p < 540
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2141 entries)
$\frac{\sigma}{\mu}$ = 12.22± 0.43 %
μ = 444± 2
σ = 54.2± 1.7
WF (2141 entries)
$\frac{\sigma}{\mu}$ = 12.64± 0.41 %
μ = 446± 2
σ = 56.4± 1.6

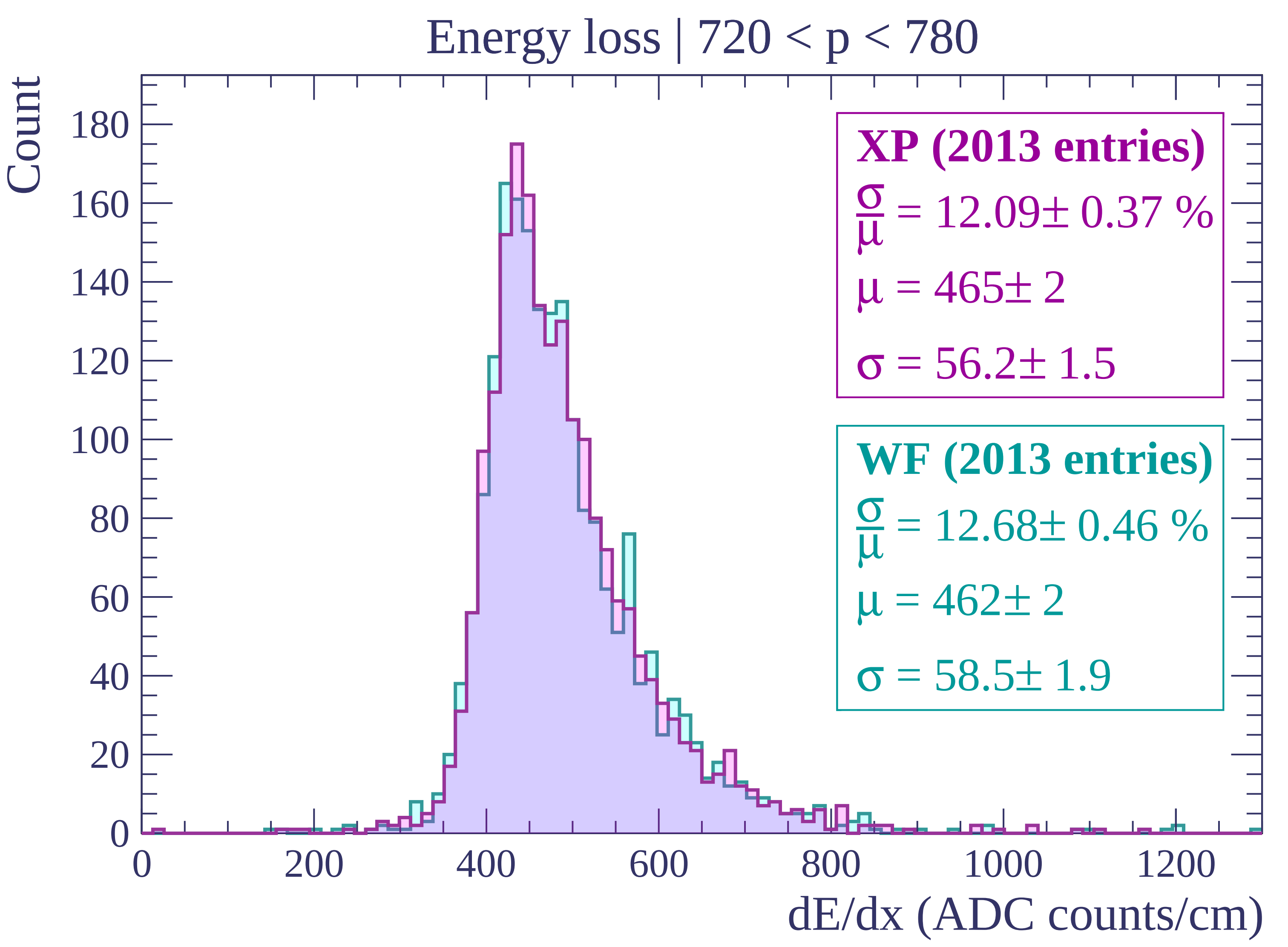


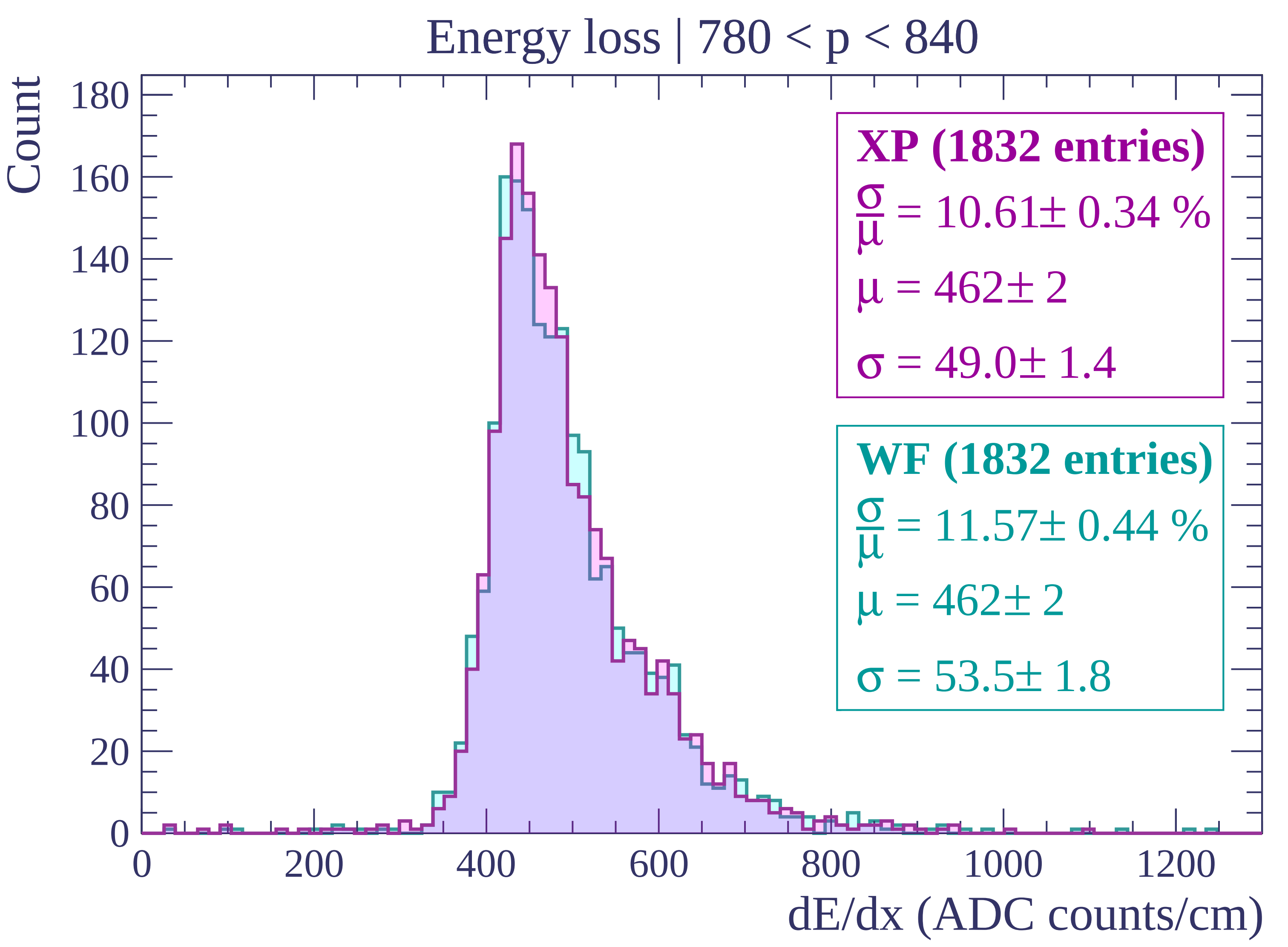
Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2119 entries)
$\frac{\sigma}{\mu}$ = 11.83± 0.39 %
μ = 446± 2
σ = 52.8± 1.6
WF (2119 entries)
$\frac{\sigma}{\mu}$ = 12.23± 0.42 %
μ = 445± 2
σ = 54.4± 1.7

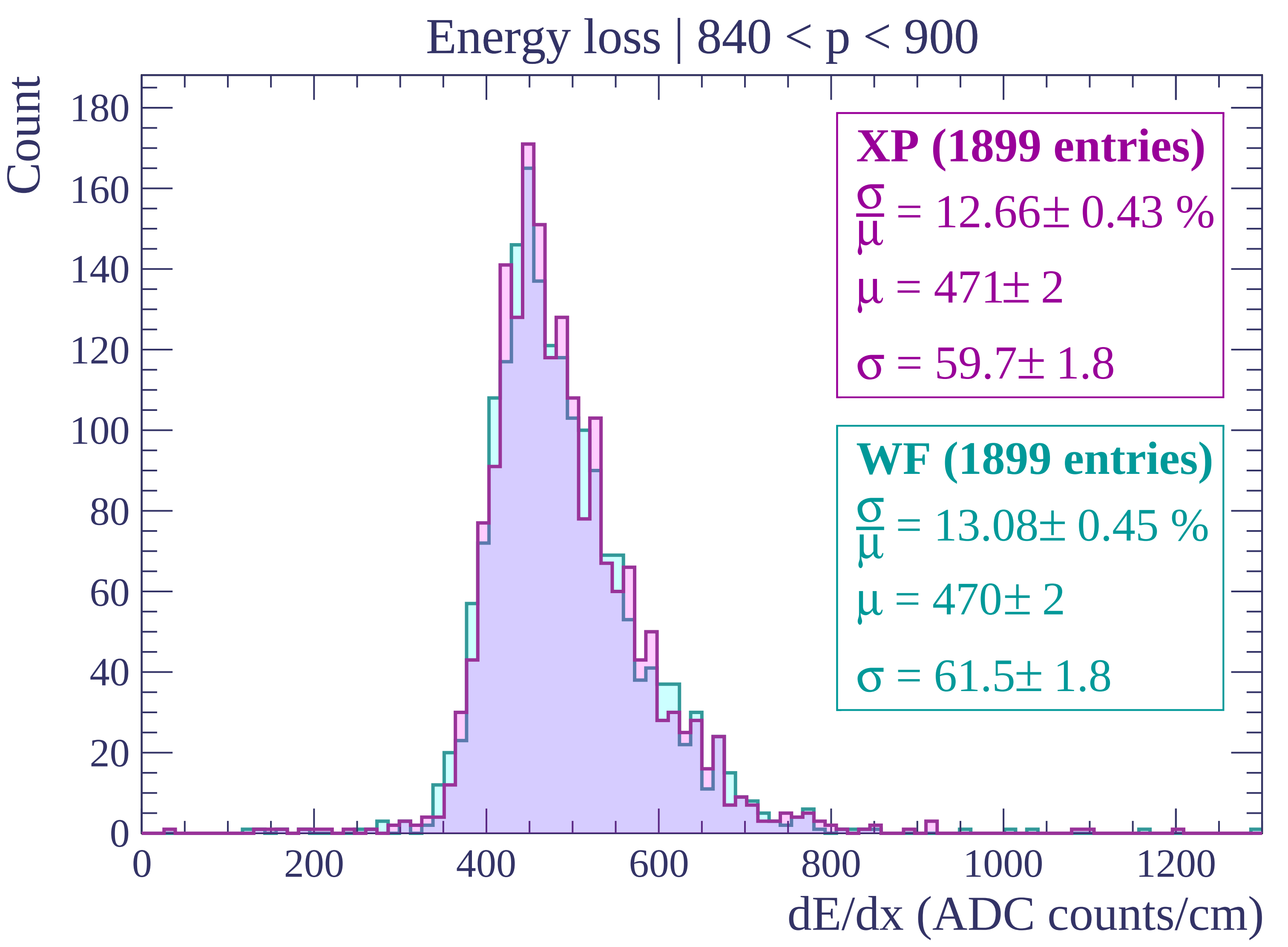


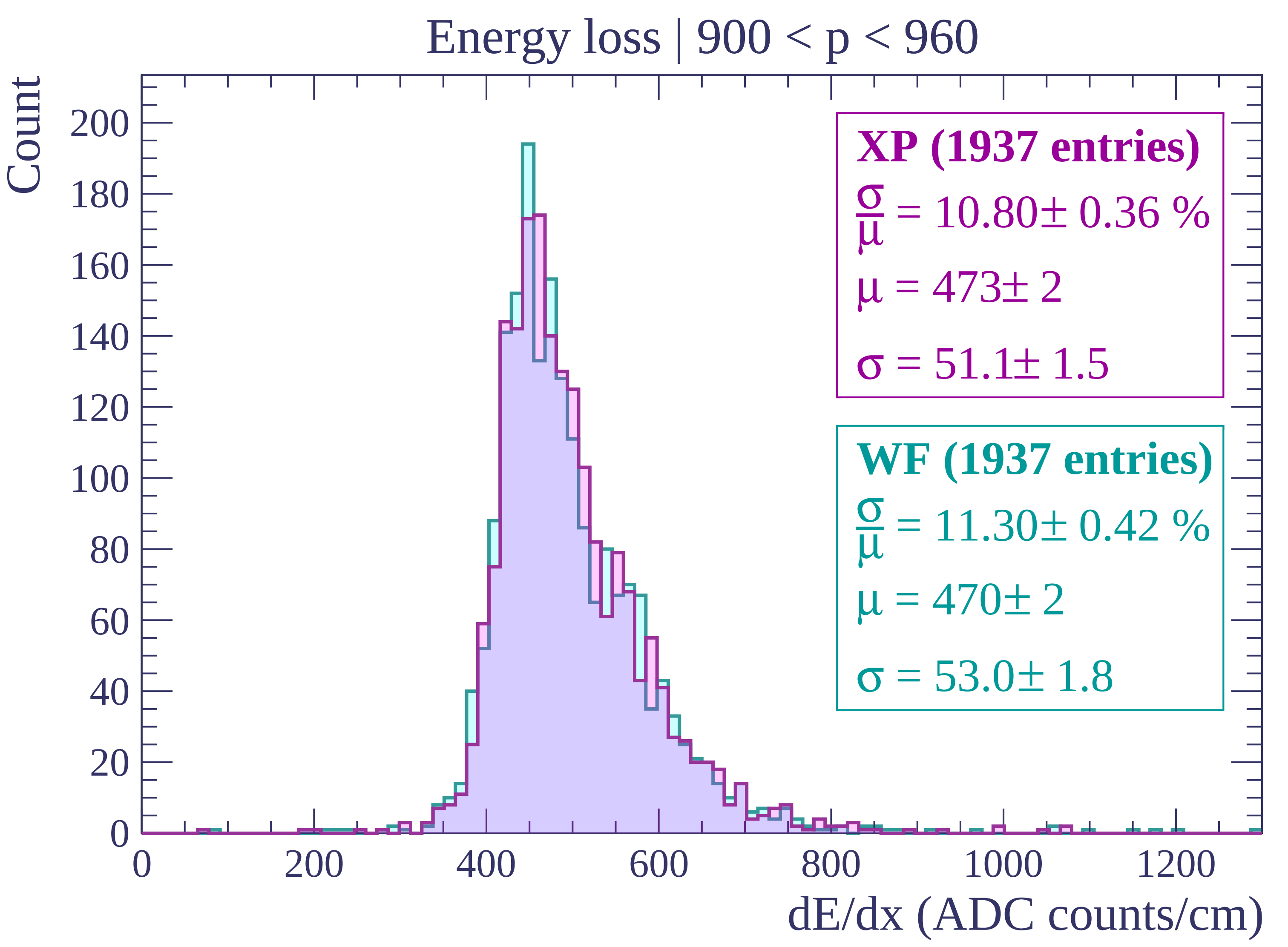
Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (2035 entries)
σ/μ = 11.19± 0.42 %
μ = 448± 2
σ = 50.2± 1.7
WF (2035 entries)
σ/μ = 11.94± 0.42 %
μ = 449± 2
σ = 53.6± 1.7


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (2049 entries)
σ/μ = 11.14± 0.33 %
μ = 456± 2
σ = 50.9± 1.3
WF (2049 entries)
σ/μ = 12.55± 0.48 %
μ = 456± 2
σ = 57.2± 1.9

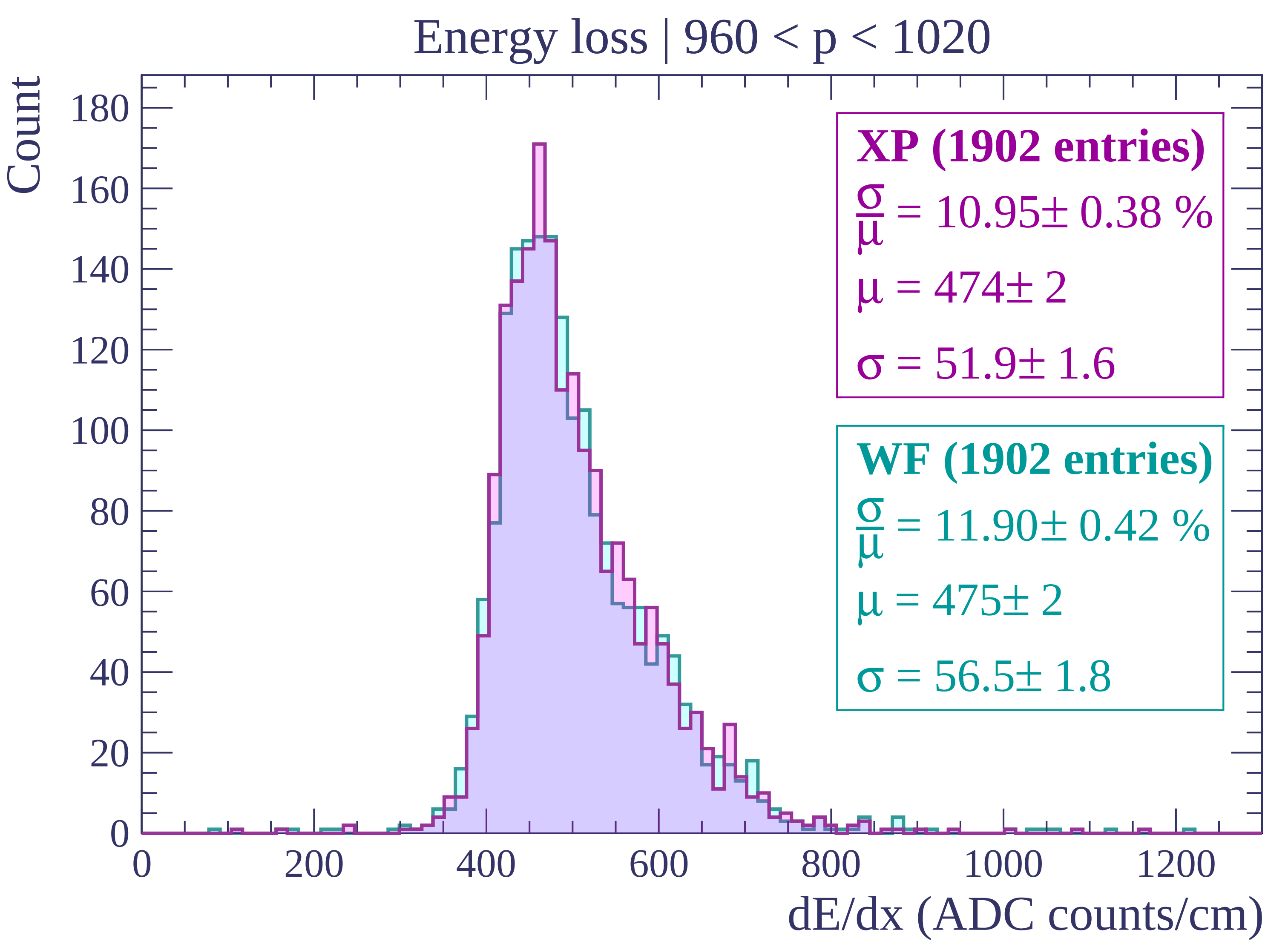

Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (2013 entries)
σ/μ = 12.09± 0.37 %
μ = 465± 2
σ = 56.2± 1.5
WF (2013 entries)
σ/μ = 12.68± 0.46 %
μ = 462± 2
σ = 58.5± 1.9

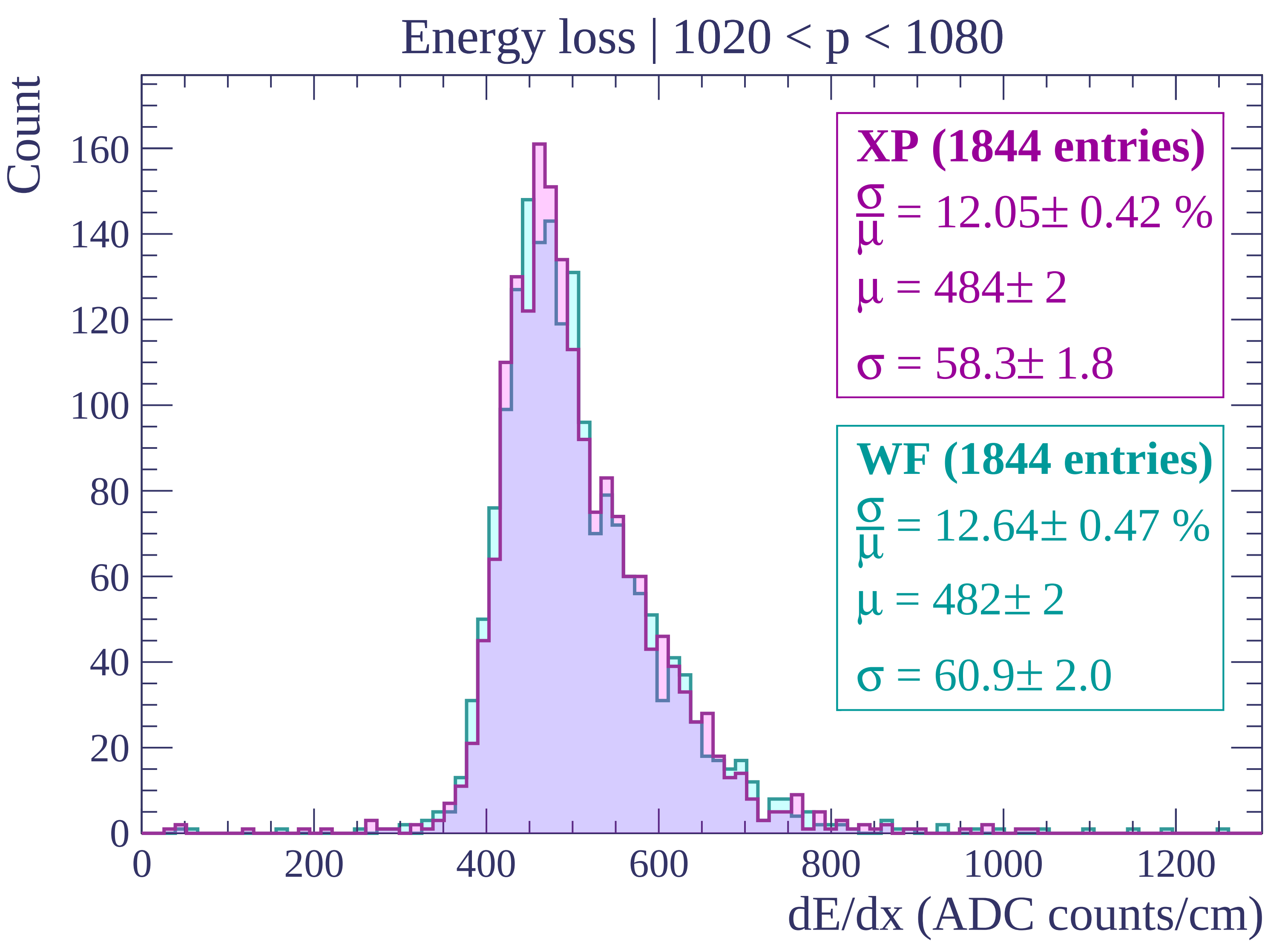

Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1832 entries)
$\frac{\sigma}{\mu}$ = 10.61± 0.34 %
μ = 462± 2
σ = 49.0± 1.4
WF (1832 entries)
$\frac{\sigma}{\mu}$ = 11.57± 0.44 %
μ = 462± 2
σ = 53.5± 1.8


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1899 entries)
$\frac{\sigma}{\mu}$ = 12.66± 0.43 %
μ = 471± 2
σ = 59.7± 1.8
WF (1899 entries)
$\frac{\sigma}{\mu}$ = 13.08± 0.45 %
μ = 470± 2
σ = 61.5± 1.8

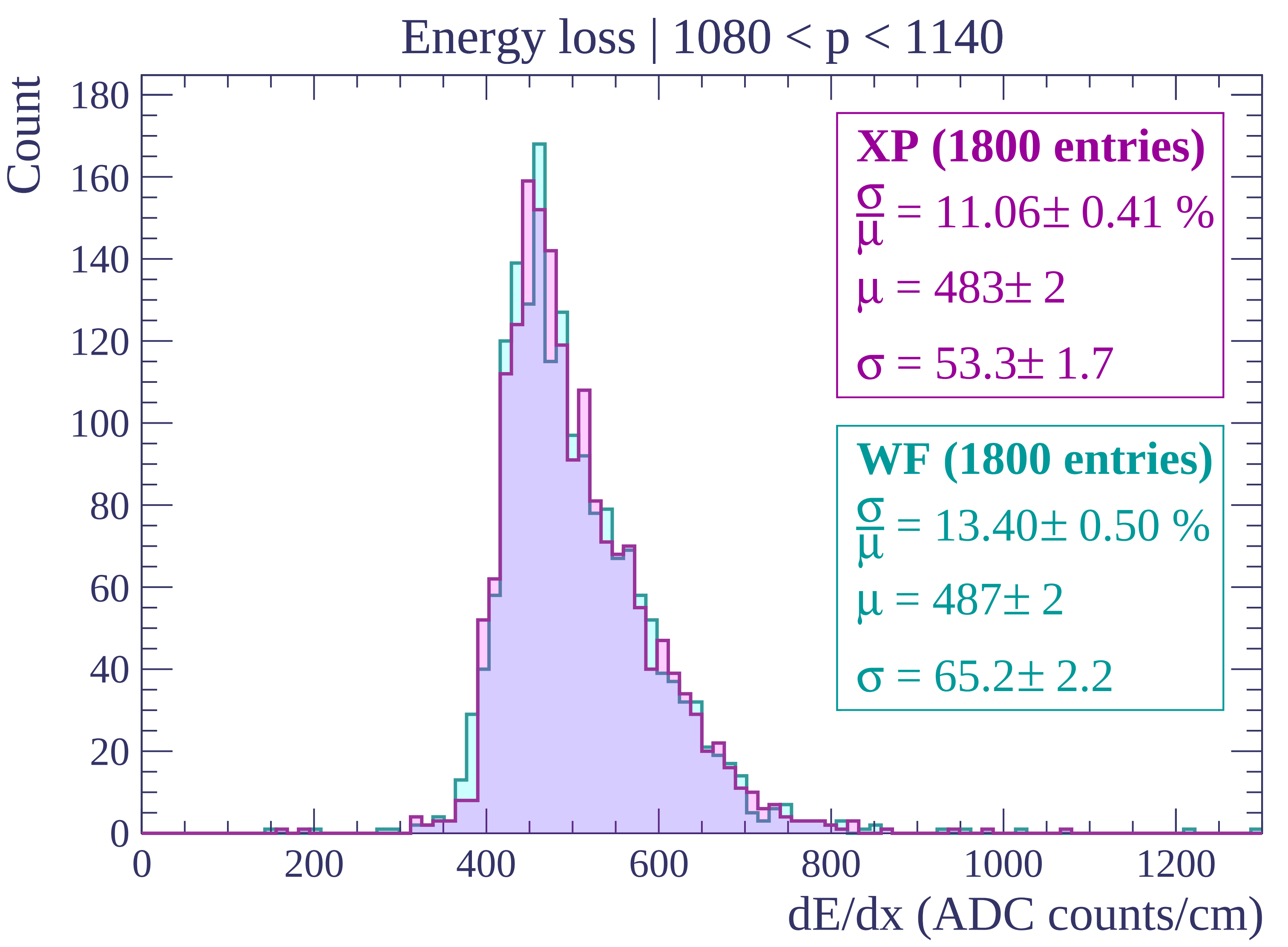


Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1937 entries)
σ/μ = 10.80± 0.36 %
μ = 473± 2
σ = 51.1± 1.5
WF (1937 entries)
σ/μ = 11.30± 0.42 %
μ = 470± 2
σ = 53.0± 1.8

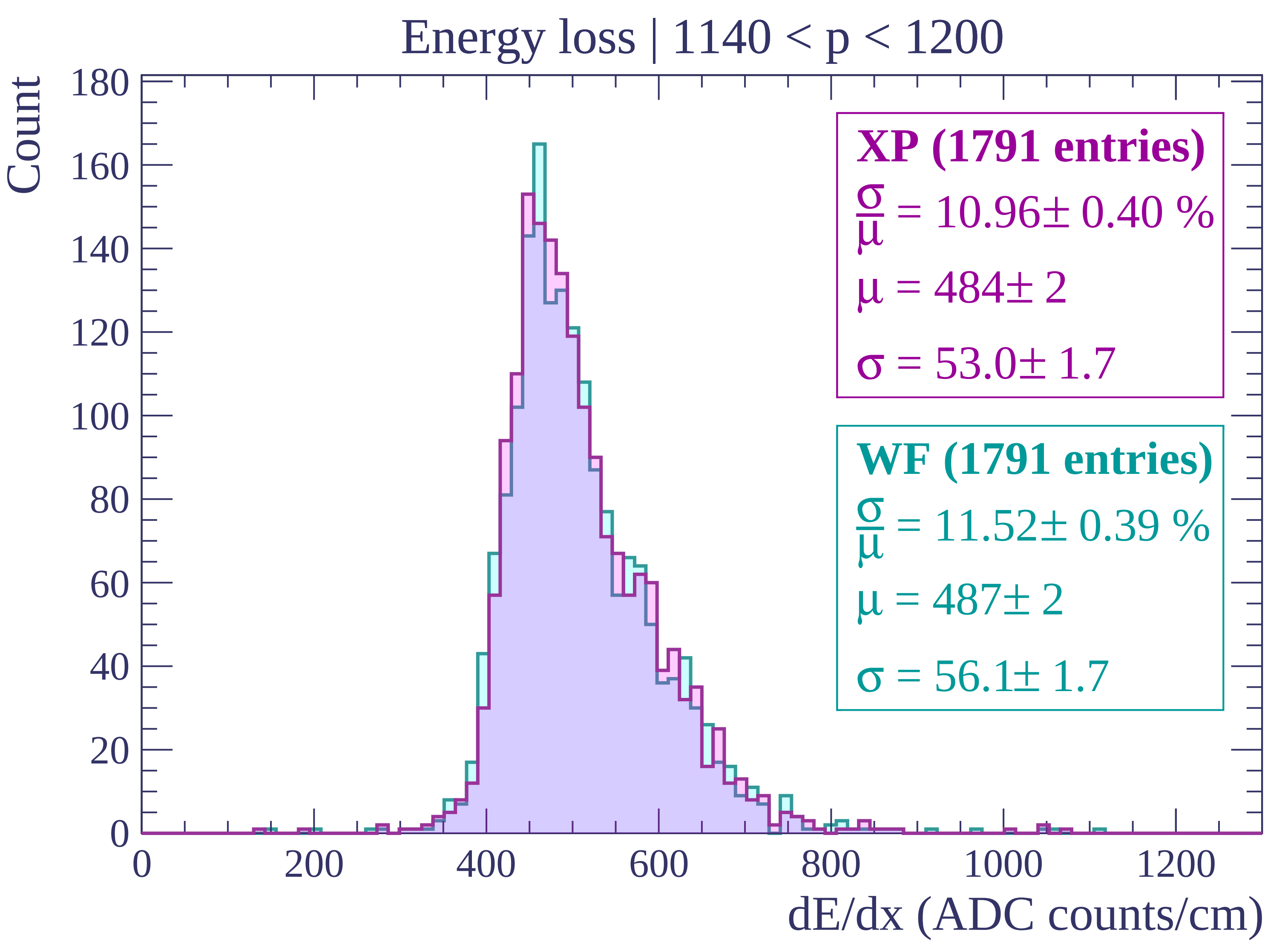

Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1902 entries)
σ/μ = 10.95± 0.38 %
μ = 474± 2
σ = 51.9± 1.6
WF (1902 entries)
σ/μ = 11.90± 0.42 %
μ = 475± 2
σ = 56.5± 1.8



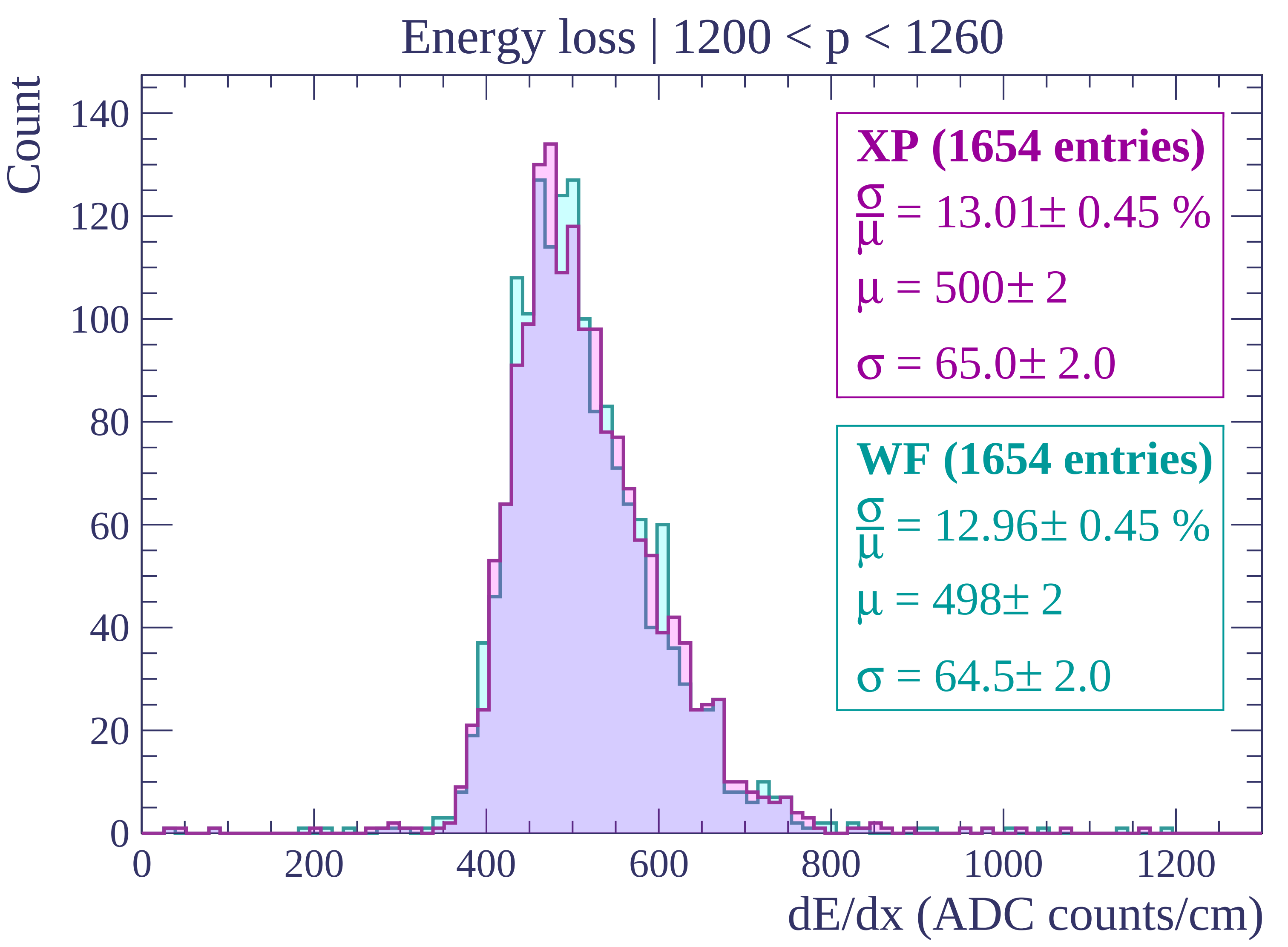
Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1844 entries)
$\frac{\sigma}{\mu}$ = 12.05± 0.42 %
μ = 484± 2
σ = 58.3± 1.8
WF (1844 entries)
$\frac{\sigma}{\mu}$ = 12.64± 0.47 %
μ = 482± 2
σ = 60.9± 2.0

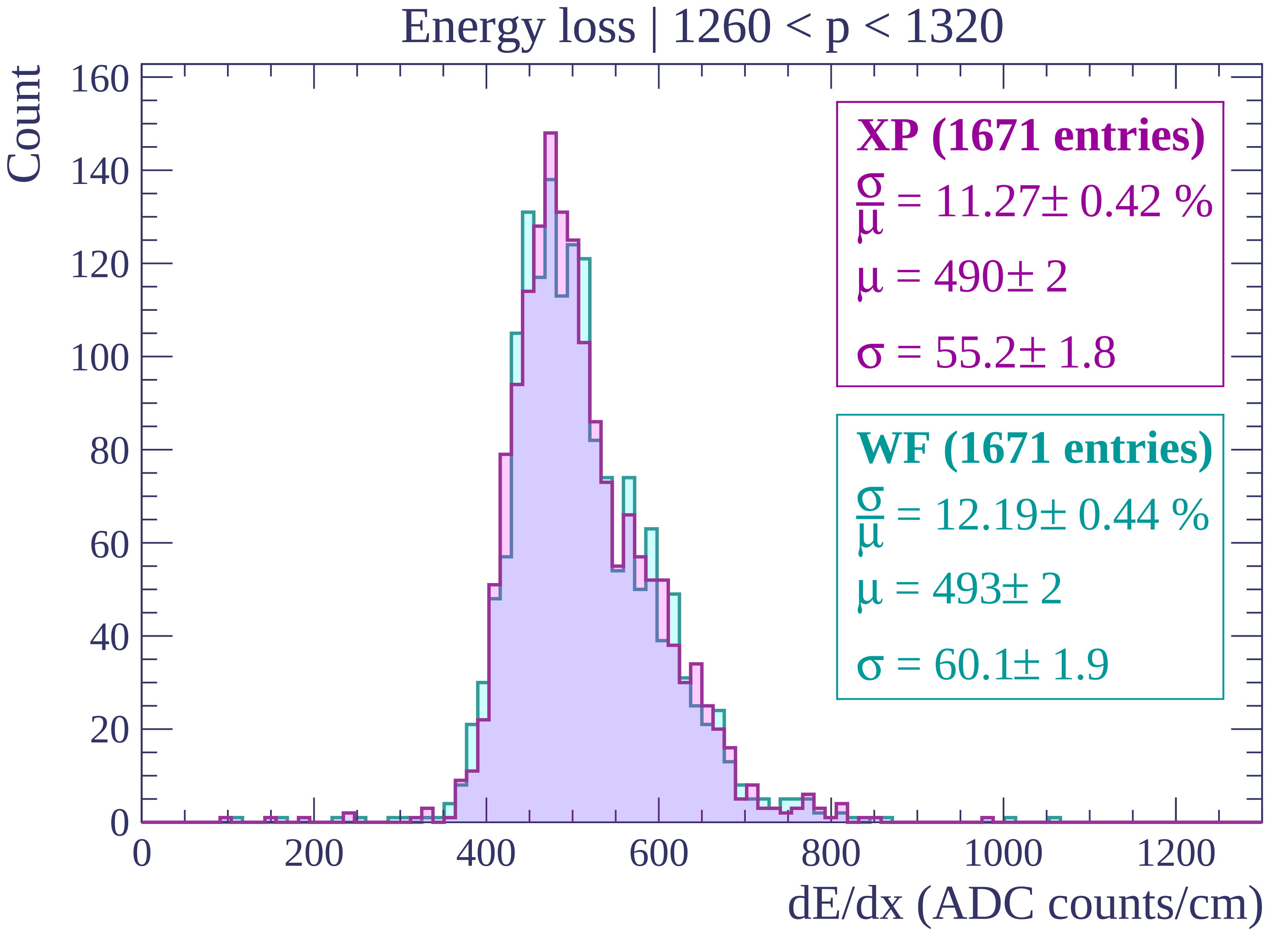


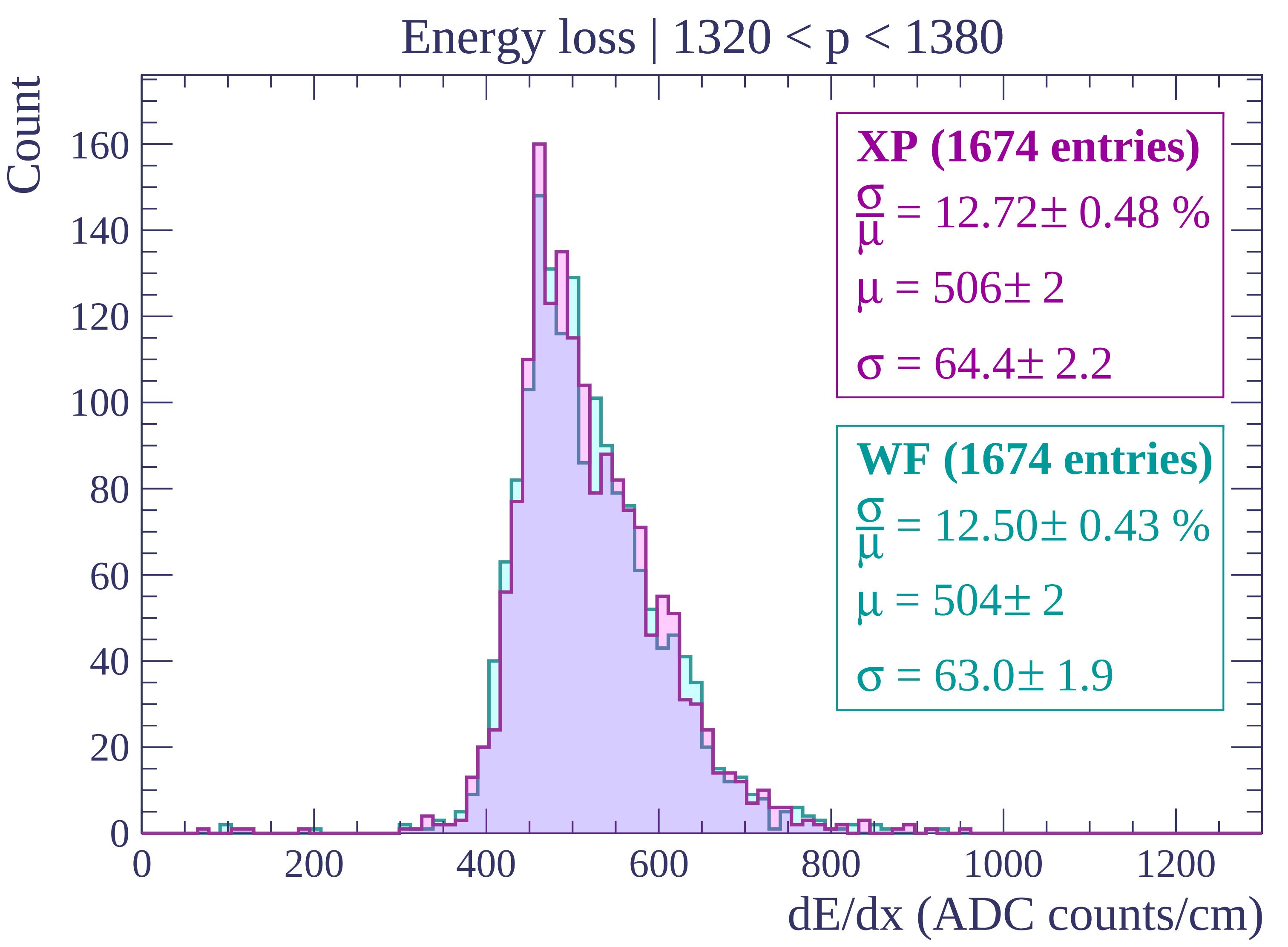
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1800 entries)
σ/μ = 11.06± 0.41 %
μ = 483± 2
σ = 53.3± 1.7
WF (1800 entries)
σ/μ = 13.40± 0.50 %
μ = 487± 2
σ = 65.2± 2.2

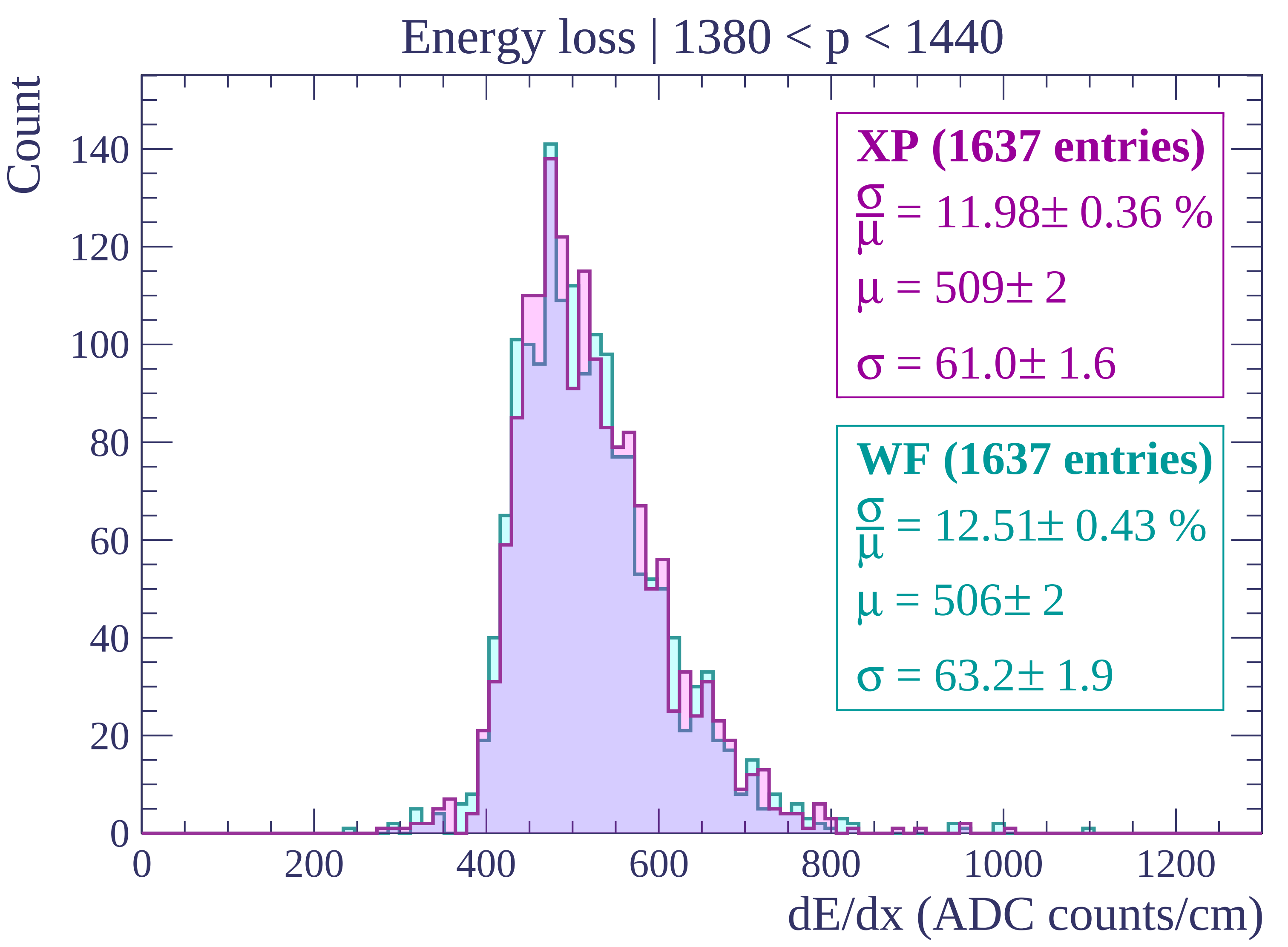


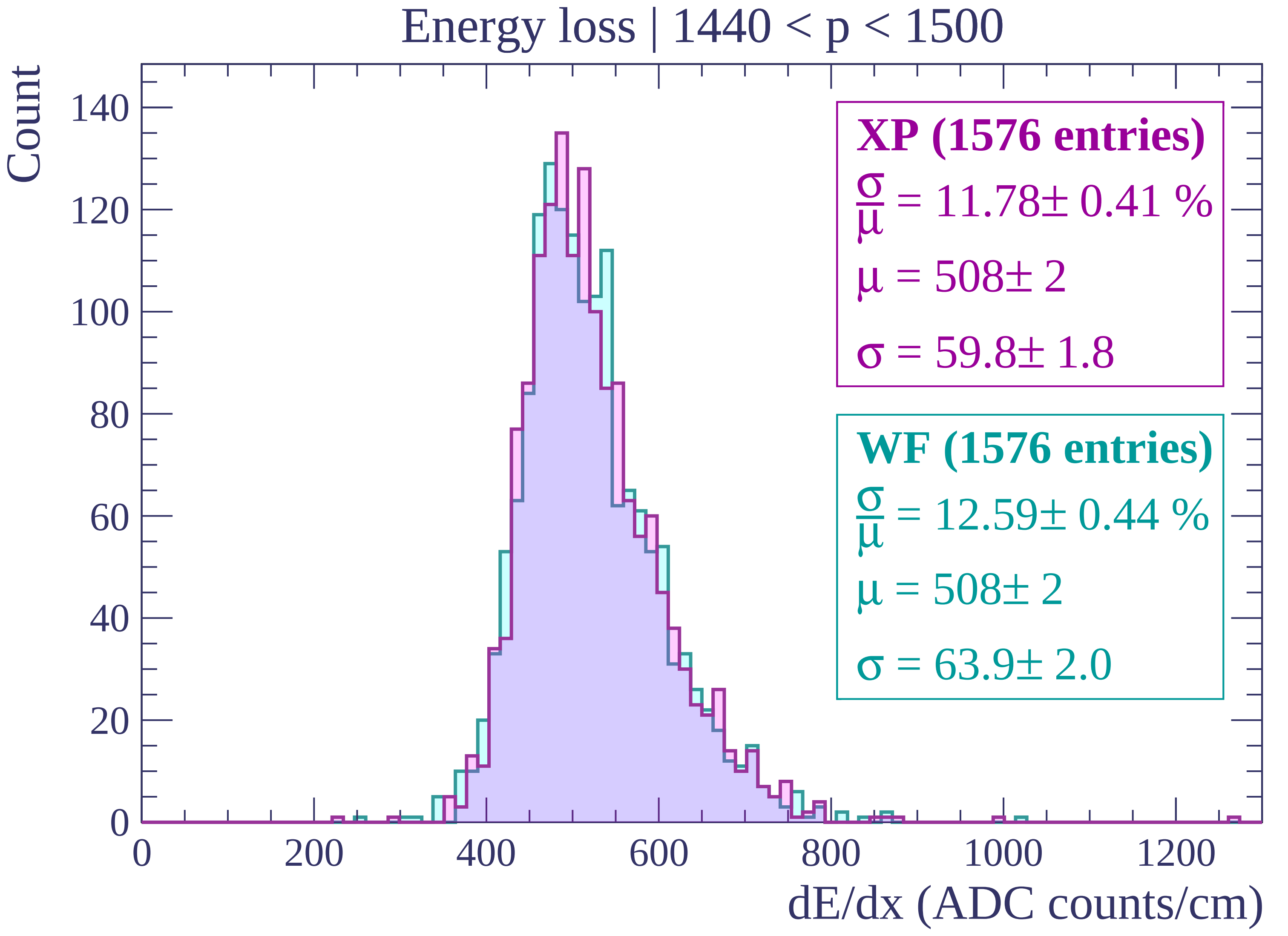
Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (1791 entries)
σ/μ = 10.96± 0.40 %
μ = 484± 2
σ = 53.0± 1.7
WF (1791 entries)
σ/μ = 11.52± 0.39 %
μ = 487± 2
σ = 56.1± 1.7
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1654 entries)
σ/μ = 13.01± 0.45 %
μ = 500± 2
σ = 65.0± 2.0
WF (1654 entries)
σ/μ = 12.96± 0.45 %
μ = 498± 2
σ = 64.5± 2.0

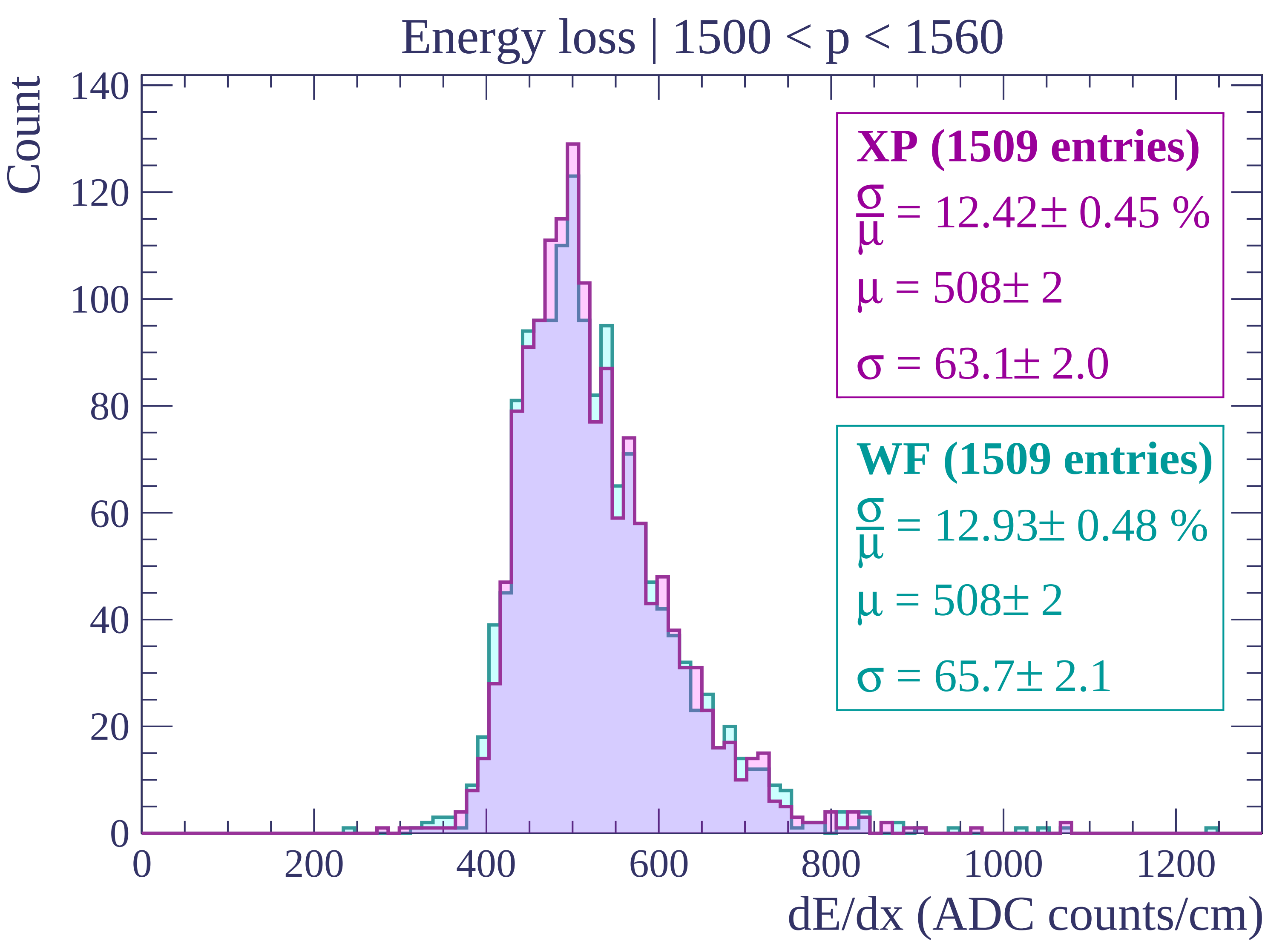

Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
XP (1671 entries)
$\frac{\sigma}{\mu}$ = 11.27± 0.42 %
μ = 490± 2
σ = 55.2± 1.8
WF (1671 entries)
$\frac{\sigma}{\mu}$ = 12.19± 0.44 %
μ = 493± 2
σ = 60.1± 1.9
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160


Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1674 entries)
σ/μ = 12.72± 0.48 %
μ = 506± 2
σ = 64.4± 2.2
WF (1674 entries)
σ/μ = 12.50± 0.43 %
μ = 504± 2
σ = 63.0± 1.9

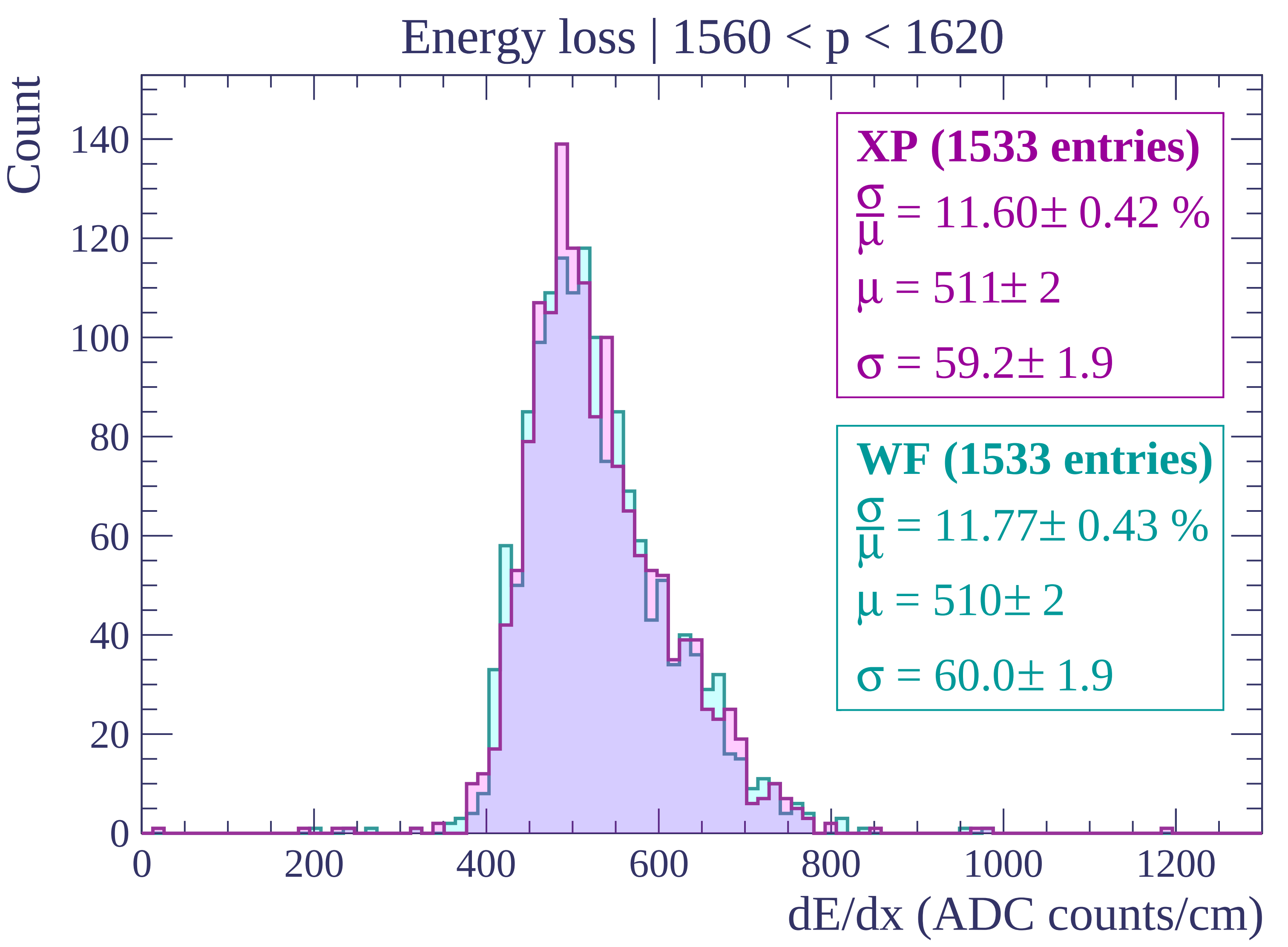

Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1637 entries)
σ/μ = 11.98± 0.36 %
μ = 509± 2
σ = 61.0± 1.6
WF (1637 entries)
σ/μ = 12.51± 0.43 %
μ = 506± 2
σ = 63.2± 1.9

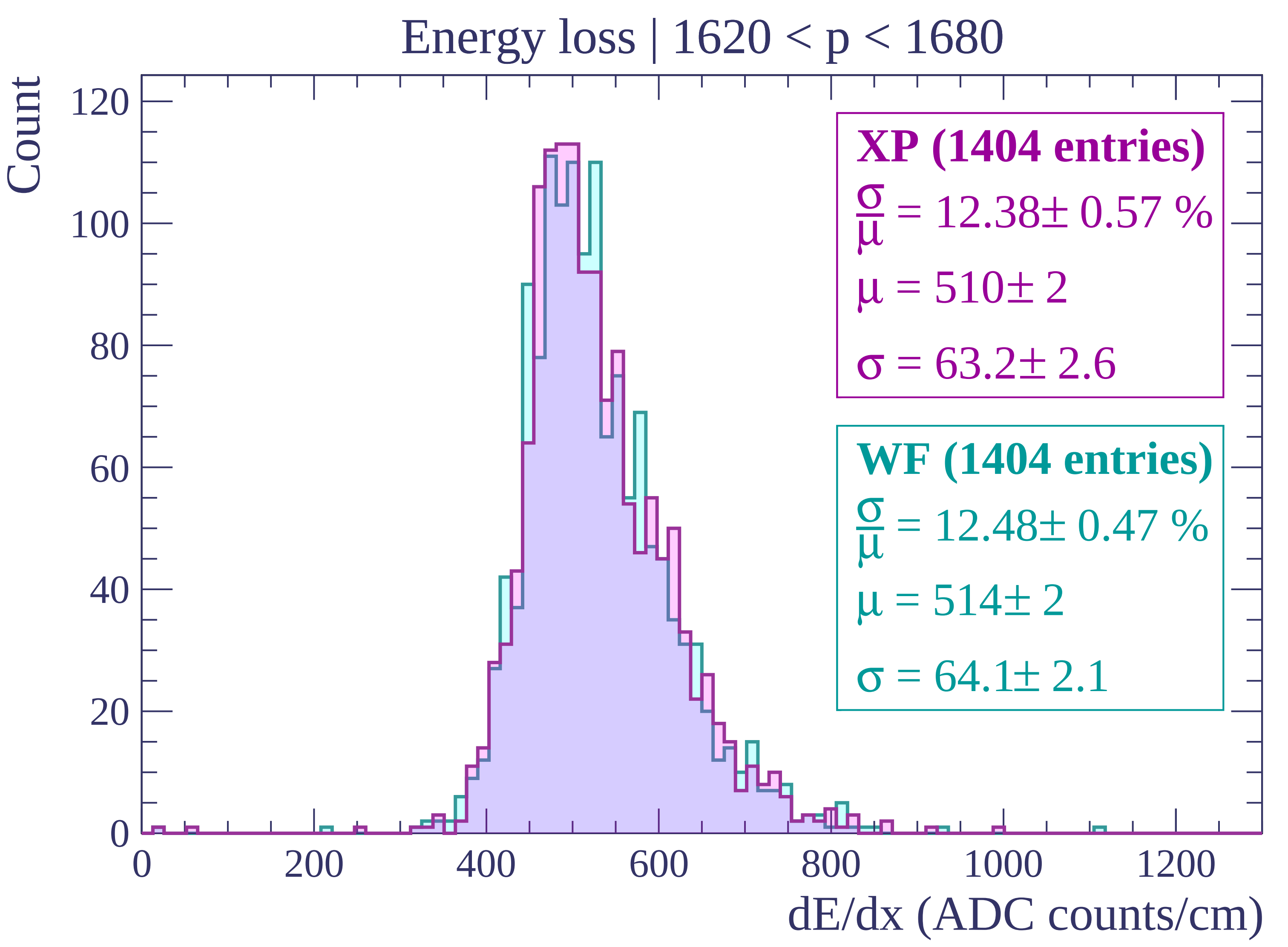


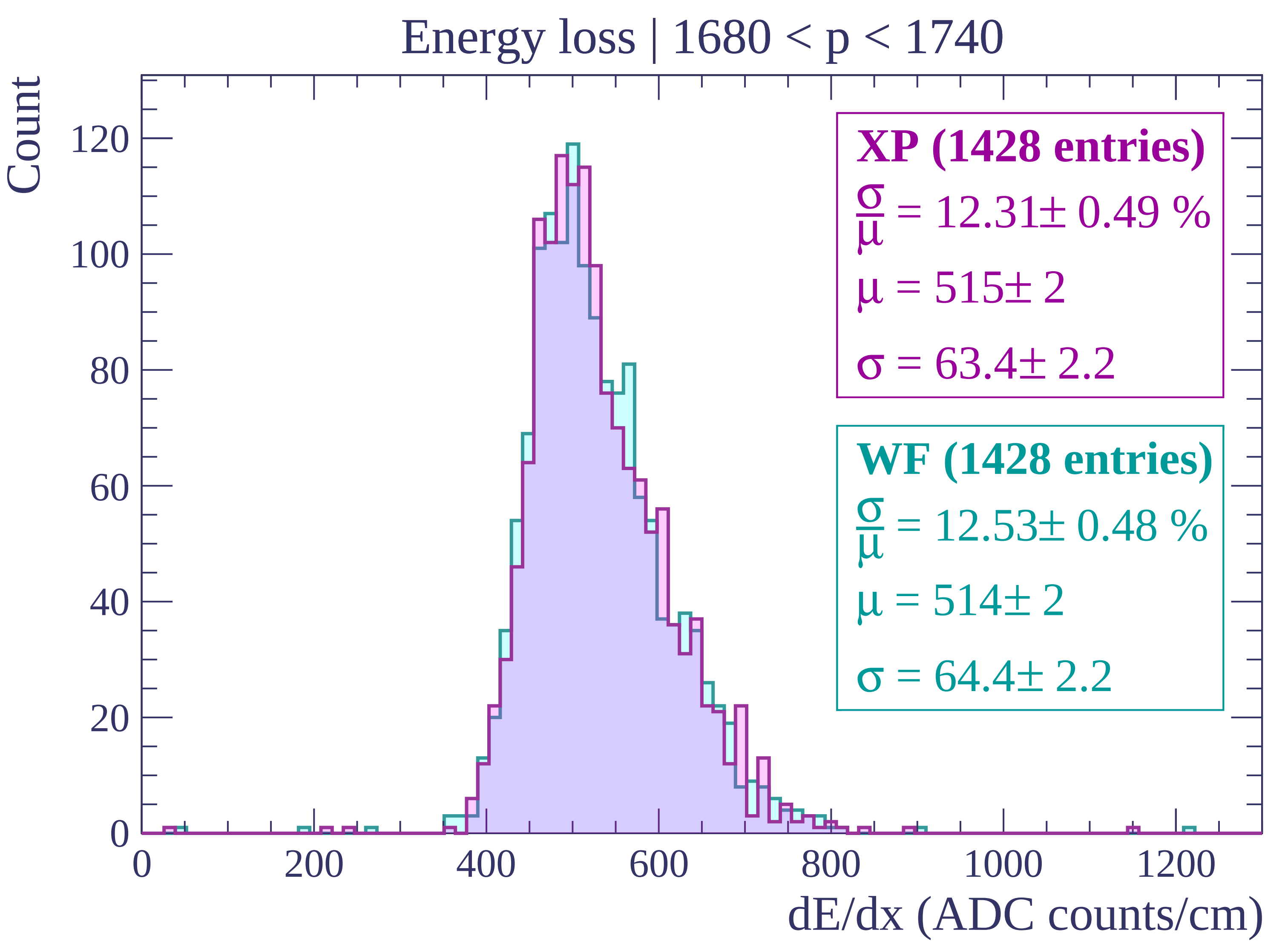
Energy loss | 1440 < p < 1500
Count
XP (1576 entries)
$\frac{\sigma}{\mu}$ = 11.78± 0.41 %
μ = 508± 2
σ = 59.8± 1.8
WF (1576 entries)
$\frac{\sigma}{\mu}$ = 12.59± 0.44 %
μ = 508± 2
σ = 63.9± 2.0
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
200
400
600
800
1000
1200



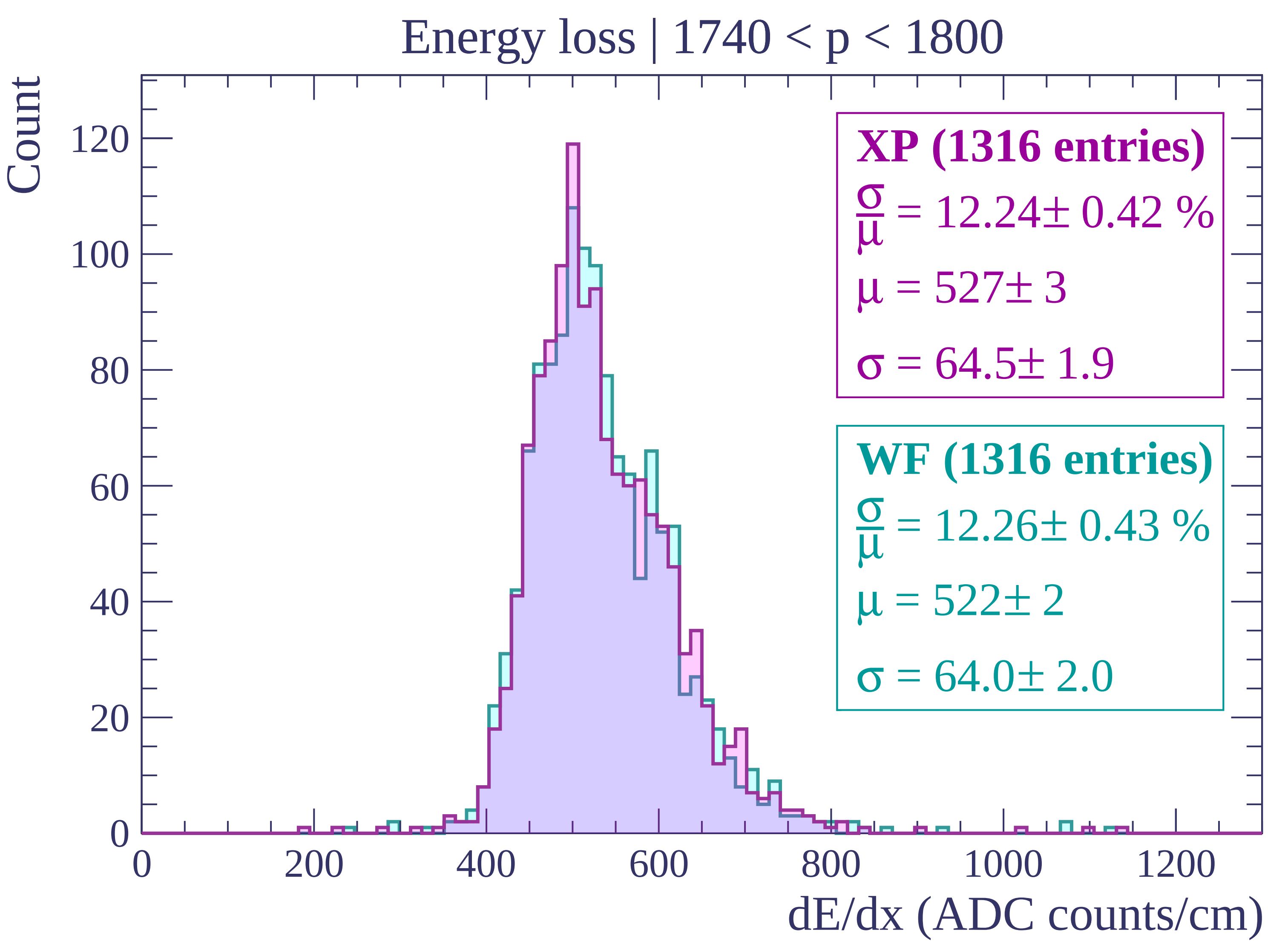
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
XP (1509 entries)
$\frac{\sigma}{\mu}$ = 12.42± 0.45 %
μ = 508± 2
σ = 63.1± 2.0
WF (1509 entries)
$\frac{\sigma}{\mu}$ = 12.93± 0.48 %
μ = 508± 2
σ = 65.7± 2.1


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1533 entries)
σ/μ = 11.60± 0.42 %
μ = 511± 2
σ = 59.2± 1.9
WF (1533 entries)
σ/μ = 11.77± 0.43 %
μ = 510± 2
σ = 60.0± 1.9

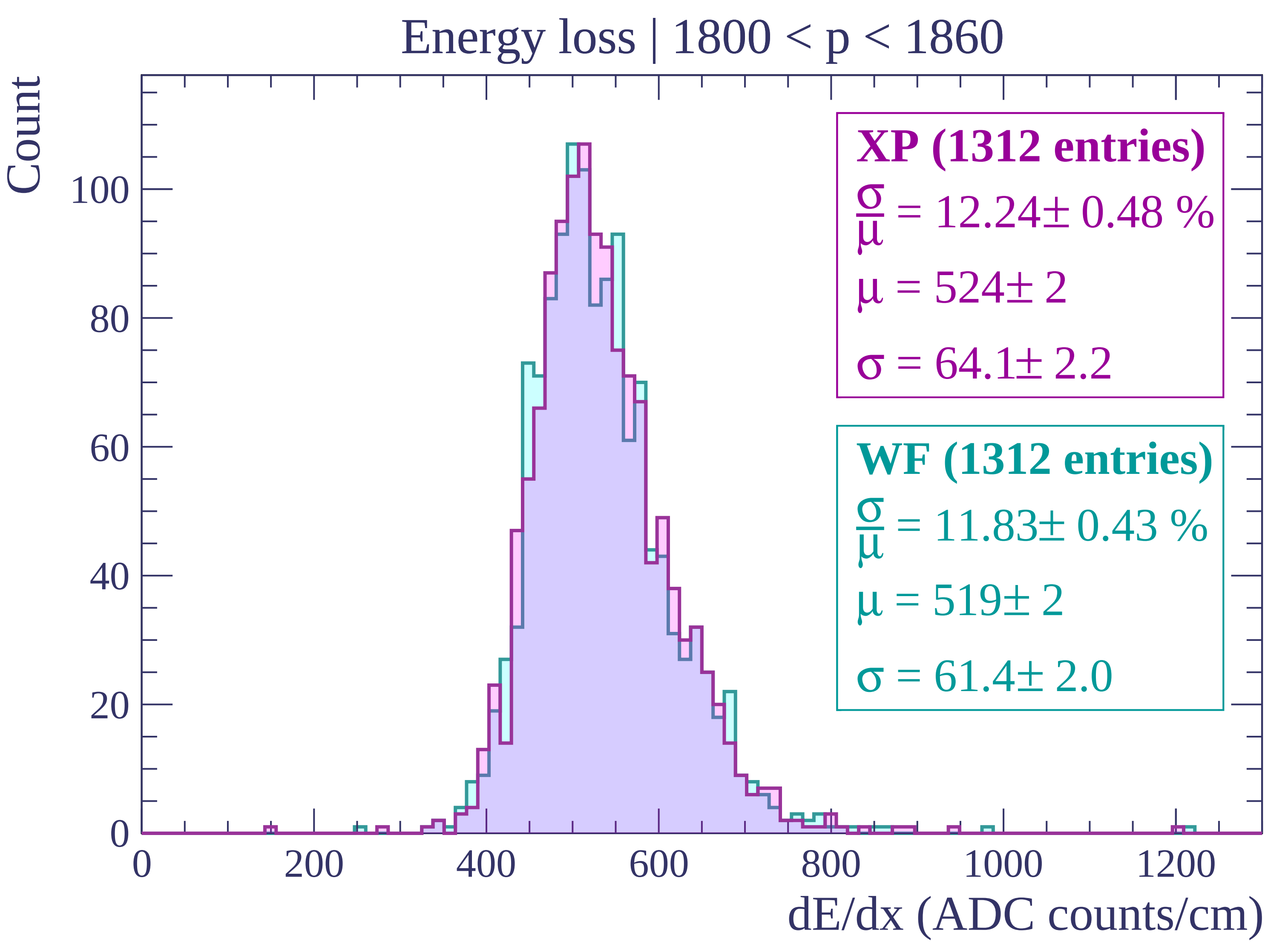

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1404 entries)
σ/μ = 12.38± 0.57 %
μ = 510± 2
σ = 63.2± 2.6
WF (1404 entries)
σ/μ = 12.48± 0.47 %
μ = 514± 2
σ = 64.1± 2.1



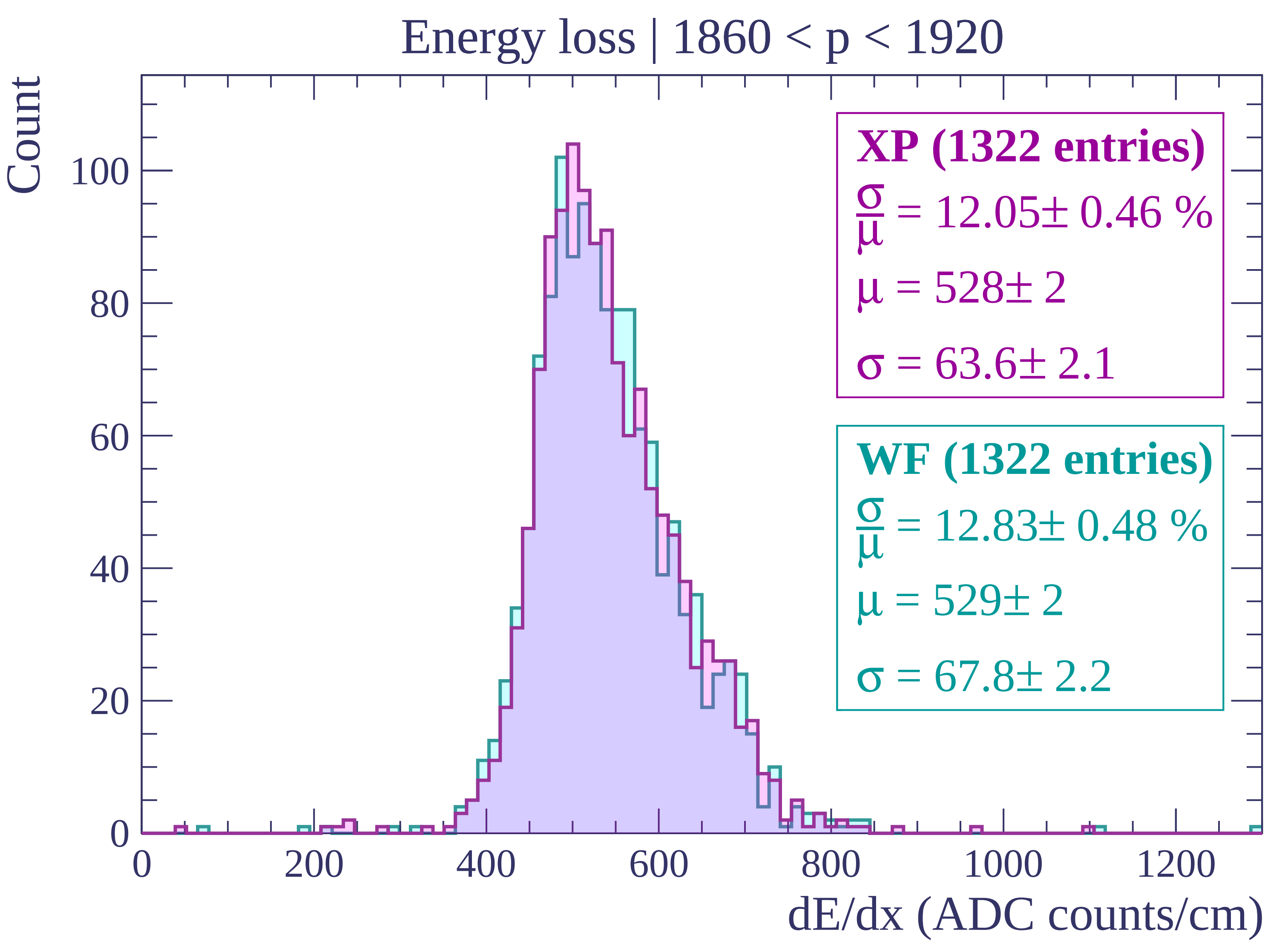
Energy loss | 1680 < p < 1740
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1428 entries)
σ/μ = 12.31± 0.49 %
μ = 515± 2
σ = 63.4± 2.2
WF (1428 entries)
σ/μ = 12.53± 0.48 %
μ = 514± 2
σ = 64.4± 2.2



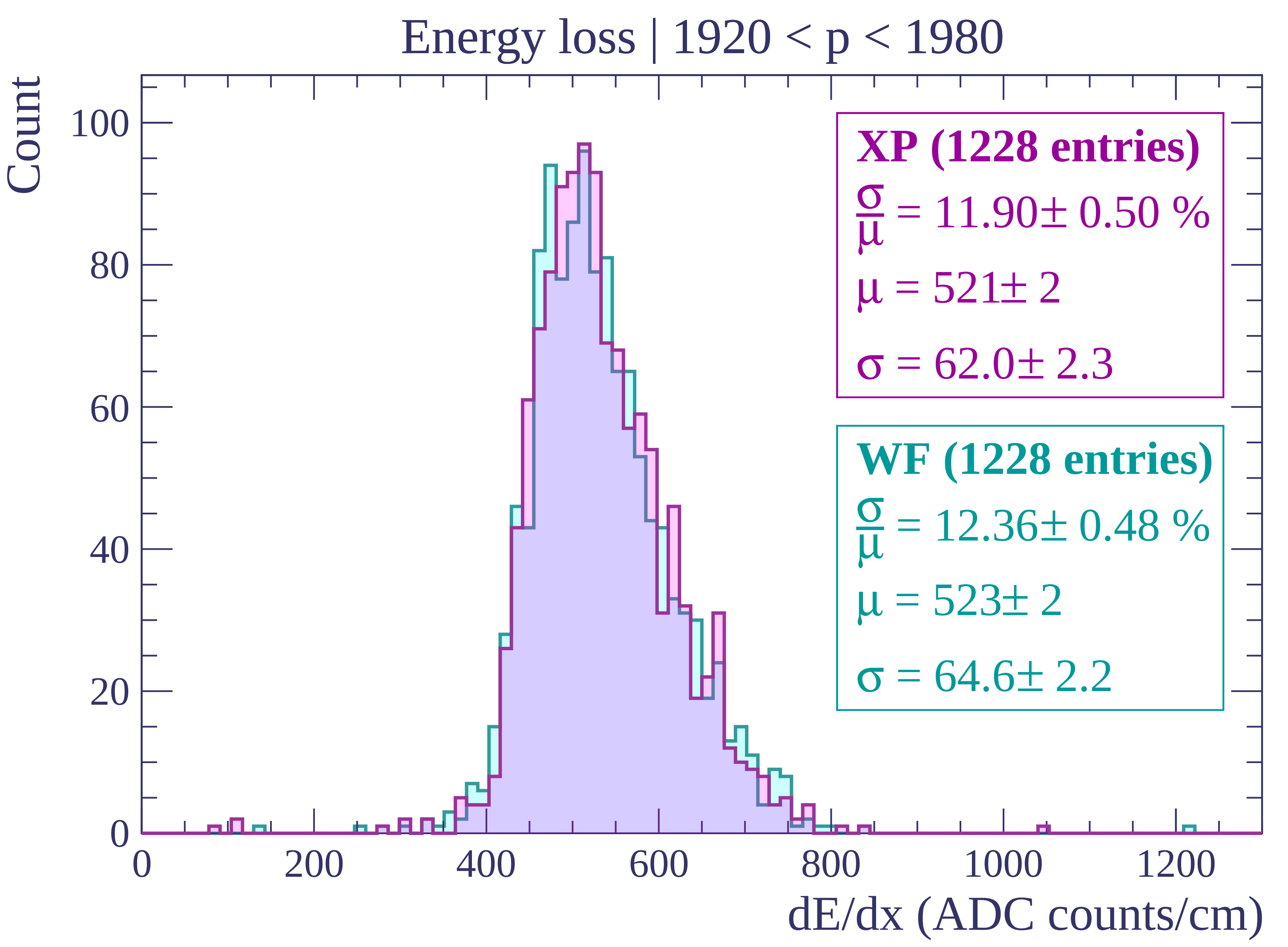
Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1316 entries)
$\frac{\sigma}{\mu}$ = 12.24± 0.42 %
μ = 527± 3
σ = 64.5± 1.9
WF (1316 entries)
$\frac{\sigma}{\mu}$ = 12.26± 0.43 %
μ = 522± 2
σ = 64.0± 2.0



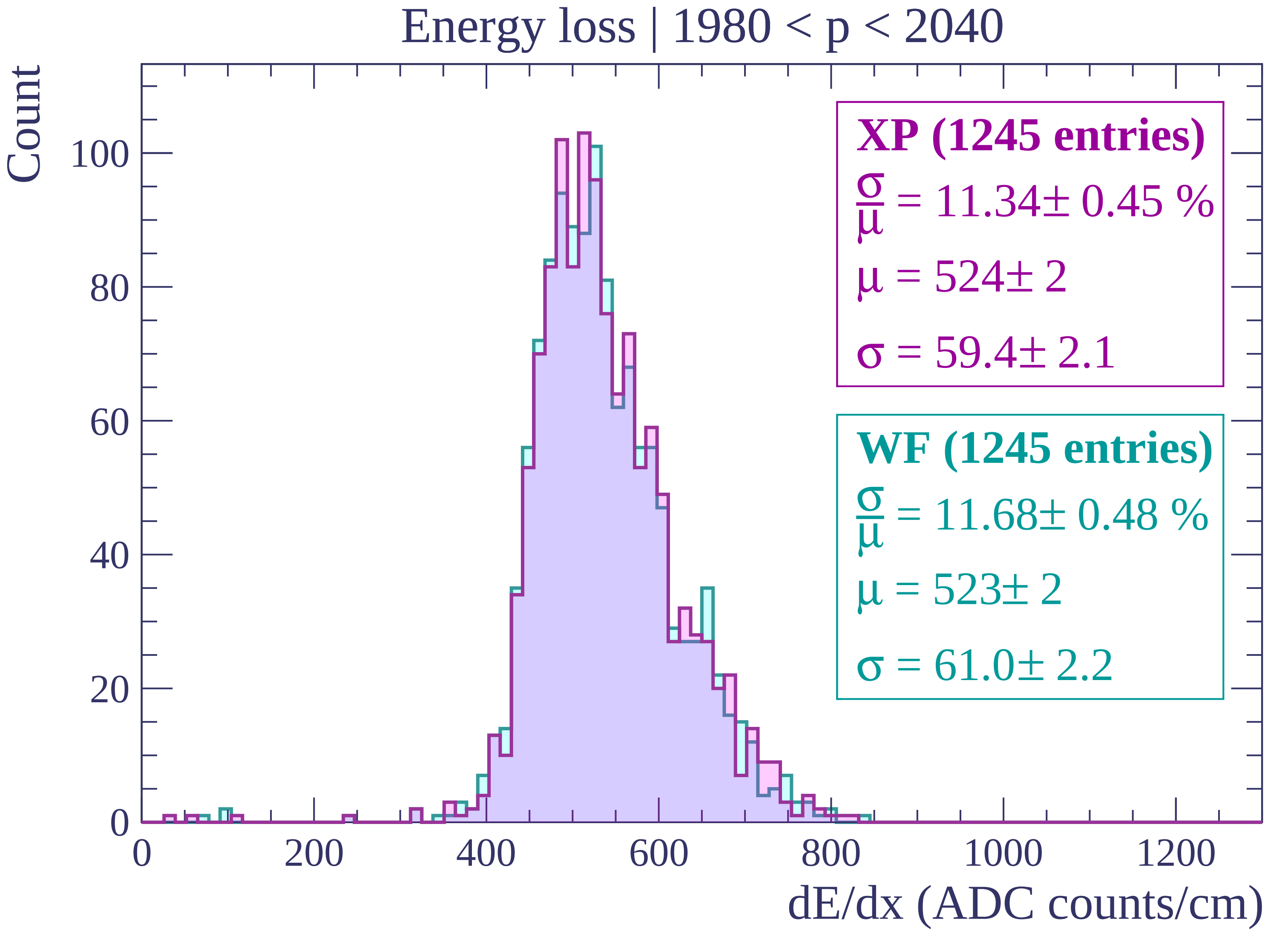
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1312 entries)
σ/μ = 12.24± 0.48 %
μ = 524± 2
σ = 64.1± 2.2
WF (1312 entries)
σ/μ = 11.83± 0.43 %
μ = 519± 2
σ = 61.4± 2.0



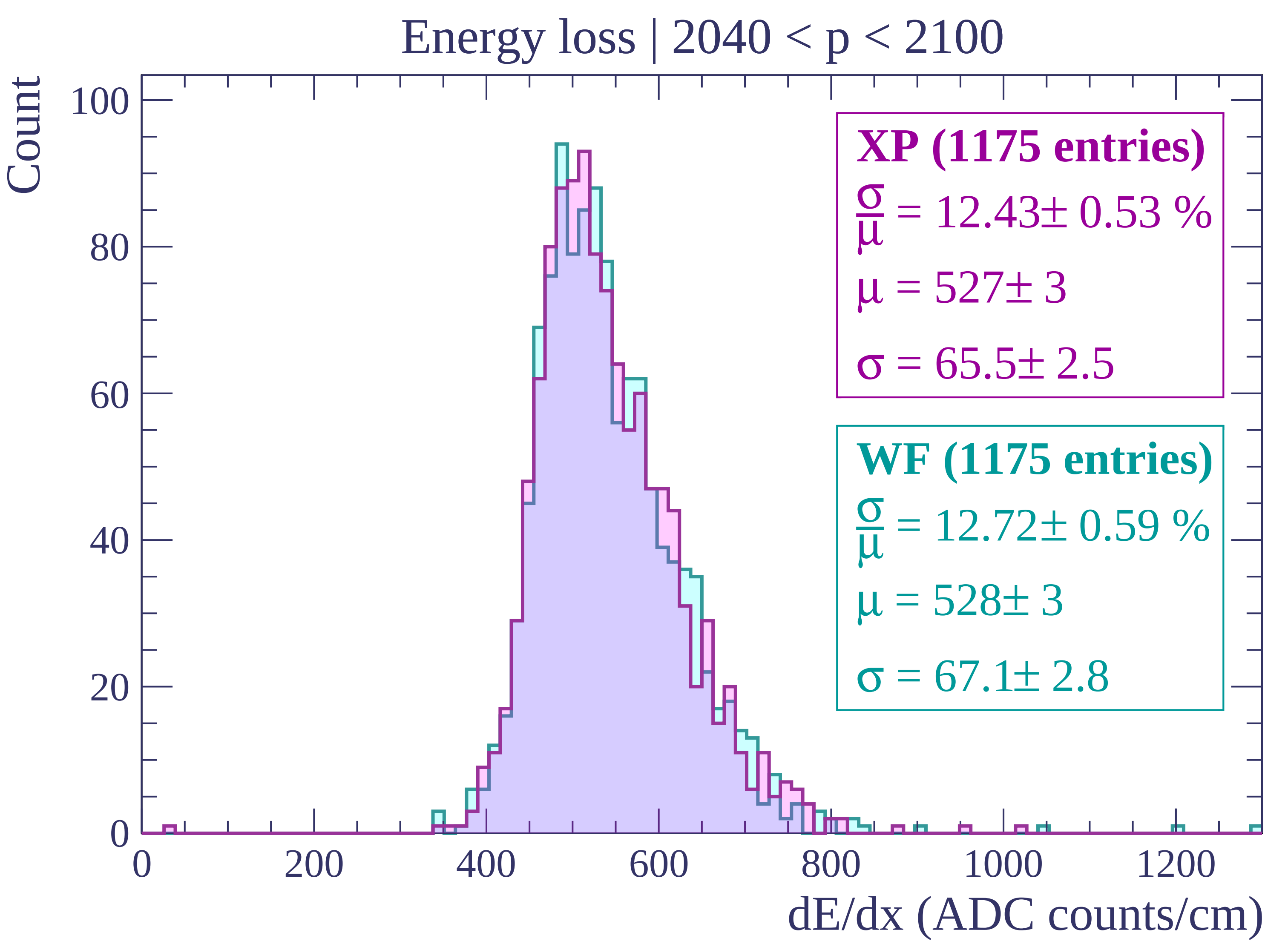
Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1322 entries)
σ/μ = 12.05± 0.46 %
μ = 528± 2
σ = 63.6± 2.1
WF (1322 entries)
σ/μ = 12.83± 0.48 %
μ = 529± 2
σ = 67.8± 2.2



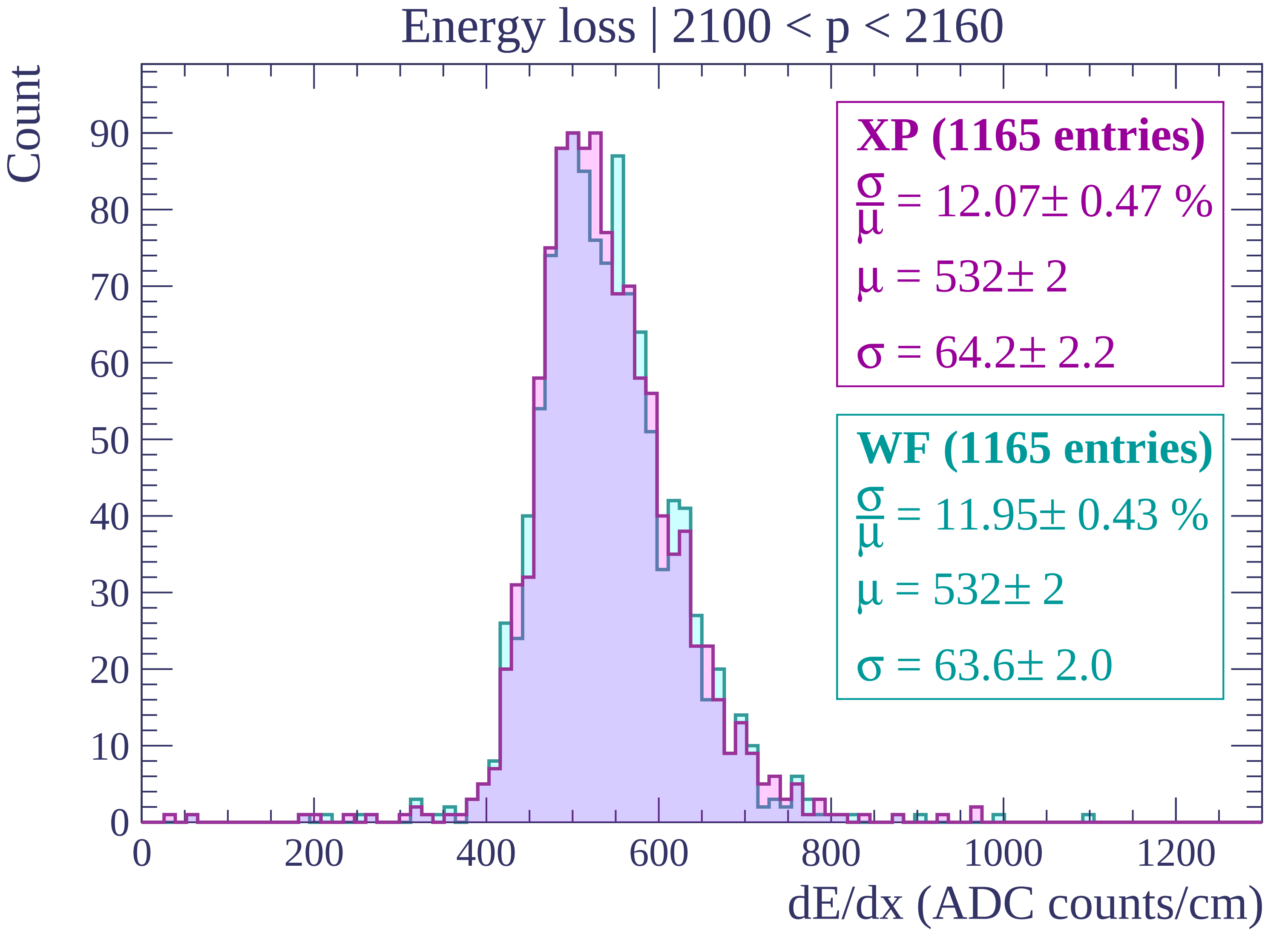
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1228 entries)
$\frac{\sigma}{\mu}$ = 11.90± 0.50 %
μ = 521± 2
σ = 62.0± 2.3
WF (1228 entries)
$\frac{\sigma}{\mu}$ = 12.36± 0.48 %
μ = 523± 2
σ = 64.6± 2.2



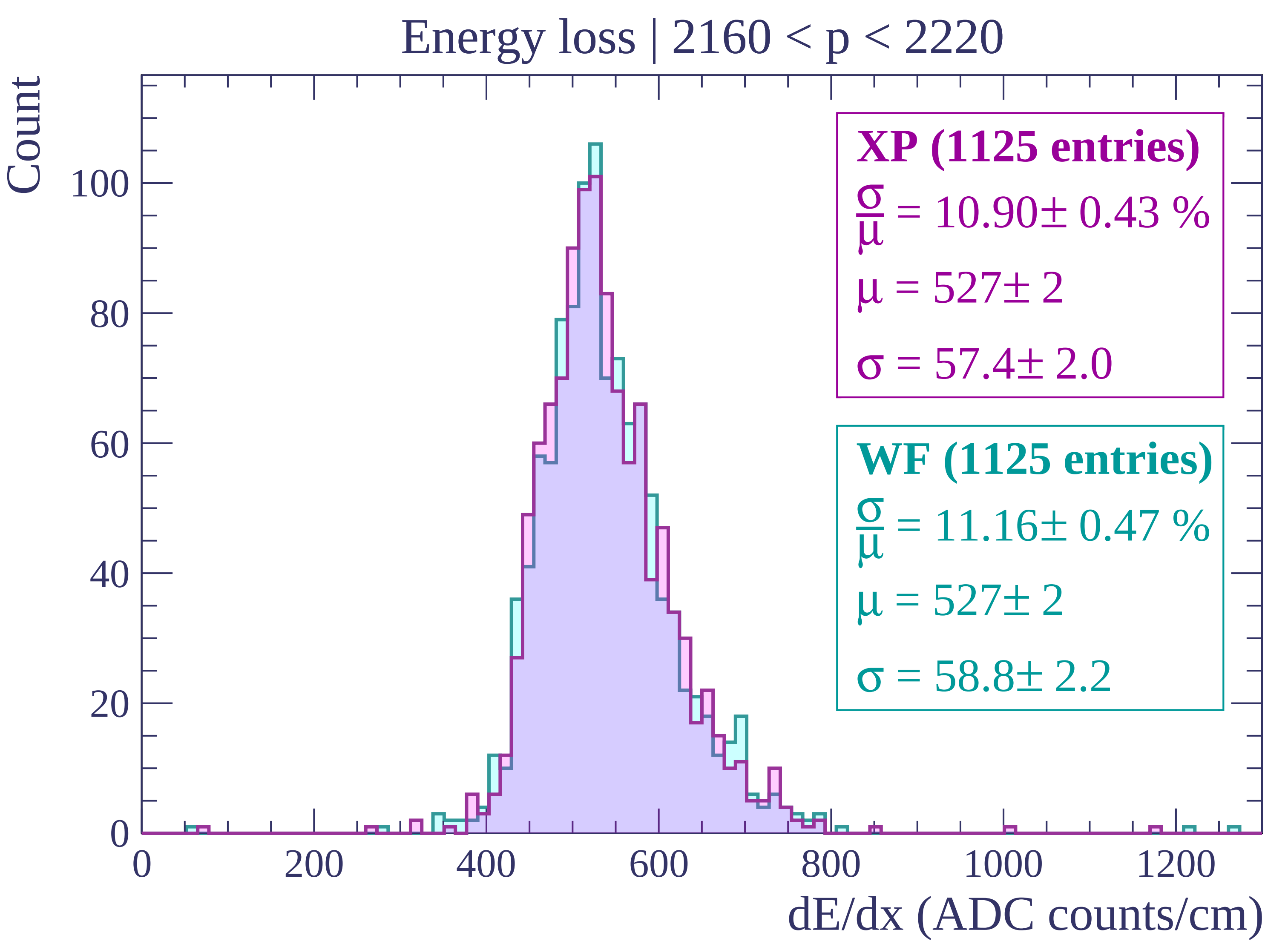
Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
XP (1245 entries)
σ/μ = 11.34± 0.45 %
μ = 524± 2
σ = 59.4± 2.1
WF (1245 entries)
σ/μ = 11.68± 0.48 %
μ = 523± 2
σ = 61.0± 2.2
0
20
40
60
80
100
0
200
400
600
800
1000
1200



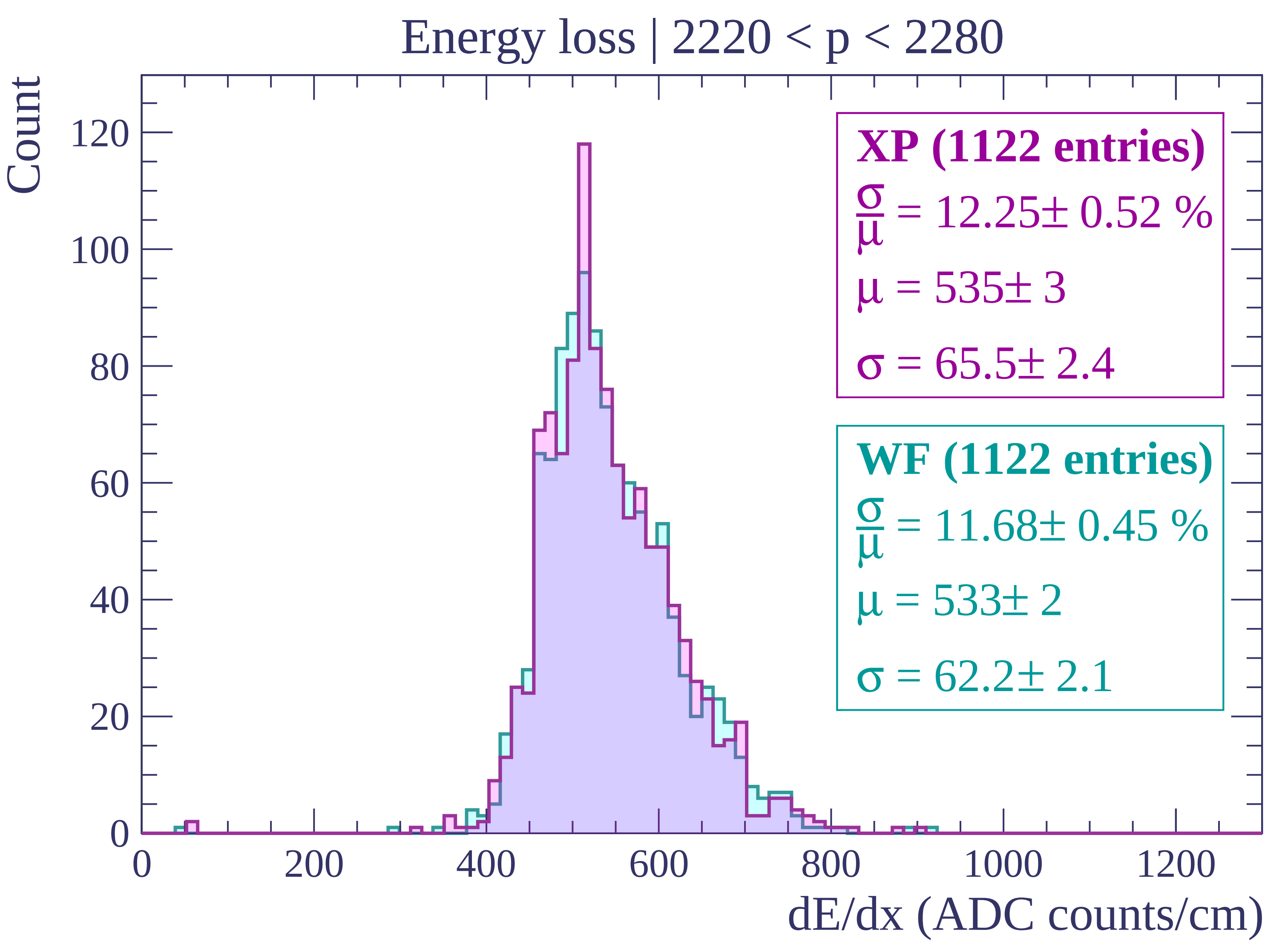
Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1175 entries)
$\frac{\sigma}{\mu}$ = 12.43± 0.53 %
μ = 527± 3
σ = 65.5± 2.5
WF (1175 entries)
$\frac{\sigma}{\mu}$ = 12.72± 0.59 %
μ = 528± 3
σ = 67.1± 2.8



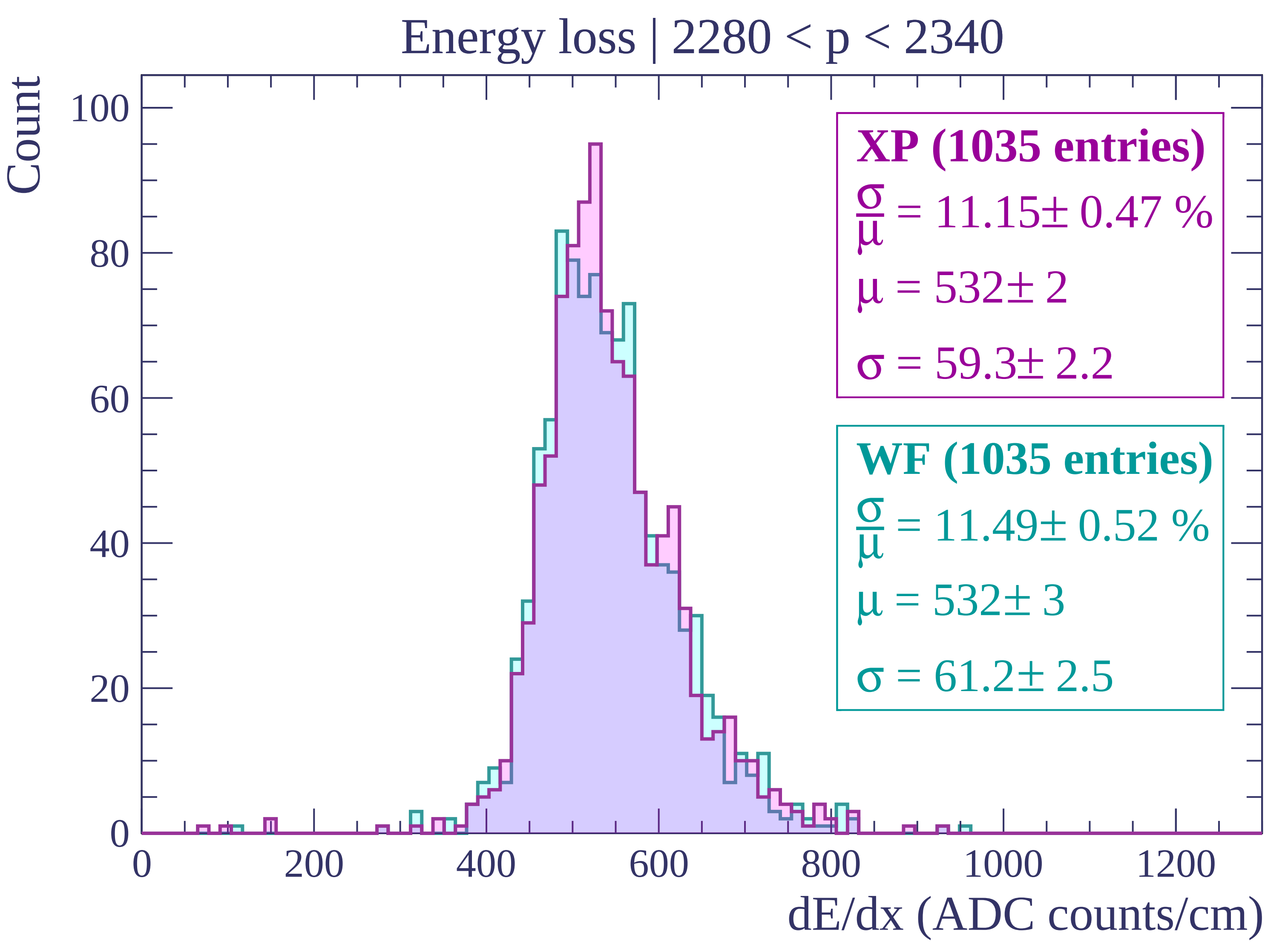
Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (1165 entries)
$\frac{\sigma}{\mu}$ = 12.07± 0.47 %
μ = 532± 2
σ = 64.2± 2.2
WF (1165 entries)
$\frac{\sigma}{\mu}$ = 11.95± 0.43 %
μ = 532± 2
σ = 63.6± 2.0



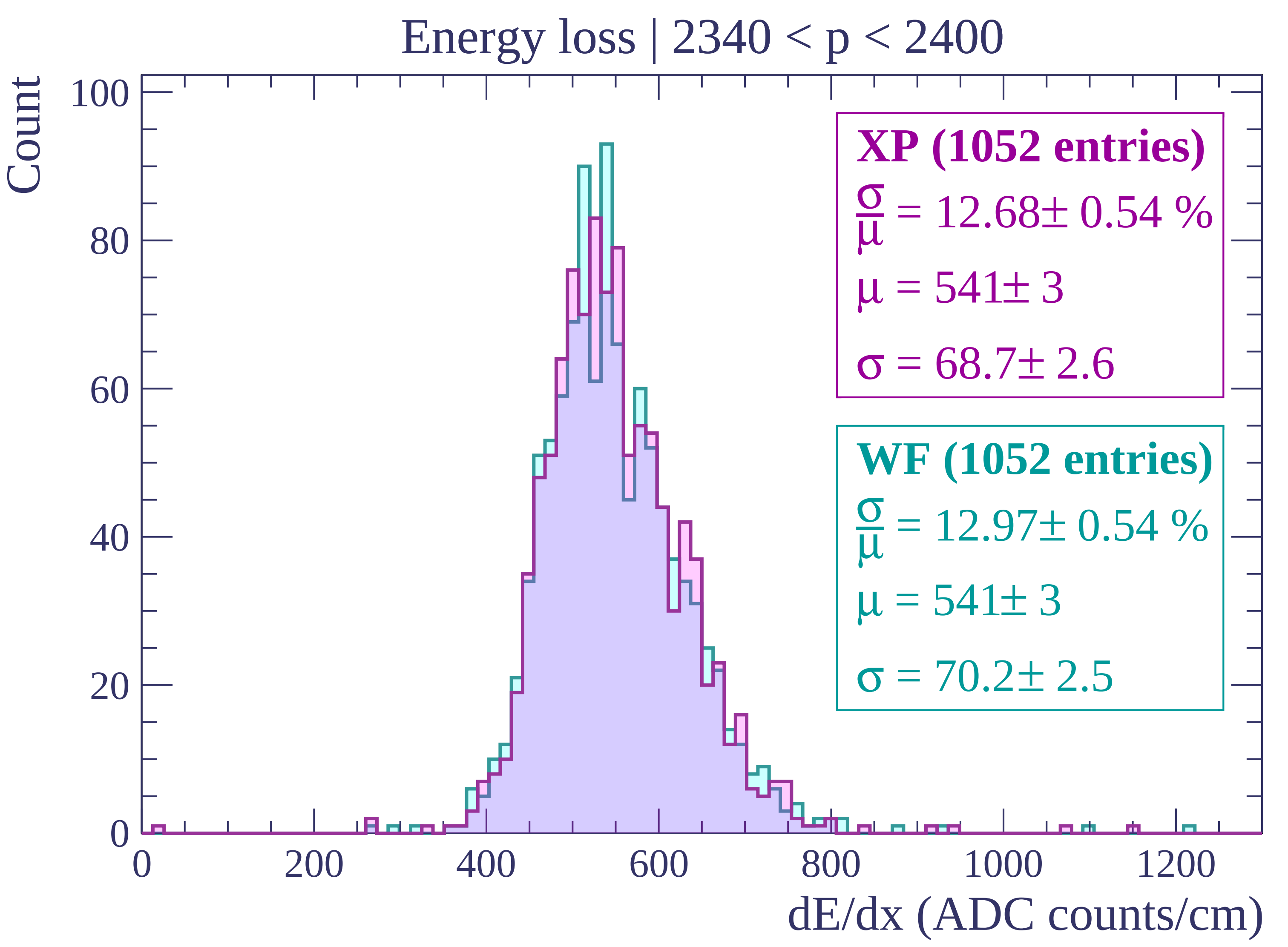
Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (1125 entries)
σ/μ = 10.90± 0.43 %
μ = 527± 2
σ = 57.4± 2.0
WF (1125 entries)
σ/μ = 11.16± 0.47 %
μ = 527± 2
σ = 58.8± 2.2

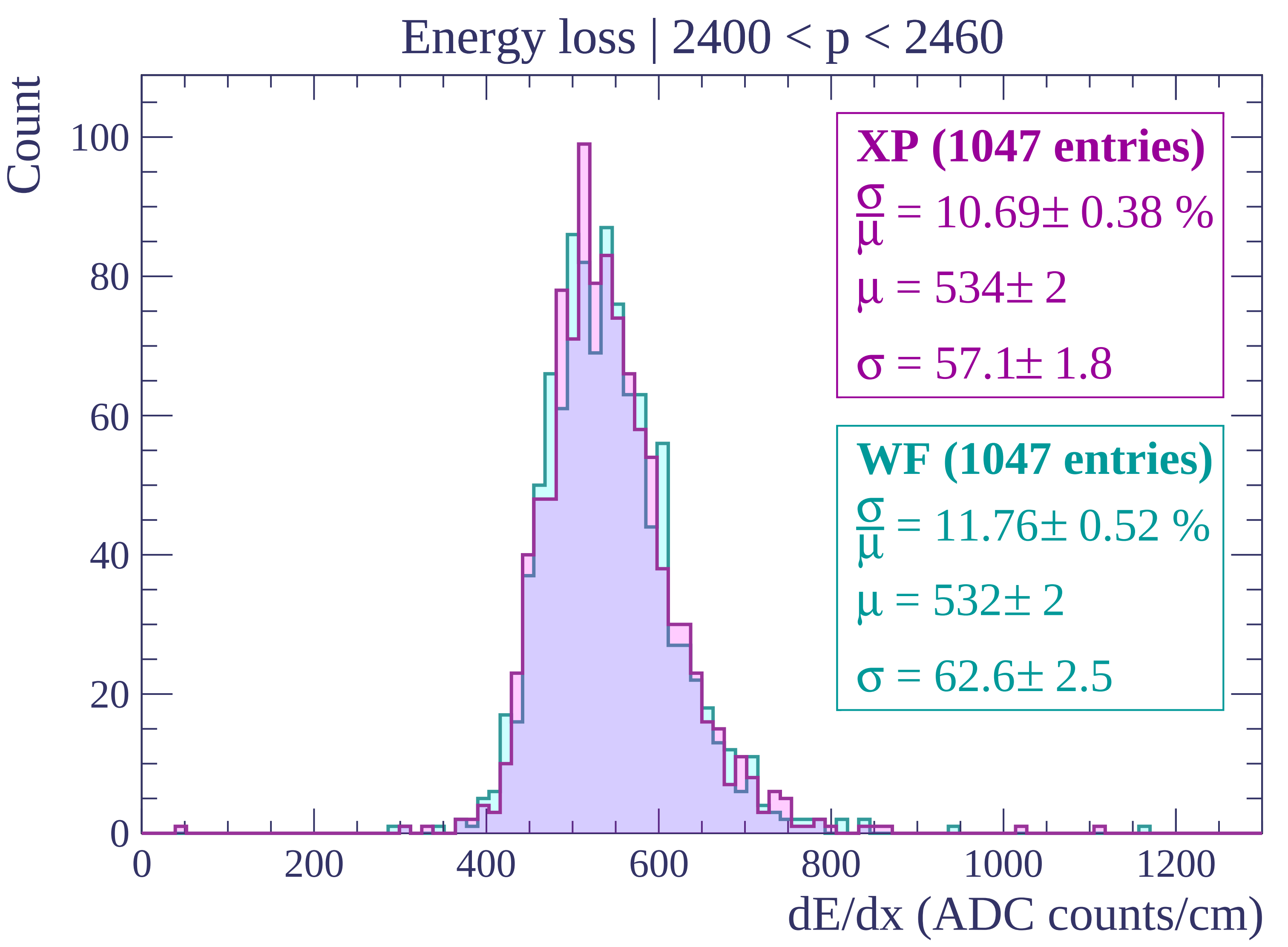


Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP (1122 entries)
σ/μ = 12.25± 0.52 %
μ = 535± 3
σ = 65.5± 2.4
WF (1122 entries)
σ/μ = 11.68± 0.45 %
μ = 533± 2
σ = 62.2± 2.1

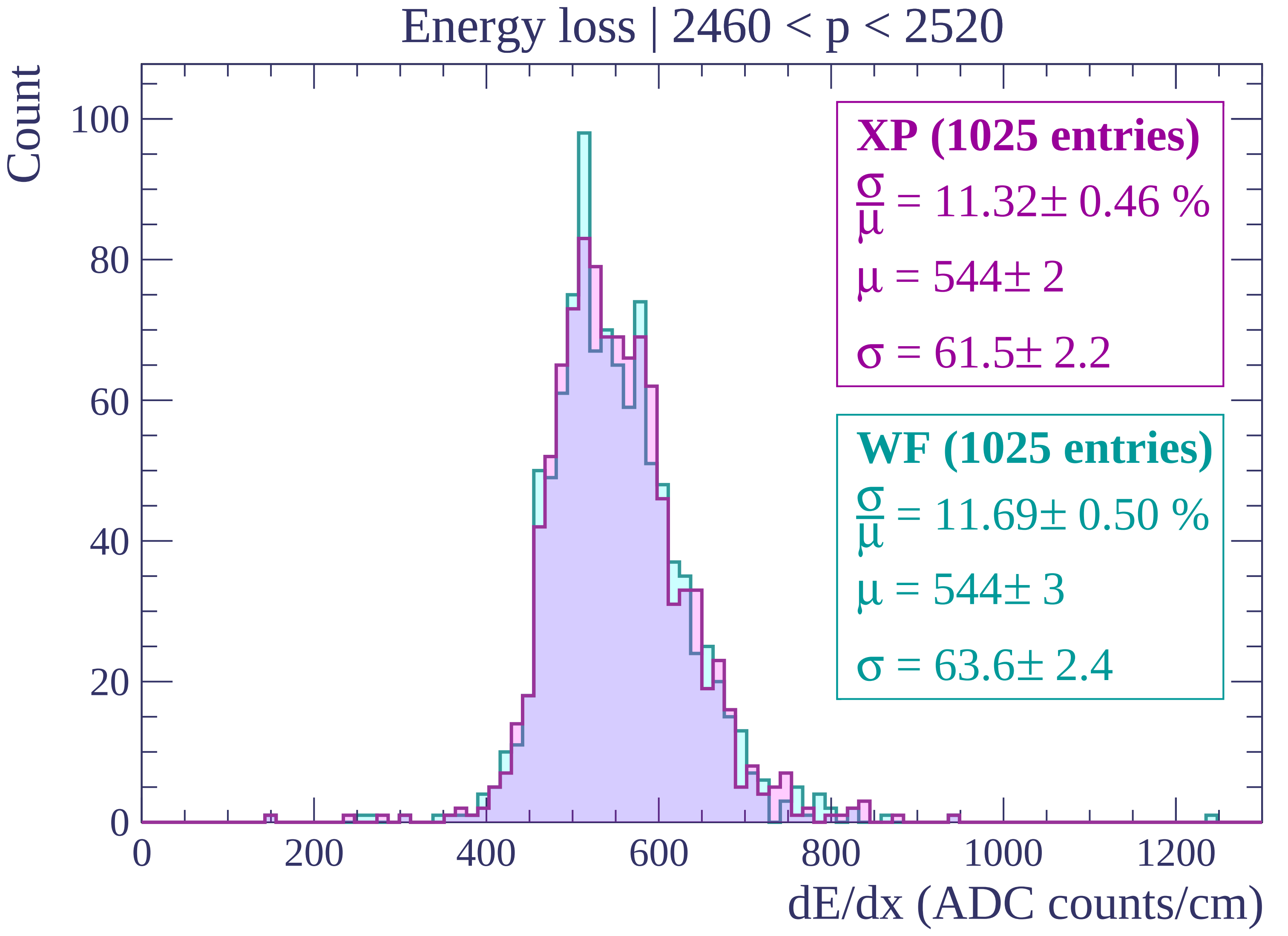


Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1035 entries)
σ/μ = 11.15± 0.47 %
μ = 532± 2
σ = 59.3± 2.2
WF (1035 entries)
σ/μ = 11.49± 0.52 %
μ = 532± 3
σ = 61.2± 2.5

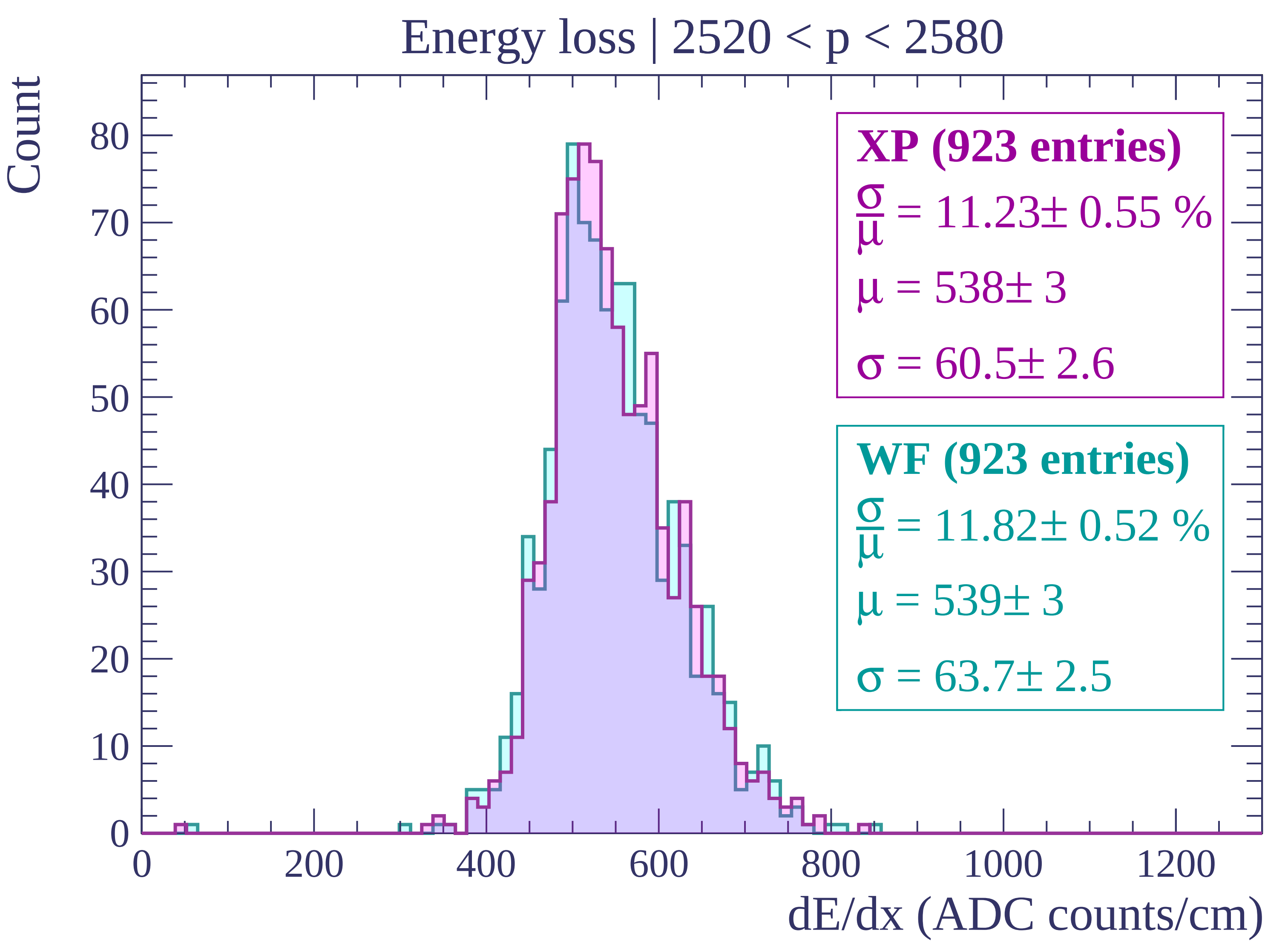

Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1052 entries)
$\frac{\sigma}{\mu}$ = 12.68± 0.54 %
μ = 541± 3
σ = 68.7± 2.6
WF (1052 entries)
$\frac{\sigma}{\mu}$ = 12.97± 0.54 %
μ = 541± 3
σ = 70.2± 2.5



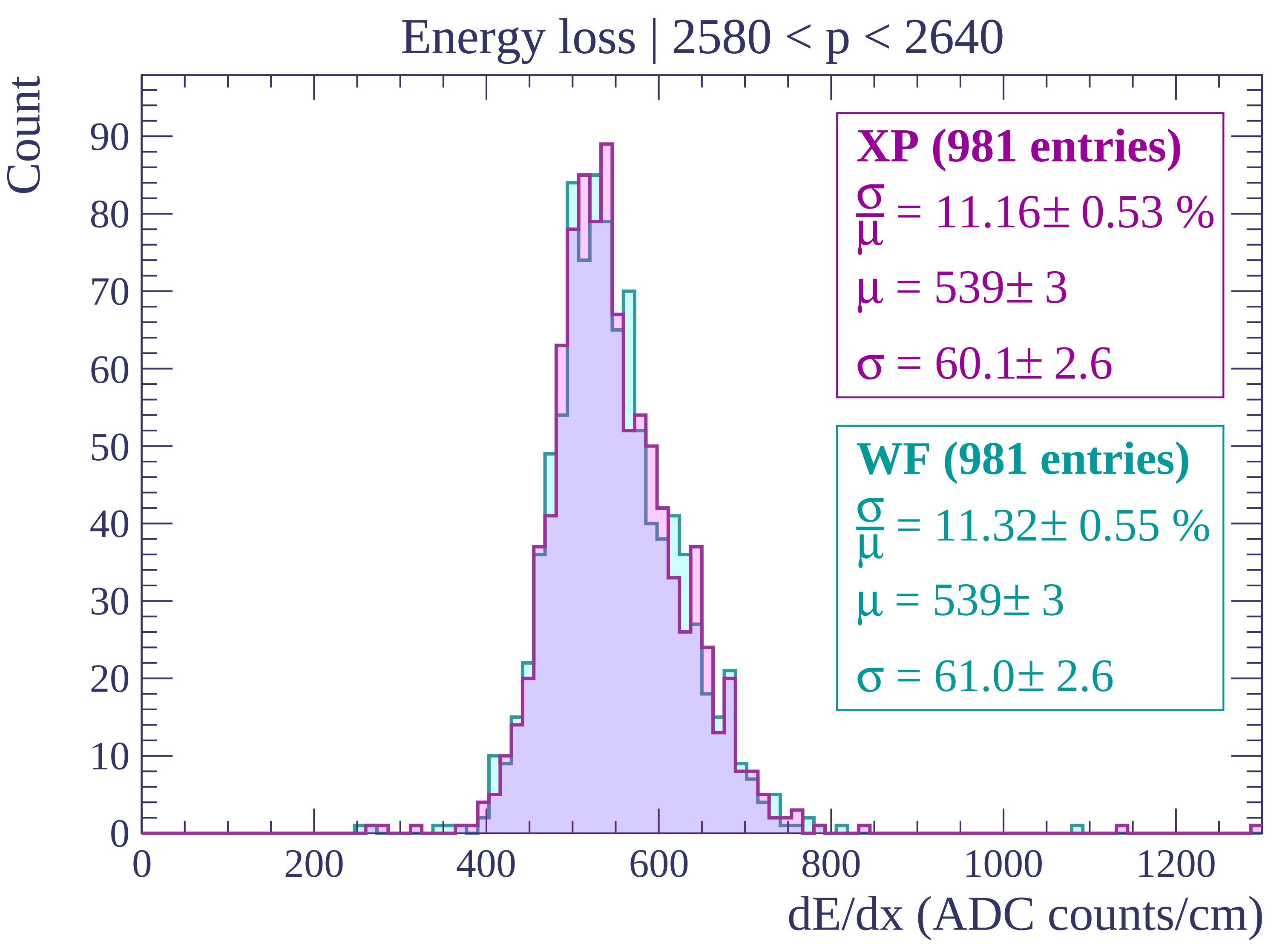
Energy loss | 2400 < p < 2460
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1047 entries)
σ/μ = 10.69± 0.38 %
μ = 534± 2
σ = 57.1± 1.8
WF (1047 entries)
σ/μ = 11.76± 0.52 %
μ = 532± 2
σ = 62.6± 2.5

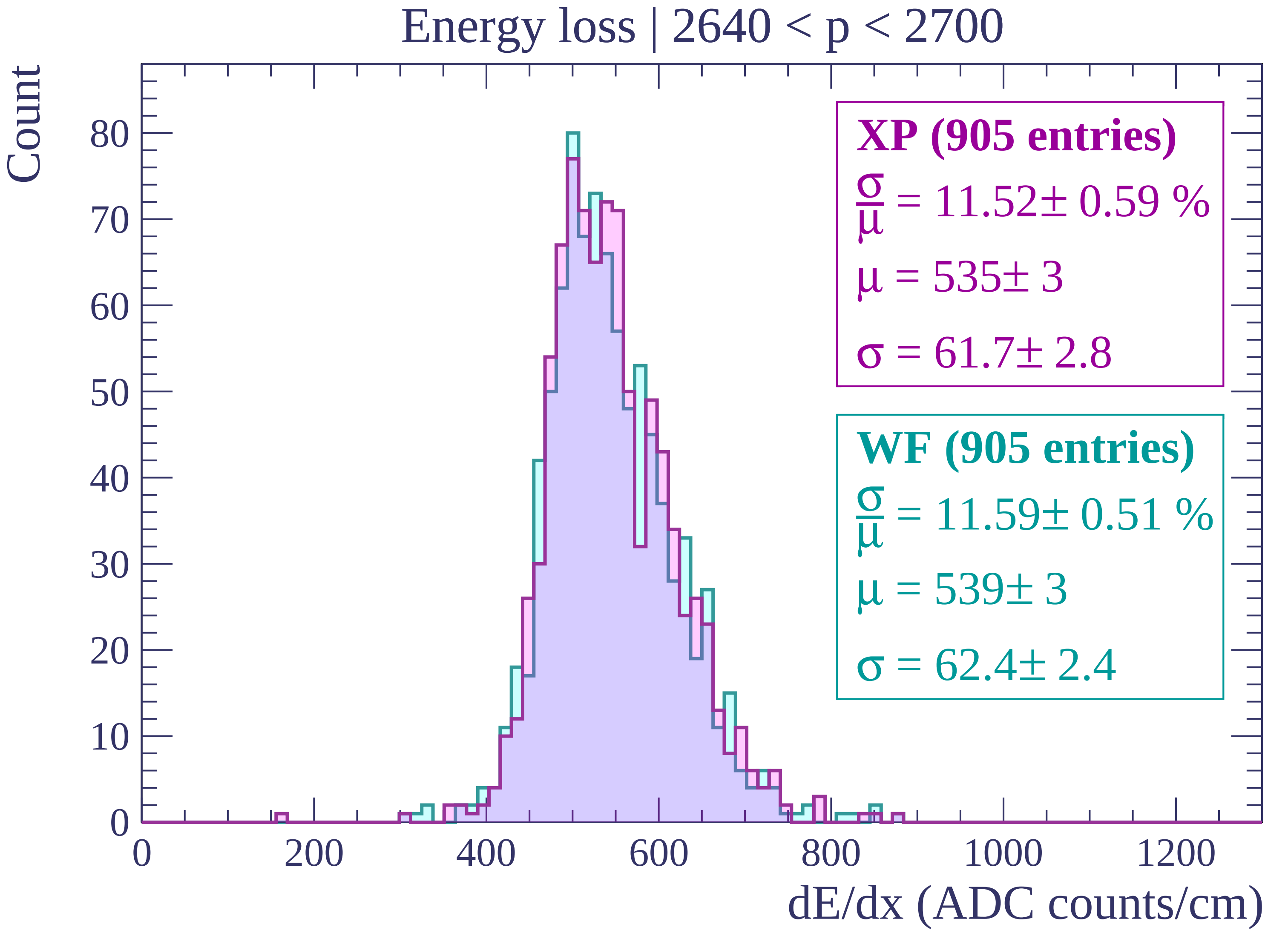


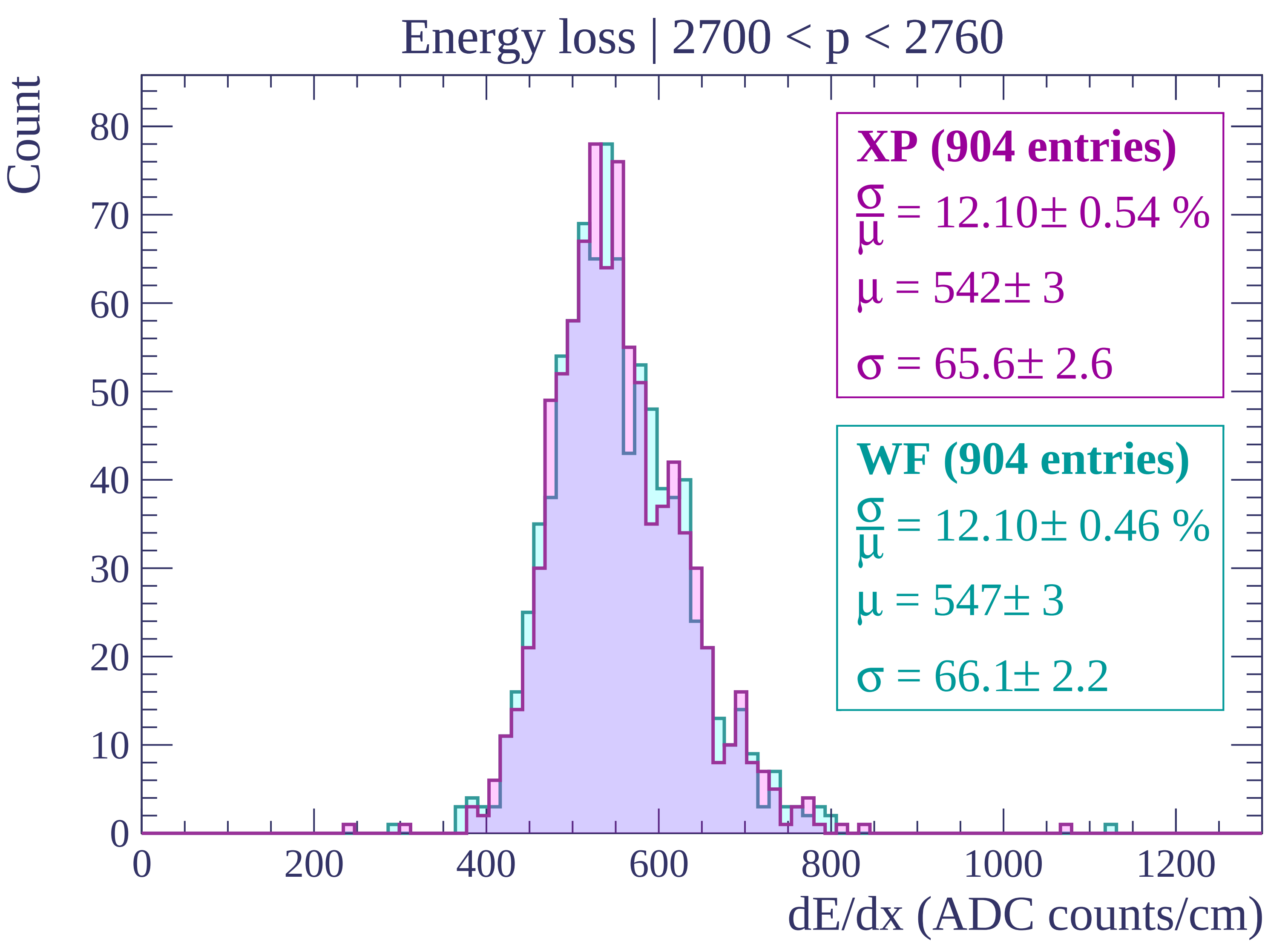
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (1025 entries)
σ/μ = 11.32± 0.46 %
μ = 544± 2
σ = 61.5± 2.2
WF (1025 entries)
σ/μ = 11.69± 0.50 %
μ = 544± 3
σ = 63.6± 2.4

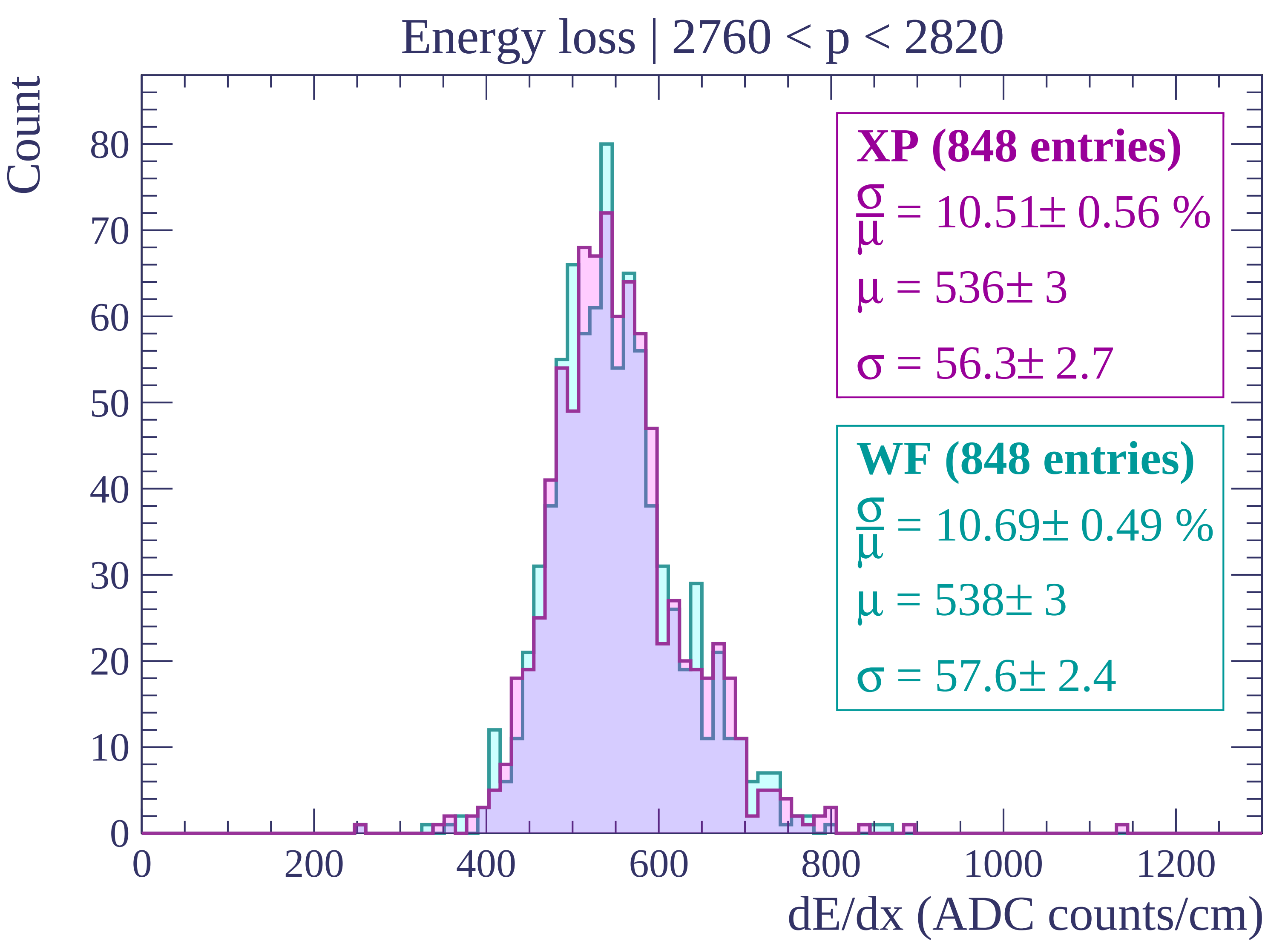


Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (923 entries)
σ/μ = 11.23± 0.55 %
μ = 538± 3
σ = 60.5± 2.6
WF (923 entries)
σ/μ = 11.82± 0.52 %
μ = 539± 3
σ = 63.7± 2.5

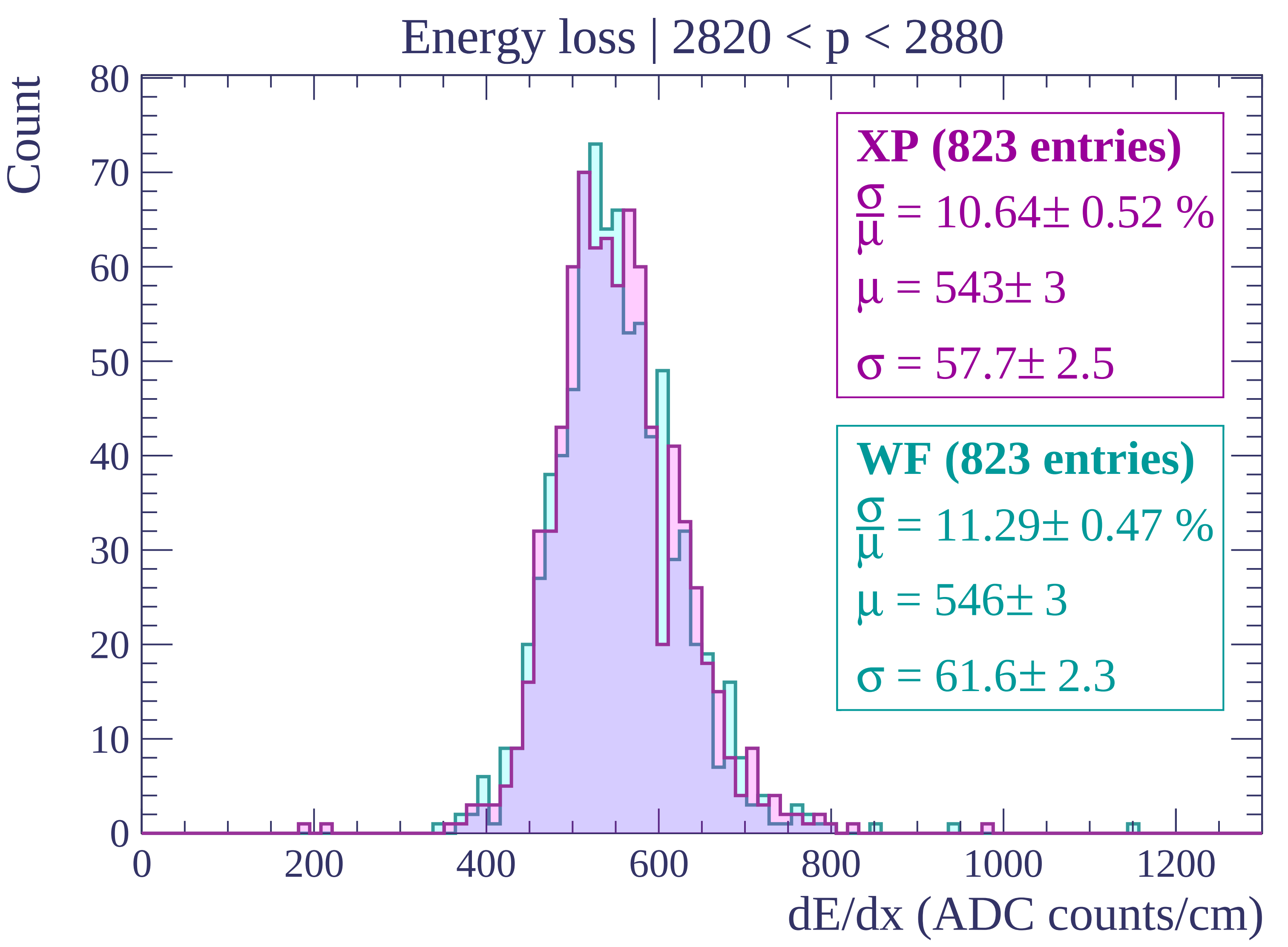

Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (981 entries)
$\frac{\sigma}{\mu}$ = 11.16± 0.53 %
μ = 539± 3
σ = 60.1± 2.6
WF (981 entries)
$\frac{\sigma}{\mu}$ = 11.32± 0.55 %
μ = 539± 3
σ = 61.0± 2.6


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP (905 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.59 %
μ = 535± 3
σ = 61.7± 2.8
WF (905 entries)
$\frac{\sigma}{\mu}$ = 11.59± 0.51 %
μ = 539± 3
σ = 62.4± 2.4

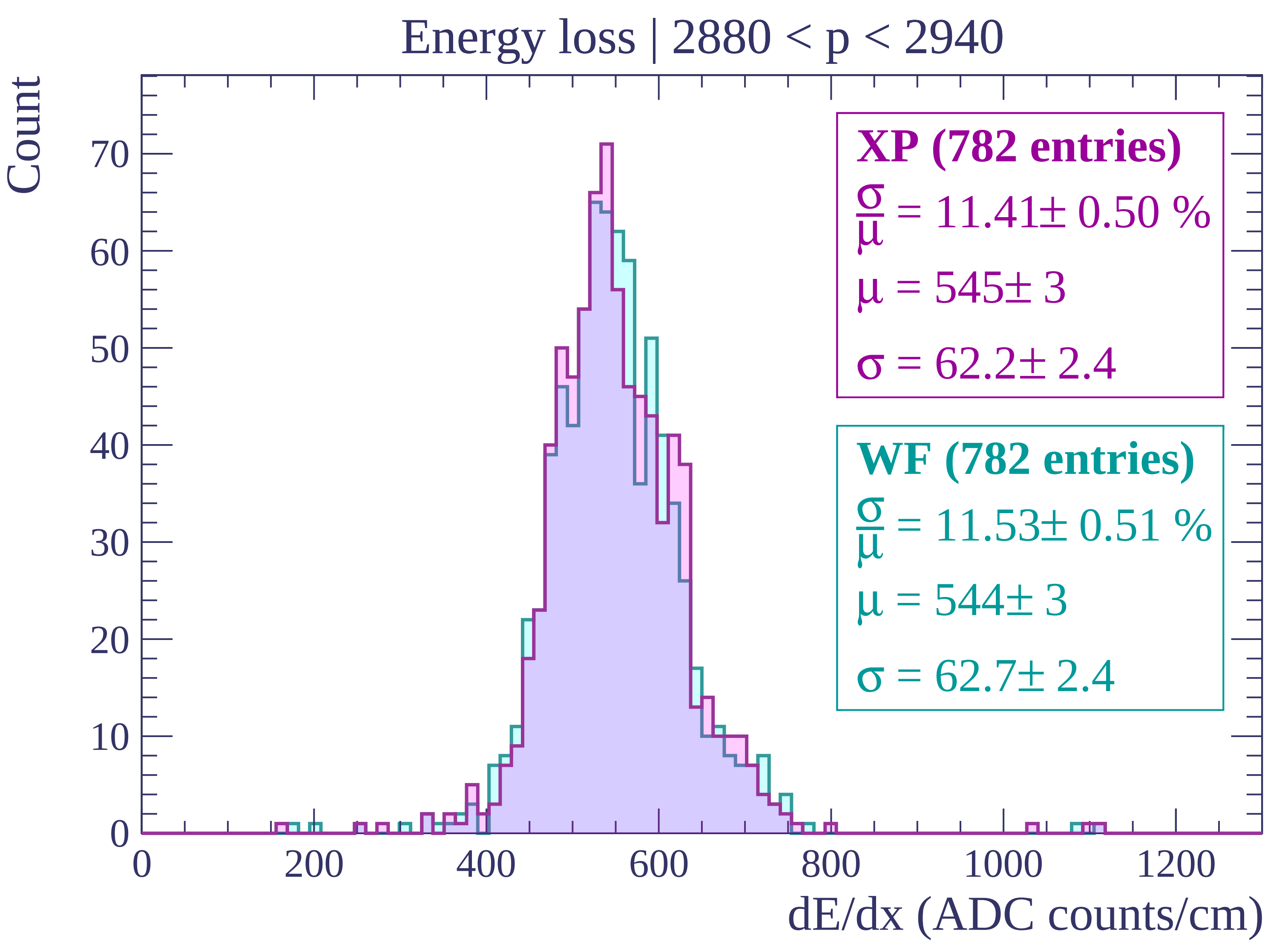

Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP (904 entries)
σ/μ = 12.10± 0.54 %
μ = 542± 3
σ = 65.6± 2.6
WF (904 entries)
σ/μ = 12.10± 0.46 %
μ = 547± 3
σ = 66.1± 2.2



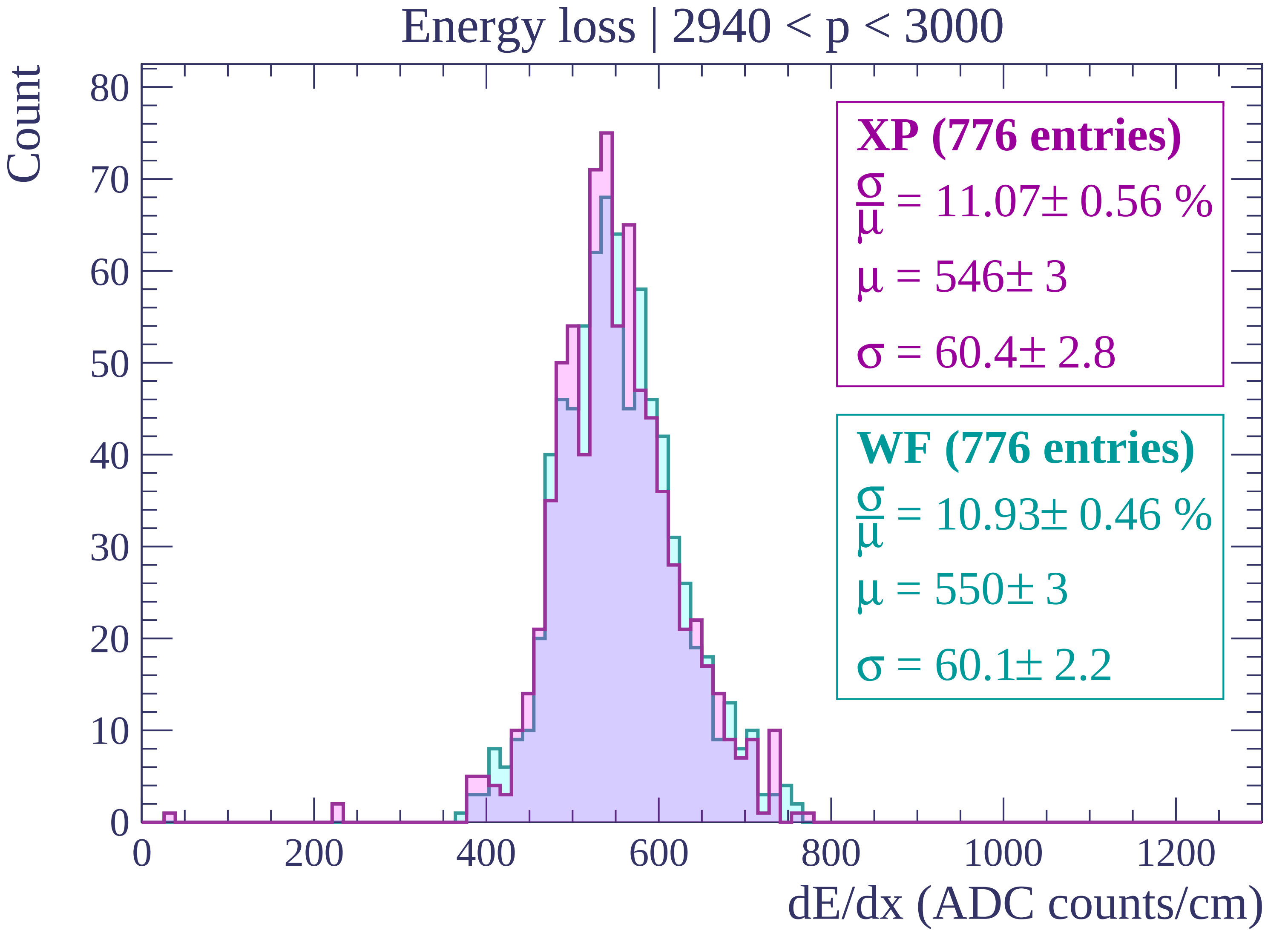
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (848 entries)
σ/μ = 10.51± 0.56 %
μ = 536± 3
σ = 56.3± 2.7
WF (848 entries)
σ/μ = 10.69± 0.49 %
μ = 538± 3
σ = 57.6± 2.4

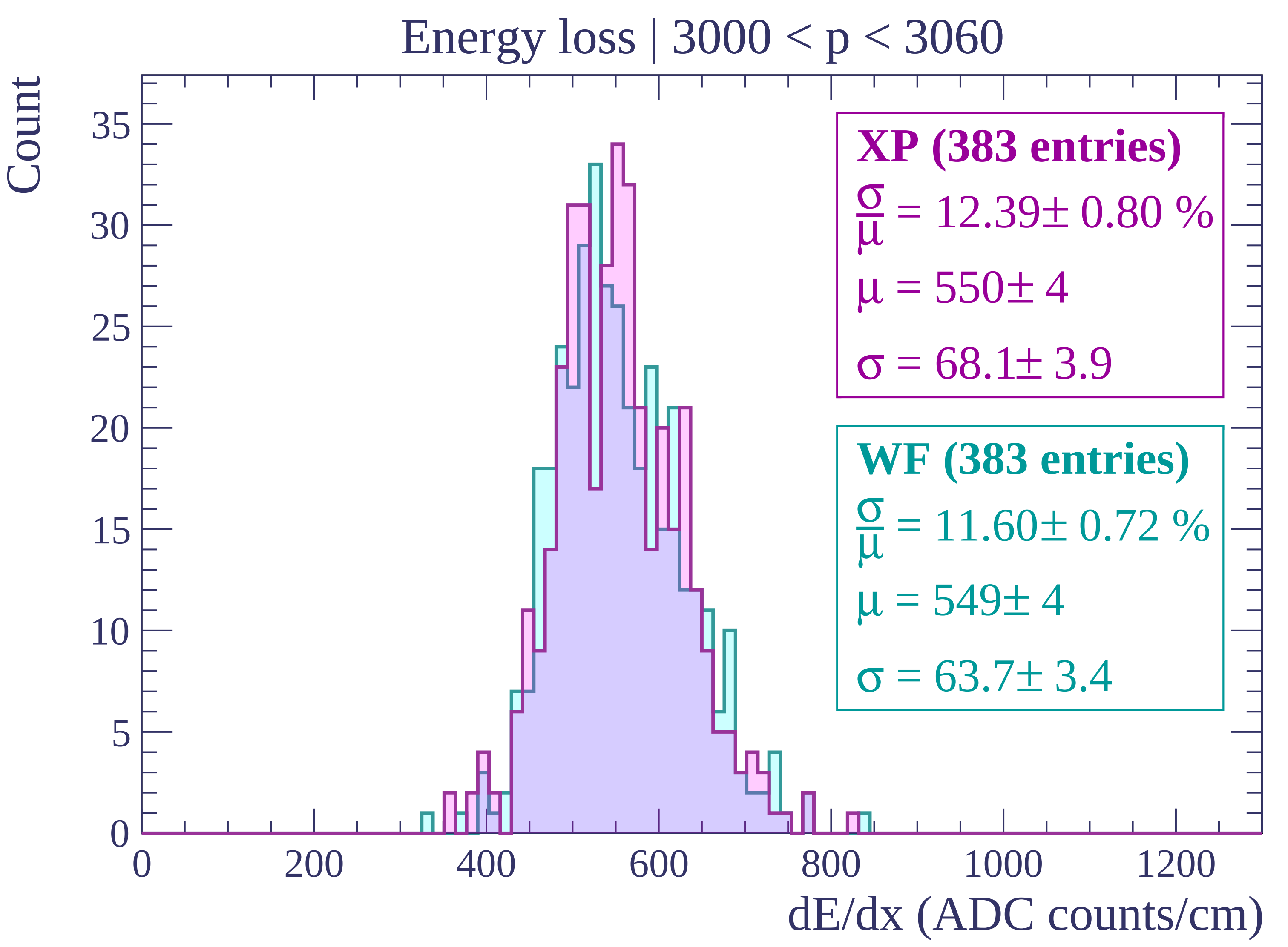


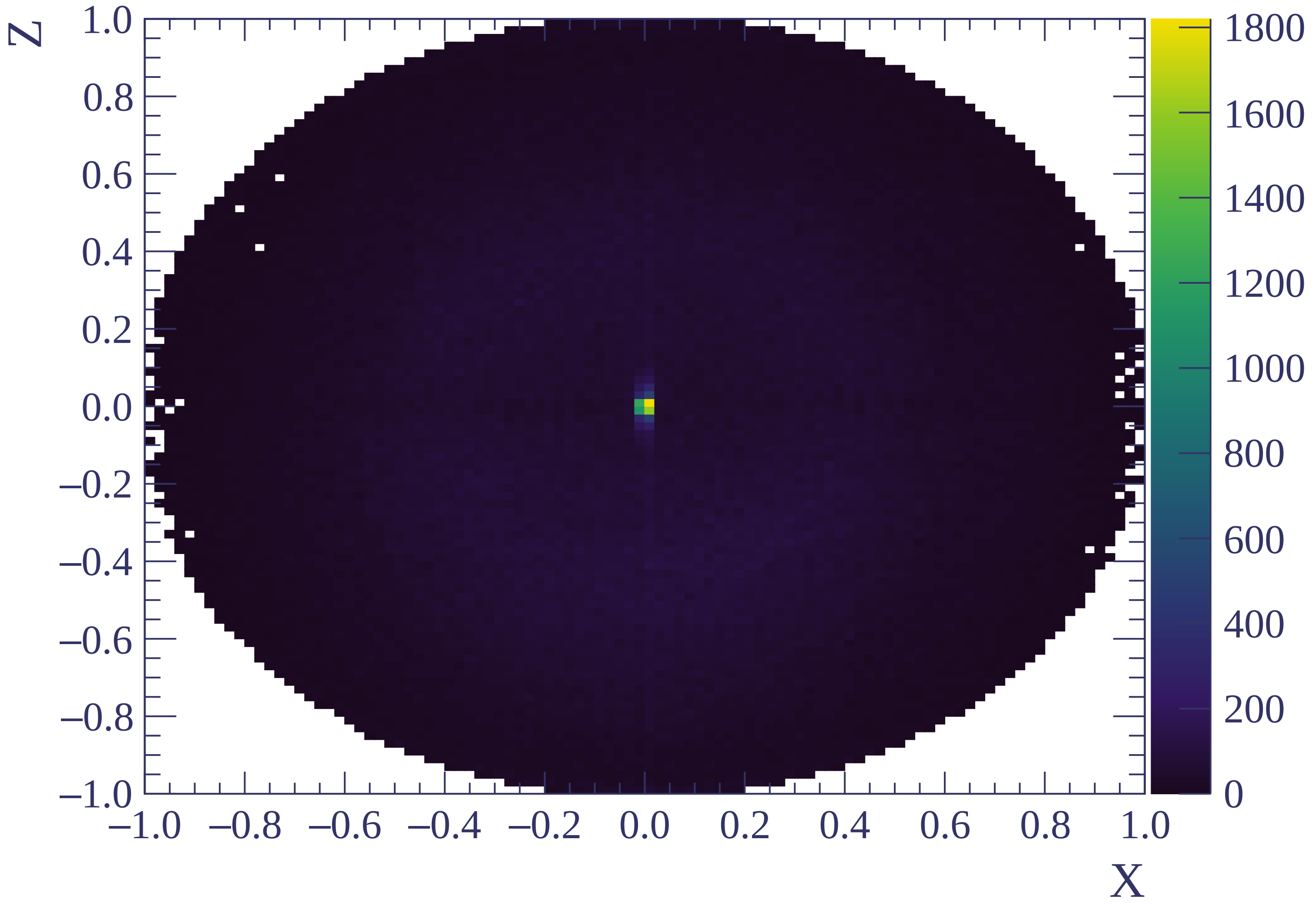
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (823 entries)
$\frac{\sigma}{\mu}$ = 10.64± 0.52 %
μ = 543± 3
σ = 57.7± 2.5
WF (823 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.47 %
μ = 546± 3
σ = 61.6± 2.3



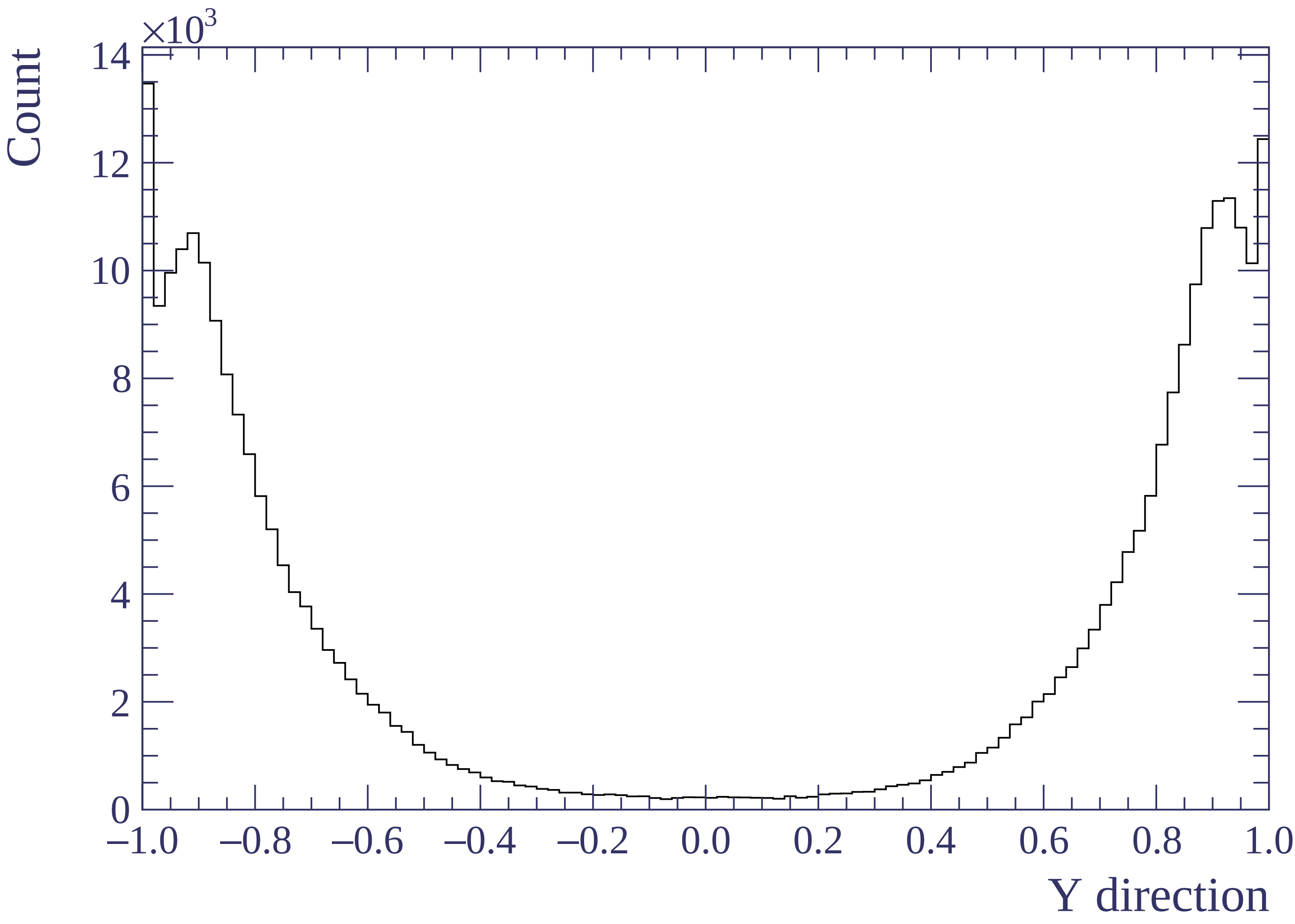
Energy loss | 2880 < p < 2940
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (782 entries)
$\frac{\sigma}{\mu}$ = 11.41± 0.50 %
μ = 545± 3
σ = 62.2± 2.4
WF (782 entries)
$\frac{\sigma}{\mu}$ = 11.53± 0.51 %
μ = 544± 3
σ = 62.7± 2.4


Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (776 entries)
$\frac{\sigma}{\mu}$ = 11.07± 0.56 %
μ = 546± 3
σ = 60.4± 2.8
WF (776 entries)
$\frac{\sigma}{\mu}$ = 10.93± 0.46 %
μ = 550± 3
σ = 60.1± 2.2

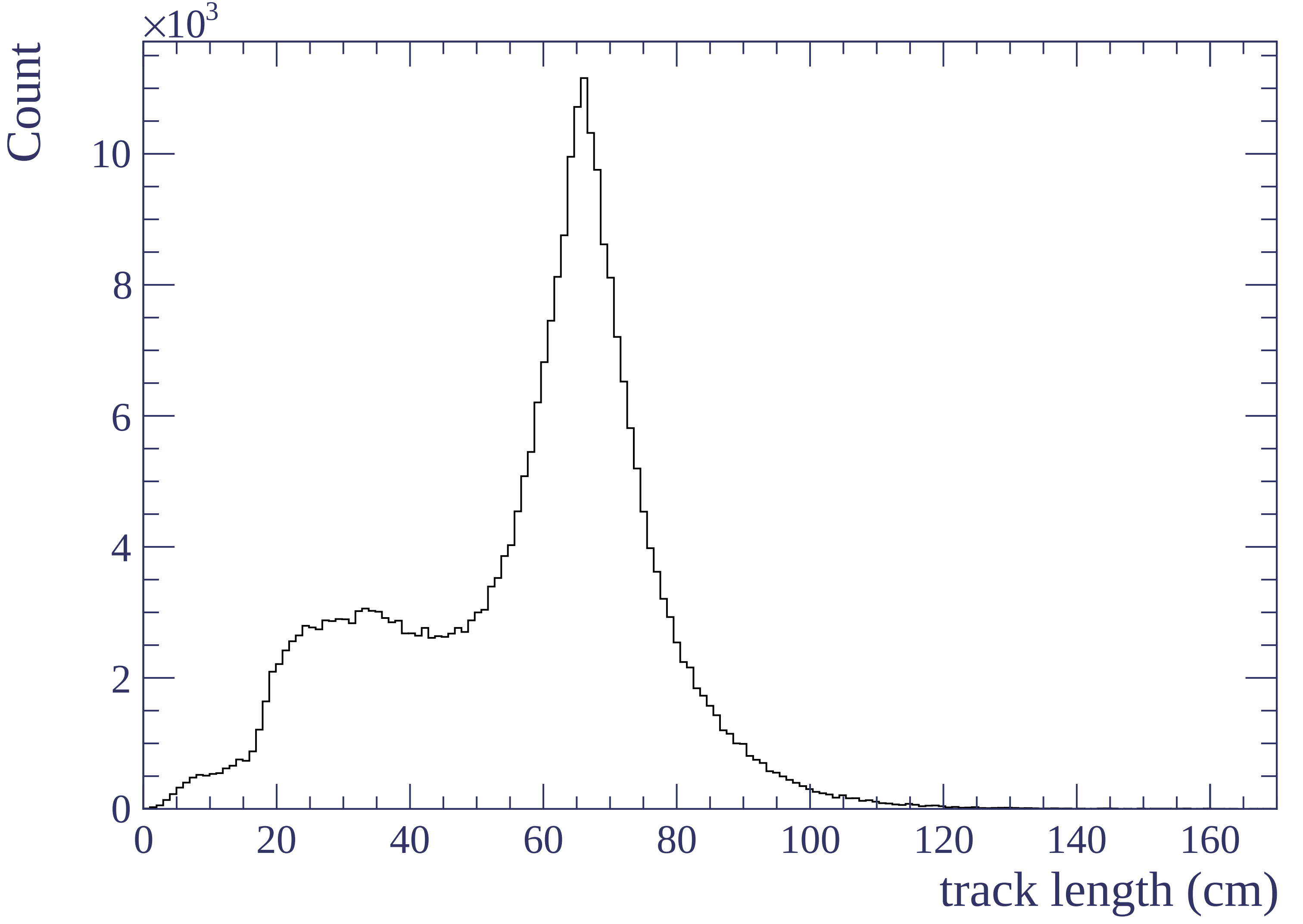

Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
XP (383 entries)
$\frac{\sigma}{\mu}$ = 12.39± 0.80 %
μ = 550± 4
σ = 68.1± 3.9
WF (383 entries)
$\frac{\sigma}{\mu}$ = 11.60± 0.72 %
μ = 549± 4
σ = 63.7± 3.4
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35



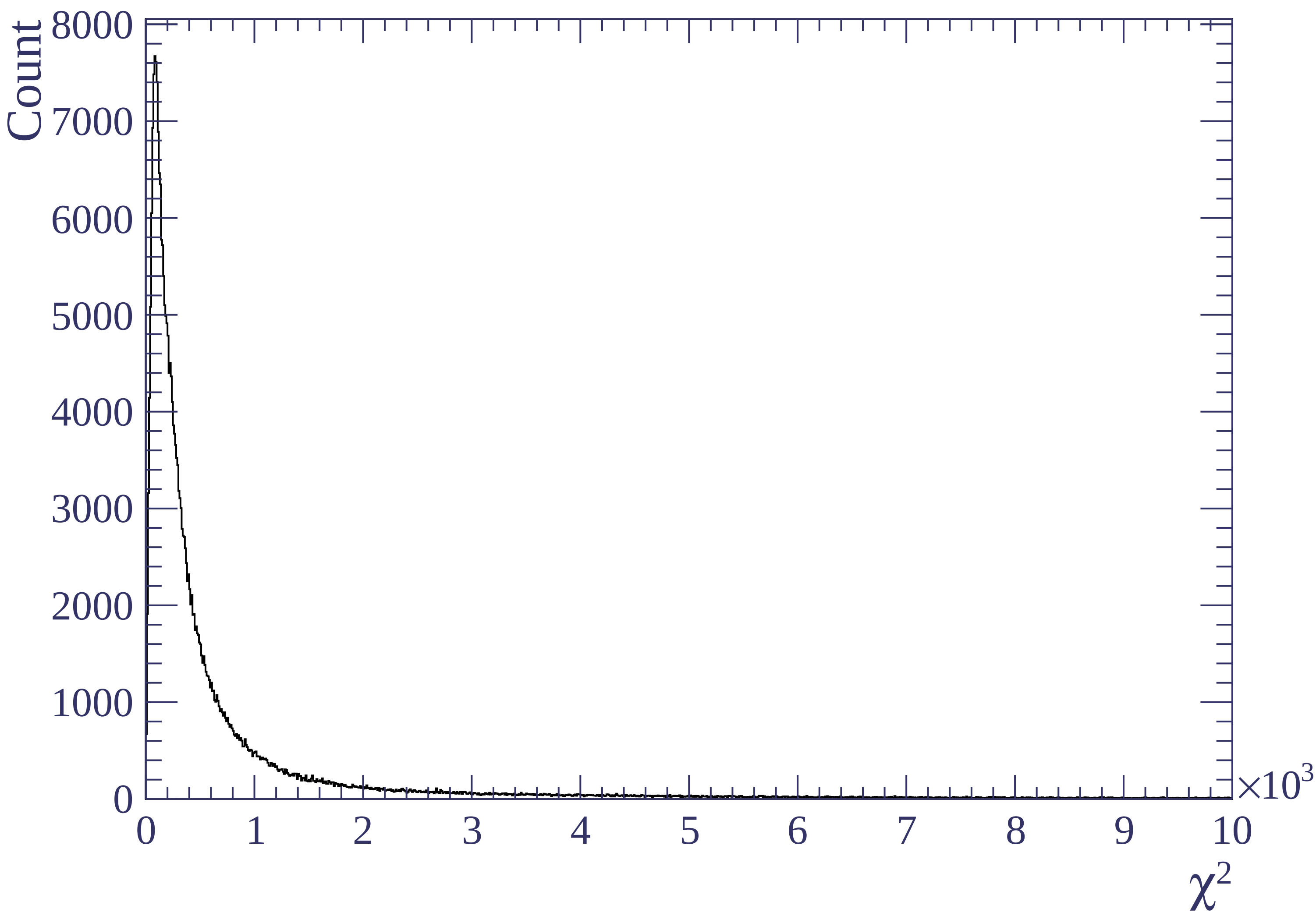
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
1800
1600
1400
1200
1000
800
600
400
200
0



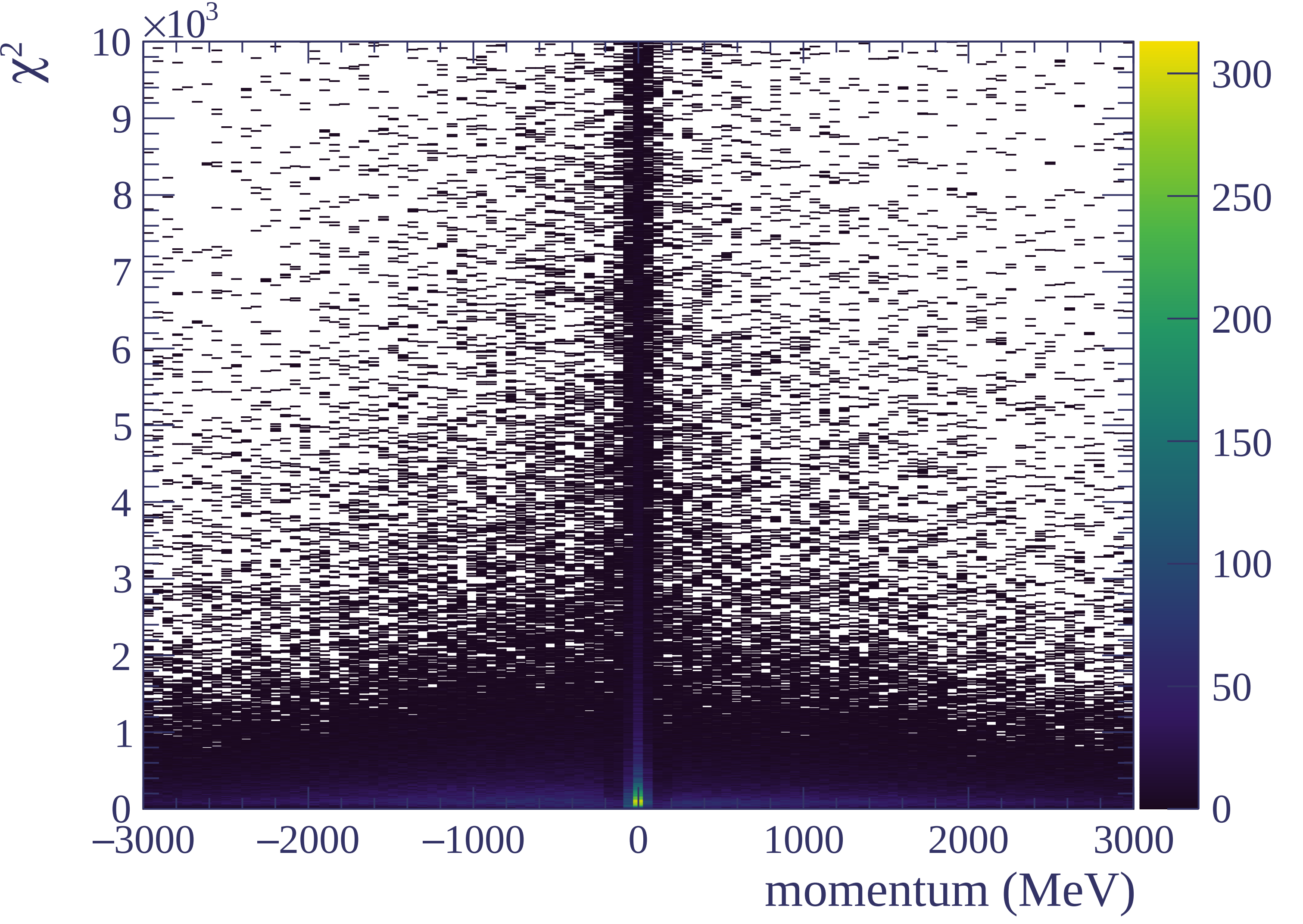
×10³
Count
14
12
10
8
6
4
2
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



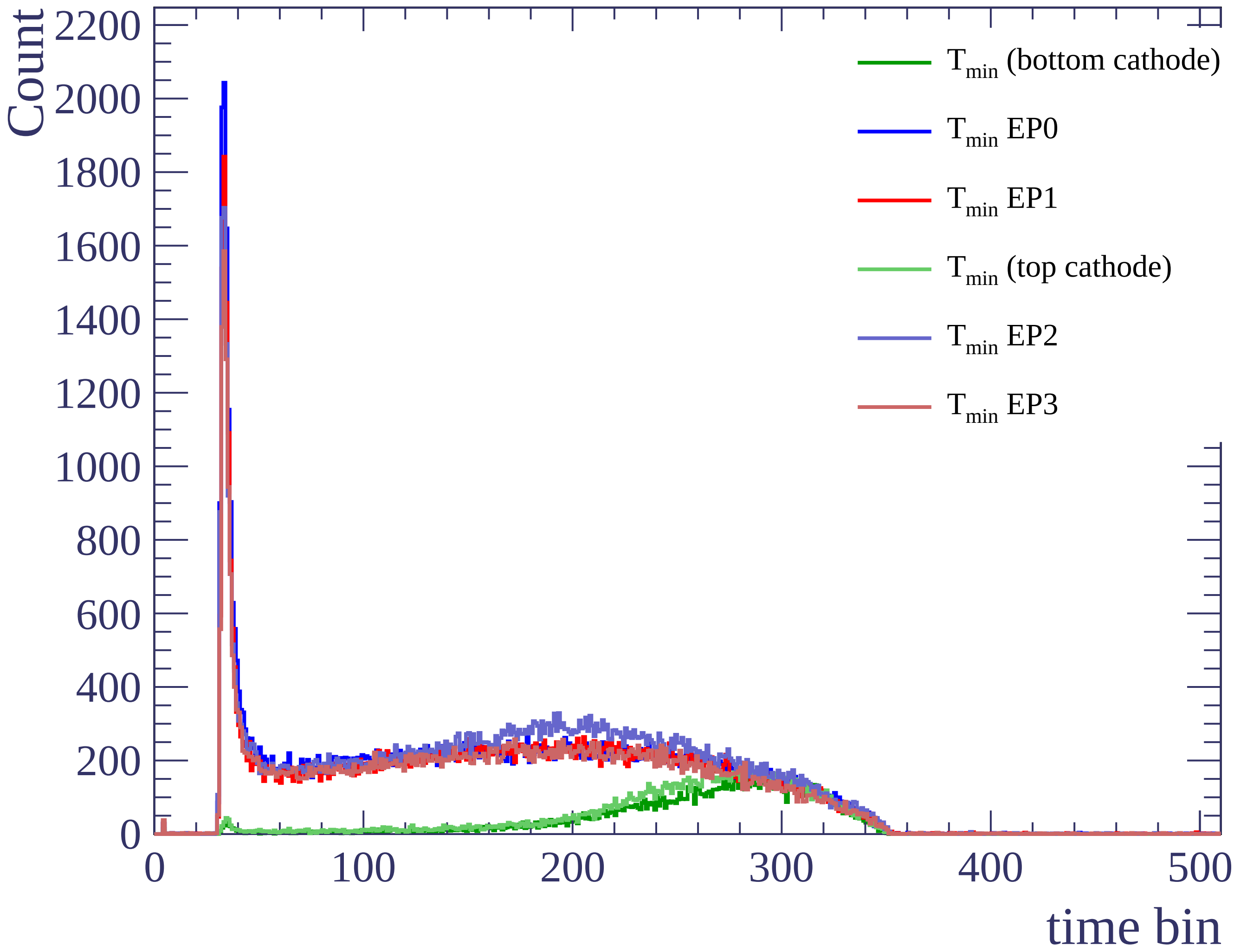
×10^3
Count
0
2
4
6
8
10
0
20
40
60
80
100
120
140
160
track length (cm)

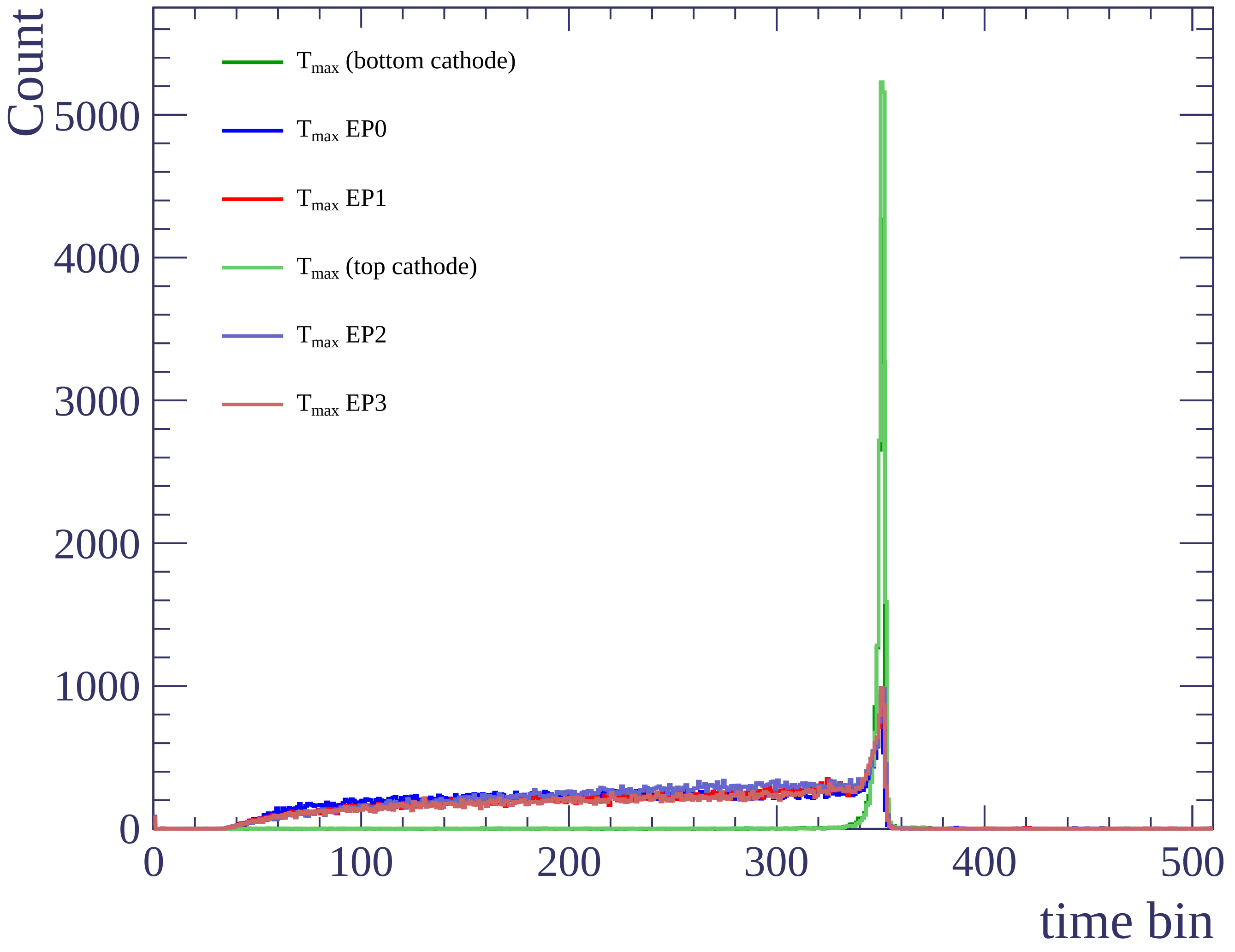


Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
×10³
χ²



χ^2
$\times10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
300
250
200
150
100
50
0


Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
100
200
300
400
500


Count
time bin
0
1000
2000
3000
4000
5000
0
100
200
300
400
500
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3